Universitat de València

Instituto de Ciencia Molecular (ICMol)

Doctoral Programme in Nanoscience and Nanotechnology

Volatile Organic Memristive Devices for Temporal Computing: Compositional Control of Fading Memory in Polymer–Electrolyte Composites

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Doctoral thesis presented by

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in partial fulfilment of the requirements for the degree of

Doctor of Philosophy (PhD)

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2026

Abstract

Data movement between separate memory and processing units accounts for an increasing share of the energy and latency of modern computing. One physical approach is to use device dynamics to store and transform temporal state. This thesis develops a *volatile organic memristive device*: a two-terminal element made from an ion-containing polymer–electrolyte composite whose conductance retains a decaying record of recent electrical history. The work asks whether this *fading memory* can be controlled and what it can be used for.

Synaptic measurements on assay-specific junctions of a proof-of-concept platform made from Super Yellow, Hybrane, and lithium triflate cover current–voltage hysteresis, potentiation and depression, an excitatory post-synaptic current, distinguishable short- and long-term retention, and spike-timing-dependent plasticity. A fast-triflate/slow-lithium account is retained as a candidate interpretation, but the electrical data do not identify the moving species. The fabrication archive records a campaign-scale loss of reproducibility in the Hybrane line. Changing to a poly(ethylene oxide) host restores a usable switching window and leads to a fabrication-record methodology and an automated feature pipeline. On the PEO platform, a replicated composition series tunes the fading-memory time over roughly 2–20 s. Host, anion and cation substitutions remain a sample-limited survey, and the data do not support the hypothesised $Li^+ > Na^+ > K^+$ ordering. Each composition is summarised as a component card

with its time constant, corner frequency, and a within-cell tolerance of $\sigma(\ln t_{1/2}) \approx 0.26$. Behavioural models that follow the measured relaxation are then tested in simulation in two roles. As a designed filter, the fastest composition removes the tonic level of electrodermal signals as well as a digital filter (synthetic response-detection F1 0.713 versus 0.701; WESAD stress-versus-baseline AUC 0.952 versus 0.951) and is equivalent to an RC element with the same half-life. As reservoir banks, mixed compositions raise memory and information-processing capacity in every random seed, but on recorded physiology a linear filter bank at the same time constants matches or exceeds them, except on sustained three-class affect labels (macro-F1 0.562 versus 0.508).

The replicated PEO/lithium-triflate series identifies composition as a control of fading memory, while the sample-limited chemistry survey does not support the proposed cation ordering. Read as component values, the compositions give a printable two-terminal element whose seconds-scale time constant is set at formulation, without a separate capacitor. Its practical value now depends on the bench tests that the simulations motivate: filtering a recorded signal with a fabricated device and comparing it with a printed RC element.

Resumen

El movimiento de datos entre unidades separadas de memoria y procesamiento representa una proporción creciente del consumo energético y la latencia de la computación moderna. Una estrategia física consiste en utilizar la dinámica del dispositivo para almacenar y transformar un estado temporal. Esta tesis desarrolla un *dispositivo memristivo orgánico volátil*: un elemento de dos terminales fabricado a partir de un compósito polímero–electrolito con iones móviles cuya conductancia conserva un registro decreciente de su historia eléctrica reciente. El trabajo examina si esta *memoria evanescente* puede controlarse y para qué puede utilizarse.

Las medidas sinápticas realizadas en distintas uniones de una plataforma de prueba de concepto fabricada con Super Yellow, Hybrane y triflato de litio comprenden histéresis corriente–voltaje, potenciación y depresión, corriente postsináptica excitatoria, retención a corto y largo plazo distinguibles, y plasticidad dependiente del tiempo de disparo. Una asignación de triflato rápido y litio lento se conserva como interpretación candidata, pero las medidas eléctricas no identifican las especies móviles. El archivo de fabricación documenta una pérdida de reproducibilidad a escala de campaña en la línea de Hybrane. El cambio a una matriz de poli(óxido de etileno) restaura una ventana de conmutación utilizable y da lugar a una metodología de registro de fabricación y a una canalización automatizada de extracción de características. En la plataforma de PEO, una serie de composiciones replicada ajusta el tiempo de memoria evanescente entre

aproximadamente 2 y 20 s. Las sustituciones de matriz, anión y catión siguen siendo un estudio limitado por el tamaño muestral, y los datos no respaldan el orden catiónico hipotético $Li^+ > Na^+ > K^+$. Cada composición se resume en una ficha de componente con su constante de tiempo, su frecuencia de corte y una tolerancia entre dispositivos de $\sigma(\ln t_{1/2}) \approx 0{,}26$. Los modelos de comportamiento que reproducen la relajación medida se evalúan después en simulación en dos funciones. Como filtro diseñado, la composición más rápida elimina el nivel tónico de la actividad electrodérmica con la misma precisión que un filtro digital (F1 de detección de respuestas sintéticas 0,713 frente a 0,701; AUC de estrés frente a reposo en WESAD 0,952 frente a 0,951) y equivale a un elemento *RC* con la misma semivida. Como bancos de reservorio, las composiciones mixtas aumentan la capacidad de memoria y de procesamiento de información en todas las semillas aleatorias, pero sobre fisiología registrada un banco de filtros lineales con las mismas constantes de tiempo las iguala o supera, salvo en las etiquetas afectivas sostenidas de tres clases (macro-F1 0,562 frente a 0,508).

La serie replicada PEO/triflato de litio identifica la composición como variable de control de la memoria evanescente, mientras que el estudio químico limitado por el tamaño muestral no respalda el orden catiónico propuesto. Leídas como valores de componente, las composiciones definen un elemento imprimible de dos terminales cuya constante de tiempo, del orden de segundos, se fija en la formulación, sin un condensador separado. Su valor práctico depende ahora de los ensayos de laboratorio que motivan las simulaciones: filtrar una señal registrada con un dispositivo fabricado y compararlo con un elemento *RC* impreso.

Resum

El moviment de dades entre unitats separades de memòria i de processament representa una proporció creixent del consum energètic i la latència de la computació moderna. Una estratègia física consisteix a utilitzar la dinàmica del dispositiu per a emmagatzemar i transformar un estat temporal. Aquesta tesi desenvolupa un *dispositiu memristiu orgànic volàtil*: un element de dos terminals fabricat a partir d'un compòsit polímer–electròlit amb ions mòbils la conductància del qual conserva un registre decreixent de la seua història elèctrica recent. El treball examina si aquesta *memòria evanescent* es pot controlar i per a què es pot utilitzar.

Les mesures sinàptiques realitzades en diferents unions d'una plataforma de prova de concepte fabricada amb Super Yellow, Hybrane i triflat de liti comprenen histèresi corrent–voltatge, potenciació i depressió, corrent postsinàptic excitatori, retenció a curt i llarg termini distingibles, i plasticitat dependent del temps de dispar. Una assignació de triflat ràpid i liti lent es conserva com a interpretació candidata, però les mesures elèctriques no identifiquen les espècies mòbils. L'arxiu de fabricació documenta una pèrdua de reproductibilitat a escala de campanya en la línia de Hybrane. El canvi a una matriu de poli(òxid d'etilè) restaura una finestra de commutació utilitzable i dóna lloc a una metodologia de registre de fabricació i a una canalització automatitzada d'extracció de característiques. En la plataforma de PEO, una sèrie de composicions replicada ajusta el temps de memòria evanescent entre aproximadament

2 i 20 s. Les substitucions de matriu, anió i catió continuen sent un estudi limitat per la grandària mostral, i les dades no confirmen l'ordre catiònic hipotètic $Li^+ > Na^+ > K^+$. Cada composició es resumeix en una fitxa de component amb la seua constant de temps, la seua freqüència de tall i una tolerància entre dispositius de $\sigma(\ln t_{1/2}) \approx 0{,}26$. Els models de comportament que reprodueixen la relaxació mesurada s'avaluen després en simulació en dues funcions. Com a filtre dissenyat, la composició més ràpida elimina el nivell tònic de l'activitat electrodèrmica amb la mateixa precisió que un filtre digital (F1 de detecció de respostes sintètiques 0,713 enfront de 0,701; AUC d'estrés enfront de repòs en WESAD 0,952 enfront de 0,951) i equival a un element *RC* amb la mateixa semivida. Com a bancs de reservori, les composicions mixtes augmenten la capacitat de memòria i de processament d'informació en totes les llavors aleatòries, però sobre fisiologia registrada un banc de filtres lineals amb les mateixes constants de temps les iguala o les supera, excepte en les etiquetes afectives sostingudes de tres classes (macro-F1 0,562 enfront de 0,508).

La sèrie replicada PEO/triflat de liti identifica la composició com a variable de control de la memòria evanescent, mentre que l'estudi químic limitat per la grandària mostral no confirma l'ordre catiònic proposat. Llegides com a valors de component, les composicions defineixen un element imprimible de dos terminals la constant de temps del qual, de l'ordre de segons, es fixa en la formulació, sense un condensador separat. El seu valor pràctic depèn ara dels assajos de laboratori que motiven les simulacions: filtrar un senyal registrat amb un dispositiu fabricat i comparar-lo amb un element *RC* imprès.

List of Publications

The following output arose directly from the work presented in this thesis.

Peer-reviewed article

1. C. D. Prado-Socorro, S. Giménez-Santamarina, L. Mardegan, L. Escalera-Moreno, H. J. Bolink, S. Cardona-Serra and E. Coronado, "Polymer-Based Composites for Engineering Organic Memristive Devices," *Advanced Electronic Materials* **8**, 2101192 (2022). DOI: 10.1002/aelm.202101192.
   This article is the basis of chapter 2.

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List of Abbreviations

Abbreviation	Meaning
ATR(-FTIR)	Attenuated total reflectance (Fourier-transform infrared spectroscopy)
AUC	Area under the receiver-operating-characteristic curve
CCC	Concordance correlation coefficient
CMOS	Complementary metal–oxide–semiconductor
DRAM	Dynamic random-access memory
EDA	Electrodermal activity
EIS	Electrochemical impedance spectroscopy
EMA	Exponential moving average (linear first-order filter)
ENODe	Electrochemical neuromorphic organic device
EPSC	Excitatory post-synaptic current
HSAB	Hard–soft acid–base (theory)
IPC	Information-processing capacity
ITO	Indium tin oxide
I–V	Current–voltage (characteristic)

Abbreviation	Meaning
KOTf	Potassium triflate (potassium trifluoromethanesulfonate)
LiOTf	Lithium triflate (lithium trifluoromethanesulfonate)
LOSO	Leave-one-subject-out (cross-validation)
LTM	Long-term memory
MC	(Linear) memory capacity
NaOTf	Sodium triflate (sodium trifluoromethanesulfonate)
NARMA	Nonlinear autoregressive moving average (benchmark task)
OTf	Triflate, $CF_3SO_3^-$ (trifluoromethanesulfonate)
PEO	Poly(ethylene oxide)
RC	Resistor–capacitor (first-order filter element, $\tau = RC$)
ReRAM	Resistive random-access memory
SCR	Skin-conductance response
SI	Supporting information
SRAM	Static random-access memory
STDP	Spike-timing-dependent plasticity
STM	Short-term memory
SY	Super Yellow (PDY-132 poly(*p*-phenylene-vinylene) semiconductor)
T_g	Glass-transition temperature
TFSI	Bis(trifluoromethanesulfonyl)imide

Abbreviation	**Meaning**
TMPE	Trimethylolpropane ethoxylate (hyperbranched polyether host)
UV–Vis	Ultraviolet–visible (spectroscopy)
VFT	Vogel–Fulcher–Tammann (relaxation)
WESAD	Wearable Stress and Affect Detection (physiological corpus)
XRD	X-ray diffraction
1T1M	One-transistor–one-memristor (array cell)
τ_S, τ_L	Short- and long-term fading-memory time constants
β	Stretched-exponential (Kohlrausch) exponent

Motivation, Objectives and Structure of the Thesis

In many modern workloads, moving data between separate processing and memory units consumes more energy than the arithmetic operation itself. A single off-chip memory access can cost three to four orders of magnitude more energy than an arithmetic operation [1], and large machine-learning models and continuous sensor streams require frequent memory access [2]. As gains from transistor scaling slow, data movement accounts for an increasing share of system energy and latency [3]. This thesis examines one physical response: devices whose internal dynamics store and transform temporal state within the same substrate [4, 5].

The devices studied here are *volatile organic memristive devices*: two-terminal elements made from ion-containing polymer composites whose conductance retains a decaying record of recent electrical activity. Their relaxation is treated as a device function. A controlled decay makes each device a physical *leaky integrator*, and the decay time acts as a component value: it sets the corner frequency of the element when it filters a slow signal, and it sets the memory of the element when it serves as a node of a *temporal* (reservoir-style) computer. Long time constants are costly to build in analogue CMOS [6], which is where a material with a formulation-set relaxation time of seconds could be useful. The thesis tests two linked propositions: composition controls the time constant in

the replicated PEO/lithium-triflate series, and the measured relaxation is usable for temporal processing, either as a designed filter element or as a node of a reservoir bank.

The work begins with a fully characterised Super Yellow/Hybrane/lithium-triflate exemplar and then follows the loss of reproducibility that prompted a change to PEO. On the PEO platform, a replicated composition series tests how formulation controls the dynamics; the smaller host, anion, and cation series remain chemical hypotheses rather than material laws. Behavioural models fitted to these measurements are then evaluated as filter elements and as reservoir banks on synthetic and recorded physiological signals.

Open Questions and Objectives

Realising and understanding an exemplar. Can a fully reversible two-terminal organic memristive platform be made from a solution-processed polymer-electrolyte composite using standard organic-electronic fabrication? Can that common formulation and architecture support analogue switching, distinguishable short- and long-term memory, an excitatory post-synaptic current response, and spike-timing-dependent plasticity? Can the fast and slow memory components be explained by the composite's cation–anion mobility asymmetry?

Composition as the quantitative control variable. Within a common host and a common alkali-metal triflate family, how do the device dynamics depend on composition when the polymer-to-semiconductor and salt-to-semiconductor mass ratios are varied around a reference formulation? This composition dependence is the axis the thesis sets out to *replicate*. Host, anion, and cation substitutions pose hypotheses about

the same dynamics, in particular whether moving the cation along the Li^+, Na^+, K^+ series shifts the measured fading-memory timescale as coordination chemistry suggests.

Utility for temporal processing. What can the composition-set time constant be used for? Does a device chosen from the measured grid perform a standard slow-signal filtering step as accurately as a conventional filter, and how does it compare with an *RC* element of the same half-life? Does a bank of mixed compositions, with its measured device-to-device spread, compute anything that linear filtering does not?

These questions motivate six objectives:

1. Design, fabricate, and characterise a proof-of-concept two-terminal organic memristive device based on a Super Yellow, Hybrane, and lithium-triflate composite, processed from a single solution onto a standard ITO/composite/Ag stack, and demonstrate the basic memristive and synaptic functionality in a single platform.

2. Provide a mechanistic account of the device's two memory regimes in terms of the displacement of mobile triflate anions and lithium cations within the oxygen-rich ion-conducting host, supported by hard–soft acid–base reasoning on the cation–oxygen interaction.

3. Build a comparative experimental corpus on a poly(ethylene oxide)-based platform loaded with alkali-metal triflate salts, fabricated under a common protocol, and characterise it through three common dynamical measurements: I–V hysteresis, variable-number-of-pulses potentiation, and variable-delay depotentiation.

4. Quantify the composition dependence through a replicated PEO and lithium-triflate concentration series, and survey host, anion, and cation substitutions on the same three measurements.

5. Construct experimentally grounded behavioural models from those measurements, using composition-indexed relaxation kernels, nonlinear pulse-update rules, and device-to-device variability, and summarise each composition as a component card with its time constant, corner frequency, and tolerance.

6. Evaluate these models in simulation for two uses. As designed filters, compare single devices and device pairs with RC and digital filters on synthetic electrodermal signals with known ground truth and on WESAD recordings. As reservoir banks, test memory-capacity, NARMA, and information-processing benchmarks, WESAD physiological-context reconstruction and affect classification, and a PhysioNet Non-EEG robustness replication, each against linear-filter controls.

Structure of the Thesis

Chapter 1 develops the conceptual and materials background, from the limits of conventional computing to the polymer-electrolyte composite used in the experimental work. Chapter 2 presents the Super Yellow/Hybrane/lithium-triflate proof of concept and its complete synaptic measurement suite; it is based on Prado-Socorro et al. [7]. Chapter 3 reconstructs the later loss of reproducibility, the change to a PEO host, and the fabrication-record methodology adopted for the remaining devices.

Chapter 4 tests composition on a replicated PEO/lithium-triflate grid and surveys the smaller host, anion, and cation series with the same three dynamical measurements. Chapter 5 turns those measurements into component cards and behavioural models, then evaluates them as designed filters and as reservoir banks on synthetic and physiological

signals. Chapter 6 compares the work with conventional filters and with temporal and event-driven neuromorphic hardware, sets the limits of the evidence, and identifies the experiments still required.

Chapter 1

Introduction

Data movement between processing and memory imposes growing energy and latency costs on modern computing. Large machine-learning models and continuous sensor streams place particularly heavy demands on that movement [2]. Section 1.1 describes the architectural origin of this limit. The approach taken here uses the dynamics of a physical device to perform part of the computation. The biological synapse motivates this approach because a single element transmits a signal, modifies its strength according to recent activity, and retains that activity over several timescales [8].

This thesis asks whether ion motion in a soft polymer host can give a two-terminal organic memristive device the fading memory needed for temporal computation. The question connects solid-state electronics, polymer physical chemistry, and computational neuroscience.

1.1 Architectural Limits of Modern Computing

General-purpose digital computers use a stored-program architecture in which processing and memory are physically separate. On contemporary hardware, moving operands between those units can cost substantially more energy than the arithmetic operation [1, 9]. Smaller transistors and faster clocks do not remove this transfer cost. The resulting limits are architectural as well as technological: the *memory wall*, the *energy wall*, and the slowing of classical transistor scaling all arise from the separation between stored data and the circuitry that acts on it [3, 10].

1.1.1 Von Neumann Architecture and Its Limitations

The architectural model of modern digital computers is the *stored-program* concept set out by von Neumann in 1945 [11]: instructions and data share one memory and are fetched word by word by a central processing unit across a connecting bus. This is what makes a single machine general-purpose, and engineering refinements (registers, caches, branch prediction, memory controllers) have left the topology essentially unchanged. The same topology is the source of the limitation. Because the processor and the memory are *physically distinct* subsystems, every instruction fetch, operand load, and result write-back must cross the channel between them; no arithmetic can begin before its operands arrive, and no result is preserved until it is moved back. This is the *processor–memory dichotomy*.

Because only a bounded amount of data can cross that channel per cycle (the *von Neumann bottleneck* [12]), performance is set by the slowest interface, not by the arithmetic. Pipelining, multi-level caches, speculation, prefetching, and out-of-order issue all mitigate this, but they preserve the

separation between computation and storage, so they shift the bottleneck without removing it. Wulf and McKee [10] named the resulting imbalance the *memory wall*: with processor speed long outpacing DRAM latency and bandwidth, an ever-larger share of each cycle is spent waiting on memory. Modern microprocessors accordingly devote much of their silicon area, and an even larger fraction of their power, to caches, prefetch logic, memory controllers, and interconnect, whose only purpose is to keep the arithmetic units supplied with data [9]. The bandwidth and energy aspects of this imbalance are summarised in fig. 1.1.

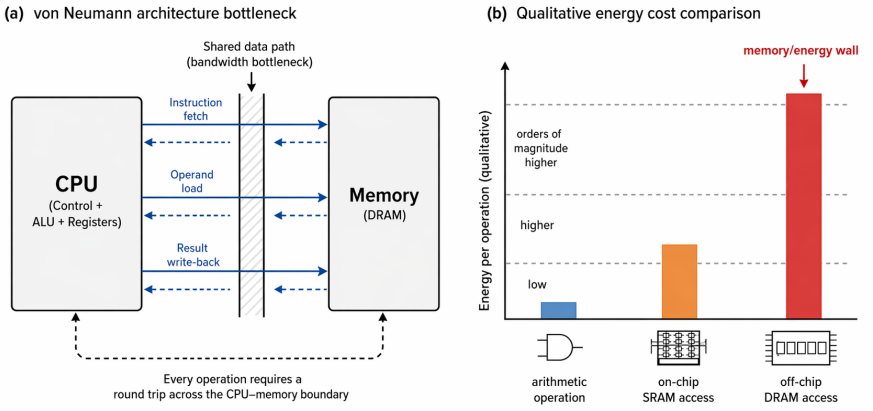


Figure 1.1: Schematic of the von Neumann architecture and the associated bandwidth and energy costs of data movement. (a) In the stored-program paradigm, the central processing unit and the memory unit are physically separated and communicate through a shared bus, so that every instruction fetch and every operand access must traverse that bus. (b) The energy cost of moving a word across this boundary exceeds, by orders of magnitude, the energy cost of the arithmetic operation performed on it, and this asymmetry has increased as technology has scaled [1].

Horowitz [1] compared the costs of elementary operations in a representative CMOS process: a 32-bit integer addition costs on the order of 0.1 pJ, an SRAM access costs tens of picojoules, and a DRAM access costs several hundred picojoules to nanojoules (table 1.1). The system-level contribution depends on how often each operation occurs. A neural-

network inference performs many multiply–accumulate operations, each requiring access to a stored weight, and training repeats these accesses across large datasets. Operand transport can therefore account for a substantial share of the energy budget. This *energy wall* arises from interconnect capacitance and from the total wire length of the memory subsystem. Spatial separation of computation and storage preserves these physical transfer costs.

Table 1.1: Approximate energy cost per operation in a representative 45 nm CMOS process, after Horowitz [1]. Memory accesses cost between one and four orders of magnitude more energy than the arithmetic they enable, and the asymmetry widens with the size of the memory addressed and with the distance the data has to travel.

Operation	**Energy**	**Class**
8-bit integer add	0.03 pJ	Arithmetic
32-bit integer add	0.1 pJ	Arithmetic
32-bit floating-point add	0.9 pJ	Arithmetic
32-bit floating-point multiply	3.7 pJ	Arithmetic
Register-file read	0.1 pJ	On-chip storage
SRAM read (8 kB)	5 pJ	On-chip storage
SRAM read (1 MB)	100 pJ	On-chip storage
Cross-chip wire (∼10 mm)	100 pJ	Interconnect
DRAM read	1.3 nJ	Off-chip storage

For several decades, Moore's increase in transistor counts [13] was accompanied by Dennard scaling [14], so successive generations provided more transistors with improved speed and energy efficiency. Dennard scaling ended around 2005, when leakage currents prevented proportional voltage scaling [3]. Transistor counts continued to rise, but performance and energy gains slowed, and power density became a limiting constraint (the *post-Dennard* or *dark-silicon* regime [15]). Process scaling alone now provides smaller reductions in the cost of the von Neumann model. At the same time, machine-learning workloads rely heavily on repeated matrix operations over memory-resident parameters [2]. Hennessy and Patterson [3] identify this period as an inflection point and argue for

domain-specific hardware that places memory and computation closer together.

1.1.2 In-Memory and Event-Driven Computing Paradigms

Two responses to this separation have emerged. The first keeps the digital workload largely intact but performs operations directly within, or very close to, the memory array; this route is described as *in-memory*, *near-memory*, or *processing-in-memory* computing. The second uses temporally structured event streams and treats the intrinsic dynamics of the physical device as part of the computation itself; this route is known as *event-driven* or *temporal* computing. The two responses, set against the von Neumann baseline they address, are summarised in fig. 1.2.

In-memory computing bypasses the classical *fetch–compute–write-back* cycle by executing operations directly inside the memory array, as in analogue matrix–vector multiplication on a resistive crossbar:

1. Each memory cell contains a conductance value that reflects a matrix entry.

2. Voltages are applied across the rows, representing the input vector.

3. The resulting column currents represent the sum of products, in accordance with Kirchhoff's laws.

This changes the array from a passive storage device into an active computing tool. Demonstrations using metal-oxide resistive memories [16], phase-change materials [20, 21], and organic ionic devices [22–24] have achieved matrix–vector operations with energy efficiencies that surpass those of traditional digital accelerators, and similar principles can be

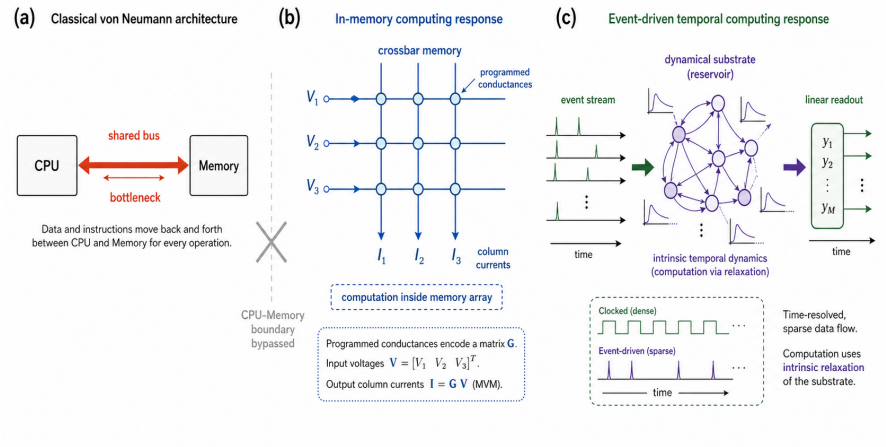


Figure 1.2: Two architectural responses to the von Neumann bottleneck. (a) In the classical stored-program architecture, every operation traverses a CPU–memory bus that fixes the bandwidth and energy ceilings of the system. (b) In the in-memory paradigm, computation is performed inside the memory array: a resistive crossbar with programmed conductances executes an analogue matrix–vector multiplication by Ohm's and Kirchhoff's laws, so the dot product no longer crosses the CPU–memory boundary [16, 17]. (c) In the event-driven paradigm, dense clocked data are replaced by a sparse stream of asynchronous events that drive a dynamical substrate whose intrinsic relaxation is itself the computation; only a linear readout layer downstream is trained [5, 18, 19].

extended to convolutional [25] and attention-like processes (the matrix-multiplication kernels at the core of transformer architectures [26]).

At the device level, however, this version of in-memory computing still requires memory properties. Each cell should hold a quasi-static conductance that encodes a trained weight, and which should be set once, read many times, and remain stable enough to meet the application's numerical precision requirements. Non-volatility over seconds to years becomes a design goal.

Event-driven computing is motivated because many signals of interest carry information in time. Natural sounds, visual transients, motor commands, biomedical waveforms, and streaming sensor outputs are all

non-static examples. Their information is distributed across amplitude, waveform evolution, and event timing, so the computational system must evolve with the input. At the algorithmic level, this is the domain of spiking neural networks [27]. At the circuit level, it corresponds to event-driven asynchronous logic. At the device level, which is the level of this thesis, it requires materials whose intrinsic relaxation dynamics are matched to the signals they process.

Biological nervous systems handle sensorimotor and cognitive tasks at substantially lower power than current artificial systems. Relevant architectural features include sparse event-based signalling, synaptic responses that depend on recent activity, and electrochemical processes that store and modify state. Neuromorphic engineering applies selected principles from these systems, including sparse-event coding [28], the use of slow device dynamics as a computing element, and functional diversity supplied by device heterogeneity [4].

Boyd and Chua [29] showed that any time-invariant nonlinear operator with *fading memory*, in an appropriate function space, can be approximated by a finite Volterra series (a power-series expansion in time-shifted inputs). This means that a system whose current state retains a decaying record of past inputs can perform various temporal computations when paired with a trainable linear readout. This concept was later formalised through algorithms such as Jaeger's *echo state network* [18, 30] and Maass, Natschläger, and Markram's *liquid state machine* [19], which were combined under the umbrella of *reservoir computing* [31, 32]. In this framework, an untrained dynamical system (the reservoir) is driven by the input stream, with only the readout being trained. When the reservoir is realised in a physical substrate, the approach is known as *physical reservoir computing* [5], with implementations in optical, mechanical, spintronic, volatile-memristive, and organic electrochemical systems [33–36].

The relevant device metrics depend on the application. Trained-weight storage requires long retention. A reservoir instead requires a finite interval over which previous inputs influence the current output: the fading-memory timescale. Controlled variation between device timescales can then give a common readout access to temporal structure over a wider range than any individual device could cover [5].

1.2 The Biological Synapse as the Engineering Template

In neural circuits, a single physical element both transmits a signal and stores part of the history that shapes future transmissions: the synapse. The same molecular event that relays information from one neuron to the next also alters its future strength, so signalling and memory share a substrate. The device target is therefore synaptic signalling and plasticity, not full neuron emulation.

1.2.1 Neural Signal Transmission: A Brief Overview

A neuron is a polarised cell specialised to receive, integrate, and transmit electrical signals [37]. Dendrites collect inputs from other cells, the soma integrates them, and when the membrane potential at the axon hillock crosses a threshold the axon propagates an outgoing signal whose terminals release a chemical messenger onto downstream cells. A mature animal neuron typically receives 10^3–10^4 such contacts [38], and each is a *synapse*. How a neuron computes is therefore governed far more by the number and strength of its synapses, and by how they change over time, than by the soma itself.

The axonal signal is the *action potential*: a stereotyped, millisecond-scale

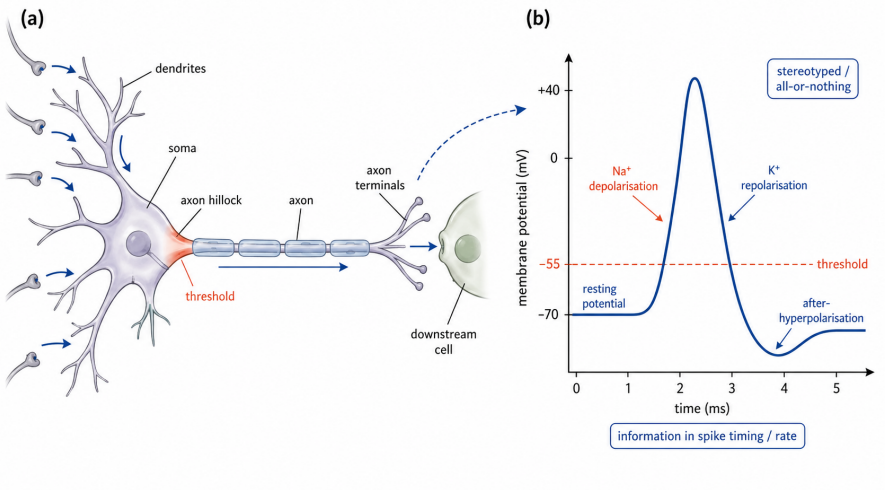


Figure 1.3: Anatomy of a neuron and the action-potential waveform. (a) A neuron collects synaptic inputs through its dendrites, integrates them at the soma, and fires an outgoing electrical event when the membrane potential at the axon hillock crosses a threshold; the resulting waveform propagates unidirectionally along the axon to the synaptic terminals, where it triggers chemical transmission onto downstream cells (the zoom shown in fig. 1.4). (b) The action potential is a stereotyped millisecond-scale depolarisation produced by the opening of voltage-gated Na^+ channels, followed by a K^+-mediated repolarisation and a brief after-hyperpolarisation before the return to the resting potential; its all-or-nothing character means that information about the input is carried by the timing and rate of these events, not their amplitude, which is the cellular basis of the event-driven view of neural computation introduced in section 1.1.2.

depolarisation that is all-or-nothing, so that information is carried not in its amplitude but in the timing, rate, and coincidence of spike trains [39, 40]. Neural computation is in this sense event-driven (section 1.1.2) and arises from the pattern of signals arriving at many synapses. The neuron's anatomy and the action potential waveform are summarised in fig. 1.3.

Synaptic transmission is largely chemical. An action potential arriving at the terminal opens voltage-gated Ca^{2+} channels, and the Ca^{2+} influx triggers neurotransmitter-filled vesicles (glutamate at excitatory cortical synapses) to fuse with the presynaptic membrane [41]. The transmitter

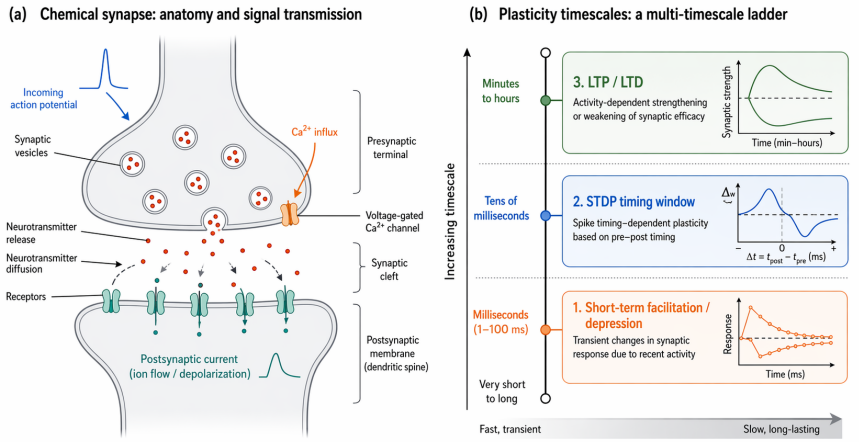


Figure 1.4: Anatomy and plasticity regimes of the chemical synapse. (a) A presynaptic terminal releases neurotransmitter vesicles into the synaptic cleft in response to a calcium-triggered action potential, and postsynaptic receptors convert the released signal into an excitatory postsynaptic current. (b) The same synapse supports a graded hierarchy of plasticity regimes (short-term facilitation and depression on millisecond timescales, spike-timing-dependent plasticity on tens of milliseconds, and long-term potentiation and depression on timescales from minutes to hours), each associated with distinct microscopic mechanisms.

diffuses across the ∼20 nm cleft and binds postsynaptic receptors [38], producing an excitatory or inhibitory postsynaptic current (EPSC or IPSC) whose amplitude and kinetics define the *strength* of the synapse at that moment. The same molecular machinery carries the signal and sets the strength of future transmissions.

The synapse combines transmission, modulation, and memory: it passes a signal, adjusts its amplitude and time course, and stores changes produced by previous activity. This co-location provides a biological reference for electronic devices that combine state and computation. The relevant anatomy and plasticity regimes are summarised in fig. 1.4.

1.2.2 Synaptic Plasticity: The Physical Basis of Learning

Experience leaves a physical trace in the nervous system, stored mainly at synapses. Hebb's 1949 postulate, that a presynaptic cell which repeatedly helps fire a postsynaptic cell strengthens its connection to it [42] ("cells that fire together, wire together"), captures two features that matter for hardware: the rule is *local* to the two cells a synapse joins, and *sensitive to correlation*. Synaptic plasticity implements Hebbian or Hebbian-like behaviour across several timescales.

On the shortest timescales, *short-term plasticity* [43, 44] makes the immediate strength of a synapse depend on its recent activity. *Facilitation* is a transient increase in EPSC amplitude when a second spike follows the first after a short delay, usually attributed to residual presynaptic Ca^{2+}; *depression* is a transient decrease caused by vesicle depletion or receptor desensitisation. With time constants from tens of milliseconds to several seconds, short-term plasticity makes the synapse a nonlinear, history-dependent temporal filter, so that identical spike sequences produce different responses depending on their timing.

On longer timescales, *long-term potentiation* (LTP) provides persistent, activity-dependent strengthening. First demonstrated by Bliss and Lømo in the rabbit hippocampus [45] and later shown to last days to weeks [46, 47], hippocampal LTP relies on NMDA receptors that open only when the postsynaptic membrane is already depolarised (a voltage-dependent Mg^{2+} block [48, 49]), making them Ca^{2+}-permeable coincidence detectors; the resulting Ca^{2+} influx drives additional AMPA receptors into the postsynaptic membrane [50, 51]. The complementary *long-term depression* (LTD) [52, 53] weakens synapses under low-frequency or anti-correlated activity, principally through reverse AMPA-receptor trafficking [54]. Together, LTP and LTD give the bidirectional, long-lasting amplitude

changes considered a primary physiological basis of learning.

Spanning these regimes, *spike-timing-dependent plasticity* (STDP), described in cortical pyramidal cells [55] and quantified in hippocampal cultures by Bi and Poo [56], makes the sign and magnitude of the change depend on the delay between pre- and postsynaptic spikes within a window of a few tens of milliseconds: a presynaptic spike before the postsynaptic one potentiates, the reverse depresses, and large delays in either direction produce no lasting change. The synapse thus holds a short memory of recent presynaptic activity, compares it with postsynaptic firing, and converts that timing relationship into a longer-lasting amplitude change.

Across these mechanisms, the synapse operates over several timescales: milliseconds for individual spikes, tens to thousands of milliseconds for short-term facilitation and depression, and minutes to hours and beyond for LTP and LTD. These partially overlapping molecular mechanisms led Abbott and Regehr [8] to describe the synapse as a computational unit. Both long-lasting and decaying state changes are relevant to hardware implementations.

1.2.3 Why Hardware Emulation of the Synapse?

The motivation for hardware synapses follows from section 1.1. In traditional computing, a software neural network retrieves data from a memory hierarchy physically separate from its arithmetic units. Each forward pass incurs data-movement costs that increase with the number of simulated synapses. A typical human brain contains approximately 10^{14}–10^{15} synapses [57] and consumes about 20 W [58]. This comparison motivates hardware that co-locates analogue state and computation; biological synapses provide one reference architecture. The order-of-magnitude envelopes of these implementation regimes are compared in fig. 1.5.

This research programme is known as *neuromorphic engineering*, whose

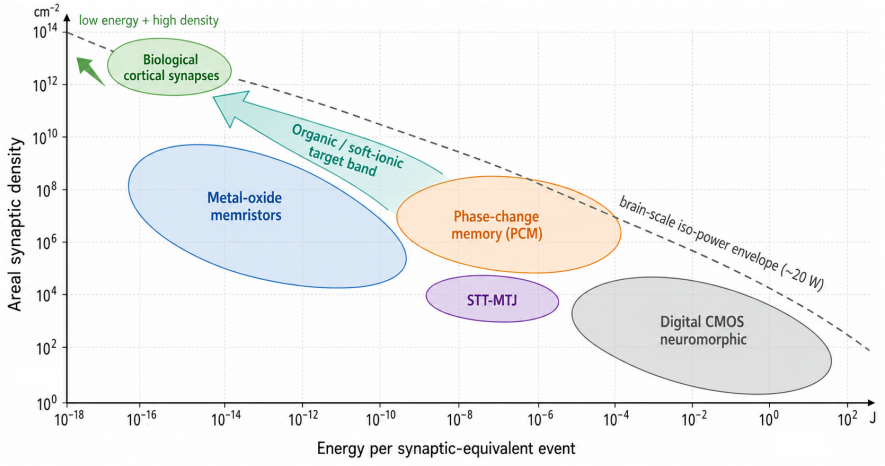


Figure 1.5: Approximate energy and density envelopes of synaptic-equivalent operations across implementation regimes. Each cluster represents a class of physical realisation: biological cortical synapses, digital CMOS neuromorphic implementations in which a synapse occupies an SRAM cell plus an arithmetic unit, the inorganic memristor families surveyed in section 1.4, and the soft-ionic composite target region forward-referenced to section 1.6. The diagonal reference line marks the brain-scale iso-power envelope ($\sim 20\,\mathrm{W}$ for $\sim 10^{14}$–10^{15} synapses) introduced at the start of section 1.2.3. Numerical bounds are conservative orders of magnitude drawn from the reviews cited in the surrounding text; individual devices may sit outside the envelopes shown.

programmatic origin is usually attributed to Mead [4]. It develops circuits in which device physics, topology, and dynamics implement selected computational characteristics of nervous systems. The synapse is a principal device-level reference.

A hardware synapse should satisfy several requirements:

- It should provide an analogue, multi-state conductance that can be set reversibly by electrical stimuli and read non-destructively with a small probe current.
- It should also be compact enough for large arrays, so that many thousands of elements can be addressed with a manageable number

of input and output lines [28, 59].

- Its response should be nonlinear in a meaningful way, for example by displaying a pulse-number or pulse-amplitude threshold below which little change is induced [17, 59].

- Finally, its dynamics should match the signals it is meant to process, whose relevant timescales span from milliseconds to seconds and beyond.

Analogue, multi-state conductance allows one device to encode several distinguishable values; a binary element would need several devices per connection to represent the same graded weight. In ionic devices, the write energy can also be low because the state is set by moving ions over short distances instead of charging large digital circuits [23, 24]. Reversibility is equally necessary. A device that can be potentiated but not reliably depressed cannot follow the bidirectional rules of section 1.2.2 and so cannot fully replicate synaptic learning [17]. These requirements define the device target used in the rest of the thesis: analogue, reversible, multi-state conductance changes set by low-energy electrical pulses.

Biological synapses combine quasi-stable memory through LTP and LTD with temporal filtering through short-term plasticity and STDP. Hardware can implement either a persistent or quasi-persistent programmable weight, as in crossbar arrays for in-memory computing [17], or a dynamic fading-memory element. This thesis focuses on the latter role through the reservoir-computing approach to event-driven temporal computing introduced in section 1.1.2 [5, 33–35].

This leads to a materials question. In biology, the timescales of short-term plasticity are determined by processes such as Ca^{2+} buffering, vesicle recycling, and receptor desensitisation, which vary across synapses [44]. A hardware implementation requires a corresponding means of adjusting

its electrical timescale. Memristive devices provide one two-terminal implementation of analogue, history-dependent behaviour.

1.3 The Memristor: Theory and History

A memristor or memristive device shows a current–voltage (I–V) response that depends on an internal state, which is influenced by the device's previous electrical activity. The ideal form is the *memristor*; the broader experimental category is the family of *memristive devices*. A relaxing internal state remains consistent with the memristive-system definition. Its decay supplies the fading-memory behaviour used later in the thesis.

1.3.1 Memristive Systems as History-Dependent Devices

A memristor is, intuitively, a resistor with memory: a two-terminal element whose resistance depends on the charge that has previously passed through it. Chua [60] introduced it in 1971 on symmetry grounds, completing the set of fundamental passive elements (resistor, capacitor, inductor) with a fourth that relates charge $q = \int i \, \mathrm{d}t$ to flux linkage $\varphi = \int v \, \mathrm{d}t$. It is described by $v = M(q)\, i$, where the *memristance* $M(q)$ has the units of resistance but, unlike an ordinary resistor, carries a record of the device's past.

If M is constant the element is an ordinary resistor; if M varies with q, the present response depends on the history of the device. Under periodic excitation this produces a *pinched hysteresis loop* in the i–v plane [61]: the loop passes through the origin (zero voltage gives zero current instantaneously) but its shape reflects the internal state left by prior stimulation. Pinched hysteresis is the standard experimental

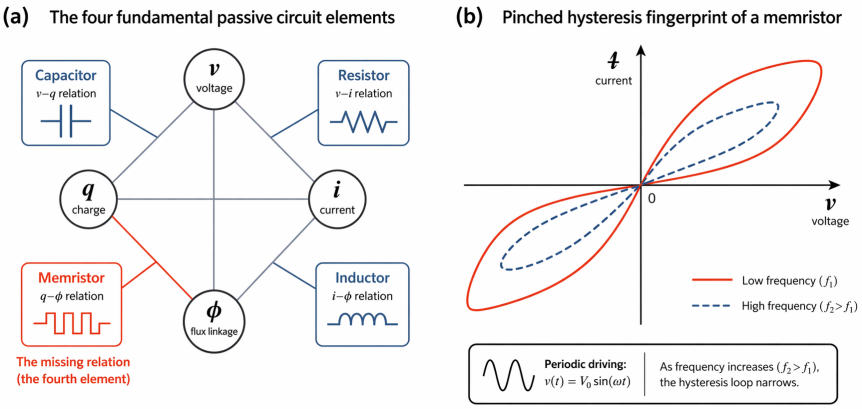


Figure 1.6: The memristor as a constitutive element and its experimental fingerprint. (a) The four fundamental two-terminal passive circuit elements, arranged by the pair of constitutive variables they relate: the resistor (v–i), the capacitor (v–q), the inductor (i–φ), and the memristor (q–φ), which completes the quadrant [60]. (b) A memristive device driven by a periodic voltage exhibits a pinched current–voltage hysteresis loop that narrows with increasing driving frequency and that passes through the origin at each zero-crossing; this loop is the canonical experimental fingerprint of memristive operation and is observed in all of the device families surveyed in the remainder of the chapter [61–63].

signature of memristive behaviour [62], and it is collected together with Chua's constitutive argument in fig. 1.6.

Few real devices satisfy the ideal relation $\varphi = \hat{\varphi}(q)$. The broader notion, introduced by Chua and Kang [63] in 1976, is the *memristive system*: an input-driven internal state $\mathbf{x}$ evolves under generally nonlinear dynamics and gates the instantaneous electrical response. This model allows $\mathbf{x}$ to include relaxation terms that drive the device back toward equilibrium once the input is removed. A volatile device whose state decays is therefore still a memristive system, even though it is not an ideal memristor. This is the framework adopted for the polymer-composite devices reported here.

1.3.2 The First Physical Realisation: TiO_2 Memristor

For more than three decades after Chua's proposal, the memristor remained mostly a theoretical construct. Thin-film metal-oxide resistive switches with hysteretic current–voltage characteristics had already been reported since the late 1960s [64–66], and many of them can now be recognised as memristive systems. At that time, however, they were not interpreted within Chua's formalism. The field still lacked a standard physical device that connected the theoretical idea to a tangible thin-film structure.

In 2008, that connection was made possible by Strukov, Snider, Stewart and Williams [67] at HP Laboratories. Their device consisted of a thin titanium dioxide (TiO_2) layer sandwiched between two platinum electrodes. Within the film, a reduced TiO_{2-x} region acted as an oxygen-deficient conductive layer, while the neighbouring TiO_2 region served as a resistive barrier. When a voltage was applied, oxygen vacancies moved, shifting the boundary between these two regions [68]. Because the total resistance depends on the relative thickness of the conductive and resistive parts, it could be modulated continuously by moving this boundary back and forth. The boundary position is the internal state, oxygen-vacancy drift updates it, and the resulting i–v response reproduces the pinched hysteresis predicted by the theory.

This result transformed the memristor from a formal circuit element into a concrete experimental object and launched a large body of work on resistive switching and memristive devices. A two-terminal resistance that can be programmed by electrical pulses, tuned over several conductance values, and read at low voltage is close to the analogue weight element required by neuromorphic hardware [69]. Crossbar demonstrations of in-memory matrix–vector multiplication followed soon after [16, 17]. The

2008 TiO_2 device is, however, only one mechanism (oxygen-vacancy drift in a transition-metal oxide), while the memristive formalism is far broader.

1.3.3 Classification of Memristive Mechanisms

In device physics, the term *memristor* describes phenomenology independently of a particular microscopic mechanism. A device is "memristive" if its current–voltage response exhibits the pinched hysteresis that Chua described and if its internal state can be both set and read electrically. This can be achieved through many different physical processes, commonly grouped into broad mechanistic families [17, 69].

These taxonomies combine two different classification axes. The first is the *spatial pattern* of the state change, whether filamentary or interface/distributed, depending on whether the conductance change is localised to a narrow conductive path or extended across the active region. The second is the *physical nature* of the state variable itself: ionic distribution, structural phase, magnetic configuration, or redox population. The two axes are orthogonal: an ionic device can also be non-filamentary, so the families considered here locate a device along both axes.

Filamentary switching. This is historically the most visible family. When a voltage is applied, a narrow conductive path forms through a previously insulating thin film, which can later be ruptured or reformed. In *conductive-bridge* random-access memories (CBRAM), the filament is built from metal atoms displaced electrochemically from an active electrode (like silver or copper) through a solid electrolyte [70, 71]. In *valence-change* memories, which are generally based on binary transition-metal oxides, the filament is instead associated with oxygen vacancies or partially reduced metal-cation states inside the oxide lattice [68].

These devices offer high ON/OFF ratios, fast switching, and non-volatile retention; their main limitation is stochasticity, since filament formation,

positioning, and rupture vary across devices and cycles [72]. They therefore work well as binary or coarse multi-level memories but are poorly suited to fine analogue representation. Panel (a) of fig. 1.7 contrasts the CBRAM and VCM variants of these devices, while panels (b)–(e) display the other four families.

Interface (non-filamentary) switching. In these devices, conductance changes over a distributed region. Some implementations modulate the height or width of a Schottky barrier by accumulating or depleting mobile ions near a metal–semiconductor interface [73]. Other implementations, especially in mixed ionic–electronic systems, drive a distributed electrochemical doping of the semiconductor [22, 23]. Because the state change is spatially extended, these devices can support a quasi-continuous range of conductance values. The ON/OFF ratio is often smaller than that of filamentary devices, but variability is usually lower and the current tends to scale more predictably with device area. This is the closest *spatial-pattern* class for several organic electrochemical devices, including the polymer-composite devices studied here, whose state variable is itself described under the ionic family below.

Phase-change memories. The active material is a chalcogenide alloy, usually $Ge_2Sb_2Te_5$ or a related composition, whose resistance depends on whether the material is in a crystalline or amorphous configuration. Joule-heating pulses are then used either to melt and quench the material into the amorphous state, or to anneal it back toward the crystalline state [20, 21, 74]. Phase-change memories provide long retention, large ON/OFF ratios, and high endurance. Their write energies are relatively large, and devices tend toward binary or few-state operation unless specifically engineered for intermediate states. They are mature storage-class memories used in analogue in-memory computing demonstrations.

Spintronic (magnetic) devices. Their resistance depends on the magnetic configuration of a tunnel junction, typically determined by the

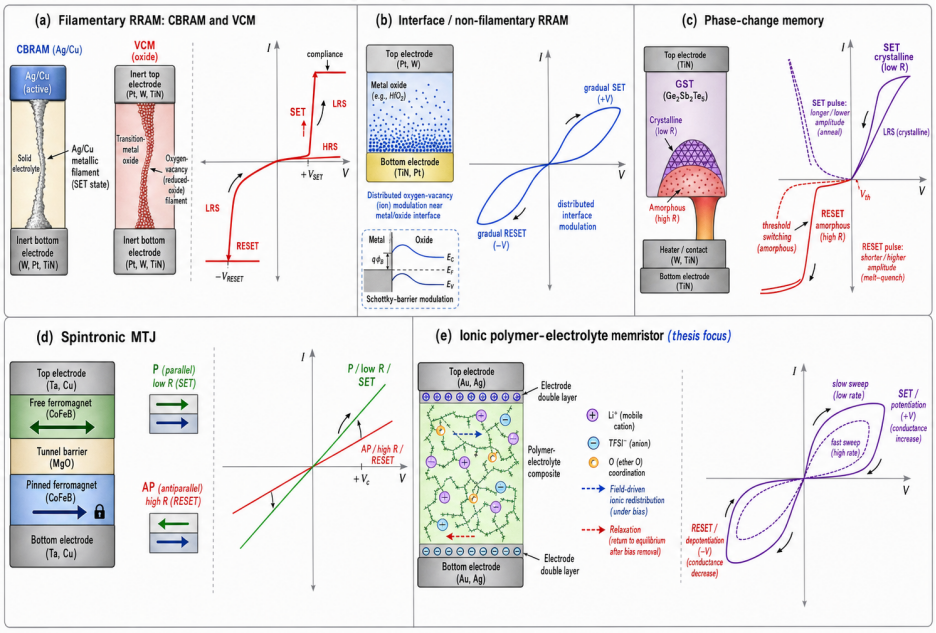


Figure 1.7: Mechanism gallery for the five memristive device families. (a) Filamentary devices, in their conductive-bridge (active-electrode metal filament) and valence-change (oxygen-vacancy filament) variants, switch abruptly between a high-resistance and a compliance-limited low-resistance branch. (b) Interface or non-filamentary devices switch through a spatially distributed change in the active layer, producing smooth, gradual, area-scaling hysteresis without an abrupt set or reset. (c) Phase-change devices toggle between amorphous and crystalline states of a chalcogenide alloy through Joule-heating melt–quench and anneal pulses, with a sharp threshold-switching event preceding the crystalline branch. (d) Spintronic magnetic tunnel junctions switch the relative orientation of two ferromagnetic layers, producing two nearly-linear branches separated by a small (order of two) ON/OFF ratio. (e) Ionic devices, in their polymer-electrolyte composite form most relevant to this thesis, redistribute mobile ions across the active layer under bias and relax back toward equilibrium when the bias is removed, producing continuous, sweep-rate-dependent hysteresis without abrupt transitions.

relative orientation of two ferromagnetic layers [75]. Spintronic devices switch very quickly at low write energy and form the basis of spin-transfer-torque magnetic random-access memory. Their limitations for analogue synaptic use are a small ON/OFF ratio (often of order two) and few stable intermediate states. Even so, their intrinsic stochastic dynamics have been put to use as a computational resource within appropriately designed architectures [76, 77].

Ionic memristors. The devices developed here belong to this family. Their internal state is the distribution of one or more mobile ionic species inside a solid matrix. The host can be an inorganic oxide, a chalcogenide, a halide perovskite, a mixed ionic–electronic conductor, or a polymer electrolyte. Likewise, the mobile species can be a small cation such as Li^+, Na^+, or Ag^+, a proton, or a small anion; in halide perovskites it is typically a migrating halide vacancy, which has been used to build energy-efficient memristive and synaptic devices [78]. The state variable is a chemical distribution of mobile species, distinguished from structural, morphological, or magnetic configurations of the host [71]. Its kinetics are governed by ionic transport, coordination, and relaxation within the matrix.

Ionic systems are intrinsically history-dependent: the distribution of mobile species at any moment reflects the electric fields previously applied. They also relax once the field is removed, as intermolecular forces and concentration gradients drive the biased distribution back toward equilibrium, so volatility is intrinsic to their operation. The associated timescales are set by ion transport and binding, both tunable chemically through the host matrix and the mobile species, the link between chemistry and relaxation that leads this thesis to the organic polymer-electrolyte subfamily.

1.3.4 From Device Metrics to Fading Memory

Memristive devices are often compared through a compact set of practical figures of merit: high- and low-resistance states, ON/OFF ratio, set/reset voltage, write energy, switching speed, retention, endurance, update linearity and symmetry, and device-to-device or cycle-to-cycle variability [17, 69, 72]. These quantities are useful only after the computational role of the device has been fixed.

For analogue in-memory computing, the device stores a *trained weight*: a conductance programmed into a crossbar cell and read repeatedly during matrix–vector multiplication [16, 17]. Long retention, high endurance, low variability, and linear symmetric updates are therefore valuable because they preserve numerical accuracy. Judged by that standard, an organic or polymer-composite device with finite retention can look weak.

For temporal computing, the same finite retention becomes the operating principle. The state of a fading-memory device should relax on a timescale matched to the input signal [19, 29]. A device used as a filter element needs a known, reproducible relaxation time; a device used as a reservoir node can also exploit nonlinearity and device-to-device differences in its temporal kernel. Figure 1.8 sketches the two modes schematically, and table 1.2 sets the corresponding parameter targets side by side.

Variability must also be separated into two kinds. Intra-device instability reduces repeatability under identical conditions. Inter-device heterogeneity is read differently by each use: it can broaden a reservoir's response by supplying different time constants and nonlinear transfer functions [5], whereas for a filter element it is the tolerance on the designed time constant, as it is for any passive component. It also complicates trained-weight arrays.

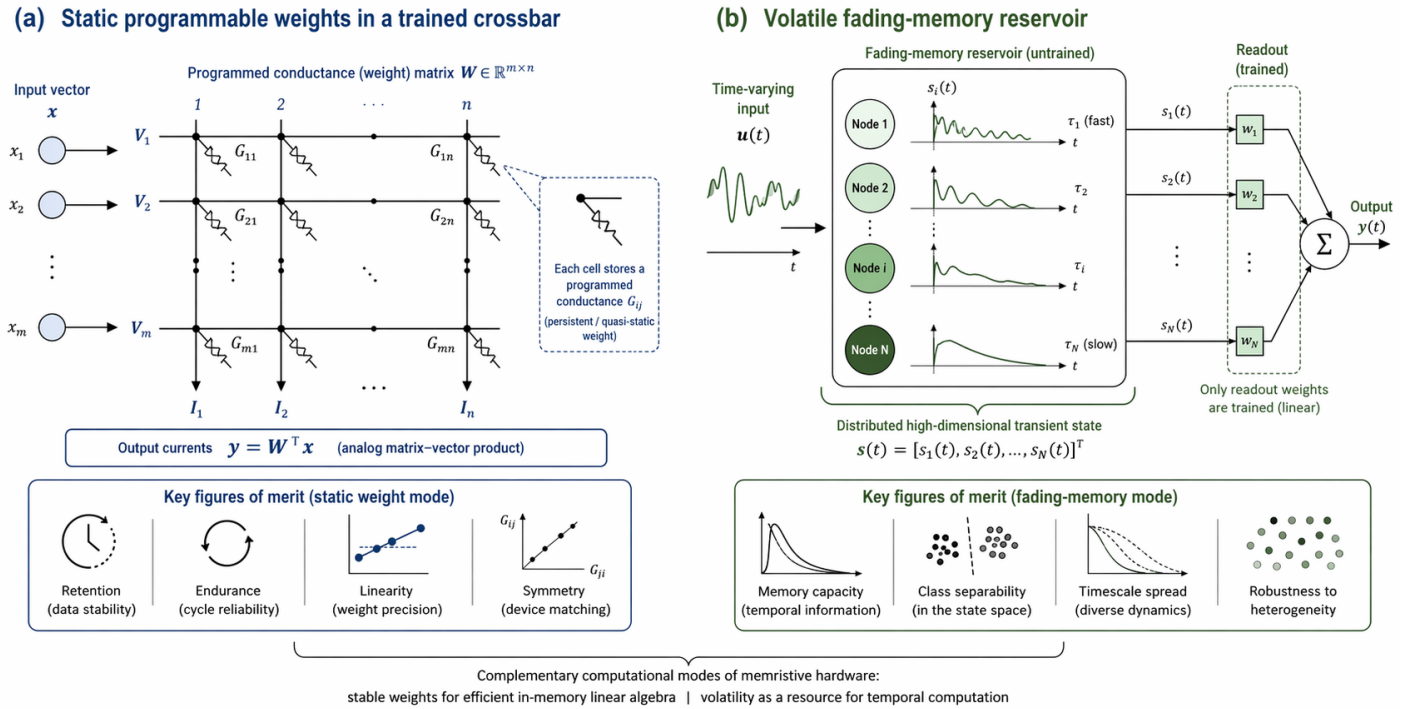


Figure 1.8: Two complementary computational roles for memristive devices. (a) A trained crossbar holds programmed conductances as non-volatile weights and performs matrix–vector multiplication by Ohm's and Kirchhoff's laws; here retention, endurance, linearity and symmetry are the relevant figures of merit. (b) A reservoir of volatile fading-memory nodes, driven by a time-varying input, projects temporal signals into a high-dimensional dynamical state; only a linear readout layer is trained, and the relevant figures of merit become memory capacity, class separability, and the spread of available time constants.

Table 1.2: Direction in which each device parameter should be optimised under the trained-weight regime (section 1.3.4) and under the fading-memory regime (section 1.3.5). For several parameters, the optimum direction reverses between regimes, so a single device cannot in general satisfy both simultaneously.

Parameter	**Trained-weight optimum**	**Fading-memory optimum**
Retention	As long as possible (s–years)	Matched to the input-signal timescale
Switching speed	As fast as possible	Matched to the input event rate
Write energy	Minimised (programmed once, read many)	Minimised within the fast-event budget
Conductance update linearity	Linear and uniform	Less critical; nonlinearity exploited as a feature
Update symmetry (± pulses)	Symmetric	Less critical; asymmetry can be exploited
Cycle-to-cycle variability	Minimised	Minimised (intra-device drift remains a defect)
Device-to-device heterogeneity	Minimised	A tolerance for filter elements; possible diversity for reservoirs [5]

1.3.5 Reservoir Computing with Volatile Memristors

Over the past decade, volatile memristive devices have been used as physical *leaky integrators*: their state rises under stimulation, decays after the stimulus is removed, and thereby carries a temporary record of recent input. Physical reservoir computing provides the computational setting for that behaviour [5].

In a volatile-memristor reservoir, the input is encoded as a stream of voltage pulses applied to an array of devices. Their internal states evolve in response to the stimuli and relax between them, so the array provides a nonlinear mapping of recent input history into a high-dimensional feature space. A linear readout trained on that space can then tackle temporal classification, regression, or prediction tasks. Demonstrations have been reported in diffusive metal-filament devices [33, 34], dynamic oxide memristors [35], and organic electrochemical systems [36].

Coincidence detection and filtering are closely related uses of the same physics. Pulses that arrive within the relaxation time of a device can accumulate, giving timing sensitivity and STDP-like responses [79]; banks of devices with different relaxation constants can process one input in parallel, with fast devices capturing sharp transients and slow devices integrating longer trends [28, 80].

A volatile device can therefore be used in two ways. In a reservoir, many devices with assorted time constants form a random feature bank, and a trained read-out selects what the task needs. In a designed circuit, one device is chosen for its relaxation time and performs a fixed operation, such as removing a slow baseline or smoothing a noisy signal; its time constant plays the role of the RC product of a passive filter. The second use matters most where long time constants are expensive. In analogue CMOS, time constants of hundreds of milliseconds already require large

capacitors or very small currents, and standard designs become bulky in silicon area [6]. Printed and flexible circuits lack compact time-dependent elements and have added temporal processing through printed low-pass filters [81]. A two-terminal film whose relaxation time is fixed by its formulation would supply seconds-scale time constants without a separate capacitor. Chapter 5 evaluates both uses from the measured device response.

Memory capacity measures how well the present reservoir state reconstructs past inputs at different delays; class separability on benchmark tasks and robustness to measured variability provide complementary tests [82]. These metrics are *task-matched*: they test whether the physical timescales of the device align with the temporal structure of the signal, which HRS, LRS, ON/OFF ratio, and retention alone do not capture.

The experimental target is therefore narrower than the full memristor literature: ionic memristive systems whose state is a chemical distribution of mobile species, whose relaxation can be tuned, and whose volatility can be used as fading memory.

1.4 Inorganic Memristors for Neuromorphic Computing

The inorganic families that dominate neuromorphic hardware are metal-oxide resistive memories, phase-change memories, and spin-transfer-torque magnetic tunnel junctions. Each is technologically important, and together they have made memristive hardware a credible post-von-Neumann substrate. Other inorganic candidates (ferroelectric tunnel junctions and FeFETs, Mott-insulator devices, and threshold-switching selectors) are narrower in analogue synaptic operation and are left outside this comparison.

1.4.1 Metal-Oxide Memristors

The most developed inorganic family is resistive switching in transition-metal-oxide thin films, in the VCM and CBRAM variants of section 1.3.3. After the TiO_2 device [67], the field expanded to HfO_2, TaO_x, NiO, and WO_x, many already familiar to silicon processing. Compatibility with back-end-of-line CMOS enabled large crossbar demonstrations [16, 70, 71]. Reported non-volatile devices reach retention of years, nanosecond switching, endurance beyond 10^9 cycles, and sub-10 nm dimensions [17, 69]. Diffusive metal-filament variants can also operate as volatile devices, with Ag or Cu clusters dissolving after programming [33–35, 83]. For analogue synaptic operation, filament stochasticity produces device-to-device and cycle-to-cycle variation in set/reset voltage and achieved conductance [72]. Nonlinear, asymmetric updates and a limited number of reliable conductance levels constrain the accuracy of in-place training on metal-oxide crossbars [17].

1.4.2 Phase-Change Memory

Phase-change memory encodes its state in the proportion of crystalline and amorphous material in a chalcogenide volume (section 1.3.3) [20]. It is a mature storage-class technology used in analogue in-memory computing. Demonstrations include mixed-precision arithmetic for solving linear systems [74] and neural-network training with PCM as the long-term weight store [84, 85]. Its thermal–structural mechanism imposes several costs: reset requires local melting, the quenched amorphous phase drifts with time and requires compensation [21], and crystallisation kinetics are tuned through alloy and geometry. These properties suit high-retention, high-endurance trained weights. They offer fewer chemical controls over fading-memory timescales than the soft-matter platform pursued here.

1.4.3 Spin-Transfer-Torque Magnetic Tunnel Junctions

A spin-transfer-torque device is the magnetic tunnel junction (MTJ) of section 1.3.3, here with a thin MgO barrier and switched between parallel and antiparallel magnetisation by a spin-polarised current through a shared read/write port [75]. STT–MTJs switch in nanoseconds at femtojoule energies, exceed 10^{12} cycles, and integrate in the CMOS back end [86]. Their intrinsic stochasticity has been used for probabilistic computing [76, 77]. For analogue synaptic operation, the parallel/antiparallel resistance ratio is small (often near two), stable intermediate states are few, and increasing their number requires more elaborate stacks [86, 87]. Retention and write energy trade off through thermal stability and switching current [86], while the crystalline ferromagnetic stack requires fabrication methods distinct from solution processing. These properties make STT–MTJs fast, non-volatile, few-state stochastic elements.

1.4.4 General Limitations of Inorganic Memristors

The three families differ in mechanism and maturity but offer limited control over the specific variables studied in this thesis. Their design parameters are primarily geometric, process-related, or stoichiometric within narrow windows [17, 72, 80]. They do not generally permit direct substitution of mobile ions with different binding chemistry or of hosts with different segmental motion. Volatile implementations exist through diffusive filaments [33–35, 83], amorphous-phase drift [21], and superparamagnetic fluctuations [76]. Their relaxation times are set by vacancy drift, crystallisation, or magnetic energy landscapes, so adjustment through a single chemical composition variable is limited. Vacuum deposition, high-temperature annealing, and top-down lithography suit dense CMOS-integrated crossbars but restrict substrate and process temperature [16,

84, 85]. Solution-processable layers allow direct reformulation and a wider range of substrates [80]. Deliberate composition changes may also separate controlled inter-device heterogeneity from intra-device instability more clearly than these established platforms do [5, 17, 21, 72].

These constraints motivate an organic ion-mediated platform that combines analogue response, chemical control of relaxation, and solution processing. Figure 1.9 summarises the qualitative comparison, while table 1.3 provides the corresponding quantitative parameters.

1.5 Organic Materials for Memristive Applications

The inorganic platforms each solve part of the problem, but none of them can be tuned by chemistry alone: reshaping the switching physics requires redesigning a rigid multilayer stack, not changing a molecule, a host, a salt, or a mixing ratio [24, 80]. Organic and polymeric materials lift that constraint. A semiconducting polymer can be tuned through backbone, side chains, and molecular weight [88]; an ion-transport host through its chelating groups, topology, and glass-transition temperature [89, 90]; and a mobile ion through charge, radius, and Lewis acidity [91]. Most such active layers are also solution-processable and processed below 150 °C, so a new formulation can be prepared by changing component concentrations in solution, and the mild processing is compatible with flexible, conformal, and biological substrates, as the stretchable artificial afferent nerve of Kim et al. [92] and the *nervetronic* systems reviewed by Park and coworkers [93] illustrate.

Organic switching can spread over many molecular sites, which makes graded analogue states more accessible than in a single-filament mechanism [22, 23]. Its relaxation timescale can also be engineered chemically.

Table 1.3: Typical operating parameters for the inorganic memristor families. Diffusive (volatile) metal-filament devices are listed separately because they are deliberately operated in the fading-memory regime, not the non-volatile one. Numerical ranges are conservative orders of magnitude drawn from the indicated reviews; individual devices may sit outside them. "cont." = quasi-continuous conductance.

Family	**Switching**	**Retention**	**ON/OFF**	**Endurance**	**Energy/write**	**Analogue states**
VCM [68, 69, 72]	1–100 ns	years	10–10^3	10^9–10^{12}	0.1–10 pJ	4–8
CBRAM [17, 69]	1–100 ns	years	10^4–10^6	10^8–10^{10}	0.1–10 pJ	2–4
PCM [20, 21, 74, 84]	10–100 ns	years (drifts)	10^2–10^3	10^8–10^{10}	10–100 pJ	8–16
STT–MTJ [76, 77]	1–10 ns	years	2–3	$> 10^{12}$	0.1–1 pJ	2
Diffusive (volatile) [33–35]	μs–ms	ms–s	10–10^3	$\sim 10^6$	0.1–10 pJ	cont. (volatile)

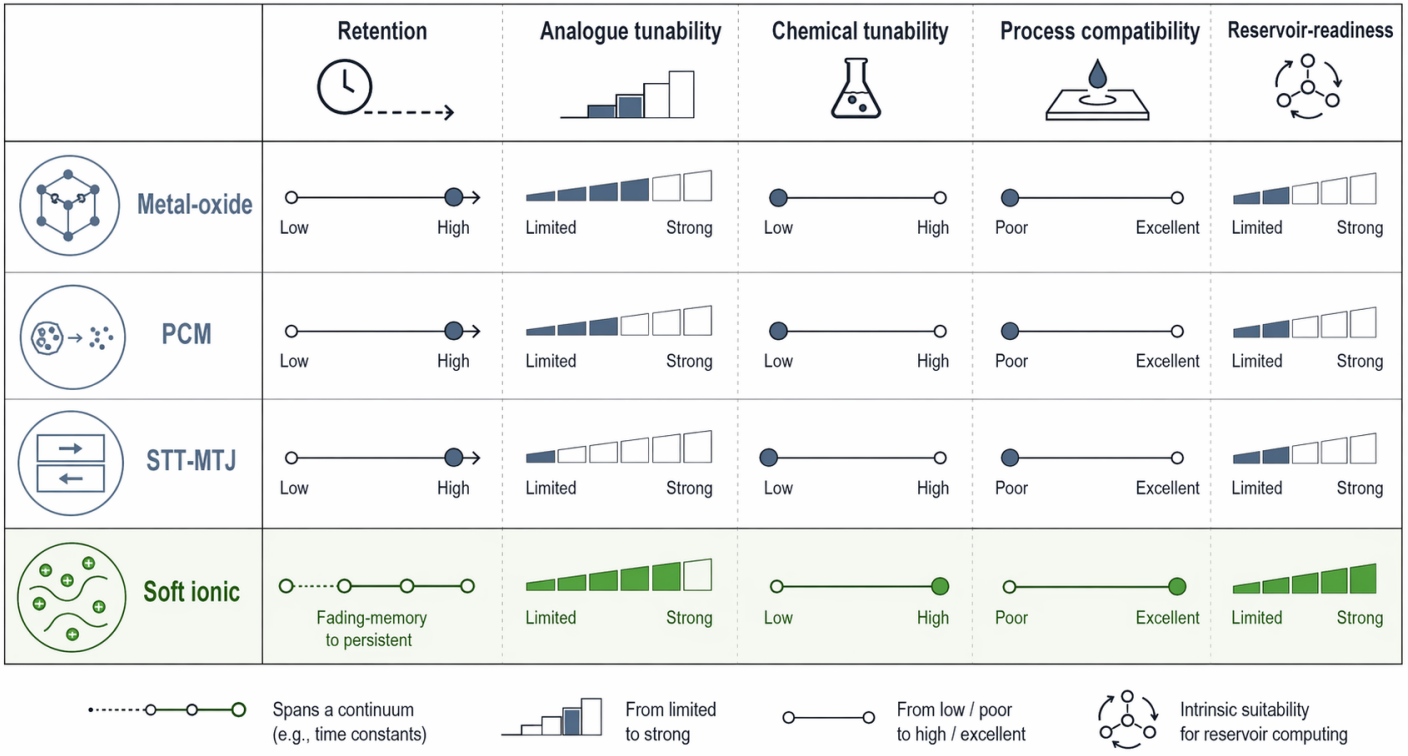


Figure 1.9: Comparison of inorganic memristor families with the soft-ionic composite route pursued in this thesis. For each family, qualitative assessments are shown along five axes: retention (the range of fading-memory to non-volatile behaviour that has been demonstrated), analogue tunability (the number of reliably programmable intermediate states), chemical tunability (the extent to which composition acts as an independent handle on the switching physics), substrate compatibility (vacuum/high-temperature workflows versus solution-processable and mild-temperature workflows), and reservoir-readiness (suitability as a fading-memory node in a temporal-computing scheme).

Ion transport in a polymer electrolyte depends on how strongly the mobile ion binds the host, on host segmental mobility, and on the number of mobile species [89, 94]. Compositional changes within one material family can therefore cover timescales from milliseconds to many seconds, the range of biological short-term plasticity and slow temporal signals [43, 44]. Cost, reproducibility, biocompatibility, and lifetime must still be established case by case. The device target is a reversible mixed ionic–electronic platform whose chemistry controls the fading-memory state.

1.5.1 Organic Mechanisms as Design Choices

As with their inorganic counterparts, the term *organic memristor* names a material family, not a single mechanism. The relevant comparison is what each mechanism (each with its own state variable, reversibility, timescale, and degree of analogue control) offers for a chemically tunable, ion-mediated, two-terminal synaptic element.

Charge trapping and detrapping. Deep or shallow traps (molecular defects, embedded nanoparticles, or boundaries between incompatible polymer phases) are filled and emptied under bias, and the trap occupation, through its local space-charge environment, sets the conductance. Trapping is common in disordered organic semiconductors and operates at modest voltages, but its analogue control is limited: the state is often binary [95], the conductance can depend non-monotonically on occupation, the trap identity is hard to establish, and the retention (set by trap depth relative to thermal energy) is not easily tuned by composition. Panel (a) of fig. 1.10 sketches this trap-occupation mechanism.

Redox switching. A redox-active species (a discrete molecule in a polymer host, a redox polymer, or a unit on a conjugated backbone) is oxidised or reduced under bias, and the distribution of oxidation states sets the conductance. The state variable is then a chemical ox-

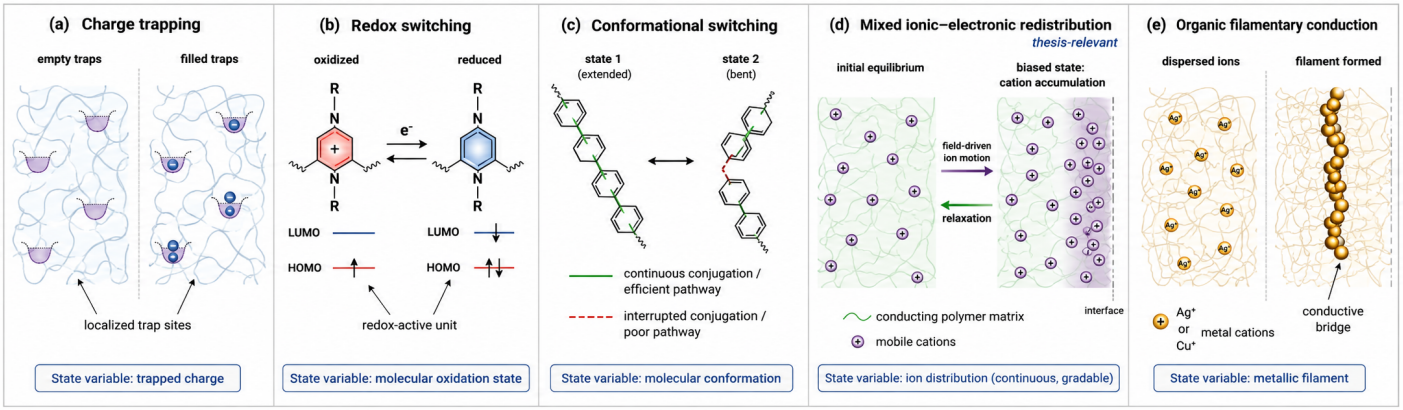


Figure 1.10: Molecular-level representation of the five mechanism families behind organic memristive behaviour. (a) Charge trapping: localised trap sites associated with defects, embedded nanoparticles, or interphase boundaries in a disordered polymer matrix, filled and emptied under bias. (b) Redox switching: a redox-active moiety alternating between its oxidised and reduced states, with the local electronic structure changing accordingly. (c) Conformational switching: a molecular unit toggling between two well-defined geometric or isomeric states with different conjugation or transport pathways. (d) Mixed ionic–electronic redistribution: a population of mobile cations within an electronically conducting polymer matrix, displaced and accumulated near an interface under bias and relaxing back toward equilibrium when the bias is removed. (e) Organic filamentary conduction: metallic cations (typically Ag^+ or Cu^+) from an active electrode migrating through the organic matrix to form a conductive bridge.

idation state with experimentally accessible potentials, reversible over many cycles in favourable cases; important examples include redox-active conjugated-polymer systems [96–98] and early electrochemical devices based on coordination complexes [99]. Not every redox process is cleanly reversible, though (side reactions, dimerisation, and slow degradation limit endurance), and purely redox devices tend to be bistable unless the response is distributed over many sites.

Conformational (structural) switching. In this case, the active layer contains a molecular species whose geometry, isomerisation state, or aggregation state can be modulated by an electric field, local heating, or coupled electrochemical processes. Examples include photochromic and electrochromic switches, as well as bistable molecular architectures whose conductance depends on molecular conformation [97, 98]. The state of these systems is related to a well-defined structural change, so they offer a high level of molecular design precision. However, the synthesis can be challenging, and the switching timescale is fixed by a particular molecular rearrangement.

Ionic redistribution in mixed ionic–electronic systems. This mechanism is the closest match to the experimental platform. The active layer combines an electronic conductor (usually a conjugated polymer) with mobile ions held in an ion-conducting host. Under bias the ions move and accumulate near interfaces, modulating electronic transport through interfacial doping, electrostatic band bending, or electrochemical doping; the conductance change is gradual, multi-state, and directly tied to the amount and arrangement of displaced ionic charge [24]. The same physics underlies light-emitting electrochemical cells (LECs) [100–106], where redistribution forms a doped junction, and organic electrochemical transistors (OECTs) [107], where a gate drives the ion exchange. In the two-terminal memristive mode the driving bias itself plays the role of the gate, and the residual ionic distribution left after the bias is removed is

what carries the memory of the stimulus.

The state variable is thus the distribution of mobile ions in the host, and relaxation after bias (governed by ion transport, the polymer binding environment, salt concentration, and morphology) sets the fading-memory process. The practical costs are that ionic motion is slower than electronic switching and that side reactions with the electrodes must be suppressed through chemistry and contact choice [108].

Organic filamentary conduction. Metallic species (typically Ag^+ or Cu^+ from an active electrode) migrate through the organic matrix and form conductive bridges similar to those found in inorganic CBRAM [70, 71]. These devices inherit the strengths of filamentary switching, such as large ON/OFF ratios and low switching voltages, but also its weaknesses, like stochastic filament formation and intra-device instability [109, 110].

The five families and their trade-offs are collected in table 1.4.

1.5.2 Organic Precedents for the Composite Strategy

The relevant precedents form a set of partial solutions to the same materials problem, not a chronological checklist: how to combine ion-mediated conductance control, analogue reversibility, and a simple device geometry. The earliest two-terminal organic memristive devices are usually taken to be the polyaniline (PANI) electrochemical junctions of Erokhin, Berzina, and co-workers [111–114], in which a PANI channel interfaced with a poly(ethylene oxide)-based electrolyte is doped electrochemically by ionic redistribution under bias. This work already paired a conjugated polymer with an ion-conducting polyether host, the same materials combination used here. It also framed the device from the start as an adaptive, analogue-tunable element supporting history-dependent conductance

Table 1.4: Mechanism families behind organic memristive behaviour.

Family	State variable	Reversibility	Analogue gradation	Timescale tunability	Refs.
Charge trapping	Trap occupation	Partial	Limited (often binary)	Low	[95]
Redox	Oxidation state of a moiety	Variable	Bistable to few-state	Low–medium	[96, 99]
Conformational	Molecular geometry / isomer	Good (per design)	Bistable / few-state	Low (per molecule)	[97, 98]
Mixed ionic–electronic	**Mobile-ion distribution**	**Good**	**Continuous**	**High (host chemistry)**	[22–24]
Filamentary (organic CBRAM)	Metallic filament	Limited (stochastic)	Few states	Low	[109, 110]

changes.

The LEC route is especially close to the platform used here. Zakhidov's *light-emitting memristor* [99], in which an ionic $[Ru(bpy)_3]^{2+}$ complex supplied both the emitter and the mobile ions, showed that redistribution in a polymer host yields memristive behaviour, while resistive switching in simple films such as poly(vinyl alcohol) [95] confirmed that the phenomenon is accessible in cheap matrices but typically binary and mechanistically opaque. The best analogue control came from the electrochemical-transistor route: Gkoupidenis' PEDOT:PSS OECTs reproduced short-term plasticity, paired-pulse facilitation, and long-term potentiation and depression [107, 115], Xu's core–sheath nanowire transistors [116] and the Van de Burgt–Fuller organic electrochemical neuromorphic device (ENODe) [22, 23] delivered linear, symmetric updates through ion exchange between electrolyte and channel [24], and halide-perovskite-gated OECTs extended the accessible timescales [117]. Each such synaptic element, however, requires a third terminal and a contacted electrolyte volume, which makes a self-contained two-terminal device an attractive complement for the reservoir paradigms of section 1.3.5.

The two-terminal line developed in parallel. Liu's biomimicking redox organic memristor [96] demonstrated coexisting short- and long-term memory in a passive two-terminal device, but also showed partial irreversibility in the redox route: the long-term regime sometimes involved permanent chemical change that limited controlled state evolution [118]. Molecular memristors, especially Goswami's metal-coordinated azo-aromatic layers [97, 98], achieved contrast, endurance, and multi-state operation through a demanding synthesis; stretchable and flexible devices [92, 93, 119, 120] demonstrated mechanical compliance; and perovskite–organic hybrids [78, 121] and organic electrochemical reservoir computing [36] broadened the field further, the latter showing that mixed ionic–electronic devices can serve as volatile physical-reservoir elements.

By the early 2020s the field had matured into a subfield consolidating around shared physical principles [24, 122], with the individually demanding features each demonstrated (ion-mediated operation in two- and three-terminal forms, analogue symmetric programming, short-to-long-term transitions, mechanical compliance, molecular precision) but not yet combined in one simple platform. The three-terminal route bought reversibility and analogue control at the cost of architectural complexity, the two-terminal redox route simplicity at the cost of full reversibility, and the molecular route precision at the cost of synthesis. In that 2022 publication context, the specific gap addressed by the proof-of-concept chapter was a two-terminal, reversible, multi-state, ion-mediated device with interpretable chemistry built from standard organic-electronic components. The route pursued here fills that gap with a composite of a conjugated semiconducting polymer, an ion-conducting polymer host, and a dissociated alkali-metal salt. These parallel developments are collected in fig. 1.11.

Contemporary physical-reservoir computing spans several node, input, output, and network architectures, so comparison requires those layers to be stated separately [123]. Since the proof-of-concept publication, organic hardware has advanced from electrochemical networks to gate- and wavelength-tuned vertical transistors with measured temporal characteristics from 0.005 s–13.2 s [124], and to reconfigurable organic electrochemical transistors that implement both reservoir and read-out functions [125]. Electrical control of relaxation is also no longer unique to multi-terminal structures: a two-terminal HfO_2 memristor has used an inter-pulse base voltage to tune its fitted relaxation time over more than two orders of magnitude [126]. No priority is claimed here for the first tunable two-terminal physical reservoir. The narrower contribution is replicated composition control of seconds-scale fading memory in a solution-processed, self-contained two-terminal organic composite, followed by model-based

rather than wired-reservoir task evaluation.

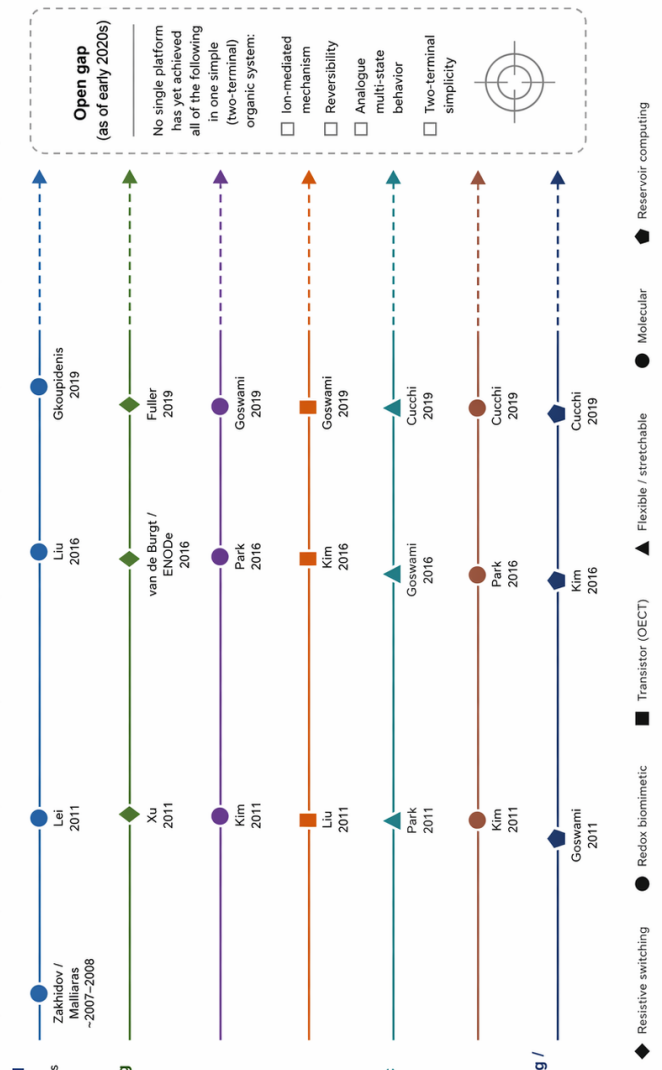


Figure 1.11: Historical map of organic memristive and organic synaptic devices, arranged by mechanism family and by year of representative publication. The map frames the literature context of the 2022 proof-of-concept publication: by the early 2020s, several demanding features of a neuromorphic organic device had each been demonstrated, but the specific combination targeted here was a self-contained two-terminal platform combining reversibility, multi-state analogue behaviour, and an interpretable ion-mediated mechanism.

1.6 Polymer Electrolyte Composites: The Enabling Strategy

The composite platform follows directly from the mixed ionic–electronic precedents above. It places a semiconducting conjugated polymer, an oxygen-rich ion-conducting polymer host, and a dissociated alkali-metal salt in one solution-processed active layer, then operates that layer in a standard two-terminal stack. Electronic transport, ion solvation, and mobile-charge supply are therefore assigned to separate components that can be changed independently.

1.6.1 Ion-Conducting Polymer Chemistry

An ion-conducting polymer is a macromolecular host in which dissolved or dispersed ions can move under an applied electric field without the help of a liquid solvent phase. Poly(ethylene oxide) (PEO) established the model system: alkali-metal salts were first shown to form solid complexes with PEO in 1973 [127], ionic conductivity was measured in those complexes in 1975 [128], and the resulting solid polymer electrolytes became practical ion conductors through the late 1970s and early 1980s [89, 90, 94]. Their relevance here is control of cation coordination, segmental motion, host crystallinity, and composition (the same variables that set memristive relaxation), not battery performance.

The polyether host has a high density of ether oxygens along the backbone or side chains. In PEO, those oxygens appear regularly along the –CH_2–CH_2–O– repeat unit, and the chain conformation in the amorphous phase allows each cation to coordinate several ether oxygens from one or more polymer chains [94, 129]. Hybrane-type hyperbranched polyester-amide hosts arrange the oxygen density differently, through ether and ester groups attached as side chains on a branched polymer core [130], but the

coordination chemistry is of the same kind.

In both hosts, oxygen donor sites can solvate alkali cations, stabilise salt dissociation, and define local coordination environments. Charge-delocalised anions such as triflate ($CF_3SO_3^-$) or bis(trifluoromethanesulfonyl)imide (TFSI, $N(SO_2CF_3)_2^-$) are commonly treated as weakly coordinating counter-ions [131]. Neither description fixes the relative macroscopic mobility: ion pairing, host segmental motion, concentration, morphology, and field also contribute.

Ion transport in these hosts is usually described through three related microscopic phenomena:

- From the *vehicular* side, a cation and its solvation shell move together through the polymer matrix, so long-range transport is coupled to polymer motion [132].
- From the *hopping* or site-to-site side, the cation instead transfers between coordination sites through a local rearrangement of oxygens, without macroscopic motion of the host. This last view is conceptually close to Grotthuss-type transport in proton conductors [133], even though the chemistry of the transfer itself is different.
- Finally, from the *segmental-motion-assisted* side, short polymer segments reorganise around the cation, creating and destroying coordination sites that allow effective diffusion [134, 135].

These three phenomena are not mutually exclusive, although the weight of each contribution during transport depends on host chemistry, cation identity, salt concentration, and temperature [89, 94].

The macroscopic ionic mobility of a polymer electrolyte generally correlates with the segmental mobility of the host, which is set by the glass transition temperature T_g and by whether the host remains amorphous

during operation. Below T_g, segmental motion is essentially frozen and ionic conduction drops. Above T_g, it is activated, and ionic transport becomes measurable, with the temperature dependence in this regime typically following the Vogel-Fulcher-Tammann form characteristic of fragile glass-formers [89, 136].

Even above T_g, however, not all of the host contributes equally. Crystalline regions contribute little to conduction and can even trap ions in ordered environments [129, 135]; the textbook picture is that conduction proceeds essentially through the amorphous fraction, although it has since been shown that specific crystalline stoichiometric phases can themselves support measurable ionic motion [137]. Low-crystallinity or fully amorphous hosts are therefore preferred for ion transport. A low T_g and a low crystalline fraction can be pursued through copolymerisation, branching, or plasticisation; hyperbranched architectures such as the Hybrane family are attractive here because branching itself suppresses crystallinity [138]. For room-temperature memristive operation the host should remain amorphous and mobile under ambient conditions, ideally without added plasticisers.

The strength of cation binding depends on cation identity, which hard–soft acid–base (HSAB) reasoning [91] makes explicit. Ether and ester oxygens in polyether and polyester-amide hosts are hard Lewis bases: their donor pairs are localised, their polarisability is low, and their coordination to alkali cations is primarily electrostatic. Among the alkali cations, Li^+ is the hardest Lewis acid (small, with a high charge density), and should therefore interact most strongly with hard oxygen donors. Na^+ and K^+ are progressively softer, with larger radii, lower charge densities, and weaker electrostatic coupling to the same oxygen environment.

In a common polyether host and at fixed composition, the residence time of the cation at a coordination site might be expected to follow the coordination-strength order $Li^+ > Na^+ > K^+$. This provides a hypothesis

for the devices' fading-memory timescale: harder cations should relax more slowly and softer cations more quickly. The experimental chapters test this hypothesis. In the archive used here, composition is the replicated quantitative variable, while cation order is not robust across host and anion. The coordination geometry, the three transport mechanisms, and the cation–oxygen binding hierarchy that motivate the hypothesis are summarised in fig. 1.12.

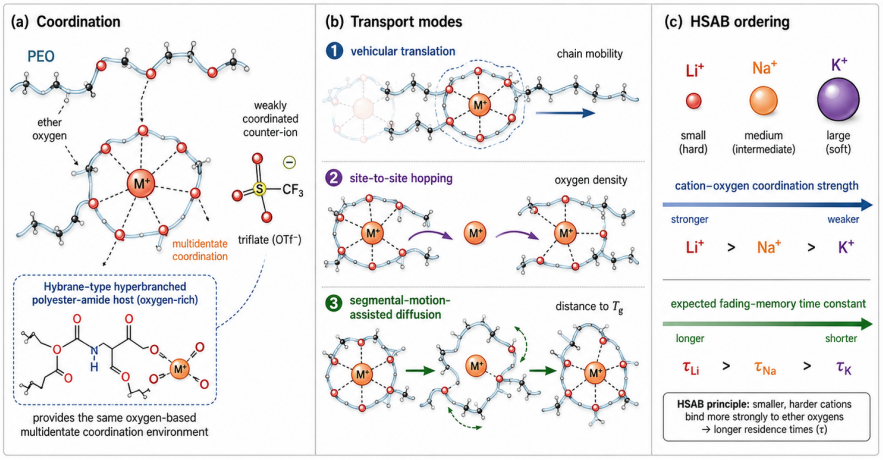


Figure 1.12: Molecular-level chemistry of ion transport in the polyether-hosted polymer-electrolyte composite. (a) A short poly(ethylene oxide) segment coordinates an alkali cation through four to six ether oxygens, with triflate shown nearby as a weakly coordinating counter-ion [94, 129, 131]; oxygen coordination in the Hybrane host of chapter 2 is a qualitative analogy, not a measured coordination structure. (b) Long-range ionic transport can combine vehicular translation, site-to-site hopping, and segmental-motion-assisted diffusion [132, 134, 135]. (c) Within a common polyether host, hard–soft acid–base reasoning motivates a $Li^+ > Na^+ > K^+$ cation–oxygen coordination-strength hypothesis [91]. Coordination strength alone does not determine mobility, and the thesis does not assume this ordering as a host- and anion-independent law.

A second set of design parameters is provided through composition. A polymer electrolyte is defined by the host, ion identity, and salt-to-host ratio; in a composite device, the polymer-to-semiconductor ratio also

contributes. The salt-to-host ratio sets the number and spatial density of occupied coordination sites, while the host-to-semiconductor ratio sets the volume available for ionic transport and the connectivity of the electronic phase. The two ratios are therefore treated as independent composition axes in the later chapters.

These considerations motivate the host choices made in this thesis. Hybrane DEO750 8500 is a hyperbranched polyester-amide with a high density of ether and ester oxygens. It provides an amorphous, low-crystallinity ion-transport host and is used in chapter 2 as the host of the proof-of-concept Super Yellow (SY)/Hybrane/lithium triflate (LiOTf, $LiCF_3SO_3$) composite. The synaptic responses were measured on assay-specific junctions sharing this formulation and architecture; they do not form a device-population study.

PEO is the host used for the broader comparative work. Like Hybrane, its ether oxygens coordinate alkali-metal cations, segmental motion enables transport, and composition can be varied while the electronic semiconductor is held fixed. PEO also has an extensive solid-polymer-electrolyte literature covering its phase diagrams, coordination geometries, conduction mechanisms, and salt-dependent dynamics [89, 94, 128, 129, 135]. The replicated composition series therefore uses a host whose structural and transport behaviour is independently constrained. A hyperbranched polyether, trimethylolpropane ethoxylate (TMPE), provides a comparison between linear and branched polyether architectures. Host, anion, and cation substitutions are surveyed at smaller sample sizes and treated as illustrative contrasts rather than material laws.

1.6.2 LEC-Derived Composite Operation

Polymer light-emitting electrochemical cells (LECs) provide an established example of stable, reversible coupling between ionic redistribution in a

polymer host and electronic transport in a conjugated semiconductor [139]. An LEC is a simple two-terminal sandwich [100]: a conjugated light-emitting polymer is mixed from a common solution with an ion-conducting polymer and a dissociated salt. Under bias, cations migrate toward the cathode and anions toward the anode, forming doped regions that inject carriers into the undoped bulk, and the resulting dynamic p-i-n profile produces electroluminescence in the intrinsic region between the doped edges [103].

Two models were historically used to describe this operation. In the electrochemical-doping model [100, 140], the dominant effect is bulk doping of the conjugated polymer; in the electrodynamic model [101, 102], it is the lowering of charge-injection barriers by ion accumulation at the contacts. Later experiments [103, 105] and a unifying framework [104] showed these to be limiting cases of the same ion redistribution under different injection regimes, not contradictory mechanisms. The two models, their reconciliation, and the bias-off configuration that the memristive composite reuses are compared on a common cathode–anode position axis in fig. 1.13; studies of electrochemical side reactions [108] additionally identify the degradation pathways that must be suppressed.

LECs operate over useful windows with lifetimes of hundreds to thousands of hours [106, 141], provided the contacts and operating protocol support repeated charge injection and withdrawal without rapid destructive side reactions [142–144]. Their electrolyte supplies mobile ions whose redistribution modulates the conjugated polymer. Sustained bias in an LEC maintains a p-i-n profile for emission; in a memristive device, a pulse instead writes an ionic distribution whose bias-off residue provides the stored state. Ionic redistribution can therefore serve as the internal state variable of a two-terminal memory device.

The most direct precedent is the Hybrane-based composite that underlies the proof-of-concept device of chapter 2. Mardegan and co-workers [106]

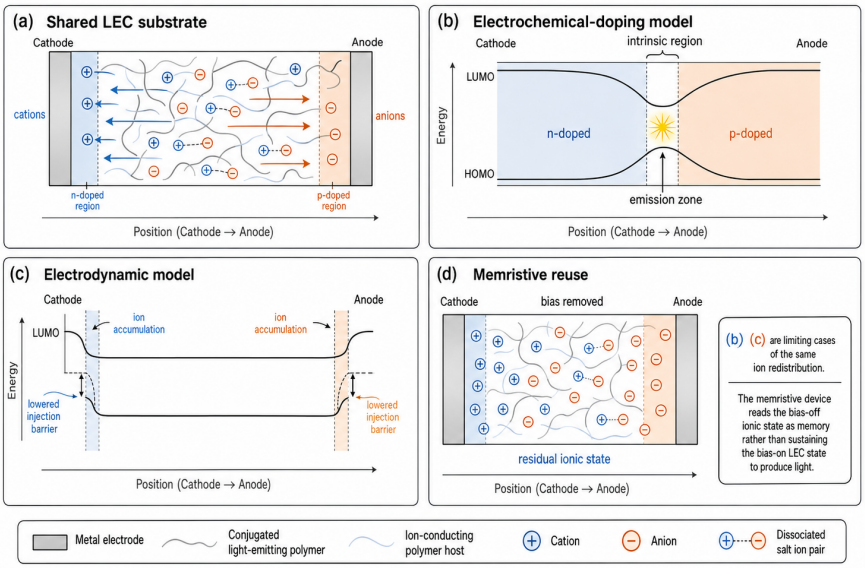


Figure 1.13: Light-emitting electrochemical cell operating models and their memristive reuse, drawn on a common cathode–anode position axis. (a) Shared substrate: under sustained bias, cations drift toward the cathode and anions toward the anode, forming doped regions at the two contacts. (b) Electrochemical-doping model: the dominant effect of the ionic redistribution is *bulk doping* of the conjugated polymer, producing a spatially wide p-i-n profile with emission localised at a narrow intrinsic region. (c) Electrodynamic model: the dominant effect of the same redistribution is *injection-barrier lowering* at the metal/polymer interfaces, producing thin ion-accumulation layers at the contacts and an essentially flat-band bulk. The two pictures were later reconciled as limiting cases of the same redistribution under different injection regimes. (d) Memristive reuse: with the bias removed, the residual ionic distribution left behind by the previous bias is the *stored internal state* of the device.

demonstrated stable LECs based on Hybrane hyperbranched polyesteramide hosts blended with a conjugated light-emitting polymer, reporting long operational lifetimes under continuous bias. Their work validates the semiconducting-polymer / ion-conducting-host / salt architecture, the solvent system, and the processing protocol that chapter 2 then reuses under pulsed bias to operate the same composite as a memristive device.

1.6.3 The Ionic Response in Time and Frequency

The ionic redistribution that stores the device state has a characteristic relaxation time, set by the host segmental mobility discussed in section 1.6.1. This timescale has two equivalent descriptions. In the time domain, a written ionic profile decays toward equilibrium over a period that the proof-of-concept and comparative chapters measure in seconds (chapters 2 and 4). In the frequency domain, it appears as a cutoff. At low drive frequencies, ionic redistribution produces the pinched hysteresis of fig. 1.6. Above the redistribution rate, the junction approaches a passive dielectric response and the hysteresis loop narrows [62, 63]. For a slow polymer-electrolyte host, this crossover falls in the sub-kilohertz range and overlaps the frequency bands of peripheral physiological signals examined in chapter 5.

The standard small-signal probe of this frequency response is electrochemical impedance spectroscopy (EIS). A small alternating voltage is applied across a range of frequencies and the complex impedance recorded; plotted as a Nyquist or Bode spectrum, it separates the bulk resistance to ion transport through the host from the capacitance of the interfacial layers that accumulate at the contacts. An equivalent-circuit model assigns these features to discrete circuit elements: for polymer light-emitting electrochemical cells and the related composites used here, the circuit of Munar and co-workers resolves a bulk ionic resistance together with geometric and chemical (double-layer) capacitances [145]. That bulk ionic resistance is the equilibrium counterpart of the driven relaxation (a more ion-rich, more mobile matrix has both a lower ionic resistance and a faster fading memory), so impedance spectroscopy gives an independent, small-signal test of the link between composition and timescale that large-signal switching alone cannot supply [89, 131]. Chapter 4 uses both probes, equilibrium impedance spectra and biased frequency sweeps, as microscopic corroboration of the composition control established from

the driven dynamics.

1.6.4 Rationale for the Composite Approach

The composite separates electronic transport, ion solvation and transport, and mobile-charge supply across three components [100]: a conjugated polymer, an oxygen-rich host, and a salt. Salt, cation, and blend ratio can therefore be changed independently of the rest of the stack [24]. The salt-to-semiconductor ratio changes the nominal supply of potentially mobile ions. HSAB reasoning motivates stronger cation–oxygen binding for harder alkali cations, while charge-delocalised triflate is weakly coordinating [131]; it does not, on its own, rank their drift mobilities. A fast/slow species asymmetry is therefore one candidate origin for multiple memory timescales [111, 113, 114]. Its effect must be tested within each host, anion, electrode, concentration, and drive protocol.

Composition is a second control variable. The salt-to-semiconductor mass ratio sets the relative supply of mobile ions and hence the achievable size of the conductance change [89, 90], while the host-to-semiconductor mass ratio sets the relative amount and connectivity of the segmentally mobile phase in which they rearrange; in principle these vary the amplitude and the rate of the response separately, though the comparative chapter treats their independence as an experimental question. The three components share a common solvent, so changing the formulation requires only different solution ratios on the same electrode geometry. This makes a replicated composition grid practical, while host, anion, and cation substitutions remain lower-power chemical contrasts.

Devices with the same polymers but different blend ratios should exhibit a spread of ionic relaxation times, with sample-limited chemistry and device-to-device variation adding further diversity. Each formulation would then define an element with its own time constant, selected at

fabrication in the way a component value is selected from a catalogue. Together, such elements form a bank of temporal kernels with distinct decay constants [5, 82]. This spread complicates precise trained-weight storage but can cover a range of fading-memory timescales [33–36]. The application chapter evaluates both the individual elements and the banks from the measured device behaviour.

The experimental chapters test this rationale first with the Hybrane/LiOTf proof-of-concept device of chapter 2, and then with the broader PEO-based corpus in which composition is varied as the replicated axis and chemistry is surveyed as illustrative side evidence. Figure 1.14 summarises this concept used throughout the thesis, linking the two-terminal stack to the qualitative timescale ladder originally expected from cation and composition chemistry.

From Background to Device

The resulting device target is a solution-processed composite of a conjugated semiconducting polymer, an oxygen-rich ion-conducting host, and a dissociated alkali-metal salt. It is operated as a two-terminal element whose residual ionic distribution forms the internal state. Chapter 2 introduces the proof-of-concept SY/Hybrane/LiOTf device, its memristive and synaptic response, and the ion-transport mechanism proposed to explain its two retention regimes.

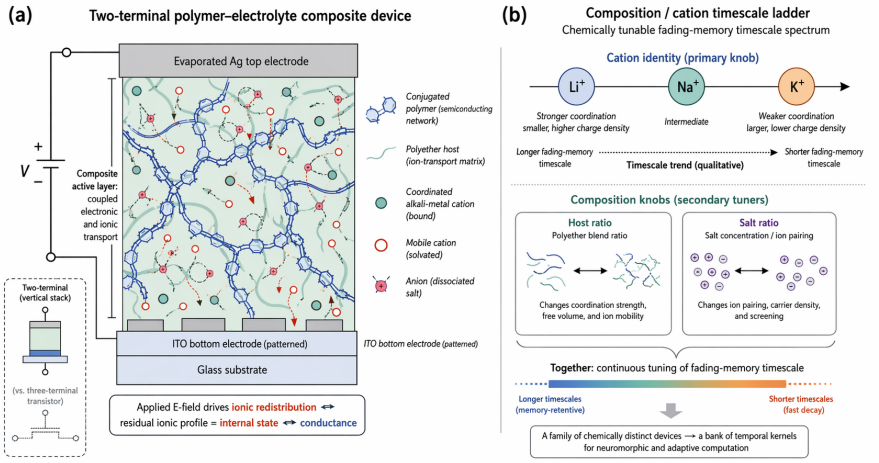


Figure 1.14: The thesis-specific polymer-electrolyte composite device and its chemically tunable dynamical timescale. (a) Vertical two-terminal stack: a solution-processed composite active layer combines a conjugated semiconducting polymer network with an oxygen-rich ion-conducting host and alkali-metal salt, between a patterned ITO bottom electrode and an evaporated silver top electrode; the working model assigns memory to a residual ionic distribution after bias. (b) Timescale ladder *as anticipated at the design stage*, with cation identity drawn as the primary control and the host-to-semiconductor and salt-to-semiconductor mass ratios as secondary ones, placing the fading-memory time along the $Li^+ > Na^+ > K^+$ coordination order. The panel records the starting hypothesis, not the result. Chapter 4 inverts this hierarchy: composition (the PEO/SY and salt/SY mass ratios) is the replicated quantitative control of the timescale, whereas the cation is sample-limited and its anticipated ordering is not borne out (the triflate retention series runs in reverse).

Chapter 2

Proof of Concept: Polymer-Based Composite Organic Memristive Devices

2.1 Introduction and Motivation

This chapter reports the proof-of-concept polymer-electrolyte memristive platform introduced at the end of chapter 1. The composite combines a semiconducting poly(phenylene vinylene) (PPV) derivative (*Super Yellow*, SY), an oxygen-rich hyperbranched polyester-amide ion-conducting host (*Hybrane*® DEO750 8500), and a lithium salt (lithium triflate, LiOTf, $LiCF_3SO_3$), all processed from a common solvent into a standard two-terminal (2-T) sandwich on indium tin oxide (ITO) and silver. The experiments test whether this common formulation and architecture support current–voltage hysteresis, potentiation and depression, excitatory post-synaptic current, distinct retention regimes, and spike-timing-dependent plasticity. The assays were not all performed on one physical junction: the available specimen mapping is stated in table A.1. The chapter also

develops a candidate cation–host coordination interpretation. Chapter 4 evaluates its cation-ordering prediction but does not confirm a transferable ordering.

At the start of this work the literature already contained several partial demonstrations of these functions in different architectures (reviewed in section 1.5.2): ion-mediated analogue switching in multi-layer polyaniline junctions [111, 113, 114], bistable ionic redistribution in a light-emitting memristor [99], tunable short-to-long-term transitions in a partially-irreversible redox device [96, 118], and reversible analogue control in three-terminal organic electrochemical transistors [22, 23, 107, 116]. Each covered part of the specification developed in section 1.2.3 (analogue, reversible, multi-state, electrically driven conductance changes), but their combination in one self-contained two-terminal organic platform with a chemically testable ion-mediated hypothesis remained, at that point, an open problem.

The active-layer logic adopted here is the composite logic of section 1.6.4. The three functional roles (electronic transport, ionic transport, and mobile-charge supply) are assigned to three different components, so that each role can be tuned chemically without redesigning the rest of the stack. *Super Yellow* supplies the electronic-conductance pathway. *Hybrane* supplies an oxygen-rich ion-transport host. LiOTf supplies a small hard cation and a charge-delocalised, weakly coordinating counter-anion. This chemical contrast motivates the candidate mechanism of section 2.5; hard–soft acid–base (HSAB) reasoning alone does not determine mobility in the composite.

The measurements fall into two groups. The first comprises three dynamical protocols: current–voltage hysteresis characteristic of memristive switching, potentiation as a function of the number of applied voltage pulses, and depotentiation as a function of the delay before read-out. These three measurements recur, on the same 2-T device geometry,

throughout the comparative work of Chapters 4 and 5. The second group comprises extended synaptic protocols specific to this chapter: an excitatory post-synaptic current (EPSC) sequence with four numerically separated readouts; the resolution of the depotentiation measurement into short- and longer-lived regimes, denoted short-term memory (STM) and long-term memory (LTM) following the convention of the artificial-synapse literature [24, 92]; and a spike-timing dependent plasticity (STDP) protocol that produces an asymmetric timing kernel. Re-fitting the archived five-junction mean gives branch time constants of approximately 89 ms and 150 ms; these are a waveform-level analogy to biological STDP timescales, not evidence of a biological learning mechanism [56, 59]. The published interpretation associates the fast response with triflate redistribution and the slow response with lithium displacement from oxygen coordination sites. The cation survey of Chapter 4 does not reproduce the resulting simple cation order, so that species-level assignment remains a candidate rather than a conclusion.

Within the thesis the chapter is the only one in which the full synaptic suite is the main evidence; only the three dynamical measurements named above carry forward to the comparative corpus.

2.2 Materials Design and Composite Formulation

2.2.1 Design Rationale

The operating principle of the device is the same one developed in section 1.6.4: ionic migration within a solid-state polymer matrix under an applied electric field redistributes mobile ions near the electrodes, which in turn modifies the energy barriers for charge injection and the

doping level of the semiconducting polymer, and therefore the electronic conductance of the device. The composite active layer is built to satisfy three functional requirements simultaneously:

1. **Electronic-conductance pathway:** a semiconducting polymer provides the medium through which the electronic current flows, and whose conductivity is susceptible to modulation by ionic redistribution and electrochemical doping.

2. **Ion-transport medium:** a polymer electrolyte provides a structural environment in which dissolved ions remain mobile under an applied field, yet are retarded enough by intermolecular interactions that, once the field is removed, the conductance change persists for seconds (long enough to be read out) before fading back toward equilibrium.

3. **Ion source:** a salt provides cations and anions whose field-driven redistribution can modify the electronic current; a species-specific assignment to the two retention regimes is considered as a hypothesis below.

The composition adopted here allows a chemically motivated comparison between the two ions of the salt. Li^+ is a small, hard Lewis acid with a high charge-to-radius ratio ($\lambda = z/r \approx 1.3\,\text{Å}^{-1}$–$1.7\,\text{Å}^{-1}$ across four- to six-fold oxygen coordination, with the corresponding Shannon radii in the range 0.59 Å–0.76 Å [146]). Ether and ester oxygens in *Hybrane* can coordinate alkali cations, while the negative charge of $CF_3SO_3^-$ is delocalised and triflate is commonly treated as a weakly coordinating counter-anion. This contrast motivates a fast-anion/slow-cation hypothesis under bias. It does not establish that assignment: mobility also depends on ion pairing, salt concentration, polymer segmental motion, electrode boundary conditions, and field. The two-regime interpretation below is therefore a testable model, not a direct consequence of HSAB classification.

2.2.2 Super Yellow: The Semiconducting Polymer

Super Yellow (commercial PDY-132, Merck) is the semiconducting PPV-family copolymer in the blend. Its role in the proof-of-concept device is to provide the electronic-conductance path whose injection barriers and local doping state can be modulated by mobile ions. The relevant material facts for the main argument are its solution processability in cyclohexanone, its established use in organic light-emitting devices [88], and frontier levels of approximately $E_{\mathrm{HOMO}} = -5.4\,\mathrm{eV}$ and $E_{\mathrm{LUMO}} = -3.0\,\mathrm{eV}$ [147]. The full copolymer name and component-level notes are collected in section A.1.

2.2.3 Hybrane DEO750 8500: The Ion-Transport Polymer

Hybrane is an oxygen-rich hyperbranched polyester-amide [130]. In this chapter it is used as the ion-transport host: ether and ester oxygen atoms provide hard Lewis-base coordination sites for alkali cations, while the hyperbranched architecture suppresses crystallisation and keeps the blend amorphous. Ion motion is therefore treated as segmental-motion-assisted hopping between local coordination environments, in the polymer-electrolyte picture introduced in section 1.6.1. The relevant migration activation scale is of order $50\,\mathrm{kJ\,mol^{-1}}$ for polymer electrolytes near room temperature [89]; the longer discussion of host structure and transport is in section A.1.

2.2.4 Lithium Triflate: The Ionic Salt

LiOTf (lithium trifluoromethanesulfonate, $\mathrm{LiCF_3SO_3}$) supplies the mobile ions. It is compatible with the cyclohexanone-processed *Super Yellow*/*Hybrane* blend and is widely used in polymer-electrolyte and LEC

systems [100, 105]. Its small, hard Li^+ cation can coordinate the host oxygens, whereas the triflate charge is delocalised. The chapter uses that contrast to formulate, and later delimit, a candidate explanation of the two relaxation regimes; supporting material notes are in section A.1.

2.2.5 Composite Formulation and Preparation

The active layer uses a mass ratio of *Super Yellow*:*Hybrane*:LiOTf = 1:0.30:0.09, processed from cyclohexanone under dry nitrogen. After mixing, the final solution concentrations are $8.72\,\mathrm{mg\,mL^{-1}}$ (*Super Yellow*), $2.62\,\mathrm{mg\,mL^{-1}}$ (*Hybrane*), and $0.78\,\mathrm{mg\,mL^{-1}}$ (LiOTf). The ratio is the optimised formulation reported in the published proof-of-concept work [7]: higher salt contents degraded film homogeneity, whereas lower salt contents reduced the available mobile-carrier density. Detailed solution preparation and optimisation notes are given in section A.1. The same formulation had also been validated in related *Super Yellow*/*Hybrane*/lithium-salt LECs with long electroluminescent operation [106].

2.3 Device Architecture and Fabrication

2.3.1 Device Geometry: The Two-Terminal Vertical Structure

The device adopts a vertical sandwich geometry, in which the active layer is placed between a bottom transparent electrode and a top metallic electrode (see fig. 2.1). This 2-T architecture is the simplest memristive configuration: it has no gate electrode and needs no transistor to address the device, in contrast to the *three-terminal* (3-T) organic synaptic transistors discussed in section 1.5.2 [107, 115, 116].

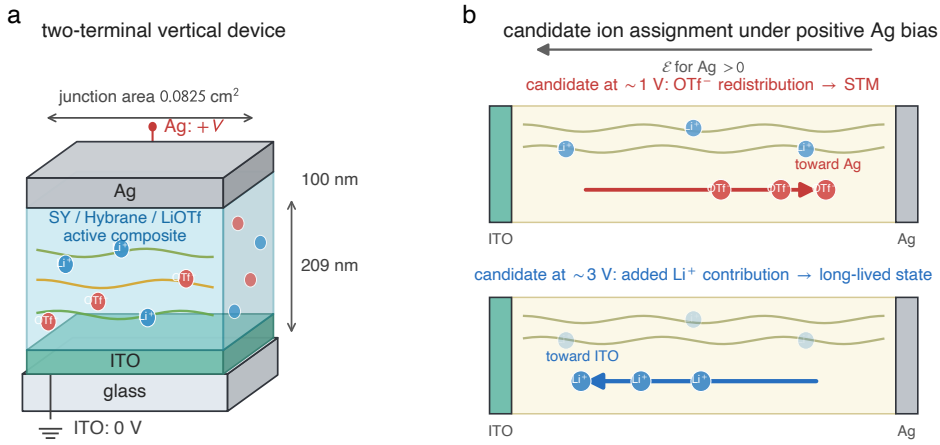


Figure 2.1: Schematic of the two-terminal vertical synaptic device and candidate active-layer mechanism. The device stack is ITO-glass / *Super Yellow*/*Hybrane*/LiOTf composite / Ag; the composite separates the electronic pathway, ion-transport host, and mobile-ion supply. The species labels show the working fast-anion/slow-cation assignment reported in Ref. [7], not a direct measurement of ion motion; the comparative cation data do not establish that assignment as a general law.

The 2-T design reduces fabrication complexity: the device is made by three depositions (ITO substrate, spin-coated active layer, evaporated metal), which makes the same stack compatible with the comparative corpus of Chapter 4. The coexistence of short-term memory and long-term memory without an independent gating signal also supports attribution of the measured memory to the active layer rather than to a gate field acting on a separate channel.

The bottom electrode is a pre-patterned layer of ITO deposited on borosilicate glass. ITO combines optical transparency with adequate electronic conductance ($\sigma \approx 10^4\,\mathrm{S\,cm^{-1}}$) [148], allowing optical inspection of the active layer and characterisation of the device stack by techniques such as UV-Vis spectroscopy and electroluminescence imaging. The work function of ITO ($\Phi_{\mathrm{ITO}} \approx 4.7\,\mathrm{eV}$) is well matched to the ionisation potential

of PPV-family semiconductors [149], facilitating hole injection into the active layer.

The top electrode is silver (Ag), deposited by thermal evaporation. Ag was selected over gold for reasons of cost and practical feasibility; its work function ($\Phi_{Ag} \approx 4.3\,\text{eV}$) creates a modest electron injection barrier with the *Super Yellow* conduction band, consistent with asymmetric injection conditions often observed in LEC-type structures [101]. The junction active area defined by the shadow mask geometry is $0.0825\,\text{cm}^2$.

2.3.2 Fabrication and Active-Layer Morphology

ITO-patterned glass substrates were cleaned by sequential ultrasonication, dried under nitrogen, and transferred into a nitrogen glovebox. The composite solution was spin-coated at 2000 rpm for 60 s and annealed at 75 °C for 3 h. Contact profilometry gave an active-layer thickness of 209 nm at this spin speed, and tapping-mode AFM showed smooth films with RMS roughness in the range 1 nm–3 nm across the investigated processing window [7]. The spin-speed optimisation, profilometry method, AFM parameters, and supporting morphology evidence are reported in section A.3.

The top electrode was 100 nm of Ag deposited by thermal evaporation through a shadow mask. The mask defines a junction area of $0.0825\,\text{cm}^2$. Detailed cleaning and evaporation parameters are collected in section A.1. The complete stack therefore requires solution processing followed by one metal evaporation step.

2.4 Electrical Characterisation

Electrical measurements used a Keithley 2450 SourceMeter in a custom probe station. Unless otherwise stated, measurements were made under nitrogen with voltage applied to the Ag top electrode relative to grounded ITO. Protocol-level details are given in section A.2.

The characterisation follows the two-family distinction of section 2.1. Sections 2.4.1 to 2.4.3 report the three dynamical measurements carried forward to Chapters 4 and 5; sections 2.4.4 and 2.4.5 then report the device-specific extended protocols, the EPSC multi-state readout and spike-timing dependent plasticity. The short- and long-term memory regimes (short-term memory/long-term memory) are read off the variable-delay depotentiation measurement and so appear with it in section 2.4.3.

2.4.1 Current-Voltage Hysteresis

A defining electrical response of a memristive device is a current–voltage (I–V) hysteresis loop pinched at zero voltage [60–62]. At $V = 0$, the net current is zero regardless of device history, while the *slope* of the I–V curve at zero voltage (i.e., the instantaneous conductance) depends on the history of applied voltage.

Figure 2.2 shows the I–V curves obtained by applying successive triangular voltage sweeps to the device at a sweep rate of $0.25\,\mathrm{V\,s^{-1}}$, with the voltage ramped from 0 V to a maximum of 1.2 V and back to 0 V (single polarity, positive). A waiting period of 10 s was observed between successive sweeps to allow partial ionic relaxation. Ten sequential sweeps are shown.

Progressive conductance trajectory. The current at any fixed voltage *increases* with each successive sweep, in separated increments across this sequence. At 1 V the forward current grows from 0.022 µA on the

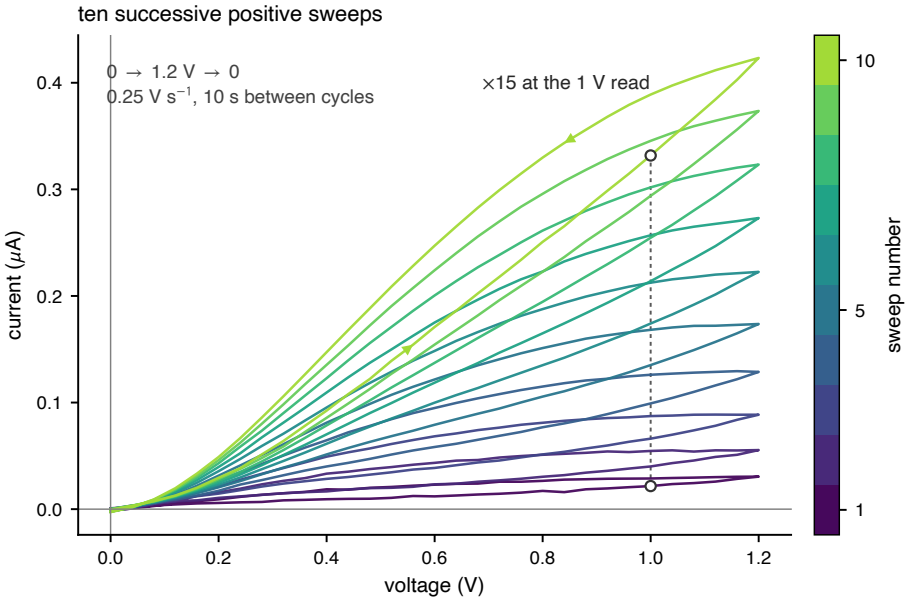


Figure 2.2: Current–voltage hysteresis of the composite memristive device. Ten successive positive triangular sweeps (0 V–1.2 V, 0.25 V s^{-1}, 10 s between cycles) fan out in near-uniform increments, the forward current at 1 V growing roughly 15-fold from the first to the tenth cycle. The clockwise loop closes at the origin (pinched hysteresis) without abrupt switching. Representative SY/Hybrane/LiOTf device (NM_v025) of the published 1:0.30:0.09 formulation, reproducing the hysteresis of Ref. [7].

first sweep to 0.33 µA on the tenth (a ratio of approximately 15 across ten cycles). The increase is +85 % from cycle 1 to cycle 2 and +13 % from cycle 9 to cycle 10. The trace demonstrates graded analogue evolution within one junction and protocol. Without repeated read distributions it does not establish a count of reliably addressable levels. Progressive ionic redistribution is one physical interpretation of the trajectory.

Hysteresis without current rectification. In all cycles, the return sweep (decreasing voltage) produces lower current than the forward sweep (increasing voltage) at the same voltage, generating the characteristic clockwise hysteresis loop for positive voltage sweeps. This is distinct

from simple rectification (diode behaviour): a diode would produce the same forward and reverse I–V curves; the hysteresis here reflects a time-dependent internal state variable (the ionic configuration) that lags the applied voltage.

Absence of abrupt switching. Unlike digital resistance switching (SET/RESET transitions), in which the resistance changes abruptly by orders of magnitude at a threshold voltage, the conductance here evolves continuously and smoothly with each cycle. This *analogue* trajectory is compatible with tuning between the endpoints, but the number, stability, and error rate of independently readable levels were not measured.

Behaviour under negative polarity. Application of negative voltage sweeps (not shown individually in the main figure) produces a qualitatively different hysteresis: the current decreases with successive negative cycles instead of increasing. Electrochemical side reactions near ITO, analogous to insulating-layer formation reported in MEH-PPV-based LECs [105, 108], are one candidate explanation; no chemical or interfacial measurement identifies that process in this device. The quantitative hysteresis comparison therefore uses the positive-polarity protocol.

2.4.2 Variable-Number-of-Pulses Potentiation

Potentiation by a train of write pulses is the second of the three dynamical measurements introduced at the start of this section. Two complementary experiments are reported: a reversible potentiation/depression cycle that establishes the analogue, multi-state character of the conductance, and a measurement of the conductance change as a function of the number of applied pulses.

Reversible Potentiation and Depression

The conductance can be set to a specific value by applying a controlled number of voltage pulses, and decreased back toward the initial value by pulses of opposite polarity, as the potentiation/depression experiment in fig. 2.3a shows.

A pulsed voltage protocol was applied: 50 consecutive write events at 1 V, followed by 50 at −2 V. In the archived waveform each event has an energised dwell of approximately 51 ms followed by a comparable zero-voltage interval, giving a cadence of approximately 0.103 s. The conductance was monitored at the peak of each pulse (i.e., at the 1 V and −2 V levels, respectively). This protocol is analogous to repeated pre-synaptic stimulation and subsequent inhibition.

The conductance is continuously potentiated from its initial value by more than 200 % during the 1 V pulse train, and subsequently depotentiated toward the initial value during the −2 V train. The potentiation and depression curves are smooth and monotonic, without abrupt transitions. Return to a value near the initial level after the depression protocol is consistent with reversible ionic redistribution within the measured protocol.

The write energy is state-dependent because the current rises as the junction potentiates. Estimating each active dwell as $E_k \simeq |V_k I_k| \Delta t_k$ from the archived voltage, current, and time arrays gives 0.084 µJ–1.55 µJ across the 50 positive events (median 1.08 µJ, mean 0.983 µJ, total 49.1 µJ). The first 50 negative events have a median of 0.567 µJ and total 37.3 µJ. The positive-event median corresponds to approximately 131 fJ µm^{-2} over the 0.0825 cm^2 junction. These are junction-level electrical estimates under a zero-order-hold interpretation of the sampled waveform; they exclude source, sensing, and control overhead. They also place this unoptimised large-area proof of concept above the picojoule-to-nanojoule

switching budgets of more mature organic and inorganic devices [69, 78]. The calculation and every event value are regenerated by `scripts/ch2_figures.py`; the summary is in `handouts/ch2_write_energy.csv`.

Quantitative Pulsed Read–Write Protocol

To quantify the potentiation as a function of pulse number, and to separate the act of writing from the act of reading, a non-perturbing read protocol was adopted; the same protocol is reused for the depotentiation measurement of section 2.4.3. The read–write–wait sequence, detailed in section A.2.2, comprises three steps:

1. **Read.** A reading pulse ($V_{\text{read}} = 0.5\,\text{V}$) is applied to establish the initial conductance G_0. This reading voltage is intentionally low: the activation threshold for ionic redistribution in this system was determined by independent experiments to lie above 0.7 V [7], so $V_{\text{read}} = 0.5\,\text{V}$ measures the conductance state without perturbing it.

2. **Write.** A train of N_{pulses} write pulses (V_{write}, duration 0.1 s each) is applied.

3. **Wait and read.** After a waiting time t_{wait} under open-circuit (high-impedance) conditions, a second reading pulse is applied to measure the final conductance G_f.

The conductance change is reported as the percentage ratio $\Delta G = (G_f/G_0) \times 100$.

Between experiments, the device is fully reset by connecting the junction to a short circuit for 300 s, which discharges the ionic double layers and restores the homogeneous equilibrium ion distribution. The suitability of this reset protocol was verified by confirming that G_0 returns to its original value within the measurement uncertainty before each new experiment.

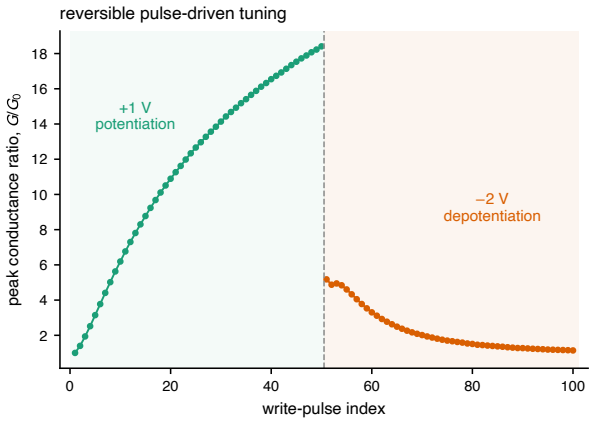
Potentiation as a Function of Pulse Number

Figure 2.3b shows the dependence of ΔG on N_{pulses} for $V_{\text{write}} = 1\,\text{V}$ and $t_{\text{wait}} = 150\,\text{ms}$. The conductance ratio increases monotonically with N_{pulses}, reaching values in excess of $10^3\%$ for $N_{\text{pulses}} > 10^3$. The relationship is approximately linear in log-log space for $N_{\text{pulses}} < 100$ and saturates at large pulse numbers, consistent with a scenario in which the degree of ionic asymmetry that can be built up under a given write voltage is bounded by the total number of mobile ionic carriers.

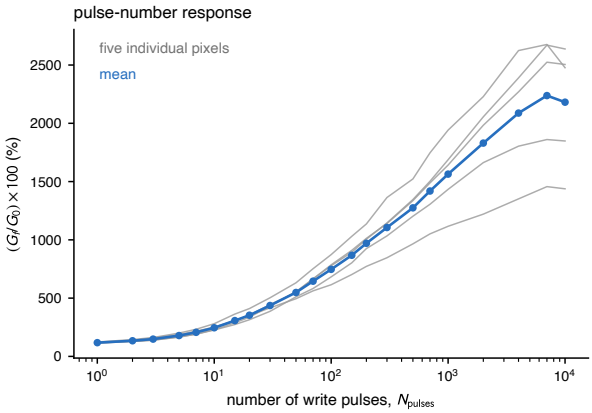
Within this archived pulse-train dataset, the low-pulse-count response varies by approximately 10 % across junctions on one substrate: for the five pixels in fig. 2.3b, the median relative standard deviation is 8.9 % and the largest value is 11.8 % over the points with $N_{\text{pulses}} < 100$. The dispersion increases modestly at large pulse numbers. These values establish within-substrate junction repeatability only; they do not estimate independent-device or batch variability.

2.4.3 Variable-Delay Depotentiation: Short- and Long-Term Retention

The third dynamical measurement carried through the thesis probes how a written conductance state fades once the drive is removed: the depotentiation of the device measured as a function of the delay before read-out. This protocol determines whether the device can emulate short-lived and longer-lived synaptic states on experimentally accessible timescales. In the artificial-synapse literature these regimes are commonly labelled short-term memory and long-term memory; in the present system, the longer-lived regime remains on the sub-minute scale and should therefore be understood operationally, not as a claim of indefinitely persistent storage.



(a) Reversible potentiation/depression.



(b) Pulse-number response.

Figure 2.3: Pulse-driven conductance integration in the composite memristive platform. **(a)** Fifty consecutive 1 V write pulses continuously potentiate junction L5 on NM_v055; subsequent −2 V pulses depotentiate it toward the initial state. **(b)** Dependence of the conductance ratio $(G_f/G_0) \times 100\%$ on the number of write pulses for $V_{\text{write}} = 1\,\text{V}$ and $t_{\text{wait}} = 150\,\text{ms}$. Thin grey lines are five crossbar junctions on the same NM_v055 substrate; the bold line is their mean. The spread therefore measures within-substrate junction variability, not device- or batch-level reproducibility. Data regenerated from the local archive associated with Ref. [7].

The measurement reuses the pulsed read–write protocol of section 2.4.2, with the waiting time t_{wait} now the swept variable and the pulse number held fixed. Figure 2.4 shows the dependence of ΔG on t_{wait} for three excitation conditions: (i) $V_{\text{write}} = 1\,\text{V}$, $N_{\text{pulses}} = 10$ (STM, blue); (ii) $V_{\text{write}} = 1\,\text{V}$, $N_{\text{pulses}} = 50$ (STM, red); and (iii) $V_{\text{write}} = 3\,\text{V}$, $N_{\text{pulses}} = 10$ (LTM, black).

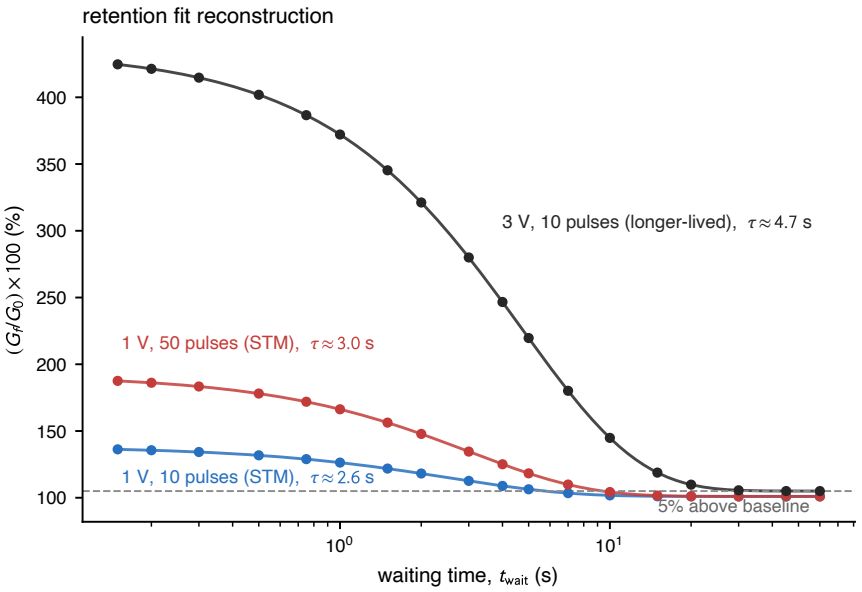


Figure 2.4: Memory retention kinetics. Conductance ratio $(G_f/G_0) \times 100\%$ versus waiting time t_{wait} for STM conditions (1 V; 10 and 50 pulses, blue/red) and the longer-lived regime denoted LTM (3 V, 10 pulses, black). Solid lines are standard-exponential reconstructions of the form in eq. (2.1); the fit parameters are summarised in section A.2.4. Data and fit parameters from Ref. [7].

For short waiting times ($t_{\text{wait}} < 500\,\text{ms}$ for STM conditions, $t_{\text{wait}} < 2\,\text{s}$ for LTM conditions), the conductance first *increases* slightly relative to the value immediately after the write pulse train. This counterintuitive behaviour is *not* ascribed to ionic inertia: ion transport in the polymer matrix is strongly overdamped, so any momentum imparted by the field relaxes on a timescale orders of magnitude shorter than the observed

enhancement. It is instead attributed to the continued redistribution of ions after the drive is removed. At the end of the pulse train the ionic configuration and its self-consistent space-charge field have not yet reached their quasi-steady profile; the residual internal field, together with the slow advance of the electrochemically doped region near the electrode, continues to raise the conductance for a brief interval before the diffusion-driven relaxation sets in. An analogous transient enhancement following a stimulus train has been described as "post-tetanic potentiation" in biological synaptic preparations [43, 44].

For longer waiting times, the expected exponential relaxation of conductance is observed as the ionic configuration relaxes back toward equilibrium by thermal diffusion. The fit record summarised in section A.2.4 reports least-squares fits in IGOR Pro 8 to a standard exponential:

$$\frac{G_f(t_{\mathrm{wait}})}{G_0} = R_\infty + A\exp\!\left(-\frac{t_{\mathrm{wait}}}{\tau}\right) \tag{2.1}$$

where τ is the characteristic relaxation time, A is the decaying amplitude, and R_∞ is the long-waiting-time conductance ratio. Equivalently, the published proof-of-concept fit corresponds to a stretched-exponential model with the stretch exponent fixed at $\beta = 1$. No free β, confidence interval, or goodness-of-fit statistic was reported for these three traces, so they establish only the two experimentally resolved relaxation times, not a fitted distribution of relaxation barriers; the free-exponent Kohlrausch descriptor is reserved for chapter 4, where the DELAYTIME archive is refitted consistently across the corpus.

The fitted characteristic times are $\tau_S = 2.57\,\mathrm{s}$ for the 1 V, 10-pulse STM trace, $\tau_S = 3.01\,\mathrm{s}$ for the 1 V, 50-pulse STM trace, and $\tau_L = 4.73\,\mathrm{s}$ for the 3 V, 10-pulse LTM trace. The corresponding offsets and amplitudes are tabulated in section A.2.4. The retention time (defined operationally as the waiting time at which G_f/G_0 returns to within 5 % of the baseline,

$G_f/G_0 < 1.05$) is 10 s–15 s for STM and exceeds 45 s for the longer-lived regime labelled LTM. At event intervals shorter than these windows, the device state can retain information from multiple sequential inputs. The measured retention remains far shorter than that of a conventional non-volatile memory.

The molecular mechanism underlying the STM/LTM distinction is discussed in detail in section 2.5.

2.4.4 Excitatory Post-Synaptic Current

The EPSC measurement is a standard protocol for reporting pulse-conditioned current states in artificial synaptic devices [24, 92]. Here it establishes the response of one reported sequence; the recovered archive contains no repeated state distributions from which to estimate read-error rates, stability, or level capacity. The nomenclature is a functional analogy to a biological experiment, not an assertion of biological equivalence.

The measurement protocol is as follows (fig. 2.5a):

1. **Initialisation.** The device is initialised to its ground state S_0 by applying a fixed read-out bias $V_0 = 1\,\mathrm{V}$ for 1 s. This bias differs in purpose from the sub-threshold probe ($V_{\mathrm{read}} = 0.5\,\mathrm{V}$) used for the non-perturbing read-out of sections 2.4.2 and 2.4.3: the EPSC states are defined operationally by the current response at a single common read-out bias against which all states S_0–S_4 are referenced, not by a passive probe required to leave the state untouched.

2. **Excitation.** A short positive write pulse of 2 V for 50 ms is applied; the device is subsequently held at 1 V for 1 s, during which the conductance spontaneously relaxes to a new excited steady state S_1. The same sequence is repeated to produce a second excited state S_2.

3. **Inhibition.** Two negative write pulses of $-2.5\,\mathrm{V}$ for $50\,\mathrm{ms}$ are applied, producing inhibitory states S_3 and S_4.

The four excited-state readouts are numerically separated from each other and from the ground-state value in the reported sequence. Defining the EPSC readout ratio $R_n = I_{S_n}/I_{S_0}$ as the signed current response of state S_n normalised to that of the ground state S_0 under the fixed read-out bias used in the experiment, the values obtained are: $R_1 = 7.33$, $R_2 = 15.55$, $R_3 = -3.72$, $R_4 = -16.97$. (The ratio is formed from signed currents, not conductances, so the inhibitory states carry a negative sign; conductance itself remains non-negative throughout.) These separations demonstrate the programmed signed response in that sequence. They do not establish reliable state identification under noise or imply additional resolvable levels without repeated writes and reads, retention distributions, and an explicit decision criterion.

The states S_1 and S_2 are reached by positive-polarity stimulation, while S_3 and S_4 are reached by negative-polarity stimulation. Calling these responses excitatory and inhibitory is an operational analogy to long-term potentiation and depression at biological synapses. Both signs are programmed in the same 2-T architecture without an independent gate terminal.

2.4.5 Spike-Timing Dependent Plasticity

Spike-timing dependent plasticity is a timing-dependent Hebbian learning rule: it implements Hebb's local correlation principle [42] in a form that was later observed experimentally in biological preparations [55, 56]. Here the timing delay is defined as $\Delta t = t_{\mathrm{post}} - t_{\mathrm{pre}}$. With this convention, if the presynaptic neuron fires shortly *before* the postsynaptic neuron (causal ordering, $\Delta t < 0$), the synapse is potentiated (LTP); if the presynaptic neuron fires shortly *after* the postsynaptic neuron

(anti-causal ordering, $\Delta t > 0$), the synapse is depressed (LTD). The magnitude of the change decays exponentially with $|\Delta t|$. This temporal asymmetry is essential for the synaptic encoding of causal relationships in neural circuits and underlies the ability of STDP-based learning rules to implement unsupervised feature extraction and pattern recognition [79].

Experimental Protocol

The STDP experiment emulates pre- and postsynaptic spikes by programming their signed resultant across the two-terminal junction. With each template referenced to its own nominal firing time, the applied device voltage is

$$V_{\text{device}}(t, \Delta t) = V_{\text{pre}}(t) + V_{\text{post}}(t - \Delta t) \tag{2.2}$$

Each signed spike template has maximum amplitude $V_{sp} = 1.55\,\text{V}$ and duration 1.2 s. The delay Δt was varied from $-600\,\text{ms}$ to $600\,\text{ms}$. Their overlap makes the resultant depend on timing; the archived applied waveform reaches a transient magnitude of $3.1\,\text{V}$ at $|\Delta t| = 50\,\text{ms}$. Thus $1.55\,\text{V}$ is the template amplitude, not a bound on the voltage experienced by the device.

For $\Delta t < 0$ (pre fires before post), the two spike waveforms partially overlap in time with additive polarity, producing a composite pulse whose peak voltage exceeds V_{sp} and which is directed in the positive sense. The ions then redistribute toward the ITO electrode, the conductance increases, and the device records a net synaptic potentiation.

For $\Delta t > 0$ (post fires before pre), the overlap produces an anti-causal summation. Depending on the exact waveforms, the effective net voltage can be of reduced amplitude or of reversed polarity relative to the causal case, so the ionic redistribution is either weaker or reversed and the

conductance decreases, in the form of a net synaptic depression.

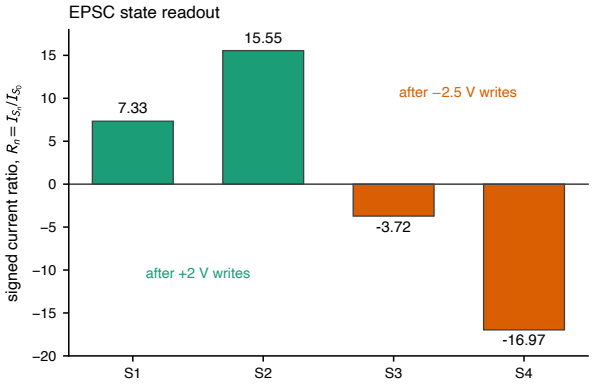
STDP Curve and Fitting

Figure 2.5b shows the measured STDP function: the percentage conductance change ΔG plotted against the timing delay Δt. Under the sign convention above, the measured kernel is asymmetric Hebbian: negative Δt (causal) produces positive ΔG (potentiation), and positive Δt (anti-causal) produces negative ΔG (depression). Each branch (potentiation for $\Delta t < 0$, depression for $\Delta t > 0$) is fitted separately, as a function of the absolute delay $|\Delta t|$, by a decaying exponential:

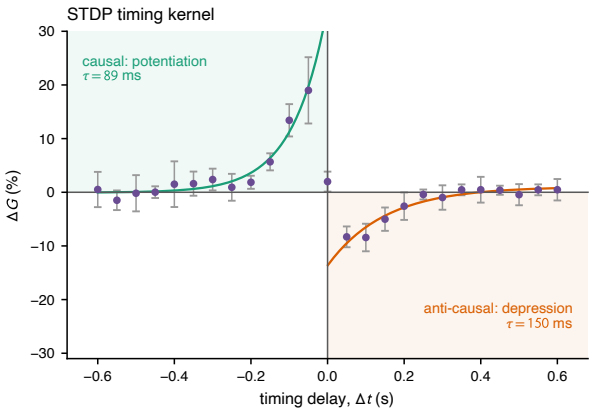
$$\Delta G(\Delta t) = A_{\pm} \exp\left(-\frac{|\Delta t|}{\tau_{\pm}}\right) + G_0 \tag{2.3}$$

where $A_{\pm}$ and $\tau_{\pm}$ are the branch-specific amplitude and time constant ($+$ for the potentiation branch, $-$ for the depression branch) and G_0 is a constant offset. A direct least-squares refit of the archived mean across five junctions gives $\tau_{+} = 89\,\text{ms}$ for potentiation and $\tau_{-} = 150\,\text{ms}$ for depression. Resampling those five same-substrate junctions gives 95% bootstrap intervals of 78 ms–95 ms and 134 ms–172 ms, respectively. These intervals quantify junction spread on one substrate; they are not confidence intervals for an independently fabricated device population. Both branches occupy the tens-to-hundreds-of-milliseconds scale reported in biological STDP experiments [56, 59], but matching a timescale does not make the artificial waveform or ionic mechanism biological.

That the learning rule is *asymmetric Hebbian* reflects the specific voltage waveform design, the stated definition of Δt, and the polarity conventions of the electrode configuration. Different waveform designs or an opposite sign convention would be expected to produce differently labelled STDP kernels, offering an additional degree of tunability in the design of



(a) EPSC state ratios.



(b) STDP timing kernel.

Figure 2.5: Compact summary of the extended synaptic measurements. **(a)** Signed EPSC readout ratios $R_n = I_{S_n}/I_{S_0}$ for the four demonstrated states, reconstructed from the published values; the recovered archive does not resolve their specimen identifier. **(b)** Spike-timing dependent plasticity: mean percentage conductance change ΔG across five junctions on the NM_v055 substrate as a function of the pre/post spike-timing delay Δt, with each branch re-fitted to the decaying exponential of eq. (2.3); error bars are one standard deviation across junctions. The resulting time constants are 89 ms (potentiation) and 150 ms (depression). Data and reported state ratios from Ref. [7]; STDP values and fits regenerated from the local NM_v055 Day19 archive.

hardware neural network architectures built from these devices.

2.5 Theoretical Framework: Ion Transport and the Origin of Memristance

Two limiting-case models from the polymer light-emitting electrochemical cell (LEC) literature provide candidate descriptions of injection and bulk transport: an *electrodynamic* (injection-limited) model [101, 102] and an *electrochemical-doping* (ohmic) model [100, 103, 140], which van Reenen *et al.* [104] reconciled as regimes of field-driven ion redistribution. Their assumptions are set out in section A.4; neither was discriminated experimentally in the present junctions.

For the present SY/Hybrane/LiOTf system, both limiting cases could contribute. The published working interpretation maps the rapid response to anion redistribution and an injection-limited contribution, and the larger, slower response to Li^+ redistribution and bulk electrochemical doping. The electrical measurements in section 2.4 identify neither the moving species nor the relative ED/ECD contributions. Such discrimination would require, for example, ion-selective or spectroelectrochemical measurements and matched temperature-, thickness-, or sweep-rate dependence. Moreover, the later cation series shifts the dynamics but runs opposite to the simple coordination-strength order under triflate and reorders with the anion (chapter 4). The mapping is therefore retained as a candidate interpretation, not an established mechanism.

2.5.1 Candidate Coordination Rationale and Its Limits

The candidate explanation for the two memory regimes starts from the coordination reasoning developed in sections 1.6.1 and 1.6.4, applied to the specific salt and host of this device.

Within the HSAB framework, Li^+ is a small hard Lewis acid and ether oxygen is a hard Lewis base [91]. In crystalline polyether complexes, Li^+ can be coordinated by four to six oxygen donors (five in PEO_3:$LiCF_3SO_3$: three ether oxygens and two triflate oxygens), with oxygen–lithium distances of 1.9 Å–2.1 Å [129]. Those structural facts support the plausibility of cation–oxygen coordination. They do not provide the migration barrier in the amorphous ternary *Super Yellow*/*Hybrane*/LiOTf film, and a local binding preference must not be equated with a macroscopic drift mobility.

Triflate is charge-delocalised and commonly described as weakly coordinating in polymer electrolytes [131, 150]. That makes greater anion mobility plausible, but does not determine it in this blend: ion pairing, polymer segmental motion, concentration, morphology, electrode reactions, and field can change both transference and activation barriers.

The resulting working hypothesis assigns the two drive regimes as follows:

- A voltage of 1 V across a 209 nm film corresponds to approximately $4.8\,\mathrm{MV\,m^{-1}}$. The candidate assignment associates the resulting 10 s–15 s retention with predominantly triflate redistribution; ion motion was not observed directly.

- A voltage of 3 V corresponds to approximately $14.4\,\mathrm{MV\,m^{-1}}$. The candidate assignment associates the retention beyond 45 s with an added cation contribution, consistent with stronger Li^+–oxygen coordination but not uniquely implied by it [129, 131].

This account yields a specific prediction: at fixed host, anion, composition, electrode, and protocol, the slow fading-memory timescale should shorten along $Li^+ \rightarrow Na^+ \rightarrow K^+$. Chapter 4 evaluates that prediction in a PEO-based survey. The measured triflate retention runs in the opposite order, and the ordering changes with the anion; the relevant chemistry cells are also sample-limited. The comparative evidence therefore does not corroborate a general cation–coordination law. It leaves the species assignment in this proof-of-concept device unresolved and identifies the matched, replicated experiment required to test it.

2.6 Device Stability and Reproducibility

2.6.1 Ambient Shelf Stability

Two distinct notions of device longevity must be kept separate. *Cycling endurance* (the number of write/erase cycles a device sustains before its performance parameters drift outside specified tolerances) governs suitability for the millions-to-billions of weight updates of on-chip training; it is not characterised for the present devices and is left for future work. *Shelf (calendar) stability* (the retention of performance during storage and intermittent operation under ambient conditions) is what is reported here, the more immediately relevant metric at the proof-of-concept stage, where the question is whether the unencapsulated composite survives realistic handling at all.

The devices reported here were characterised at regular intervals over a period of two weeks ($\sim$300 h) after fabrication, without encapsulation, while stored in ambient air at room temperature. The monitored metrics were the baseline conductance G_0, the maximum achievable conductance change $\Delta G_{\max}$, and the STM retention time τ_S; the supporting stability record is collected in section A.3.

All three metrics remain within 20 % of their initial values over the full two-week measurement period [7]. This stability compares favourably with other organic memristive devices reported in the literature, which typically exhibit significant degradation within hours to days under ambient conditions [121]. The enhanced stability is attributed to the solid-state nature of the composite: unlike devices based on liquid electrolytes or gel polymers, the Hybrane matrix provides a mechanically cohesive environment that protects the ions from evaporation and the polymer films from delamination, consistent with the stability of related Hybrane-based LEC blends [106].

2.6.2 Within-Substrate Junction Repeatability

Reproducibility is a persistent challenge for memristive devices, particularly those based on filamentary switching in inorganic materials, where the stochastic nucleation of conducting filaments produces large device-to-device and cycle-to-cycle variability [72]. Three evidence units must be distinguished here: repeated cycles on one junction, different junctions on one patterned substrate, and independently fabricated devices or batches. The archived proof-of-concept data quantify only the first two.

The strongest repeatability statement available from the archived proof-of-concept data is the one attached to the pulse-number experiment in fig. 2.3b. That dataset contains five crossbar junctions on the NM_v055 substrate, measured under the same 1 V write and 150 ms wait protocol. The shaded band is one standard deviation across those five junctions. For $N_{\mathrm{pulses}} < 100$, the median RSD is 8.9 % and the largest RSD is 11.8 %; the appropriate claim is therefore within-substrate junction repeatability at about the 10 % level in the low-pulse-count regime, not an independent-device reproducibility estimate.

The EPSC readout and STDP measurements are retained as protocol

demonstrations, but the recovered archive does not contain a normalized EPSC population table comparable to the PULSES and DELAYTIME tables used later in the thesis. They are therefore not used to support an independent EPSC RSD claim here. Likewise, the published proof-of-concept data establish same-substrate junction repeatability and two-week shelf stability; they do not constitute a powered device-to-device or batch-to-batch reproducibility study. Campaign-scale reproducibility (a separate metric on a separate axis) and its resolution by a change of ion-transport host are taken up in chapter 3.

2.7 Comparison at the Time of Publication

The historical comparison table reproduced in table A.3 places this platform alongside representative organic and hybrid memristive devices available at the time of the 2022 publication. None of the devices in that set combines fully reversible operation with both short-term memory- and long-term memory-labelled regimes in the same 2-T formulation and architecture. The closest prior report, Liu *et al.* [96], achieved short-to-long-term transitions through irreversible redox chemistry [118], whereas here the conductance returns to G_0 after the reset protocol. The table also contains no other example of STDP in a 2-T architecture, where the timing rule emerges from the superposition of pre- and postsynaptic pulses at the Ag terminal rather than from the gate of a 3-T device [92, 107, 117]. On the operating-regime axes the platform is competitive in unencapsulated shelf lifetime ($\sim$300 h, roughly an order of magnitude beyond comparable organic devices [121]) but not in energy per event: the archived positive train has a median of 1.08 µJ, above CMOS and mature inorganic-memristor budgets [28, 69]. The full table and a per-axis discussion are given in section A.5.

The comparison above retains the device-level metrics conventional in the

organic memristive literature of the early 2020s: reversibility, retention window, energy per event, and operational lifetime. Subsequent work has expanded the field through organic electrochemical neurons and neuromorphic circuits [151, 152] and the wider organic-iontronic memristor field [153], so the table locates the 2022 result historically. Within the thesis argument those metrics play a narrower role: they establish that the platform is reversible, dynamically addressable, and stable enough to support the comparative study of Chapter 4 (a PEO-based triflate composition and cation sweep, characterised through the same three common dynamical measurements) and the temporal-computing reinterpretation of Chapter 5. The extended synaptic protocols reported here remain specific to this SY/Hybrane/LiOTf device.

2.8 Summary

The fully reversible 2-T organic memristive platform supports two sets of synaptic measurements with the same polymer-composite formulation and stack. I–V hysteresis, variable-number-of-pulses potentiation, and variable-delay depotentiation carry through the rest of the thesis. EPSC, the separated short-term memory/long-term memory reading of the retention data, and STDP remain specific to this proof-of-concept formulation.

A ternary *Super Yellow*/*Hybrane*/LiOTf composite satisfies the three functional requirements of section 2.2.1 (electronic pathway, ion-transport medium, ion source), fabricated as a 2-T ITO/composite/Ag sandwich by solution processing and one evaporation step, with a 209 nm active layer below 3 nm RMS roughness. On this common platform the three dynamical measurements carried forward to the comparative corpus were established: analogue I–V hysteresis (forward current at 1 V growing ∼15-fold over ten cycles); fully reversible pulse potentiation/depression exceeding

200 %, growing past 10^3% for $N_{\text{pulses}} > 10^3$; and variable-delay depotentiation resolving two regimes (short-term memory, $\tau_S = 2.57\,\text{s}$–$3.01\,\text{s}$, retention 10 s–15 s; long-term memory, $\tau_L = 4.73\,\text{s}$, retention $> 45\,\text{s}$, from $\beta = 1$ fits). The archived positive-pulse train consumes 0.084 µJ–1.55 µJ per event as the state changes (median 1.08 µJ). The extended protocols add one numerically separated four-state EPSC sequence (signed ratios −17 to +16, without a population-level read-error estimate) and an asymmetric-Hebbian STDP window whose re-fitted branch constants are 89 ms and 150 ms. A fast-triflate/slow-lithium account and the ED/ECD limiting cases provide a candidate interpretation of the two retention regimes [104] (section A.4); the electrical data do not identify the mobile species, and the later cation survey does not confirm the predicted simple ordering. The published shelf record spans ∼300 h unencapsulated in ambient conditions (cycling endurance not characterised), while the NM_v055 pulse data show ∼10 % low-pulse-count variability across five same-substrate junctions (median RSD 8.9 %); campaign-scale reproducibility is taken up in chapter 3.

The SY/Hybrane/LiOTf formulation and 2-T architecture provide the proof-of-concept platform for the replicated test of cation–polyether coordination in chapter 4. Composition is the replicated quantitative axis in that PEO-based corpus, while host, anion, and cation substitutions remain sample-limited; the data do not establish a host- and anion-independent Li/Na/K law. The EPSC, separated short-term memory/long-term memory retention, and STDP measurements reported here serve only as formulation-level lithium priors in the later chapters. Chapter 5 uses behavioural models fitted to the comparative corpus to test the computational value of the measured fading memory.

Chapter 3

Reproducibility, Degradation, and the Fabrication-Record Methodology

3.1 The Reproducibility Problem

Chapter 2 established the memristive response of the formulation comprising *Super Yellow*, *Hybrane*, and LiOTf across assay-specific junctions and reported two-week ambient shelf stability. In the subsequent fabrication campaign, memristive hysteresis became progressively harder to obtain. Successive batches of nominally identical devices shifted from the low-current, high-area, multi-level response of the proof-of-concept platform toward high conductivity, a narrow switching window, and no usable potentiation. This chapter analyses that loss of reproducibility and the subsequent change of host before chapter 4 examines a replicated composition grid.

Individual-device shelf stability and campaign-scale reproducibility measure different processes. The proof-of-concept shelf record retained its

characteristics over days to weeks, as did other finished *Hybrane* devices. The fabrication line nevertheless changed between batches over the following year. This places the degradation in the supply or fabrication process over calendar time, while the finished devices remained stable over the measured shelf interval; section 3.3 quantifies the distinction.

The fabrication and characterisation notes identify devices NM_v026–031 (2021-04-22) as a change point, recorded at the time as "the first devices that changed electrical behaviour". The contemporaneous hypotheses remained open ("could be evaporation rate … the metal used for evaporation, and possible chemical degradation of substances") and were still unresolved in a note dated 2022-01-31, after nine months of investigation. The analysis uses this documented change point as a pre/post boundary and does not assume a smooth trend. It tests the candidate causes, quantifies the degradation under four controls, and evaluates the change to PEO.

3.2 Eliminating Candidate Causes

The team tested the candidate causes one at a time in batches with the other variables held fixed. Table 3.1 lists each cause, the batches that tested it, and the resulting verdict. This process also motivated the fabrication-record library used later in the thesis, since each test required the process variables to be recorded per device.

Device NM_v067 (2021-10-21) records that a second experimenter, working independently, "has the same problems with devices". This cross-person result excludes individual technique as the cause and indicates a shared material or process source. The same note proposed the ITO substrate, but the substrate study did not support that explanation. The *Super Yellow* semiconductor was also tested because an old bottle was

Table 3.1: Methods-of-elimination summary for the degradation investigation. Candidate causes were tested in deliberate batches with variables recorded in the fabrication record. After the exclusions, the hyperbranched *Hybrane* host stock remains the candidate examined quantitatively in section 3.3.

Candidate cause	**Crossed as (batches)**	**Verdict**
Operator / technique	cross-person check (v067): "Lorenzo has the same problems with devices"	**Eliminated.** The same result occurred with a second experimenter, excluding one operator's technique.
Illumination (light vs dark)	v015–019, vt001–006, v041–044, v053–056	Excluded (no systematic recovery of the window).
Storage / measurement atmosphere (glovebox / air / vacuum)	v015–019, v088–099	Excluded.
Baby-chamber transport (full 2×2 on Au)	v083–087, v038–040	Excluded.
Substrate (ITO) lot / colour	v079–082; substrate-difference study	Not borne out (a one-off ITO guess by a colleague did not survive the substrate comparison).
Solvent (cyclohexanone) lot	v079, v091	Excluded.
Evaporation rate / silver holder	v045–048, v106–113	Recovery *attempted* by reverting to the "year-ago" holder; not reproducibly restored.
Annealing regime	late hot-anneal batches	*Partial* window recovery (section 3.3); consistent with the chemical hypothesis but insufficient as a process correction.
Super Yellow stock (old vs new)	v045–048 ("more viscous ... maybe bonds degradation in the old one")	Eliminated as primary (old vs new *Super Yellow* gave no significant change, and fresh *Super Yellow* continued to be purchased while the window loss persisted).
***Hybrane* host stock**	residual after the above	**Remaining candidate**, examined quantitatively and mechanistically below.

noticeably "more viscous … maybe because of bonds degradation in the old one". The old-versus-new *Super Yellow* comparison produced no significant change, and devices retained the reduced window while fresh *Super Yellow* was used. After illumination, atmosphere, transport, substrate, solvent, evaporation, and semiconductor were tested and excluded, the hyperbranched *Hybrane* host stock remained the candidate consumable that had aged between batches. The remainder of the chapter tests that attribution quantitatively and proposes a candidate chemistry.

3.3 The Quantitative Degradation Signal

The device archive provides a quantitative test of the degradation recorded in the laboratory notes. The analysis controls for the sampling and protocol changes within the campaign and is reproduced in full by `scripts/bridge_hybrane_peo_reproducibility.py`; it mirrors the slicing of the group's interactive feature-explorer tool.

Four controls, each shown to matter. The analysis rests on four controls, each verified to change the conclusion if dropped: *per-device weighting* (the pixels-per-device count falls with date, Spearman $\rho = -0.80$, so every feature is reduced to one value per device before any test); a *standard-protocol corpus* ($n \approx 69$ devices, lead mass ratios, excluding the invasive recovery experiments); *sweep-amplitude matching* (the window features are read in the $\sim$3 V stratum where they are expressed, the conductivity features in a tightly matched $\sim$1.2 V band); and a *pre/post-inflection step test* fixed at the documented NM_v026 inflection, with every conductivity trend reported with and without the deliberately-reverted recovery batches. The four controls are stated in full, with the supporting correlations, in section B.1.

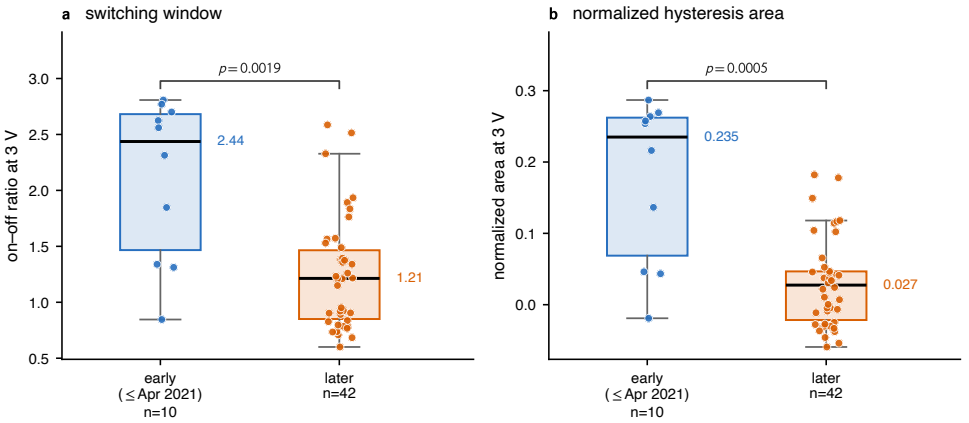


Figure 3.1: Switching-window reduction across the campaign on the standard *Super Yellow*/*Hybrane*/LiOTf/Ag corpus, read at the ∼3 V stratum where the window is expressed, one value per device. (a) On–off ratio and (b) normalized hysteresis area, both for devices fabricated before the NM_v026 change point (≤Apr 2021) versus later. Boxes are quartiles with the median annotated beside each box; points are individual devices; the bracket gives the two-sided Mann–Whitney p. The on–off ratio decreases by roughly a factor of two and the area by an order of magnitude. Reproduced by `scripts/bridge_figures.py` (F1).

Principal result: the switching window decreases at ∼3 V. On the standard corpus, per device, the switching window measured in the ∼3 V stratum decreases across the change point (fig. 3.1, table 3.2). The median on–off ratio falls from 2.44 (early, $n = 10$) to 1.21 (later, $n = 42$; Mann–Whitney $p = 0.0019$); the median normalized hysteresis area falls from 0.235 to 0.027, an order of magnitude ($p = 0.0005$); and the potentiation, measured as the per-cent change in the maximum-voltage current over a sweep, changes from +17.9 % to −2.1 % ($p = 0.013$). The on–off ratio, hysteresis area, and potentiation thus identify the same loss of switching function. The magnitude agrees with the team's recollection of an early on–off near 2.5 falling to the 1–1.7 range later, now tied to the documented change point.

Table 3.2: The degradation as a multi-feature signal on the standard corpus, per device. *Window reduction* is read at ∼3 V as an early-vs-later step at the NM_v026 change point (Mann–Whitney). *Ohmic drift* is read at matched ∼1.2 V as a per-device Spearman correlation with fabrication date, reported with all dates and with the recovery tail (≥2022-03) dropped; the correlation increases when the reverted batches are removed, as expected if they are partial recoveries. Health flags are included and discussed in the text.

Window reduction @ ∼3 V (early ≤Apr 2021, $n = 10$ vs later, $n = 42$)			
Feature	early	later	Mann–Whitney
on–off ratio	2.44	1.21	$p = 0.0019$
normalized area	0.235	0.027	$p = 0.0005$
% change in max-V current	+17.9 %	−2.1 %	$p = 0.013$
Ohmic drift @ matched ∼1.2 V (per-device Spearman vs date, $n = 64$)			
Feature	all dates	drop recovery tail	
current at max V	$\rho = +0.45$ ($p = 0.0002$)	$\rho = +0.55$ ($p < 10^{-4}$)	
hysteresis area (V·μA)	$\rho = +0.57$ ($p < 10^{-4}$)	$\rho = +0.69$ ($p < 10^{-4}$)	
current difference at on–off	$\rho = +0.51$ ($p < 10^{-4}$)	$\rho = +0.63$ ($p < 10^{-4}$)	
Health flags (per-device, ∼1.2 V, $n = 64$)			
`is_broken` *decreases* ($\rho = -0.40$, $p = 0.001$), a handling metric, *not* degradation; `is_saturated` (fails to potentiate) *rises* ($\rho = +0.27$, $p = 0.03$).			

Mechanism: ohmic drift at matched ∼1.2 V. Conductivity provides a complementary signal, read in the tightly matched ∼1.2 V band so that the sweep-amplitude confound is held fixed. The absolute current rises with fabrication date: per device, the current at maximum voltage correlates with date at $\rho = +0.45$ ($p = 0.0002$), the raw hysteresis area at $\rho = +0.57$, and the current difference at the on–off point at $\rho = +0.51$ (all $n = 64$; fig. 3.2a, table 3.2). These correlations increase when the deliberately reverted recovery batches (≥2022-03) are dropped (to $\rho = +0.55$, $+0.69$, and $+0.63$ respectively), as expected if those late batches are partial recoveries. Because the amplitude is held within a narrow band, the trend indicates more ohmic conduction independently of sweep range. The reduction in relative hysteresis and the increase in

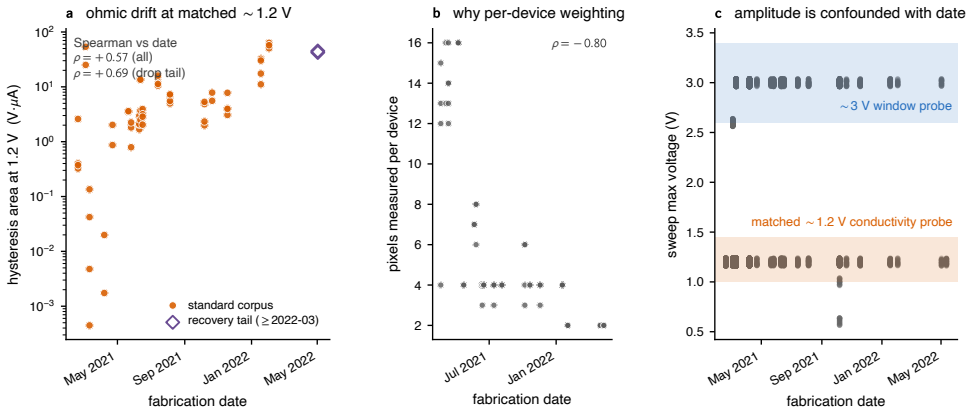


Figure 3.2: Mechanism and methodology. (a) Per-device hysteresis area read at the matched ∼1.2 V band rises with fabrication date (Spearman $\rho = +0.57$ over all dates, strengthening to $+0.69$ when the recovery-tail batches, open diamonds, ≥2022-03, are dropped; the statistics use all $n = 64$ devices, while the log axis cannot display the few non-positive areas). (b) The number of pixels measured per device falls over the campaign ($\rho = -0.80$), which is why every statistic is computed per device rather than per pixel. (c) Sweep amplitude is itself confounded with date: the matched ∼1.2 V conductivity probe and the ∼3 V window probe (shaded) are the two strata used throughout, never pooled. Reproduced by `scripts/bridge_figures.py` (F2).

absolute current are two measurements of the same degradation. The magnitudes are reported as correlations and medians because absolute current depends on sweep amplitude.

Interpretation of the health flags. The fraction of broken devices would mischaracterize the degradation. The curve-level `is_broken` flag (erratic or open-circuit behaviour caused by contact or handling problems) decreases over the campaign ($\rho = -0.40$, $p = 0.001$) as handling and contacting improved. The `is_saturated` flag (a curve whose maximum current fails to exceed the previous one, indicating loss of potentiation) rises with date ($\rho = +0.27$, $p = 0.03$). The two flags are reported separately because they measure different processes.

3.3.1 A Candidate Chemistry: Moisture-Driven Hydrolytic Aging of the *Hybrane* Stock

The available record supports attribution to the stock but cannot establish a chemical mechanism. No electrochemical impedance spectra were recorded on a *Hybrane* device; no gel-permeation chromatography, nuclear-magnetic-resonance, or Karl-Fischer analysis was performed on the reagent stock; and the aged stock no longer exists in its degraded state. The proposed moisture-driven hydrolysis mechanism therefore remains a hypothesis, accompanied by experiments that could test it.

The hypothesis. *Hybrane* is a hyperbranched polyester-amide [130, 138]: ester- and amide-rich, and strongly hygroscopic. The leading hypothesis is that, over roughly a year in the bottle, the stock takes up atmospheric water and slowly hydrolyses (chain scission), and that this process could account for the measured electrical signatures without requiring a separate finished-device ageing mechanism.

- **Ohmic drift** (current $\uparrow$ at matched 1.2 V): absorbed water together with polar scission products raises the baseline ionic and leakage conductivity and plasticises the matrix.

- **Window reduction** (on–off $\rightarrow\sim$1.2, area $\rightarrow\sim$0.03, $\%\Delta I \rightarrow -2\,\%$): in the metastable-polarisation model of chapter 2, memristance requires a matrix that retains a field-displaced ionic profile. A plasticised, lower-molecular-weight, water-laden host would relax more rapidly, reducing the loop area and potentiation across successive sweeps.

- *(secondary)* acidic hydrolysis products could partially dope the PPV-type *Super Yellow*, adding a fixed, irreversible conductance, a

second route to the "high-conductivity, undesirable" state recorded in the notes.

Internal support. Several observations are consistent with the hypothesis but do not establish it. Among late devices, annealing at $\geq$150 °C increases the median on–off ratio at $\sim$3 V to 1.95 ($n = 6$), compared with 1.22 for standard 75 °C late devices ($n = 44$; one-sided Mann–Whitney $p = 0.024$), near the early-standard reference of 1.74 (fig. 3.3). Removal of absorbed water and film densification could account for this recovery. The test remains limited by six elevated-temperature devices, co-varying process changes, and measurement-pin issues in two devices. The combination of ohmic drift, window reduction, and loss of potentiation is also consistent with a water-plasticised, ion-leaky film. Chemically, the ester and amide linkages make this material class susceptible to hydrolytic chain scission, while its polar, hyperbranched architecture is hygroscopic [130, 138].

Confirming experiments (future work). The hypothesis would be tested directly by Fourier-transform infrared spectroscopy (loss of the ester C=O band, growth of –OH) and gel-permeation chromatography (a molecular-weight drop) on *retained aged versus fresh* stock; by Karl-Fischer titration or thermogravimetry for water content; and by impedance spectroscopy to show that the ionic resistance fell. None were run on the *Hybrane* corpus, so the hypothesis stays open, and may never fully close, because it concerns a consumable that no longer exists in its aged state.

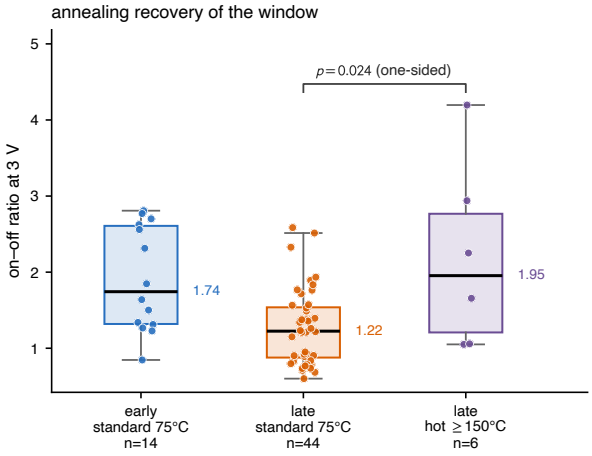


Figure 3.3: Annealing-recovery test of the hydrolysis hypothesis (*Hybrane*/Ag, on–off ratio at ∼3 V, per device). Among late devices, annealing at ≥150 °C increases the median from 1.22 to 1.95, near the early-standard reference of 1.74; one-sided Mann–Whitney $p = 0.024$. Interpretation is limited by $n = 6$ high-temperature devices, co-varying process changes, and pin issues in two devices. Reproduced by `scripts/bridge_figures.py` (F4).

3.4 Change to a PEO Host

The fabrication programme changed the ion-transport host after attempts to recover the *Hybrane* line. Replacing hyperbranched *Hybrane* with linear poly(ethylene oxide) (PEO), the host used in the comparative study of chapter 4, produced a larger switching window and multi-year reproducibility. Laboratory notes first record PEO in use by 2021-12 (NM_v083).

Switching-window comparison. At a matched ∼3 V sweep, comparing valid (non-broken, non-saturated) devices per device, the PEO host gives a larger switching window than contemporaneous *Hybrane* devices: median normalized area 0.26 (*Hybrane*) → 0.42 (PEO), Mann–Whitney $p = 0.018$; median on–off ratio 2.45 → 4.91 (fig. 3.4a,b).

Sustained temporal reproducibility. The PEO/Ag platform retained a usable switching window over roughly two and a half years (2022–2024, $n = 46$ devices): the per-device on–off ratio at $\sim$3 V had a median of 2.61, with yearly medians of 3.3/2.6/1.9 and no significant downward trend over time. The *Hybrane* platform had a median on–off ratio of 1.35 after its stock-associated window reduction (fig. 3.4c). The PEO record shows no corresponding supply-degradation trend over the measured interval.

Meaning of "reproducible". Here, reproducibility refers to the absence of a stock-associated reduction between batches over several years. It does not imply superior within-batch device uniformity: at matched $\sim$3 V, the device-to-device coefficients of variation differ only modestly (PEO $\approx$0.38 vs *Hybrane* $\approx$0.47). The evidence for PEO is its larger switching window and the absence of the supply-degradation trend over the multi-year record. Changing the suspected *Hybrane* reagent removed the trend after *Super Yellow* had been eliminated as its source (section 3.2).

3.5 Methodological Outputs

The investigation required four tools that were not available at the start of the campaign: a fabrication record, a normalized characterization procedure, an automated feature pipeline, and trend-discovery software. The same tools produce the quantitative results in the comparative and temporal-computing chapters. Five outputs are retained from this work.

- **A fabrication-record standard.** The investigation produced a per-device library schema for storage atmosphere, illumination, transport, reagent and solvent lots, operator, evaporation profile, and free-text fabrication and characterisation notes. This record

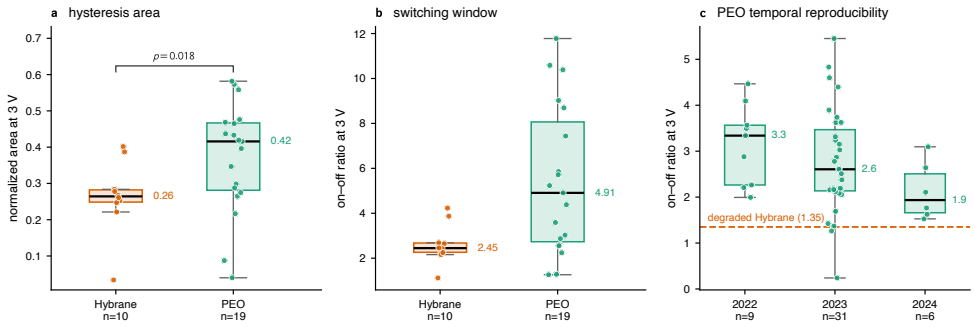


Figure 3.4: Switching-window response after changing to a PEO host (silver electrode, ∼3 V, per device). (a) Normalized hysteresis area and (b) on–off ratio are larger for PEO than for contemporaneous valid *Hybrane* devices ($p = 0.018$ for area). (c) The per-device PEO on–off ratio remains at a usable level (median ≈2.6) across 2022–2024 without a significant downward trend; the dashed line gives the degraded-*Hybrane* level (1.35). Reproduced by `scripts/bridge_figures.py` (F3).

enabled the exclusions in section 3.2 and is retained for every device in the archive.

- **A normalized electrical-characterization procedure.** Section 3.3 shows that sweep amplitude affects apparent conductivity and whether a switching window is expressed. The campaign therefore converged on fixed, documented drive and read protocols that support comparisons between devices and batches.

- **An automated feature pipeline.** The raw instrument traces are reduced to per-device, per-pixel, and per-curve feature tables by a dedicated pipeline; every feature used in this chapter's panel, and in chapter 4 and the temporal-computing chapter, is computed there.

- **Trend-discovery and curation tooling.** An interactive feature explorer identified the reported trends, whose slices are reproduced by the four controls of section 3.3; a curation tool produces the screened device set used in the analysis.

- **Figure generation.** A thesis-figure pipeline renders the quantitative figures from the same feature tables, so that every plotted number is traceable to the archive.

The comparative study of chapter 4 and the data-driven temporal-computing analysis use this same apparatus.

3.6 Reconciliation of Stability and Reproducibility

The shelf-stability result of chapter 2 and the campaign-scale loss of reproducibility concern the two different measurements defined in section 3.1.

- The published proof-of-concept shelf record was stable on the measured ∼two-week ambient scale [7]. A separate multi-day study likewise shows that finished *Hybrane* devices retain their characteristics over weeks to months. These measurements associate the campaign-scale change with the supply or fabrication process, while finished devices remained stable over the measured intervals.
- The *batch-scale reproducibility of the Hybrane platform over the fabrication campaign* is a different metric, and it declined over the following year as quantified in section 3.3.

Together, the records describe short-interval shelf stability within the proof-of-concept platform and a fabrication line over a year. The exact specimen roster of the published shelf record is not resolved in the recovered archive, and the campaign-scale reproducibility axis was not measured in the proof-of-concept study.

After the switch to PEO, replicated measurements of the volatile dynamics became possible. Chapter 4 uses the *Super Yellow*/PEO/LiOTf system

on silver as a composition grid, measured with the fabrication-record standard, normalized protocols, and feature pipeline developed during the reproducibility investigation. Because every substrate now carries its fabrication record, the substrate-to-substrate spread of each composition can be measured, and Chapter 5 uses it as the tolerance of a composition-defined element.

Chapter 4

Compositional and Chemical Control of Volatile Polymer-Electrolyte Memristive Dynamics

4.1 Introduction and Scope

Chapter 2 established the memristive response of the formulation comprising *Super Yellow*, *Hybrane*, and LiOTf and of its 2-T architecture across assay-specific junctions. The broader device corpus examined here varies the composition, electrolyte chemistry, electrode, and drive protocol of the same three-component architecture.

Three families of measurement recur on the same 2-T geometry throughout this chapter and the next: (i) current–voltage hysteresis, which reports the steady-state switching window; (ii) potentiation as a function of the number of applied write pulses, which reports how the conductance is built up; and (iii) depotentiation as a function of the delay before read-out,

which reports how a written state fades. Introduced as a named set in chapter 2, they are the shared, abundant basis of the comparative work. The richer proof-of-concept synaptic protocols of chapter 2 (excitatory post-synaptic current, separated short- and long-term memory, and spike-timing dependent plasticity) are *not* repeated here; where invoked, they serve only as formulation-level prior evidence.

Scope of the comparisons. The *Super Yellow*/PEO/LiOTf pulse and decay grids contain two to four nominally identical substrates per cell; the steady-state HYST corpus contains additional compatible batches and therefore two to thirteen substrates per cell. These replicated datasets support quantitative composition comparisons. After data-quality screening, the host, anion, and cation contrasts contain at most one or two substrates per matched cell and are used only as an illustrative survey. The electrode comparison is likewise illustrative and is available only at the lead composition (section 4.8). Drive protocol is analysed separately because it changes the apparent fading-memory timescale and therefore limits comparisons between protocols.

4.2 Materials, Devices, and Measurement Protocol

The devices retain the architecture of chapter 2: a vertical two-terminal sandwich of ITO (bottom, transparent), a solution-processed composite active layer, and a thermally evaporated silver top electrode, with a junction area of $8.25 \times 10^{-2}\,\mathrm{cm}^2$. *Super Yellow* is retained as the semiconducting component in every device of this chapter, so that the electronic-transport role is held fixed and the variation is confined to the ion-transport phase.

The experiment varies composition and chemistry. The *composition* axis

changes the mass ratios of the ion-transport polymer poly(ethylene oxide) (PEO) and lithium triflate salt (LiOTf) relative to the semiconductor: $r_{PEO} = m_{PEO}/m_{Super\ Yellow}$ and $r_{\mathrm{salt}} = m_{\mathrm{salt}}/m_{Super\ Yellow}$. These are blend-component ratios, not fractions of the total dry mass. The *chemistry* axis substitutes the ion-transport host, the salt anion (triflate or bis(trifluoromethanesulfonyl)imide, TFSI), and the alkali cation (Li^+, Na^+, K^+, as the triflate or TFSI salt). The comparative work replaces the hyperbranched polyester-amide *Hybrane* host of the proof-of-concept device (chapter 2) with two hosts: linear PEO and the hyperbranched polyether trimethylolpropane ethoxylate (TMPE). This gives a matched-chemistry contrast between linear and branched polyethers. Every device reported *quantitatively* uses the silver electrode. A separate gold-electrode corpus at the lead composition compares an active with an inert electrode, measures drive-polarity dependence on gold, and provides an independent-electrode check on the chemistry comparisons (section 4.8).

Film thickness as a measured covariate. Adding ion-transport polymer thickens the solution-processed film, so most higher-PEO cells were spun faster to reduce thickness differences. Spin speeds across the grid ranged from 1500 rpm to 3500 rpm. This compensation was incomplete and was not applied uniformly to every comparison batch. Among the 28 replicated-grid substrates with both profilometry and a dynamical measurement, median thickness rises from ≈ 227 nm at $r_{PEO} = 0.3$ to ≈ 298 nm at $r_{PEO} = 1.2$ (Pearson $r = +0.65$; fig. 4.1). The activation voltage shows no corresponding thickness trend ($r \approx -0.06$), and a 38% thickness spread within the lead cell leaves $t_{1/2}$ within 18 s–22 s. In the 23 substrates with a model-free $t_{1/2}$, the partial correlation of thickness with $\log t_{1/2}$ at fixed r_{PEO} is $+0.28$, whereas that of r_{PEO} with $\log t_{1/2}$ at fixed thickness remains -0.51. Thickness therefore does not account for the composition association, although the sample size does not exclude a secondary thickness contribution. It is retained as a measured covariate

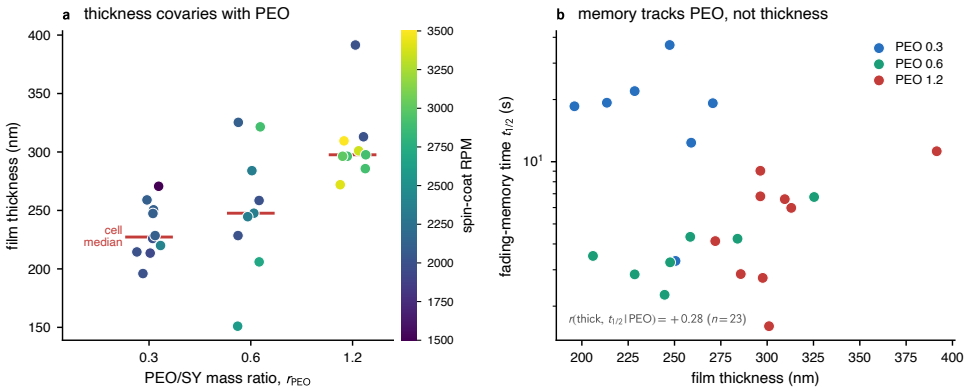


Figure 4.1: Film thickness and fading-memory time in the replicated silver composition grid. (a) Film thickness rises with r_{PEO} (red bars are cell medians; $n = 28$ substrates with profilometry and a dynamical measurement); higher-PEO cells were spun faster (colour) to reduce this difference, although residual covariation remains. (b) For the $n = 23$ substrates with $t_{1/2}$, the partial correlations are $r_{\text{partial}}(\text{thickness}, \log t_{1/2} \mid r_{PEO}) = +0.28$ and $r_{\text{partial}}(r_{PEO}, \log t_{1/2} \mid \text{thickness}) = -0.51$. Thickness does not remove the composition association, but a secondary thickness contribution is not excluded.

rather than declared fully controlled; the complete audit is given in section C.2.

Drive and read protocols. Because the comparisons below turn on them, the electrical protocols are stated explicitly, and they are *not* identical across the three measurements (table 4.1). Hysteresis uses a triangular voltage sweep. The potentiation (N-pulse build-up) was acquired with a 3 V write and 1.5 V read at a variable pulse count ($N \leq 1000$); the depotentiation (fading-memory decay) used a fixed train of 30 write pulses at a 4 V write and 2 V read. The inter-pulse interval is effectively common (0.104 s for potentiation, 0.103 s for depotentiation), but the *write amplitude differs* between the two (3 V vs 4 V), a point that matters when the two are later composed (section 4.10, Chapter 5). The read voltage is above the ionic activation threshold established in chapter 2,

Table 4.1: The three common measurements use distinct drive protocols (modern *Super Yellow*/PEO/LiOTf/Ag corpus). Potentiation and depotentiation differ in write amplitude, so the chapter never pools them and Chapter 5 composes them only as an explicit assumption.

Measurement	Write	Read	Pulse train
Hysteresis (HYST)	triangular sweep	—	—
Potentiation (PULSES)	3 V	1.5 V	variable $N \leq 1000$, 0.104 s
Depotentiation (DELAYTIME)	4 V	2 V	30 pulses, 0.103 s
Protocol overlay (v114)	3 V, 6 V	1.5 V, 3 V	30 pulses

a point returned to in section 4.6. An older 2022 device generation used a 6 V write with a 3 V read; the modern and old amplitudes are never pooled, and the same-device contrast between two amplitudes is itself the drive-protocol result (section 4.6).

4.3 Methods of Analysis

Hysteresis curves are summarised by the on–off conductance ratio and by a normalised loop area; potentiation curves by the per-decade growth of the conductance ratio with pulse number, by the peak ratio, and by whether the response turns over (decreases) at the highest pulse counts; and depotentiation curves by a fading-memory time constant.

For the depotentiation (fading-memory) decay, the comparative archive is refitted with the more general Kohlrausch (stretched-exponential) function

$$\Delta G(t_{\mathrm{wait}}) = \Delta G_0 \, \exp\left[-\left(\frac{t_{\mathrm{wait}}}{\tau}\right)^{\beta}\right] + \Delta G_{\infty}, \tag{4.1}$$

with characteristic relaxation time τ, stretch exponent $0 < \beta \leq 1$, initial

change ΔG_0, and residual ΔG_∞ [154]. The eight-to-thirteen-point decay curves often under-constrain β and τ jointly. The fitted τ is therefore reported only where the fit is well identified. In all cases, the primary timescale is the robust, model-free half-enhancement time $t_{1/2}$, the delay at which the conductance enhancement has fallen to half its initial value. Where both measures are available, they agree. Both are *apparent* timescales determined by the material and by the amplitude used to write and read the state. The composition series and chemistry comparisons each use a fixed protocol because section 4.6 shows that the timescale is protocol-dependent. Fitted β values lie predominantly in the range 0.6–0.9, consistent with stretched relaxation and the heterogeneous distribution of activation barriers invoked in chapter 2.

Data-quality control and aggregation. Two screening layers are applied before any number is reported: erratic or non-physical measurements are flagged and excluded (including the archive's `Y/N` broken flags), and visual review stores curve verdicts (used as measured, cleaned, or discarded) under the full device–measurement–day–pixel key in a curation registry applied automatically by the analysis scripts. All timescales are recomputed after this screening because the instrument-side fit is not robust to anomalous points. The fabricated substrate is the experimental unit: surviving curves are reduced to junction medians, junctions to a substrate median, and substrates to a composition-cell median before inference. A substrate with two measured junctions therefore contributes once. The representative curves of fig. 4.2 are drawn by script from substrates near their cell medians, not chosen as inferential replicates. The raw-to-summary trail, curation precedence, and stretched-exponential identification thresholds are documented in section C.1.

4.4 Composition Control of the Dynamics

The *Super Yellow*/PEO/LiOTf composition grid varies r_{PEO} over 0.3, 0.6 and 1.2 and r_{salt} over 0.045, 0.09 and 0.18. The pulse and decay datasets contain two to four nominally identical silver-electrode substrates per cell; the HYST dataset contains further compatible batches. Each measurement was acquired under its own fixed protocol (table 4.1), so every composition contrast is drawn *within* one measurement; the 3 V potentiation and 4 V depotentiation data are never pooled. Representative curves in fig. 4.2 show the measurements before they are reduced to on–off ratio, growth exponent, turnover, and fading-memory time. Across the grid, the switching window, potentiation response, and fading-memory timescale change together.

The steady-state switching window contracts as r_{PEO} rises. At fixed $r_{\text{salt}} = 0.09$, the median on–off ratio falls from ≈ 3.3 at $r_{PEO} = 0.3$ to ≈ 2.4 at $r_{PEO} = 1.2$, while the normalised loop area falls from ≈ 0.33 to ≈ 0.18. The activation voltage remains within 2.2 V–2.4 V across the grid. Composition thus changes how *much* the conductance moves, but not the voltage at which it begins to move. The widest window occurs in the low-PEO row, with the largest loop area at the low-salt corner ($r_{PEO} = 0.3$, $r_{\text{salt}} = 0.045$: area ≈ 0.44). The per-sweep on–off ratio is strongly right-skewed there: the cell median is ≈ 3.4, but individual sweeps reach ~ 40. Cell medians are therefore used throughout to prevent these few outlying sweeps from dominating the comparison.

Potentiation also weakens and changes shape with composition (fig. 4.5). The PEO and salt mass ratios affect different features of the response. The per-decade growth of the conductance ratio (the log–log slope α) falls with PEO content, most cleanly at the lowest salt ratio. It drops from $\alpha \approx 1.1$ (near-linear accumulation) at $r_{PEO} = 0.3$ to $\alpha \approx 0.3$ (strongly compressive) at $r_{PEO} = 1.2$. The usable dynamic range follows the same

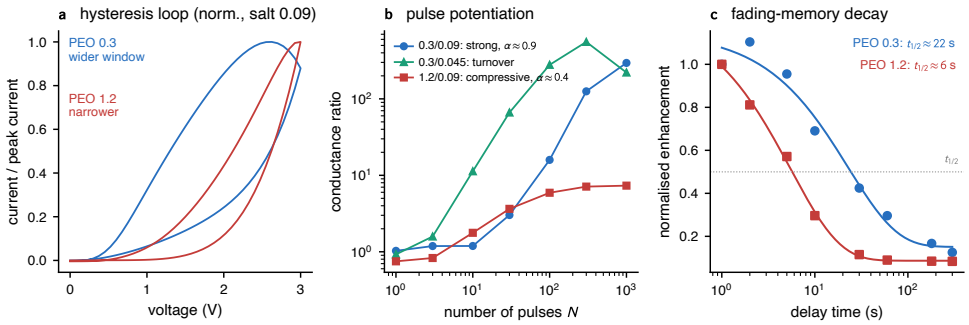


Figure 4.2: Representative raw curves of the three common measurements, on devices chosen near their cell medians (*Super Yellow*/PEO/LiOTf/Ag). (a) Hysteresis loops, each normalised to its own peak current so the comparison is the loop *openness* (the switching window), not absolute conductance: the low-PEO device (PEO 0.3, blue) opens a wider window than the high-PEO one (PEO 1.2, red), which carries more current but switches over a narrower range. (b) Pulse potentiation (conductance ratio vs pulse number, log–log): a strong near-linear accumulation ($\alpha \approx 0.9$), a low-salt cell that *turns over* and then declines, and a weak, compressive high-PEO response ($\alpha \approx 0.4$). (c) Fading-memory decay (enhancement vs delay, normalised to the first read) with stretched-exponential fits and the half-enhancement level dotted: the low-PEO device crosses $t_{1/2}$ far later ($\approx 22\,\mathrm{s}$) than the high-PEO one ($\approx 6\,\mathrm{s}$).

direction: the peak conductance ratio is largest in the low-PEO, low-salt corner ($\approx 480\times$ at $r_{PEO} = 0.3$, $r_{\mathrm{salt}} = 0.045$) and falls to a few-fold at high PEO. Salt content instead governs how long the response grows before saturation. At the lowest salt ratio (0.045), every cell *turns over* by $N \sim 100$–300 pulses; at the highest salt ratio (0.18), the conductance ratio continues to grow through the largest count tested ($N \approx 1000$), with 0.09 intermediate. The α trend is noisier at the higher salt ratios, and the sparsely sampled $r_{PEO} = 0.3$, $r_{\mathrm{salt}} = 0.18$ cell ($n = 2$, curation-recovered) departs from it. Turnover also places a practical ceiling on the usable pulse count. These composition-dependent, predominantly compressive responses provide the nonlinear input encoding used in Chapter 5; the low-salt cells additionally become non-monotone at high pulse counts.

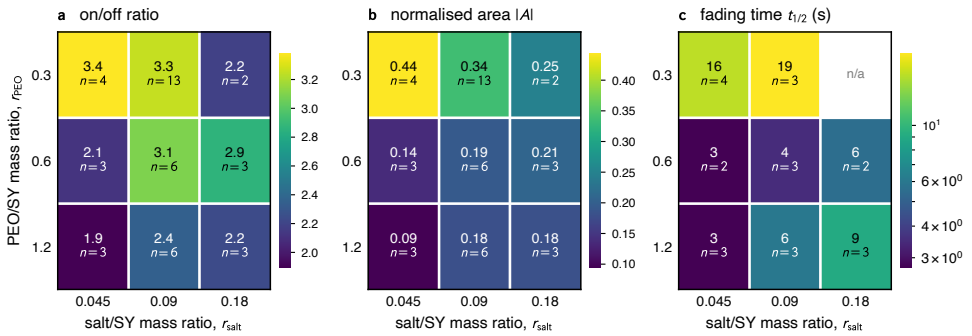


Figure 4.3: Composition control of the steady-state and dynamical response of *Super Yellow*/PEO/LiOTf devices. Across the replicated grid, increasing the PEO-to-*Super Yellow* mass ratio is associated overall with smaller switching windows (a, b) and shorter fading-memory times (c). Individual fixed-salt rows are not strictly monotone, especially in the sparsely sampled salt-rich cells.

The fading-memory timescale varies with composition over roughly 2 s–20 s. The longest retention occurs in the low-PEO row (at $r_{PEO} = 0.3$, $t_{1/2} \approx 19$ s at $r_{\text{salt}} = 0.09$, with an identified $\tau \approx 25$ s); increasing r_{PEO} to 0.6 or 1.2 shortens the timescale to a few seconds. A single curation-recovered substrate at the still-lower $r_{PEO} = 0.15$ level ($t_{1/2} \approx 22$ s) is consistent with this trend, though it lies outside the replicated grid and is not counted toward it. The independently computed instrument-side single-exponential time constant reproduces the same ordering. Within composition cells, the robust model-free timescale has $\sigma(\ln t_{1/2}) \approx 0.26$; the fit-noisier single-exponential τ has a larger spread of about 0.85 (sections C.13 and D.4). The available fabrication covariates do not account for the composition association, but secondary contributions and unrecorded batch effects remain possible (section C.2). Chapter 5 treats each cell median as a component value and the within-cell spread as its tolerance; the range across the whole device library is mapped in section C.4. The same composition axis appears in potentiation strength: the growth exponent α falls across the replicated grid, with within-cell

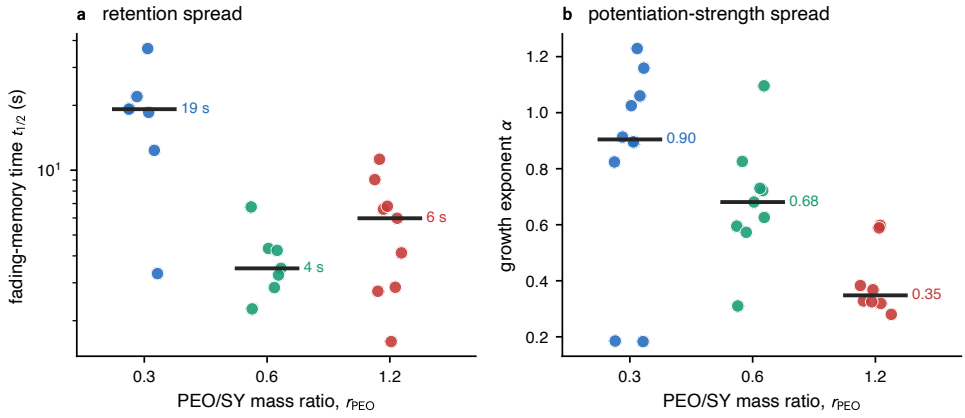


Figure 4.4: Per-substrate distributions underlying the cell medians of fig. 4.3 (*Super Yellow*/PEO/LiOTf/Ag; black bar is the cell median). (a) Fading-memory time $t_{1/2}$ and (b) potentiation growth exponent α, each against r_{PEO}. The PEO-to-SY mass ratio sets the broad level, while within-cell substrate-to-substrate variation is comparable to the steps between cells. Chapter 5 evaluates both sources of variation.

variation around each median.

Across the replicated grid, increasing r_{PEO} is associated with shorter fading-memory time and smaller switching and potentiation responses, while r_{salt} strongly separates turnover. At the substrate level, the four reported gradients are $t_{1/2}$: $\rho = -0.48$, Holm-adjusted $p = 0.022$; growth exponent: $\rho = -0.55$, $p = 0.011$; peak ratio: $\rho = -0.54$, $p = 0.011$; and on–off ratio: $\rho = -0.53$, $p = 0.003$. Permutations restricted within acquisition quarter give the same conclusion (section C.9). The axes are not fully orthogonal: a batch-adjusted factorial analysis finds no PEO×salt interaction for $t_{1/2}$ at the 5 % level, but does find interactions for growth exponent and peak ratio, so the potentiation effect of r_{PEO} depends on r_{salt}. Bootstrap intervals are reported only for cells with at least three substrates; for $n = 2$, the observed range is reported instead. This replicated composition result is the quantitative core of the chapter; the per-cell values are tabulated in section C.3. If a more ion-rich

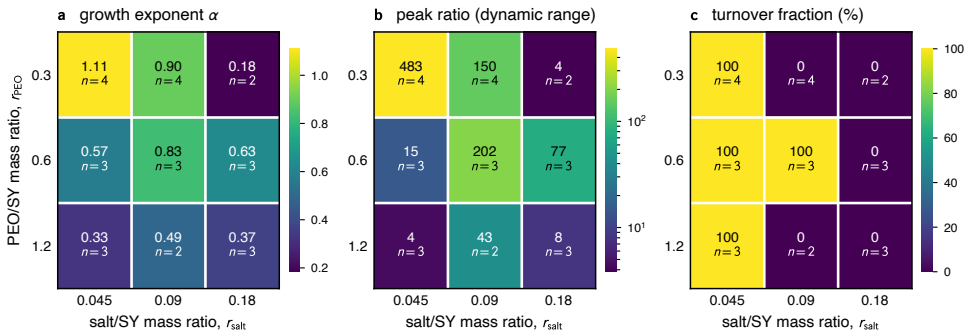


Figure 4.5: Potentiation (pulse-integration) response across the *Super Yellow*/PEO/LiOTf composition grid: (a) the log–log growth exponent α, (b) the peak conductance ratio (usable dynamic range, log scale), and (c) the fraction of substrates whose response turns over within the measured range ($N \leq 1000$). Across the replicated grid, increasing r_{PEO} is associated overall with lower α and dynamic range, although salt-dependent interactions produce non-monotone cells. Increasing r_{salt} suppresses turnover, sustaining growth to higher pulse counts. Per-cell median, with the substrate count n annotated in each cell.

matrix relaxes a displaced ionic profile more readily, its equilibrium ionic resistance should also fall. Section 4.7 tests that prediction by impedance spectroscopy.

4.5 Chemical Tuning of the Dynamics

Beyond composition, the identity of the electrolyte components shifts the dynamics further. Each matched cell in this survey (fixed composition, electrode, and protocol) contains at most one or two screened devices. The magnitudes are therefore indicative, and the sample size is given in every case. Structural and optical probes provide a separate view of the chemistry. Blend films and scratched powders at the lead composition (0.3/0.09) were examined by ATR-FTIR, X-ray diffraction, and UV-Vis absorption to probe ion association, host backbone order, crystallinity,

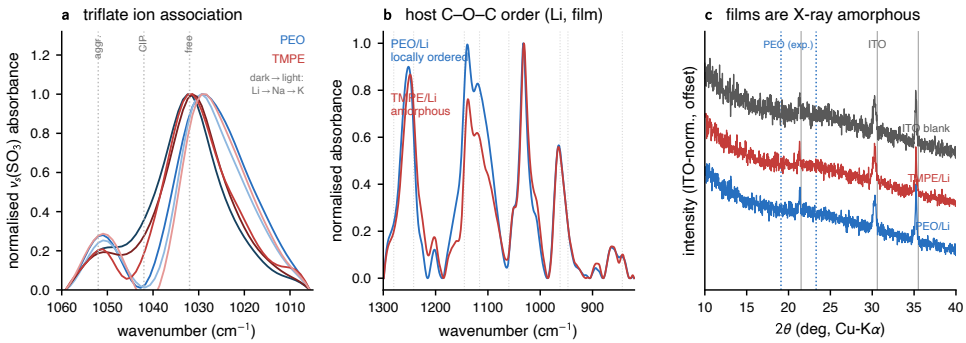


Figure 4.6: Microscopic corroboration of the chemistry axis (electrode-free *Super Yellow*-blend films/powders, 0.3/0.09, one specimen per chemistry). (a) ATR-FTIR triflate ion-association band $\nu_s(SO_3)$, whose position reports the ion-association state (spectroscopically free near 1032 cm^{-1}, contact ion pair near 1042 cm^{-1}, aggregate above; reference positions dotted [150]; PEO hosts in blues, TMPE hosts in reds, darker to lighter for Li/Na/K): it lies at 1029 cm^{-1}–1032 cm^{-1} in every host×cation sample, with no cation-resolved shift. (b) ATR-FTIR C–O–C envelope: the linear PEO host (blue) shows a resolved multiplet, indicating retained *local* (conformational) ether-backbone order, whereas the hyperbranched TMPE host (red) shows a single broad, amorphous band [150]. (c) X-ray diffraction of PEO/Li and TMPE/Li with a bare-ITO blank: all patterns show the crystalline ITO substrate reflections (grey lines; 21.5°, 30.6° and 35.5°), with no sharp PEO crystalline doublet (blue dotted; 19.1° and 23.3°) or crystalline salt phase. The composite films are therefore X-ray amorphous, and the PEO order in (b) is short-range [129, 150]. ATR spectra are normalised after a local/rubber-band baseline and XRD is normalised to the ITO peak; intensities are not comparable across specimens because sampling and thickness differ, so only band positions and shapes are interpreted.

and the semiconductor's electronic gap without driving a device (figs. 4.6 and 4.8). These measurements also use one specimen per chemistry.

Host. Substituting the linear polyether PEO with the hyperbranched polyether TMPE, at fixed lithium-triflate chemistry and composition, shortens the fading-memory time by roughly a factor of six (PEO/LiOTf: $t_{1/2} \approx 20\,\text{s}$–$25\,\text{s}$, three screened silver devices; TMPE/LiOTf: $\approx 4\,\text{s}$, one

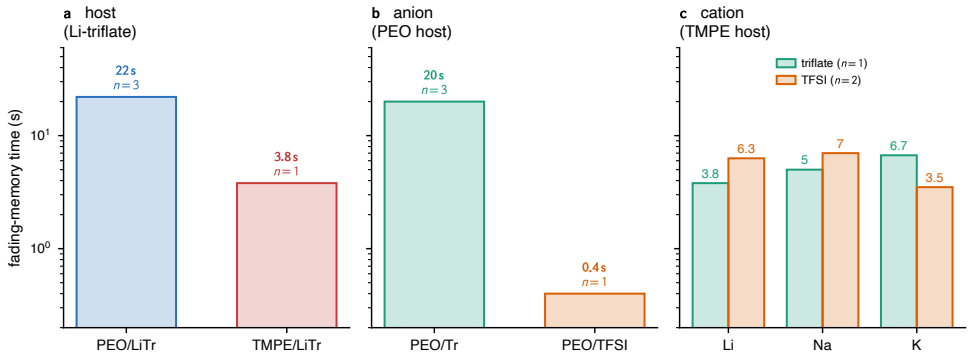


Figure 4.7: Illustrative effect of electrolyte chemistry on the fading-memory time constant (*Super Yellow*, 0.3/0.09, silver, matched 4 V/2 V protocol). Host (PEO vs TMPE, Li-triflate) and anion (triflate vs TFSI, PEO host) produce the largest shifts. The cation panel shows the same three cations in the TMPE host under each anion: potassium is longest in triflate and shortest in TFSI, so the cation produces no consistent, host- and anion-independent ordering. Sample sizes are small (n as labelled); see text.

screened silver device with an identified decay). This comparison has three PEO devices but only one TMPE device. The lithium pair from the designed host×cation batch could not be used because the PEO/LiOTf device was broken. The analysis therefore compares the TMPE device with the nearest-matched silver PEO/LiOTf devices (same operator, fresh semiconductor, glovebox), which differ in fabrication date.

An independent microscopic correlate of the host substitution is visible in the polyether backbone itself, and at two length scales. In ATR-FTIR the linear PEO film shows a resolved C–O–C stretching multiplet, with the weak order-sensitive marker bands near 843 cm^{-1}, 962 cm^{-1} and 1242 cm^{-1}, indicating retained *local* (conformational) order of the ether backbone, whereas the hyperbranched TMPE film collapses these into the single broad band of an amorphous polyether (fig. 4.6b) [150]. At the longer length scale of crystallites, however, X-ray diffraction on the same sample set finds *neither* host crystalline in the composite: every pattern

is dominated by the ITO substrate (confirmed against a bare-substrate blank), with no sharp PEO crystalline reflections and no segregated crystalline salt phase (fig. 4.6c), so the films are X-ray amorphous and PEO retains short-range order without forming long-range crystallites [129, 150]. A host that is locally more ordered, even while globally amorphous, set against a fully amorphous TMPE, is consistent with the lower segmental mobility and slower ionic relaxation that the *longer* PEO fading-memory time would require: in polyether electrolytes ion transport is carried by segmental motion of the host above its glass transition, so the relaxation of a displaced ionic profile is rate-limited by that segmental mobility [90, 134, 135]. With one specimen per host this is offered as a structural correlate, not a demonstrated cause. The host ordering is moreover not specific to the silver electrode: on the independent gold-electrode corpus at the same composition, PEO again retains far longer than TMPE and supports a markedly wider switching window and stronger potentiation (section 4.8), so the host effect survives a change of electrode. That the salt forms no crystalline phase in any chemistry additionally supports reading these blends as homogeneous, salt-dissolved ion-conducting media.

Anion. Replacing triflate with the more charge-delocalised TFSI anion in the PEO host reduces the fading-memory time from $\approx 20\,\mathrm{s}$ to approximately 1 s. This direction is consistent with the electrolyte literature: charge delocalisation in TFSI weakens ion pairing and lowers salt lattice energy, promoting dissociation and plasticisation of the polyether host and increasing ionic mobility [131, 155]. The TFSI decays also tend toward a compressed shape, whereas the triflate decays are stretched, suggesting different relaxation-time distributions. This comparison is weakly controlled: the single PEO/TFSI device belongs to a later fabrication generation and also differs in spin speed and salt filtration. The approximately twenty-fold difference is therefore reported as illustrative.

Cation. The data do not support the hypothesised $Li^+ > Na^+ > K^+$ ordering of fading-memory time derived from the hard–soft acid–base argument of chapter 2 [91]. Two cation series use the same TMPE host, composition, electrode, and protocol, and differ in the salt anion (fig. 4.7). With triflate, the timescale increases from Li^+ ($\approx 3.8\,s$) to Na^+ ($\approx 5.0\,s$) and K^+ ($\approx 6.7\,s$); each value comes from one screened device with $R^2 \geq 0.93$. With TFSI, potassium is shortest ($\approx 3.5\,s$), while lithium and sodium are longer ($\approx 6\,s$–$7\,s$, two devices per cation). Thus, cation identity shifts the timescale by roughly a factor of two within a fixed anion, but the ordering changes with the anion and within-cation variation is comparable to between-cation differences. These sample sizes cannot establish a host- and anion-independent ordering. A powered test would require a dedicated series at fixed host, anion, protocol, and electrode with at least three replicates per cell (section 4.10).

Independent probes provide additional constraints. The triflate symmetric SO_3 stretch $\nu_s(SO_3)$ lies at $1029\,cm^{-1}$–$1032\,cm^{-1}$ in all six host×cation films and powders, a spread at or below instrumental resolution with no cation ordering (fig. 4.6a). UV-Vis absorption of *Super Yellow*/TMPE films places the *Super Yellow* π–π^* peak [88] at 446 nm for Li^+, Na^+, and K^+ (fig. 4.8). These results indicate no resolved cation effect on ion-pairing state or semiconductor optical gap; neither measurement tests the ordering of retention directly. On gold electrodes, the TMPE/triflate steady-state window is also flat across the three cations (on–off ratio ≈ 1.3–1.5, no monotone trend; section 4.8).

In these illustrative comparisons, host and anion substitutions produce larger shifts in fading-memory time than cation substitution.

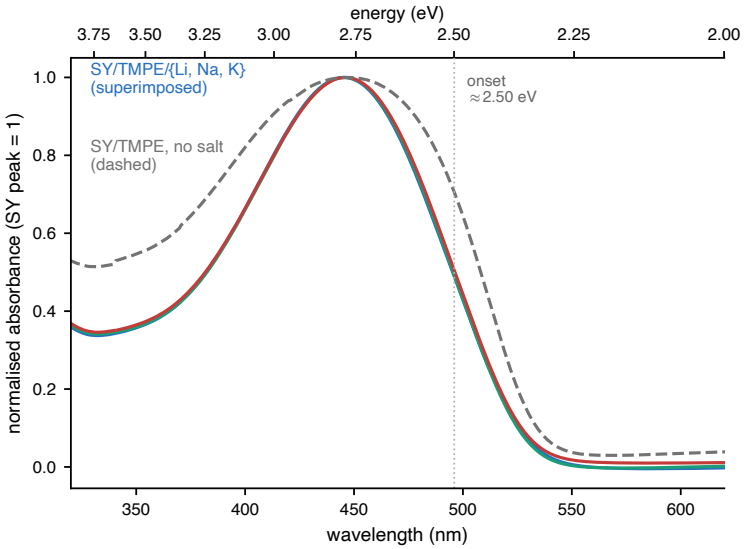


Figure 4.8: UV-Vis absorption of *Super Yellow*/TMPE blend films (lead composition 0.3/0.09; Au-electrode 2025 generation; one specimen per chemistry), normalised to the *Super Yellow* π–π^* peak. The peak ($\approx 446\,\text{nm}$) and the absorption onset ($\approx 2.50\,\text{eV}$) are superimposed for Li^+, Na^+, and K^+; the salt-free control (dashed) coincides at the band edge (its differing blue side and lower absolute absorbance are thickness/scattering, not electronic). The cation does not shift the semiconductor's optical gap. Only band *positions* are read because absorbance magnitudes are thickness-confounded.

4.6 Protocol-Dependent Timescale

The drive protocol constrains every comparison above. The read voltage exceeds the ionic activation threshold, and write amplitude controls the displaced ionic profile; both can change the apparent fading-memory time. In one device measured under both protocols, the fitted relaxation time increases from $\approx 4.6\,\text{s}$ under a $3\,\text{V}$ write and $1.5\,\text{V}$ read to $\approx 15.5\,\text{s}$ under a $6\,\text{V}$ write and $3\,\text{V}$ read (fig. 4.9).

One interpretation is that a larger write amplitude displaces more ionic charge and, at the largest amplitudes, may introduce electrode-level

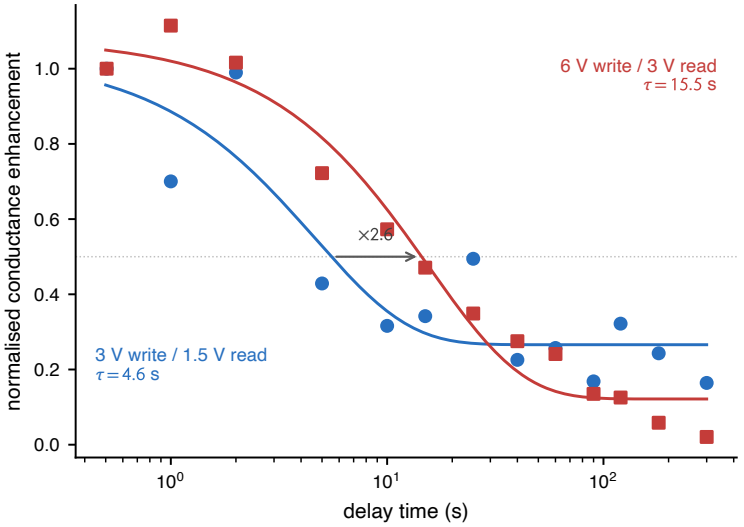


Figure 4.9: The apparent fading-memory time is set in part by the drive protocol. The same device, measured under a larger write/read amplitude, shows a markedly longer apparent relaxation time: the fitted single-exponential time constant rises from $\tau \approx 4.6\,\mathrm{s}$ to $\approx 15.5\,\mathrm{s}$, and the half-enhancement crossing (dotted, arrow) shifts by $\approx 2.8\times$.

electrochemical processes, producing a longer-lived written state. The supra-threshold read may also perturb the state during interrogation. The composition result of section 4.4 remains internally comparable because it uses one fixed protocol. Chemistry comparisons that span device generations are more exposed to protocol differences. Material and drive conditions therefore co-determine the apparent volatile timescale; cross-device or cross-chemistry comparisons must match the write/read protocol, electrode, and composition.

The same lever reappears, independently, on the gold-electrode corpus of section 4.8. There, matched-amplitude negative pulse trains sweep the pulse count and the inter-pulse interval together, with the relaxation recorded in the same run, and both halves of the protocol dependence are observed directly: potentiation grows steeply as the inter-pulse interval

shortens (rank correlation $\rho \approx -0.94$ to -0.99, $p < 0.001$, in three independent series of 9–11 matched runs), and the depth of the written state co-sets the subsequent relaxation time ($\rho \approx +0.46$ between potentiation ratio and decay half-time, $p = 0.015$, $n = 28$ runs). The drive–retention coupling that this section infers from a two-protocol contrast is therefore seen *within single experiments* on a second electrode (section C.7).

4.7 Microscopic Corroboration: Impedance Spectroscopy

Every result so far rests on the *driven* response (pulses and decays). Small-signal impedance spectroscopy offers an independent test of the mechanism proposed for the composition trend: if increasing r_{PEO} shortens the fading-memory time because a more ion-rich matrix lets the displaced ionic profile relax more readily, then the *equilibrium* ionic resistance should fall along the same composition axis. The archive contains such a measurement, acquired on the same substrates: 0 V-DC spectra (40 mV AC, 1 Hz–1×10^6 Hz) on 26 silver-electrode *Super Yellow*/PEO/LiOTf substrates spanning the composition grid.

Each spectrum is summarised in two separate ways. The first is *model-free*: the real part of the impedance at the apex of the main Nyquist semicircle, $Z_{\mathrm{real}}^{\mathrm{apex}}$, which fixes the resistance scale of the spectrum without committing to any circuit. The second is *model-based*: the bulk ionic resistance R_{ion} of an equivalent circuit adapted from the polymer light-emitting electrochemical cell analysis of Munar *et al.* [145], reported only as corroboration of the model-free quantity because the per-spectrum circuit fits are genuinely ill-conditioned. The circuit topology, its two automated fits, and the per-cell values of both observables are detailed in section C.5.

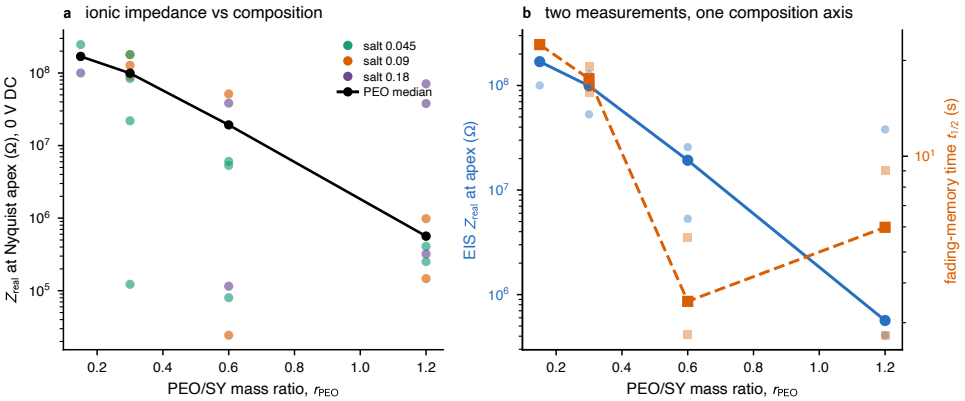


Figure 4.10: Equilibrium (0 V-DC) impedance spectroscopy corroborates the composition control of the dynamics (silver *Super Yellow*/PEO/LiOTf grid). (a) The impedance resistance scale (here the model-free real impedance at the Nyquist apex) falls by more than three orders of magnitude as r_{PEO} rises (points are substrates, coloured by r_{salt}; the black line is the r_{PEO} median). (b) The equilibrium impedance scale (left axis) and the *independently* measured fading-memory time $t_{1/2}$ (right axis) decline along the *same* PEO-to-SY composition axis (cell-level points with per-r_{PEO} median trends): the two are co-driven by composition, not independent variables.

Both observables fall steeply with r_{PEO} (fig. 4.10a). The model-free $Z_{\text{real}}^{\text{apex}}$ drops by more than three orders of magnitude across the measured series (cell medians from $\sim 10^8\,\Omega$ at $r_{PEO} = 0.15$–0.3 to $\sim 10^5\,\Omega$ at $r_{PEO} = 1.2$), with a rank correlation against r_{PEO} of -0.64 per substrate ($n = 26$) and -0.83 on cell medians; the fitted R_{ion} gives the same picture (-0.84 on cell medians). This is the direction the mechanism requires: more ion-transport polymer relative to *Super Yellow*, lower ionic resistance.

The EIS resistance scale and the *independently* measured fading-memory time fall along the *same* composition axis (fig. 4.10b): both decline together as r_{PEO} rises, and across cells $\log Z_{\text{real}}^{\text{apex}}$ covaries with $\log t_{1/2}$ at $r \approx +0.73$ ($+0.60$ per substrate, $n = 23$). Three qualifications bound this reading:

- The PEO-to-SY mass ratio is a shared covariate of both quantities. Their correlation therefore shows that equilibrium impedance and driven retention move together along the composition axis; it is not an independent causal test.

- The resistance *scale* carries the composition signal, whereas the EIS relaxation *frequency* does not: the apex frequency corresponds to a sub-second RC time and does *not* track the seconds-long fading-memory time. This separates the small-signal ionic-transport bottleneck reported by impedance spectroscopy from the slower large-signal relaxation that determines the fading-memory response. As an order-of-magnitude check, scaling the median apex time ($\approx 0.089\,\mathrm{s}$) by the low-frequency-to-geometric capacitance ratio (≈ 69) gives $\approx 6.2\,\mathrm{s}$, close to the median $t_{1/2}$; this consistency does not establish that the two measurements share one physical clock (section C.11).

- $Z_{\mathrm{real}}^{\mathrm{apex}}$ spans $\sim 10^4\times$ while film thickness spans only $2.6\times$, and the PEO–impedance association persists after adjustment for thickness (partial correlation -0.60). This makes thickness alone insufficient to explain the impedance gradient, although it does not exclude secondary geometric effects (section 4.2).

Impedance spectroscopy thus provides an independently measured equilibrium quantity consistent with the ionic-resistance part of the mechanistic picture below; it does not uniquely identify that mechanism. The same spectra bear on where the seconds-scale storage sits. Their low-frequency capacitance exceeds the geometric capacitance by a median factor of about 69 (section C.11). This is consistent with the relaxation time arising from ionic storage within the $0.0825\,\mathrm{cm}^2$ film; no external capacitor is involved, although the low-frequency estimate is model-dependent.

4.8 Electrode and Polarity Dependence

Every result so far uses the silver top electrode. The archive also contains 50 gold-electrode devices, treated here as a third illustrative axis. Every gold device uses the single lead composition (PEO or TMPE host, 0.3/0.09); this corpus therefore contains no information about the composition series of section 4.4, which remains a silver-only result. It instead contrasts electrochemically active silver with inert gold at matched chemistry and provides an independent-electrode check on the chemistry comparisons of section 4.5.

At matched chemistry and composition, and under the canonical positive-polarity protocols used everywhere above, the electrode shifts all three behavioural families, and does so coherently (fig. 4.11). In the lead PEO/LiOTf cell, replacing silver by gold narrows the steady-state switching window (median on–off ratio 3.3 $\rightarrow$ 2.0, normalised loop area 0.33 $\rightarrow$ 0.12), weakens and flattens potentiation (peak conductance ratio $\approx 124\times \rightarrow 29\times$, log–log slope $\alpha \approx 0.9 \rightarrow 0.45$), and raises the activation voltage ($\approx$ 2.4 V–2.7 V), yet *lengthens* the fading-memory time, from $t_{1/2} \approx 19$ s on silver (three devices) to ≈ 87 s on the two clean gold devices ($R^2 \approx 0.97$–0.99, $\beta \approx 0.6$). The same window-and-potentiation contraction holds in the TMPE/LiOTf cell (on–off 3.8 $\rightarrow$ 1.3; peak $54\times \rightarrow 1.9\times$), again at positive drive. The per-cell metrics for every chemistry on both electrodes, including the host-free and salt-free negative controls, are tabulated in section C.6.

This difference is consistent with the contrast between active and inert electrodes. Silver can inject and electrodeposit mobile Ag^+, the basis of electrochemical-metallisation switching, and may therefore add a large volatile component: a wider window, steeper potentiation, lower threshold, and faster relaxation [70, 71]. Gold is noble and inert, leaving the polymer-electrolyte ionic mechanism to produce a smaller, higher-threshold, and

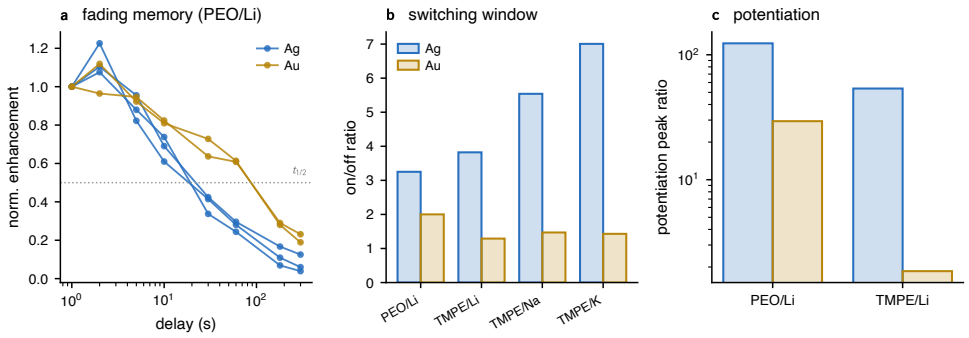


Figure 4.11: Illustrative electrode contrast at the lead composition (0.3/0.09), under the canonical positive-polarity protocols (for the negative-polarity behaviour of the gold devices see fig. 4.12). (a) Fading-memory decays of PEO/LiOTf devices, normalised to the first read: the gold curves (gold) cross half-enhancement later than the silver ones (blue), i.e. gold retains *longer*. (b) The steady-state switching window (on–off ratio) is narrower on gold for every chemistry, and is flat across the TMPE cation series on gold (no Li/Na/K ordering). (c) Pulse-potentiation dynamic range (peak ratio, log scale) is far smaller on gold. The inert electrode thus gives a weaker, higher-threshold, but more retentive response. Small n per cell, no human curve curation on the gold side, and a generation offset between the corpora bound the comparison (see text).

more persistent response. The higher activation voltage on thinner gold devices supports an interfacial electrode contribution over a film-thickness explanation [108].

Negative-polarity response of the gold devices. Additional campaigns on the same Au/TMPE/triflate generation measured negative bias. These data were recovered from outside the curated archive and analysed with a dedicated parser; their sources, validation against catalogued sweeps, and per-file tables are documented in section C.7. The devices are strongly polarity-asymmetric: within a junction, bipolar sweeps carry ≈ 2.5–$42\times$ more current at -3 V than at $+3$ V (fig. 4.12a), with the same direction across the catalogued negative-going sweeps. Under negative drive, the gold devices show a rate-dependent switching window. Across

sweep rates of $\sim$0.003–0.4 V s^{-1}, the on–off ratio rises as the sweep slows (rank correlations down to -0.99, $p < 0.001$), from ≈ 1 at the standard rate to 20–31 in the slowest -3 V sweeps (fig. 4.12b). Constant-bias holds locate an activation threshold between -1.5 and -2.5 V: current decays below this range and grows by one to three orders of magnitude above it over tens to hundreds of seconds. Negative pulse trains potentiate by one to two orders of magnitude at a matched protocol (-5 V pulses, -2.25 V read: cation medians $\approx$ 40–230$\times$), while positive pulses depress the written state. The response is therefore bidirectionally programmable under this negative-polarity protocol.

These campaigns add fading-memory measurements for the TMPE host on gold. Sparse delayed reads after the pulse trains, with the device floating between reads, follow strongly stretched exponentials ($\beta \approx 0.33$–0.5) with model-free half-decay times of ≈ 0.4 s–21 s across the three cations (fig. 4.12c). These values are on the same seconds scale as TMPE on silver and below the ≈ 87 s measured for PEO on gold, providing a gold-electrode comparison of host retention. Under the matched negative protocol, all three cations have overlapping potentiation ranges and relaxation times, with substrate-to-substrate spread comparable to any cation difference. The concentration of the response in one bias direction after removal of the active Ag^+ pathway is consistent with an interfacial role for the dissimilar ITO and gold contacts and their accumulated ion profiles, but the electrical data do not establish that mechanism.

The second electrode reproduces two observations from section 4.5: PEO retains longer than TMPE, and the switching window has no transferable cation ordering. The gold corpus also provides negative controls absent from the silver set. Host-free (semiconductor-plus-salt) and salt-free (semiconductor-plus-host) films show neither a switching window (on–off ≈ 1.0) nor potentiation (peak ≈ 1), indicating that both the ion-transport polymer and dissolved salt are required for the memristive response.

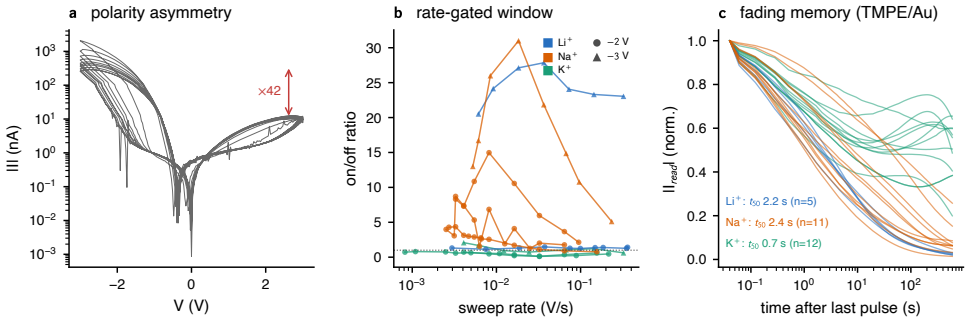


Figure 4.12: Negative-polarity response of the gold/TMPE/triflate devices (illustrative tier; section C.7). (a) Exemplar bipolar sweep (±3 V, ten cycles, Na^+ device): the negative branch carries ≈ 42× the current of the positive one. (b) On–off ratio against sweep rate for all step-count series (colour = cation, marker = amplitude): slow sweeps give ratios of 20–31, while the standard fast protocol gives ≈1. (c) Sparse-read relaxation after −5 V pulse trains, normalised to the first read, with per-cation median half-decay times. The times are on the seconds scale, as for TMPE on silver, and below those of PEO on gold.

Frequency-domain response: a bias-programmed low-frequency amplifier. A third corpus measures the same Au/TMPE/triflate chemistry in the frequency domain: 549 lock-in sweeps from 1 Hz to 500 kHz on a fresh batch, with one substrate per cation tracked from day 1 to day 23. The junction is placed in series with a 4 nF capacitor and driven by a small sinusoid superimposed on a programmable DC offset. The apparatus, channel decode, and per-file tables are documented in section C.8. Above ∼10 Hz, the unbiased junction behaves as a passive linear dielectric: the transfer ratio follows the capacitive divider and is amplitude-independent to within a median 0.2 % from 0.5 to 2.4 Vpk. This boundary is consistent with the rate dependence in fig. 4.12b. In the same plateau, junction capacitance increases Li < Na < K by ≈1.4× across the same-batch trio, compared with an ≈1 % pixel spread. Same-recipe profilometry makes a thickness origin less likely, but one substrate per salt limits this result to a candidate dielectric cation effect; profilometry on matched specimens

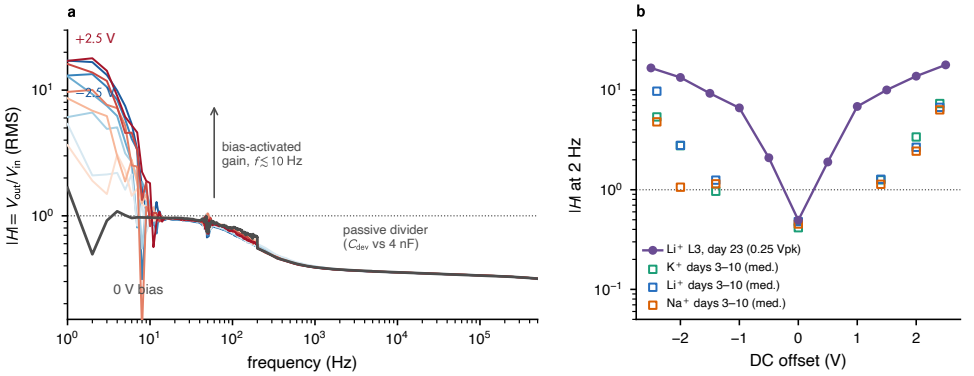


Figure 4.13: Frequency-domain response of the gold/TMPE/triflate devices (illustrative tier; section C.8). (a) Transfer ratio $|H|(f)$ of one Li^+ junction in series with 4 nF (day 23, 0.25 Vpk) at eleven DC offsets from 0 (grey) to ±2.5 V (blue negative, red positive). Above ∼1 kHz, all curves follow the passive capacitive divider; below ∼10 Hz, bias produces a fundamental-frequency gain of up to ≈18, increasing with bias magnitude and nearly symmetric in sign. (b) $|H|$ at 2 Hz against DC offset for the day-23 series (line) and per-cation medians of earlier sessions (squares; days 3–10, 0.4 Vpk–1.4 Vpk). Gain increases with bias and occurs for all three cations.

would test it directly. Under DC bias, the fundamental-frequency transfer below ∼10 Hz increases from $|H| \approx 0.5$ without bias to ≈18 at ±2.5 V (fig. 4.13). The bias-modulated ionic conductance transfers energy from the DC offset into the signal band. This gain occurs for all three cations at every age tested, is confined to the sub-10 Hz band, and is nearly symmetric in bias polarity (section C.8). A biased junction therefore operates as a voltage-programmable low-pass amplifier over a band that overlaps the physiological signals analysed in Chapter 5.

All three gold corpora share the same illustrative-tier bounds, which the SI states for each corpus (sections C.7 and C.8). Each uses the single lead composition on one or two substrates per cation; the curves are uncurated; and the supra-threshold negative read makes decay times protocol-specific, allowing only order-of-magnitude comparison with silver.

The 2024–2025 gold generation is also partly confounded with electrode because the silver chemistry was fabricated earlier and the co-fabricated silver batch was not measured under the decay protocol. Electrode and polarity are therefore treated as illustrative effects and cross-electrode checks on the chemistry axis. A matched-generation batch split between silver and gold is required for a quantitative electrode comparison.

4.9 Discussion

Composition is the only replicated axis for control of the volatile dynamics. Increasing the ion-transport-polymer-to-semiconductor mass ratio shortens the fading-memory time and is accompanied by smaller switching and potentiation responses. Host and anion substitutions shift the timescale over a wider range, but those comparisons are illustrative. Cation identity also shifts the dynamics, although the direction changes with host, anion, and protocol at the available sample sizes; the predicted $Li^+ > Na^+ > K^+$ order is therefore not transferable across the dataset.

The electrode acts jointly with drive polarity. Relative to active silver, inert gold weakens switching and potentiation and lengthens retention under the canonical positive drive. Under negative drive, the same gold devices show a full, rate-gated, bidirectionally programmable response (section 4.8). The frequency measurements locate the corresponding spectral boundary: above $\sim 10\,\mathrm{Hz}$ the gold junctions behave as passive linear capacitors at every tested amplitude, whereas a DC bias produces programmable low-frequency gain below that boundary.

The hard–soft acid–base model of chapter 2 remains a qualitative interpretation. It anticipates the cation ordering in the TFSI family, where the most weakly coordinating cation relaxes fastest, but the triflate family runs in the opposite direction. The present data therefore do not support

a quantitative cation law.

Mechanistic interpretation. The mechanism remains a hypothesis. Increasing r_{PEO} does not produce the decrease in absolute on-state conductance expected from simple dilution of the *Super Yellow* percolation pathway; across the replicated grid the association is positive ($\rho = +0.61$, $p < 0.001$, $n = 43$ substrates; section C.10). The cell-median conductance is not strictly monotone from $r_{PEO} = 0.6$ to 1.2, so this rejects a simple dilution picture rather than establishing a unique transport law. A raised conductance floor is one interpretation consistent with the contracting window: the ion-rich phase may add a parallel, weakly switchable path, with off-state conductance rising faster than on-state conductance. Faster relaxation of the displaced ionic profile could account for the accompanying decrease in fading-memory time. Salt content may set the number of mobile, chargeable states available before saturation, consistent with the later turnover at higher r_{salt}.

Host and anion effects may arise from cation–ether coordination, ion pairing, and segmental mobility [131, 135], which set the distribution of local activation barriers behind stretched-exponential relaxation [154]. Equilibrium impedance supports the ionic-resistance part of this interpretation: it falls along the composition axis and tracks the fading-memory time (section 4.7). ATR associates the host effect with local ether-backbone order, retained in PEO and absent in amorphous TMPE, while XRD shows that every composite remains X-ray amorphous and that the salt is dissolved rather than segregated. Across the cation series, ATR shows no corresponding ion-association fingerprint and UV-Vis shows no shift of the *Super Yellow* optical gap (section 4.5). These measurements are consistent with electrolyte-controlled ion transport rather than a change in the *Super Yellow* electronic backbone.

The anion and barrier-distribution links have not been established micro-

scopically. They require temperature-dependent relaxation measurements and chemistry-resolved impedance (section 4.10), so the hard–soft acid–base interpretation remains qualitative.

For circuit use, the composition grid is a set of component values. Across compositions and nominally identical substrates, the timescale spans roughly an order of magnitude. A descriptive two-σ separation heuristic on the cell-median span $\Delta \ln t_{1/2} = 1.95$, relative to within-cell scatter, gives approximately two to five composition bands. That is the number of time constants that can currently be selected by formulation and told apart despite substrate-to-substrate scatter; it is not an independently measured information-capacity bound (section C.13). The same spread could also populate a physical reservoir with distinct temporal responses [5, 19, 29]. Each junction reproduces its switching window to $\approx 15\,\%$ sweep-to-sweep (section C.12), substantially less than the diversity across the bank. Figure 4.14 places every screened silver substrate in a fading-memory ($t_{1/2}$) versus dynamic-range (peak ratio) plane. The PEO-to-SY mass ratio sets the broad region occupied by each substrate, so one composition selects a paired memory/nonlinearity operating point; the remaining within-composition scatter is the tolerance on that operating point. Chapter 5 propagates both into the simulations.

4.10 Limitations and Future Work

Statistical power is limited outside the composition axis. Only the *Super Yellow*/PEO/LiOTf composition grid is replicated; the host, anion, and cation comparisons rest on one or two screened devices per matched cell.

1. The cation hypothesis is not tested here at adequate power. The cation does shift the dynamics, but not as predicted: the triflate retention series runs K>Na>Li, the reverse of the hypothesised

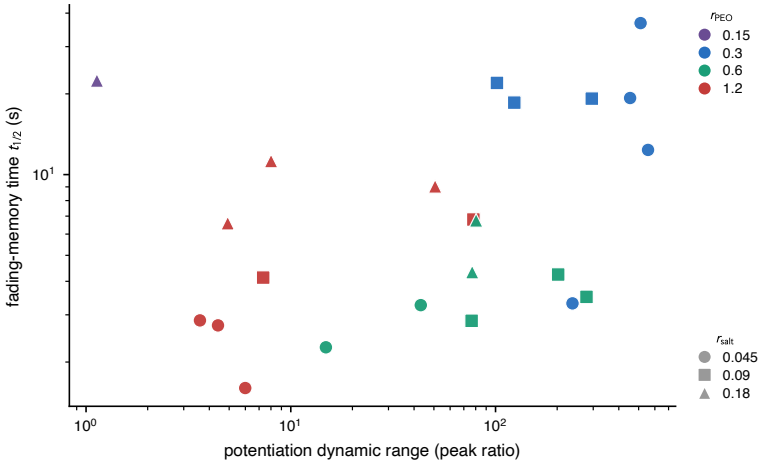


Figure 4.14: Per-substrate design space of the *Super Yellow*/PEO/LiOTf/Ag composition grid: fading-memory time $t_{1/2}$ against potentiation dynamic range (peak ratio), coloured by r_{PEO} and marked by r_{salt} (one point per fabricated substrate, median over its junctions). The PEO ratio selects the broad region a substrate occupies: low PEO (blue) in the long-memory, high-dynamic-range corner, high PEO (red) in the opposite one, so memory and nonlinearity are paired and cannot be set independently. The within-composition scatter is the substrate-to-substrate tolerance that Chapter 5 propagates into its simulations.

$Li^+ > Na^+ > K^+$, and re-orders again with the anion, so the hypothesised ordering is not what the data show and no transferable cation law can be fixed at one or two devices per cell.

2. The supra-threshold read and the amplitude dependence of the apparent timescale (section 4.6) mean that absolute retention values are protocol-specific.

3. The electrode is reported as an illustrative axis (section 4.8), but it remains partly confounded with fabrication generation, and device ageing is an uncontrolled variable that was not modelled.

4. Pulse integration and state relaxation were characterised *separately and at different write amplitudes*: the potentiation curves used a

3 V write and the fading-memory decays a 4 V write (table 4.1), at an effectively common inter-pulse interval. On the lead silver composition, the archive contains no experiment that varies input timing to probe their interaction directly. A matched-amplitude experiment exists on the gold/TMPE corpus (section 4.8), but it uses a different host, electrode, polarity, and read, so it establishes the direction of coupling rather than the lead-cell parameters.

5. The fading-memory protocol measured a single junction per device, so repeat-decay reproducibility of $t_{1/2}$ is not characterised at the population level (only one device carries two junctions; section C.12). Within-device reproducibility is established only for the steady-state switching window, from its repeated sweeps.

Because drive amplitude affects the apparent timescale (section 4.6), combining the 3 V potentiation shape with the 4 V decay in Chapter 5 requires an explicit modelling assumption that the regimes are compatible. A matched-amplitude experiment that varies both pulse count and inter-pulse interval would test this assumption. Its gold/TMPE negative-polarity counterpart shows coupling between the two measurements (section 4.6); the lead-composition silver experiment remains future work.

A dedicated cation series in one host and anion, at a fixed protocol and electrode, with a sub-threshold read voltage (as in chapter 2) and at least three replicates per cell, would turn the illustrative chemistry survey into a powered comparison. The impedance study of section 4.7 could also be extended beyond the replicated PEO/LiOTf axis at 0 V DC. The archive contains unmodelled bias-resolved spectra up to 3 V DC and chemistry-varied devices; analysing them would test the proposed host, anion, and barrier-distribution effects. A matched-generation batch split between silver and gold top electrodes would isolate the electrode effect

from fabrication generation.

Temperature-dependent relaxation measurements are needed to test the proposed role of segmental mobility and distributed activation barriers. A $\tau(T)$ series yielding an Arrhenius or Vogel–Fulcher–Tammann activation energy would connect the host and anion differences to a segmental-relaxation barrier and test the temperature dependence of the stretch exponent β. Encapsulation and systematic ageing measurements would address the remaining stability questions.

4.11 Inputs to Temporal Computing

In the replicated *Super Yellow*/PEO/LiOTf grid, composition tunes the fading-memory time over roughly 2 s–20 s, with the switching window and potentiation strength moving in the same direction. Host and anion contrasts extend the measured range, although each rests on at most one or two devices per matched cell. Cation identity also shifts the dynamics, but the available data do not establish the predicted $Li^+ > Na^+ > K^+$ order as a transferable law. Drive protocol co-sets the apparent timescale and must remain fixed in quantitative comparisons.

Chapter 5 uses the relaxation kernel of each cell ($t_{1/2}$ and, where identified, β) as a component value and the substrate-to-substrate spread as its tolerance. It also uses the pulse-update curves, including their saturation and turnover. Because r_{PEO} changes both the fading-memory time and the input nonlinearity α (fig. 4.14), each composition defines a paired operating point rather than two independent settings. The salt-to-SY mass ratio provides a partly separate control of dynamic range and turnover. The simulations use these values in two ways: single cells and pairs of cells as designed filters, and banks of one or several compositions as reservoirs.

Chapter 5

Temporal Processing with Composition-Selected Devices: Designed Filters and Reservoirs

5.1 Introduction and Scope

Chapter 4 showed that composition sets the fading-memory time of these devices over about one decade of seconds. This chapter asks what that time constant can be used for. Behavioural models fitted to the measured response are tested in two roles. In the first, one composition, or a pair of compositions, is chosen for its time constant and used as a designed filter; the references are the conventional components for the same job, an RC element with the same half-life and a digital filter. In the second, many devices form a reservoir: a bank of randomly masked nodes whose states feed a trained linear read-out. Banks drawn from one composition are compared with banks spanning several, and linear exponential-moving-

average (EMA) banks at the same half-lives serve as controls.

Peripheral physiological signals are the test bed because their timescales overlap those of the devices without rescaling: phasic electrodermal responses, respiration, and heart-rate variability vary over seconds, and tonic levels drift over minutes. The chapter uses synthetic electrodermal signals with known responses, the Wearable Stress and Affect Detection corpus (WESAD), and an independent PhysioNet corpus. Affective computing supplies the tasks and labels; the chapter does not propose a new affect classifier.

All results in this chapter are *in silico*; no new devices were fabricated. Each node is a compact behavioural model parameterised by the fits in chapter 4 (section 5.2), and every read-out is a single linear ridge regression.

5.2 Measured Behavioural Model

Each composition cell is reduced to a compact behavioural model and used as one reservoir node. The model is fitted only to the three common measurements of chapter 4: hysteresis, pulsed potentiation, and delayed depotentiation. Its parameters therefore derive from measured device data.

Each replicated composition cell is summarised by a *parameter card*: the model-free half-enhancement time $t_{1/2,c}$, any identified stretch exponent β_c, the potentiation growth exponent α_c, the measured peak ratio $R_{\mathrm{peak},c}$, and the pulse count $N_{\mathrm{peak},c}$ at that peak. The fading memory is the stretched-exponential (Kohlrausch) retention identified in chapter 4,

$$\lambda_c(\Delta t) \;=\; \exp\Big[-(\Delta t/\tau_c)^{\beta_c}\Big],$$

where $\tau_c = t_{1/2,c}/(\ln 2)^{1/\beta_c}$ is derived so that every card reproduces the primary, model-free cell-median half-time exactly. For cells without an identified stretch exponent, $\beta_c = 1$ and therefore $\tau_c = t_{1/2,c}/\ln 2$. The write envelope is

$$\frac{G(N)}{G_0} = 1 + (R_{\mathrm{peak},c} - 1)\left(\frac{N}{N_{\mathrm{peak},c}}\right)^{\alpha_c}, \qquad 0 \le N \le N_{\mathrm{peak},c}.$$

It uses the measured baseline, exponent, peak, and pulse count rather than an unbounded power law. Where a measured train turns over after its peak, the node is capped at the peak: the post-peak recovery dynamics were not measured and are not extrapolated.

Driven by a rate-coded input $u_n \in [0,1]$ sampled at interval Δt, a device node accumulates pulse progress p and relaxes along its measured kernel:

$$p_n^{(i)} = \mathrm{clip}\left[\sum_{k\ge 0} g_k^{(i)}\,\frac{\mathrm{clip}(w_i u_{n-k}, 0, 1)}{N_{\mathrm{peak},i}},\ 0,\ 1\right], \qquad g_k^{(i)} = \lambda_i (k\,\Delta t)^{1/\alpha_i},$$

$$x_n^{(i)} = (p_n^{(i)})^{\alpha_i}, \qquad \frac{G_n^{(i)}}{G_0^{(i)}} = 1 + (R_{\mathrm{peak},i} - 1)\,x_n^{(i)}.$$

Here $x_i \in [0,1]$ is the normalised enhancement, w_i is a non-negative input mask, and one full-scale sample advances the pulse-progress coordinate by at most one effective pulse. That sample-to-pulse mapping is a modelling convention, not a measured circuit timing relation. The kernel $g = \lambda^{1/\alpha}$ makes the enhancement left by a single write decay exactly as the measured λ_i, and with no relaxation and unit input the node follows the measured envelope to its peak and cannot exceed it. Because the sum runs over the whole measured relaxation, a stretched decay is reproduced at any sampling interval. Applying the one-step factor $\lambda_i(\Delta t)$ repeatedly would instead compound into a plain exponential and, for $\beta < 1$, shorten the simulated memory (for the lead cell at $\Delta t = 1\,\mathrm{s}$, to a half-life of 11.5 s instead of the measured 19.2 s). A node that reaches its peak is held

there, with its stored history rescaled (section D.2). The internal state is assumed to be read sub-threshold without disturbance, as in the proof-of-concept device of chapter 2. Every read-out is a single *linear* ridge regression on the device states.

Write nonlinearity and fading memory were measured at different write amplitudes (3 V for potentiation and 4 V for depotentiation; table 4.1). Combining them in one node assumes that the two regimes are compatible and probably underestimates the coupling between drive and retention. Both measurements also used a single, effectively common inter-pulse cadence, so the model does not resolve input timing below that cadence. The gold/TMPE negative-polarity corpus of chapter 4 measures integration and relaxation within single runs and finds both interval-dependent potentiation and written-depth-dependent relaxation. Because it uses another host, electrode, polarity, and read protocol, it does not provide parameters for the banks used here. The experiment needed for the lead composition is specified in section 5.12. Per-cell parameter cards and all simulation and read-out settings are tabulated in section D.2.

Component cards. Read as circuit elements, the cards form the component table of table 5.1. Each cell is a low-pass element whose time constant is set by composition, whose corner frequency follows from its measured kernel, and whose tolerance is the within-cell scatter. For a single-exponential cell the corner is $f_c = 1/(2\pi\tau_{RC})$. For a stretched cell it lies lower, because the long tail of the relaxation retains low frequencies, so the half-life alone does not fix the filter.

Table 5.1: Component cards of the replicated PEO/LiOTf grid, ordered by half-life. $t_{1/2}$ and β define the measured relaxation; $\tau_{RC} = t_{1/2}/\ln 2$ is the time constant of the single-exponential (RC) element with the same half-life; f_c is the $-3\,\mathrm{dB}$ corner of the low-pass formed by the measured kernel and by that RC element; α, R_{peak}, and N_{peak} describe the pulse response. n counts fabricated substrates. The median within-cell tolerance is $\sigma(\ln t_{1/2}) = 0.26$ (cell range 0.09–1.02; section D.4).

Cell	n	$t_{1/2}$ (s)	β	τ_{RC} (s)	f_c device (Hz)	f_c RC (Hz)	α	R_{peak}	N_{peak}
PEO 1.2/0.045	3	2.7	1.00	4.0	0.0404	0.0404	0.33	4	100
PEO 0.6/0.045	2	2.8	0.97	4.0	0.0375	0.0401	0.57	15	100
PEO 0.6/0.09	3	3.5	0.61	5.0	0.0085	0.0316	0.83	202	300
PEO 0.6/0.18	2	5.5	0.61	8.0	0.0054	0.0200	0.63	77	1000
PEO 1.2/0.09	3	6.0	0.91	8.6	0.0152	0.0185	0.49	43	1000
PEO 1.2/0.18	3	9.0	0.87	13.0	0.0092	0.0122	0.37	8	1000
PEO 0.3/0.045	4	15.8	0.64	22.8	0.0022	0.0070	1.11	483	300
PEO 0.3/0.09 (lead)	3	19.2	0.83	27.7	0.0038	0.0058	0.90	150	1000

5.3 Designed Filters from the Component Cards

Before any reservoir is assembled, each composition cell can be used on its own. Its relaxation kernel defines a causal low-pass filter whose corner frequency is fixed when the film is formulated (table 5.1). Two elementary circuits follow from it. A device subtracted from its own input forms a high-pass baseline remover, $y = x - \mathrm{LP}_c(x)$. Two compositions subtracted from each other form a band-pass, $y = \mathrm{LP}_{\mathrm{fast}}(x) - \mathrm{LP}_{\mathrm{slow}}(x)$, which gives no output for a constant input. Both are routine preprocessing steps for slow sensor signals.

Electrodermal activity (EDA) is the test case. Its phasic part, the skin-conductance responses (SCRs) that follow sympathetic arousal, rides on a tonic level that drifts over minutes. Conventional pipelines remove the tonic level with a high-pass filter near 0.05 Hz and then detect SCRs or measure power in the 0.045 Hz–0.25 Hz band associated with sympathetic tone [156]. Three references frame the comparison: an RC element with the same half-life as the device, which is the conventional component for

the same job; a causal digital Butterworth phasic filter (0.05 Hz–1 Hz); and the unfiltered signal.

Each device is represented by its measured kernel $\exp[-(t/\tau_c)^{\beta_c}]$, applied as a causal convolution at 4 Hz on the assumption that the conductance enhancement superposes. Every simulated device draws its own time constant from the measured within-cell scatter, $\sigma(\ln t_{1/2}) = 0.264$; the RC references are ideal. The composed nonlinear node of section 5.2 gives the same ranking once its read-out gain is set for small signals (section D.7).

Design space. All eight high-pass corners lie between 0.003 and 0.041 Hz, below the lower edge of the sympathetic band (fig. 5.1a). The two fastest cells ($t_{1/2} \approx 2.7$ s) place the corner close to the conventional 0.05 Hz tonic cut-off. No cell is fast enough to set the upper edge of the band. A two-composition band-pass therefore passes 0.002 Hz–0.047 Hz, which is the tonic range. For stretched cells, the low-pass corner also sits well below that of the RC element with the same half-life (PEO 0.6/0.09: 0.009 against 0.032 Hz), because the long tail of a stretched relaxation retains low frequencies. The half-life alone therefore does not specify a filter; its corner has to be read from the measured kernel, as in table 5.1.

Synthetic signals with known responses. Synthetic sessions supply ground truth. They contain Bateman-shaped SCRs [157] at about two and seven per minute in alternating five-minute calm and aroused blocks, a tonic rise in the aroused blocks, minutes-scale drift, and either low noise, broadband noise, or sharp motion-like artefacts. Detection thresholds and lags are calibrated on 12 sessions and scored on 30 others (fig. 5.1b,c). The fastest single-device high-pass detects SCRs with F1 0.713, 0.680, and 0.676 in the three conditions, against 0.701, 0.669, and 0.647 for the digital filter; under artefacts it is higher in 29 of 30 sessions. The RC element with the same half-life scores 0.712, 0.677, and 0.675. The best

two-composition band-pass reaches only 0.50–0.54 and passes about twice as much tonic as phasic power. The unfiltered signal scores 0.690 on clean data, because peak prominence already discounts a slow baseline; filtering helps mainly against noise and artefacts.

WESAD recordings. On WESAD, per-minute phasic power is compared between the stress and baseline segments of each subject (fig. 5.1d). The single-device high-pass gives a within-subject AUC of 0.952 on the chest and 0.921 on the wrist, against 0.951 and 0.927 for the digital filter and 0.947 and 0.922 for a filter restricted to the sympathetic band. The per-minute features of the device and digital filters are closely correlated (Spearman $\rho = 0.98$ and 0.99). The tonic level alone gives 0.915 and 0.885. The device-minus-digital difference is +0.001 on the chest ($p = 0.94$) and −0.006 on the wrist (2/15 subjects higher, $p = 0.038$). The two-composition band-pass is lower at both sites (0.903 and 0.880).

Device against *RC* element. Across all eight cells, the measured kernel never improves on the *RC* element with the same half-life (table D.7). The fastest cell relaxes as a simple exponential ($\beta = 1.00$) and is therefore the same filter as its *RC* counterpart. The two most stretched cells ($\beta = 0.61$) are slightly worse on the chest (−0.009 AUC; $p = 0.013$ and 0.038). Used as a filter, a composition-selected device is an *RC* element whose time constant is set at fabrication. Its case rests on how that time constant is obtained. A 0.0825 cm^2 two-terminal film reaches seconds with no separate capacitor, and impedance spectroscopy places the storage in the ionic response of the film (section 4.7). In analogue CMOS, time constants of hundreds of milliseconds already need large capacitors or very small currents [6], and printed circuits have used printed low-pass filters for temporal processing, including printed EDA stress detection [81, 158]. Footprint, voltage, stability, and tolerance were not compared against a

printed *RC* element realising the same time constant; that comparison would decide the practical value of the device.

5.4 Reservoir Computing and the Common Metric Language

The reservoir principle. The remaining sections test the second use of the cards, as nodes of a reservoir. Reservoir computing uses a fixed dynamical system whose state is connected to a trained linear read-out [18, 19, 31]. A physical reservoir requires fading memory, so its state retains recent input history, and a nonlinear response that maps inputs into features accessible to the read-out. Chapter 4 measures both properties in these devices: stretched-exponential depotentiation supplies fading memory, and compressive potentiation supplies nonlinearity. Memristive and organic implementations have previously used the same combination [5, 33, 35, 36]. The behavioural models test whether the measured polymer-electrolyte response supplies that combination.

Architecture. The reservoir is a spatial bank of N *independent* device nodes, each represented by the behavioural model of section 5.2 and driven by a randomly masked copy of the input. With no inter-node recurrence, differences between banks arise from their intrinsic timescales, device-to-device scatter, and input masks rather than engineered connectivity. Each reservoir result is paired with a linear control: exponential moving averages (EMAs) of the input channels at the device half-lives, $\tau = t_{1/2}/\ln 2$, read out in the same way. The EMA bank has the time constants of the device bank but none of its nonlinearity or random masking. For Demonstration A, the same single-composition node could instead be deployed as a time-multiplexed delay-feedback reservoir with virtual

nodes along a delay line [159]; the spatial bank is used throughout for consistency.

Common metrics. Linear memory capacity MC_k and its total [160] quantify the past input recoverable at each lag. NARMA-10 [161] provides a nonlinear system-identification benchmark, and the information-processing-capacity framework [82] separates linear and nonlinear contributions (fig. 5.4). Affective tasks are scored by class-balanced macro-F1. Robustness is tested by injecting the measured within-cell timescale spread into the bank (fig. 5.9); no common scatter estimate is assumed for α or β.

5.5 Benchmark Tier: Capacity and System Identification

Reservoir suitability is first evaluated without an application-specific task. Linear memory capacity measures fading memory, while information-processing capacity separates its linear and nonlinear components. The NARMA-10 system-identification task is used only to rank compositions, not to claim absolute performance. All three benchmarks use a random rate-coded input, N device nodes, and a linear ridge read-out.

Statistical units. Random input and bank seeds sample the algorithm's sensitivity to stochastic initialisation; they are not independent devices, experiments, or subjects. Seed-only benchmark comparisons therefore report means, standard deviations, and sign consistency without inferential p-values. Inferential paired tests are reserved for independent held-out subjects in the physiological-corpus analyses, or for independently fabricated devices in the experimental chapters. The complete

distinction is recorded in table D.1.

Memory capacity. The linear memory capacity of Jaeger measures how much of the past input a reservoir can reconstruct: MC_k is the squared correlation between a ridge reconstruction of the input k steps ago and its true value, and the total is the sum over all lags, bounded above by the number of nodes N [160]. Figure 5.2 compares two banks of equal size ($N = 24$): a *homogeneous* bank of jittered copies of the lead cell (PEO 0.3/0.09) and a *heterogeneous* bank spanning the replicated composition grid ($\tau \approx 4$–$30\,\mathrm{s}$). Total capacity rises from 3.22 ± 0.32 to 4.44 ± 0.45 (means $\pm$ SD over ten random input/bank seeds), a ratio of 1.38 ± 0.14 and a paired gain of 1.22 ± 0.41 that is positive in all ten seed pairs. Most of the gain sits at the shortest lags: single-step recall rises from $\mathrm{MC}_1 = 0.41$ to 0.98, whereas recall at lag 15 is similar (0.08 and 0.07). Copies of one slow cell carry strongly correlated states, and the faster cells of the heterogeneous bank supply the short-lag information they lack.

Composition sweep. The same benchmarks, run on single-composition banks, rank the compositions against one another (fig. 5.3). The strongly stretched PEO 0.6/0.09 cell ($\beta = 0.61$) attains the highest single-composition memory capacity (3.35 ± 0.36, mean $\pm$ SD over ten seeds) and the lowest error on the NARMA-10 system-identification benchmark [161] (NRMSE 0.614 ± 0.018); the lead cell PEO 0.3/0.09 gives 2.96 ± 0.26 and 0.692 ± 0.033. A stretched relaxation is a distribution of time constants, so one stretched device already spreads its memory over several lags. The differences between neighbouring cells are algorithmic rankings, not experimental evidence of a unique optimum. The NARMA-10 numbers carry only *relative* weight: the absolute error is high (≈ 0.6–0.9) because a bank of independent, non-recurrent nodes is a weak NARMA-10 reservoir.

Comparison with randomly distributed timescales. A generic echo-state bank of the same size tests whether the measured timescales are individually important or whether their range is sufficient. A random bank spanning approximately the same range ($\tau = 3$–$30\,\mathrm{s}$) has a memory capacity of 4.97 ± 0.78, compared with 4.44 ± 0.45 for the device bank (ten matched seeds; the device is higher in four). Expanding the random range to 0.5–60 s raises the capacity to 5.56 ± 0.74 (the device is higher in one). The measured values are therefore not uniquely optimal; this control attributes the available linear memory to the range of fading times rather than to their exact distribution. The broader-range result is only a modelling prompt: the corresponding slow devices have not been measured.

Linear and nonlinear capacity. Memory capacity counts only linearly recoverable information. The information-processing-capacity framework of Dambre et al. decomposes total capacity into orthogonal polynomial degrees: degree-one terms give linear capacity, while degree-two terms measure nonlinear self and cross-lag transformations; the total is bounded by the node count N [82]. Figure 5.4 reports this decomposition. The homogeneous bank has total capacity 5.18 ± 0.34, comprising 3.61 linear and 1.57 nonlinear capacity. The heterogeneous bank reaches 9.77 ± 0.73, comprising 5.06 linear and 4.71 nonlinear capacity; its nonlinear increment of 3.13 is positive in all ten seed pairs. Of the heterogeneous nonlinear capacity, 2.55 is from self terms and 2.16 from cross-lag products. Like the linear comparison, this is a sensitivity result for the composed behavioural model rather than an experimental population estimate.

Table 5.2: Characteristic timescales of the peripheral signals used in this chapter, and the composition cells whose fading-memory window covers them [163–165]. The grid of chapter 4 covers the phasic, respiration, and heart-rate-variability bands; no measured cell reaches the minutes-scale tonic band.

Affective signal	Timescale	Device coverage
Phasic EDA (SCR)	∼1–10 s	low–mid PEO ($t_{1/2}$ 3–19 s)
Respiration	∼3–5 s	mid PEO
HRV, HF band	∼2.5–7 s	mid PEO
HRV, LF band	∼7–25 s	PEO 0.3 ($t_{1/2} \approx$ 16–19 s)
Tonic SCL	minutes	not reached by the measured grid

5.6 Physiological Signals as a Test Bed

Affective computing uses physiological signals to infer emotional and physiological state [162]; here its signals and labels serve as a test bed for the device models. The device timescales, from a few seconds to tens of seconds, overlap the phasic electrodermal, respiration, and heart-rate-variability bands without time rescaling (table 5.2). These signals also contain several timescales at once: phasic electrodermal responses, respiration and heart-rate-variability rhythms over seconds, and tonic levels drifting over minutes [163–165]. Figure 5.5 places the measured relaxation time of each composition on the same axis as these physiological bands and shows the minutes-scale range not reached by composition alone.

Extending the model to tonic timescales. Raising the write amplitude lengthens fading memory in the measurements of chapter 4. Adding modelled nodes at $\tau = 120$ and 300 s changes little: random-input capacity beyond 75 s rises from 0.82 to 0.97 (9/10 seed pairs), total capacity is unchanged (4.67 and 4.64), and slow-band physiological reconstruction at delays up to 120 s rises by only $+0.003 \pm 0.022$ R^2 (5/10 draws; sec-

tion D.6). The stretched tails of the measured cells already carry much of this long-lag memory. The extension is hypothetical in any case, because no minutes-scale cell was measured and the drive-amplitude data do not identify these two time constants.

The experiments that follow use the Wearable Stress and Affect Detection corpus (WESAD) of Schmidt et al. [166]: fifteen subjects wearing a chest sensor that records electrodermal activity, respiration, temperature, and an electrocardiogram, with periods labelled baseline, stress, or amusement. WESAD is used in two complementary ways: as a source of real physiological streams whose multi-lag temporal context the reservoir must encode (section 5.7), and as a downstream affective-classification task (section 5.8).

5.7 Physiological Temporal-Context Reconstruction

The benchmarks of section 5.5 show that a spread of device timescales broadens memory on a random input. This section tests the same property on the multi-timescale physiological signals introduced in section 5.6, isolating temporal-context encoding before turning to affect labels (section 5.8).

The reservoir is run continuously over each WESAD session at $\Delta t = 1\,\mathrm{s}$ on four chest channels (electrodermal activity, respiration, temperature, and an electrocardiogram-derived heart rate), and a linear read-out reconstructs each channel at delays of 1, 3, 8, 20 and 45 s. These delays span fast phasic and beat-to-beat context, respiration- and HF-HRV-scale context, and tonic, LF-HRV-scale context [163, 164]. Evaluation is leave-one-subject-out over all fifteen subjects, with $N = 48$ nodes per bank.

Preprocessing is causal. ECG filtering and RR-interval assignment use only completed beats; resampling uses the nearest past label. Each channel is mapped to $(0, 1)$ by a monotone transform fitted to the first 60 s only: the calibration-prefix 5th and 95th percentiles map to 0.25 and 0.75, without hard clipping later values. Thus neither later held-out values nor affect labels set the scale. The reported out-of-sample R^2 uses the training-subject mean as its deployable null predictor, clips reconstructed normalised physiology to $[0, 1]$, and weights each held-out subject equally.

Table 5.3 reports the held-out reconstruction quality. A memoryless bank gives $R^2 = 0.636$, compared with 0.646 for instantaneous input. A single-timescale device bank raises R^2 to 0.679–0.684, and the heterogeneous composition bank reaches 0.707: +0.061 over instantaneous input and +0.023 over the stronger same-size homogeneous control. The heterogeneous-minus-fast and heterogeneous-minus-slow differences are each positive in 14/15 subjects (Holm-adjusted $p = 0.0067$). The fast homogeneous bank is strongest at the shortest delays and the slow homogeneous bank at the longest (fig. 5.6). A linear EMA bank at the eight cell half-lives, with no device nonlinearity or random mask, reaches 0.742 and exceeds the heterogeneous device bank in all fifteen subjects (-0.035; Holm-adjusted $p = 1.8 \times 10^{-4}$); a single EMA at the lead half-life gives 0.688. The useful content of the bank is therefore its set of time constants, and the nonlinear write and random masks lower the reconstruction.

5.8 WESAD Affective Classification

The downstream task is affective-state classification from the WESAD physiological streams with a leave-one-subject-out linear read-out. The reservoir and the 15 s causal prediction smoother run on the complete physical timeline; samples with undefined affect labels are excluded only from fitting and scoring. This prevents the smoother from joining labelled

Table 5.3: Subject-mean leave-one-subject-out R^2 for multi-lag physiological context reconstruction on WESAD ($N = 48$ nodes; means over ten bank seeds where applicable). The null prediction is the training-subject mean. Bands are reconstruction delays: fast = 1–3 s, mid = 8 s, slow = 20–45 s. The heterogeneous bank exceeds the stronger homogeneous control by +0.023 R^2 (14/15 subjects; Holm-adjusted $p = 0.0067$), and the linear EMA bank at the same half-lives exceeds the heterogeneous bank by 0.035 (15/15 subjects).

Bank	overall R^2	fast	mid	slow
Instantaneous	0.646	0.699	0.647	0.592
Memoryless	0.636	0.691	0.637	0.580
Homogeneous, fast	0.679	0.777	0.655	0.593
Homogeneous, slow	0.684	0.687	0.694	0.677
Heterogeneous	0.707	0.757	0.693	0.664
EMA, lead half-life	0.688	0.692	0.695	0.682
EMA bank, 8 half-lives	**0.742**	**0.794**	**0.724**	**0.699**

blocks separated by long protocol gaps. The study that introduced WESAD reports, with engineered window features and conventional classifiers under leave-one-subject-out validation, chest macro-F1 of 0.725–0.744 for the three classes and 0.904–0.915 for stress versus non-stress, and 0.861 for stress versus non-stress from the wrist [166]. The streaming linear readout of this chapter reaches 0.562, 0.799, and 0.756 on the same three tasks (table 5.4). The formulations differ (per-step streaming against windowed features), but a gap of 0.1–0.2 is too large to attribute to formulation alone: the device models do not approach an engineered-feature pipeline on these labels.

Demonstration A: window-level electrodermal classification. Binary stress-versus-baseline classification from electrodermal activity, read out at the window level from the lead composition alone, reaches leave-one-subject-out macro-F1 0.671 (fig. 5.7a). A static baseline using the mean, standard deviation, last value, and slope of the same signal reaches

Table 5.4: Matched comparison with the study that introduced WESAD. All rows report class-balanced macro-F1 under leave-one-subject-out validation over the same fifteen subjects; binary rows separate stress from baseline and amusement pooled. Schmidt et al. give the two best classifiers for each task. This work reports the heterogeneous device bank (three-class: seed mean; binary: clean streams). Accuracy-only and differently split studies are omitted.

Task	Study	Features and read-out	macro-F1
Three-class, chest	Schmidt et al. [166]	window features; AdaBoost, LDA	0.725, 0.744
	This work	streaming device states; linear ridge	0.562
Binary, chest	Schmidt et al. [166]	window features; RF, LDA	0.904, 0.915
	This work	streaming device states; linear ridge	0.799
Binary, wrist	Schmidt et al. [166]	window features; AdaBoost, RF	0.859, 0.861
	This work	streaming device states; linear ridge	0.756

0.714. The reservoir does not improve this 60 s-window task.

Demonstration B: streaming three-class tracking. The reservoir runs continuously over each session, and baseline, stress, or amusement is read out at every labelled step. Table 5.5 decomposes the result into four matched-size banks. A majority-class predictor reaches macro-F1 0.231 and a stratified-random predictor 0.333. Instantaneous input reaches 0.482; the memoryless bank reaches 0.513 ± 0.038; and the homogeneous fading-memory bank reaches 0.562 ± 0.038. The heterogeneous bank also reaches 0.562 ± 0.051, a $+0.080$ gain over instantaneous input and $+0.001 \pm 0.046$ relative to homogeneous, with heterogeneity higher in four of ten matched seed pairs. Linear EMA controls at the lead half-life and at all eight half-lives score 0.503 and 0.508. At the representative seed,

Table 5.5: Leave-one-subject-out macro-F1 for streaming three-class WESAD tracking (means ± SD over ten matched bank seeds; the linear controls have no seed dependence). Gains are relative to instantaneous input. The two device banks tie and both exceed linear EMA controls at the same half-lives; adding the measured timescale spread does not improve the sustained-label task.

Features	macro-F1	gain
Instantaneous (no memory)	0.482	—
Memoryless bank	0.513 ± 0.038	+0.031
Homogeneous bank (one τ)	$\mathbf{0.562 \pm 0.038}$	+0.080
Heterogeneous bank (τ spread)	$\mathbf{0.562 \pm 0.051}$	+0.080
Linear EMA, lead half-life	0.503	+0.021
Linear EMA bank, 8 half-lives	0.508	+0.026

amusement remains the weakest class (F1 0.40, versus 0.83 for baseline and 0.77 for stress). The experiment supports a memory benefit relative to an instantaneous read, and on this task the device banks also exceed the linear filters, but it shows no timescale-diversity benefit for these sustained labels.

Why heterogeneity does not improve the label scores. The affect labels are sustained states that change only at protocol boundaries, so one slow timescale already carries most of the history that separates them. A wider timescale distribution helps when the target explicitly requires several lags (table 5.3), but not here. This is also the one task in the chapter where the device banks exceed linear filters at the same time constants: 24 device states score 0.562, whereas the 32 features of the EMA bank score 0.508.

5.9 Continuous Monitoring Under Deployment Conditions

The sustained WESAD states can often be classified from the instantaneous signal level, as shown by the static-feature baseline. Wearable measurements also contain sensor noise and motion artefacts, particularly in wrist-worn electrodermal and photoplethysmographic signals. Fading memory implements temporal low-pass filtering and may therefore reduce sensitivity to transient artefacts. The tests below add synthetic noise, measured-motion gating, and an independent sensor corpus.

Figure 5.8 tests a continuous binary stress detector (stress versus baseline plus amusement, read out at every step, leave-one-subject-out, with the same linear read-out used throughout). On clean streams, macro-F1 ranges from 0.781 to 0.807. At $\sigma = 0.4$, instantaneous and memoryless reads fall to 0.641 and 0.649, while the homogeneous and heterogeneous memory banks retain 0.818 and 0.811. The heterogeneous-minus-instantaneous subject difference is +0.171 (12/15 positive, paired Wilcoxon $p = 0.015$). Clean false-alarm rates are similar (instantaneous 0.094, heterogeneous 0.089 seed-mean). Transition-window F1 changes from 0.832 to 0.840, but the subject difference is not resolved (6/15 positive, $p = 0.58$); all banks have zero median onset latency under the five-second stable-call definition. Temporal smoothing is useful under corruption, not for earlier detection in this protocol.

Accelerometer-gated motion artefact. Because the preceding corruption is synthetic, a second test gates its amplitude with the measured chest accelerometer. At the strongest gating, the instantaneous read falls to 0.702, the heterogeneous bank retains 0.811, and the EMA reaches 0.809. Both gains over the instantaneous read are +0.111; the EMA gain is resolved ($p = 0.035$) and the heterogeneous gain is not ($p = 0.055$;

section D.5). The data-driven control therefore supports temporal filtering under measured-motion corruption, but not a separate device-model advantage.

Classical-filter control. A single exponential moving average at the lead half-life reaches macro-F1 0.804 at $\sigma = 0.4$, compared with 0.641 for instantaneous input; its paired subject gain is +0.170 ($p = 0.022$). A linear EMA bank at the eight half-lives reaches 0.808, with no resolved gain over the single EMA (+0.003, $p = 0.35$). The heterogeneous device bank reaches 0.811 and does not differ from the EMA bank (−0.002, $p = 0.64$; table D.3). Robustness is therefore attributable to temporal low-pass filtering at these time constants, which the device bank performs as well as the linear control and no better.

Cross-corpus replication. The robustness test was repeated on the PhysioNet Non-EEG corpus of Birjandtalab et al. [167]: twenty subjects wearing an Affectiva Q wrist sensor for electrodermal activity and skin temperature, with pulse-oximeter heart rate. Cognitive and emotional stress are compared with relaxation; physical exertion is excluded. On clean streams, seed-mean heterogeneous and instantaneous scores are 0.657 and 0.639; the representative-seed subject difference is not resolved ($p = 0.11$). At $\sigma = 0.4$, they are 0.653 and 0.451: a +0.201 gain, positive in all twenty subjects ($p = 1.9 \times 10^{-6}$; table 5.6). The linear EMA bank retains 0.633, and the device bank exceeds it by 0.023 (16/20 subjects, $p = 0.017$). This independent corpus reproduces the noise-response ordering, while confirming that the clean-stream advantage is small.

Table 5.6: Noise-response comparison on the independent PhysioNet Non-EEG stress corpus [167] (twenty subjects; leave-one-subject-out, linear read-out). At $\sigma = 0.4$, the heterogeneous-minus-instantaneous difference is $+0.201$, positive in 20/20 subjects ($p = 1.9 \times 10^{-6}$). Clean-stream scores are comparable.

Detector	macro-F1 (clean)	macro-F1 ($\sigma = 0.4$)
Instantaneous (no memory)	0.639	0.451
Homogeneous (memory)	**0.678**	**0.669**
Heterogeneous (memory)	0.657	0.653
Linear EMA bank	0.647	0.633

5.10 Toward a Worn Device: Sensor Site and Operating Budget

The robustness of the previous section is a laboratory statement until two further questions are addressed: whether the advantage survives a consumer-grade sensor, and what can be said about an operating budget. The first is tested with the WESAD wrist recordings. For the second, only a reference calculation is possible because the simulated PEO nodes were not measured as a wired bank.

Sensor site. WESAD records each session with both a chest strap and an Empatica E4 wrist band. The wrist band has fewer and noisier channels: skin conductance, skin temperature, and photoplethysmography-derived heart rate, without respiration. On clean wrist data (table 5.7), instantaneous and heterogeneous scores are 0.724 and 0.756. At $\sigma = 0.4$ they are 0.615 and 0.769; the paired subject gain is $+0.117$ (10/15 positive, $p = 0.035$). The corresponding chest scores are 0.641 and 0.811. The linear EMA bank gives 0.773 clean and 0.755 at $\sigma = 0.4$ on the wrist, and the device bank does not differ from it (-0.013, $p = 0.38$). Clean wrist false-alarm rates are not improved (instantaneous 0.112, heterogeneous

Table 5.7: Continuous binary stress-detection macro-F1 on the WESAD chest strap versus the Empatica E4 wrist band (leave-one-subject-out, linear read-out; seed means). Under injected corruption, the temporal bank exceeds instantaneous input at both sites and does not differ from the linear EMA bank; σ follows fig. 5.8.

	Chest		Wrist E4	
Detector	clean	$\sigma = 0.4$	clean	$\sigma = 0.4$
Instantaneous (no memory)	0.781	0.641	0.724	0.615
Memory reservoir (het)	0.799	**0.811**	0.756	**0.769**
EMA bank, 8 half-lives (linear)	**0.806**	0.808	**0.773**	0.755

0.106), so the supported benefit is robustness under injected corruption rather than fewer clean-stream alarms.

Operating-budget reference. The parameter count is exact for the simulated architecture: the device dynamics are fixed, and the fitted $(N{+}1) \times K$ read-out contains 75 weights for $N = 24$, $K = 3$, or 147 for $N = 48$. The read-out also requires NK multiply–accumulates per step. Energy is less certain. Recalculation of one proof-of-concept *Super Yellow*/*Hybrane*/LiOTf junction gives 0.084 µJ–1.55 µJ across its positive write sequence (median 1.08 µJ). Multiplying that median by 24 nodes gives a reference core cost of 25.9 µJ per update, or 25.9 µW at one update per second; the corresponding 48-node value is 51.8 µW.

These products are not power predictions for the proposed reservoir. They transfer a 1 V result from one large-area *Hybrane* junction to PEO models identified at 3 V–4 V, without measured bank wiring, read energy, or peripheral circuits. They exclude the sensor analogue front-end, analogue-to-digital conversion, addressing, and control, and are neither an upper nor a lower bound. The device response remains faster than the sub-hertz physiological signals considered here, and its relaxation would perform the temporal filtering during the write/read cycle. The assumptions are

tabulated in section D.8.

5.11 Matching Device and Task Timescales

Choosing a cell. For a designed filter, the target corner frequency selects the composition (table 5.1). The two fastest cells, with corners near 0.04 Hz, are the baseline removers for electrodermal activity; slower cells smooth over tens of seconds and pass the tonic band. Because stretching lowers the corner, a stretched cell must be chosen by its corner frequency, not by its half-life. Corners above about 0.04 Hz lie outside the replicated grid; the faster host and anion variants of chapter 4 (TMPE, $t_{1/2} \approx 4\,\mathrm{s}$; TFSI, $\approx 1\,\mathrm{s}$) indicate where faster elements might come from, at survey-level evidence.

Reservoir banks. Performance depends on the overlap between device and task timescales (fig. 5.5). In the single-composition sweep, the strongly stretched PEO 0.6/0.09 cell gives the highest memory capacity for the seconds-scale benchmark (fig. 5.3). For the multi-lag physiological task, no single cell covers the full delay range, and the heterogeneous bank gives the highest device-bank score, below the linear EMA bank at the same half-lives (table 5.3).

Effect of timescale breadth. Table 5.8 separates fading memory from timescale breadth and from the device nonlinearity. On random input, breadth raises total linear memory ($+1.22 \pm 0.41$, positive in 10/10 seeds) and nonlinear information-processing capacity. On real physiology, heterogeneity gives a +0.023 context-reconstruction gain over the stronger homogeneous control, but a linear EMA bank at the same half-lives does better than either. On sustained three-class labels the two device banks

Table 5.8: Comparison of instantaneous input, homogeneous and heterogeneous device banks, and a linear EMA bank at the eight cell half-lives across the physiological tasks in this chapter. Temporal features show their largest gains under corruption. The linear bank gives the highest physiological-context score and the device banks the highest three-class score; under corruption the device and linear banks are close. Bold marks the highest score per row.

Task	Metric	Instant.	Homog.	Heterog.	EMA bank
Physiological context (multi-τ)	R^2	0.646	0.684	0.707	**0.742**
Three-class affect tracking	macro-F1	0.482	**0.562**	**0.562**	0.508
Stress monitoring, clean	macro-F1	0.781	**0.807**	0.799	0.806
Stress monitoring, σ=0.4	macro-F1	0.641	**0.818**	0.811	0.808
Wrist stress, clean	macro-F1	0.724	0.752	0.756	**0.773**
Wrist stress, σ=0.4	macro-F1	0.615	0.766	**0.769**	0.755

tie, and under corruption the homogeneous bank is marginally stronger. The task, not diversity alone, determines the useful distribution.

When one slow timescale dominates, as in the sustained stress labels and the stress-monitoring task, the homogeneous and heterogeneous banks are within 0.008 macro-F1 of each other at every site and noise level. The available variation comes from composition and device-to-device scatter; chemistry and drive amplitude provide additional, sample-limited axes in chapter 4.

Operating point and integration. The models apply in the measured sub-kilohertz regime, which covers the physiological signals used here but not fast logic. The assumed sub-threshold, state-preserving read sets the read-voltage and read-disturb limits for a circuit implementation. Figure 5.9 varies timescale-only jitter. Its primary value, $\sigma(\ln t_{1/2}) \approx 0.264$, is the median within-cell substrate-level spread on the screened

replicated grid (section D.4). The observed per-cell estimates span 0.089–1.018, so the wider sweep is a sensitivity analysis rather than a fitted uncertainty distribution. The legacy single-exponential fit gives a larger $\sigma(\ln \tau) \approx 0.85$, retained only as an upper stress-test value. No measured common jitter is applied to α or β. At zero timescale scatter, copies of one card are more redundant; the heterogeneous bank still differs through composition-dependent card parameters.

5.12 Limitations and Scope Conditions

In silico. The demonstrations are simulations. Each node is the measured $\phi \otimes \lambda \otimes f$ behavioural model of section 5.2, not a bench circuit. Every figure caption names the data behind it. Neither a wired filter nor a wired reservoir was built.

Designed filters. The filter results assume that the conductance enhancement superposes and that the measured large-signal decay also governs small signals around a written state. The composed node predicts a faster small-signal kernel when $\alpha < 1$ (section D.7); a small write superposed on a written state would decide between the two. The synthetic generator uses chosen parameter ranges rather than a fitted corpus, and the WESAD stress segment (a speaking task) may add movement-related phasic power; this affects all filters alike. The *RC* references are ideal components, and footprint, energy, and tolerance were not compared against a fabricated *RC* element.

The composition assumption. As detailed in section 5.2, write nonlinearity and fading memory were measured separately, at different write amplitudes and one inter-pulse cadence. Combining them assumes that the two regimes are compatible; even the direction and magnitude of the

omitted coupling are uncertain for the silver/PEO system. In a sensitivity analysis, each node's retention was rescaled with its write exponent over $\kappa = -0.5$ to $+0.5$. The mean heterogeneous-to-homogeneous capacity ratio stays between 1.29 and 1.42, with the heterogeneous bank higher in every seed pair (section D.6). The gold/TMPE negative-polarity measurements provide an external directional reference, $\rho = +0.46$ between write depth and decay half-time across 28 single-run traces. The nominal cards already have a stronger across-card rank association ($\rho \approx +0.57$), so the central external estimate maps to $\kappa^\star = 0$ and its upper confidence limit to about $+0.31$; the capacity ratio is 1.38 and 1.33 at these points. Because host, electrode, polarity, and protocol differ, this is a sensitivity anchor rather than validation of the composed silver model. A matched-amplitude experiment on the lead composition that varies pulse count and inter-pulse interval together is required.

Operating regime and read-out. The model is identified only in the measured sub-kilohertz regime; fast temporal tasks are outside its scope. Every read-out is a single ridge regression because a nonlinear read-out could supply the task nonlinearity independently of the reservoir. The read is assumed to be sub-threshold and state-preserving.

Heterogeneity axes. The quantitative comparisons span the replicated PEO/LiOTf composition grid; the single recovered PEO 0.15 substrate is excluded. Chemistry and drive amplitude enter only as secondary controls because their matched cells contain at most two substrates (chapter 4); Demonstration B is a proof of principle for composition heterogeneity, not a powered comparison across chemistries. The cards index composition, but thickness and other fabrication variables are not fully separable from that design. The available covariate checks do not explain the composition association, yet they cannot prove composition is the sole cause (fig. 4.1,

section 4.2).

Corpora, protocol, and negative controls. The affective results use WESAD as the main corpus, and noise robustness is replicated on the PhysioNet Non-EEG dataset (section 5.9). A broader multi-corpus study remains future work. In WESAD, the experimental blocks follow a fixed protocol rather than a counterbalanced order. Affect label, elapsed time, and slow session drift are therefore partly entangled: causal prefix calibration prevents future-sample leakage, but it cannot establish that a classifier is specific to affect rather than protocol position. The sustained block labels also do not provide independent ground truth for physiological onset, so the transition and latency analyses quantify agreement with protocol boundaries, not the onset of an internal state.

The strongest robustness results use synthetic, channel-wise corruption and should be read as controlled stress tests. The measured-motion gate is a more realistic but weaker control and does not resolve a heterogeneous-over-instantaneous advantage. The linear controls bound the device-specific contribution. A linear EMA bank at the same half-lives exceeds the device bank on physiological reconstruction and matches it under synthetic corruption on the chest and wrist (table D.3); the device banks exceed the linear controls only on the sustained three-class labels and, by 0.023, under corruption on the Non-EEG corpus. Timescale heterogeneity does not improve the final-label score (section 5.8).

5.13 Summary

Read as component cards, the replicated compositions supply time constants of about 4 to 28 s and low-pass corners of 0.002 Hz–0.04 Hz. Used as a designed filter, the fastest cell removes the tonic level of electrodermal

signals as well as a digital filter (synthetic SCR detection F1 0.713 against 0.701; WESAD chest stress-versus-baseline AUC 0.952 against 0.951) and behaves as an RC element with the same half-life. The measured stretch never improves on that element, and two-composition band-passes fall below the phasic band.

Behavioural models fitted to the measurements of chapter 4 also support reservoir simulations with a linear read-out. Each node follows its measured relaxation kernel and is bounded by the measured potentiation peak and pulse count. Under this model, the heterogeneous bank raises total linear memory from 3.22 to 4.44 (10/10 seeds) and mean information-processing capacity from 5.18 to 9.77. On WESAD, a lead-cell reservoir scores 0.671 on windowed binary EDA, below a static control (0.714). Streaming three-class macro-F1 is 0.562 for both heterogeneous and homogeneous banks, above linear EMA controls (0.508). Under synthetic corruption, the heterogeneous bank retains 0.811 while instantaneous input falls to 0.641, and a linear EMA bank at the same half-lives retains 0.808.

The additional value of a timescale distribution depends on the task and metric. It is large for random-input memory and nonlinear capacity, modest but subject-resolved for multi-lag physiological reconstruction ($+0.023$ R^2 over the slow homogeneous control), and absent for sustained WESAD labels. On reconstruction, a linear EMA bank at the same half-lives does better than any device bank (0.742 against 0.707), so the time constants themselves carry most of the benefit. Noise robustness replicates on the independent Non-EEG corpus and represents temporal filtering rather than a device-specific accuracy advantage.

The chapter therefore finds a practical role for the composition-set time constant as a designed filter element, and little added value from random banks over linear filters at the same time constants. The next experimental steps follow from that role: replaying recorded electrodermal signals

through a fabricated device, measuring the relaxation of a small write on a written state, and comparing a device with a printed RC element of the same time constant. A matched-amplitude pulse protocol and a replicated chemistry study would then test the model assumptions and extend the range of available time constants.

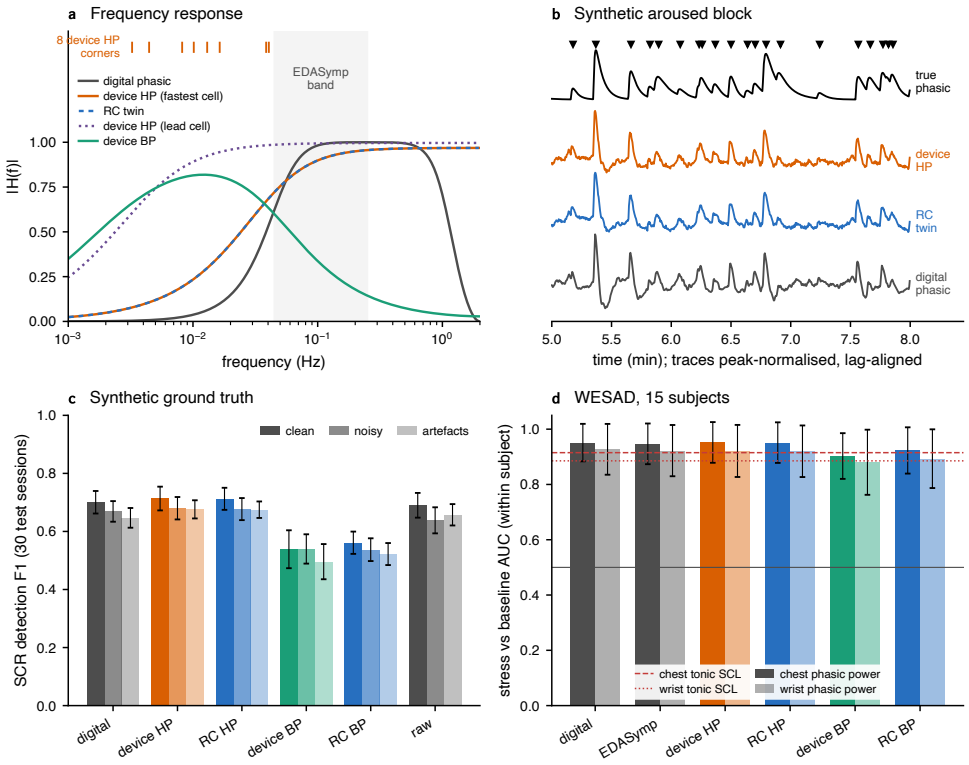


Figure 5.1: Composition-selected devices used as designed filters for electrodermal activity (simulations of the measured kernels). (a) Magnitude responses of a digital phasic filter, a single-device high-pass built from the fastest cell, the *RC* element with the same half-life, a high-pass built from the lead cell, and the best two-composition band-pass; ticks mark the high-pass corners of all eight cells, all below the sympathetic EDASymp band [156]. (b) Lag-aligned outputs during a synthetic aroused block; triangles mark the true SCR peaks. (c) SCR detection F1 on 30 synthetic test sessions (mean ± SD) for clean, noisy, and artefact-laden signals, with thresholds calibrated on 12 separate sessions. (d) Within-subject stress-versus-baseline AUC of per-minute phasic power on WESAD chest and wrist recordings (15 subjects, mean ± SD); dashed lines give the AUC of the tonic level alone. Device instances carry the measured within-cell timescale scatter; *RC* elements are ideal.

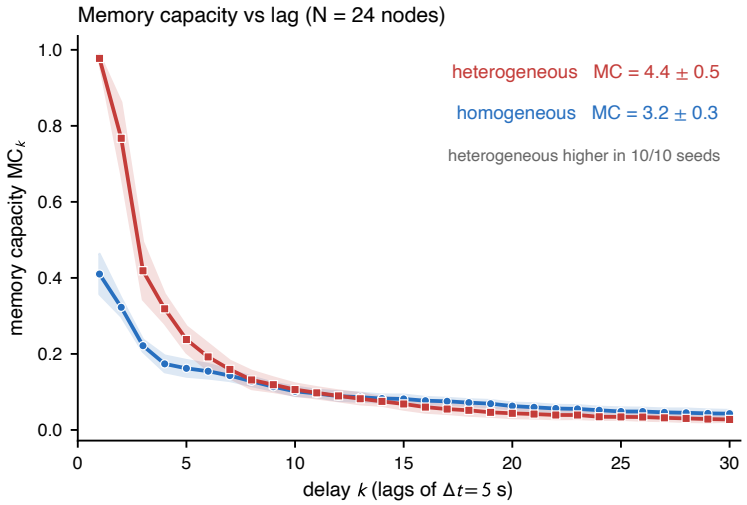


Figure 5.2: Linear memory capacity MC_k as a function of delay k for a homogeneous bank (all PEO 0.3/0.09 nodes) versus a heterogeneous composition bank ($\tau \approx 4$–$30\,\mathrm{s}$). Total capacity is 3.22 ± 0.32 and 4.44 ± 0.45, respectively; the heterogeneous value is higher in all ten seed pairs, mostly through recall at the shortest lags. Shaded bands are $\pm 1\,\mathrm{SD}$ over ten random input/bank seeds. Seeds quantify algorithmic sensitivity rather than experimental replication. $N = 24$ independent leaky nodes, random uniform input, linear ridge read-out.

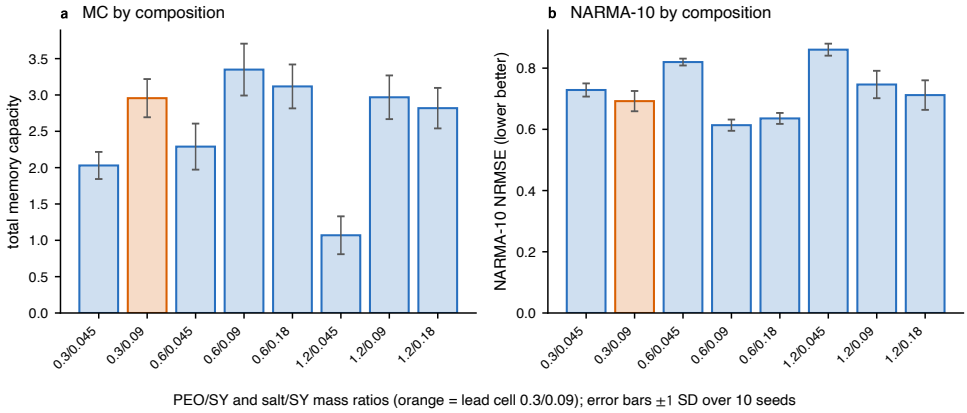


Figure 5.3: (a) Total linear memory capacity and (b) NARMA-10 reconstruction error (NRMSE, lower is better) for single-composition reservoirs across the PEO/LiOTf grid. The stretched PEO 0.6/0.09 cell gives the highest memory capacity and the lowest NARMA-10 error; the lead cell PEO 0.3/0.09 is shown in orange for reference. Only the *relative* composition ranking is claimed: absolute NARMA-10 NRMSE is high because a bank of independent, non-recurrent leaky nodes is a weak NARMA-10 reservoir.

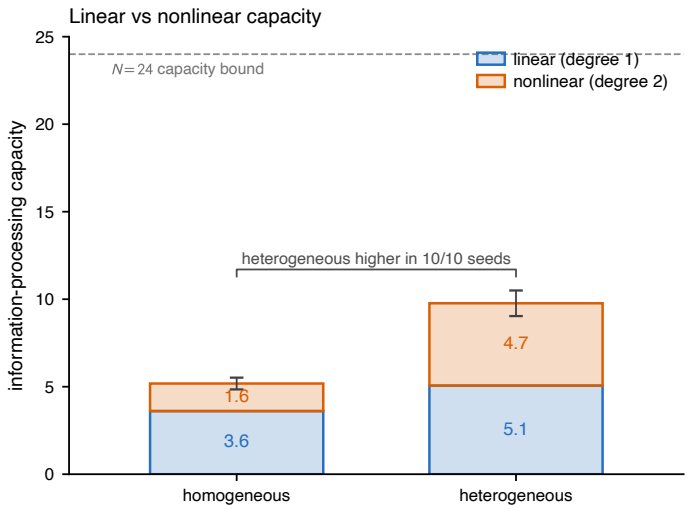


Figure 5.4: Information-processing capacity [82] split into linear (degree-one) and nonlinear (degree-two) parts for homogeneous and heterogeneous banks ($N = 24$, ten seeds). Both banks have nonzero degree-two capacity under the bounded node model. Mean total capacity changes from 5.18 ± 0.34 to 9.77 ± 0.73; the nonlinear component changes from 1.57 to 4.71 and is higher in all ten seed pairs. The dashed line is the $N = 24$ upper bound; seeds quantify algorithmic sensitivity rather than experimental replication.

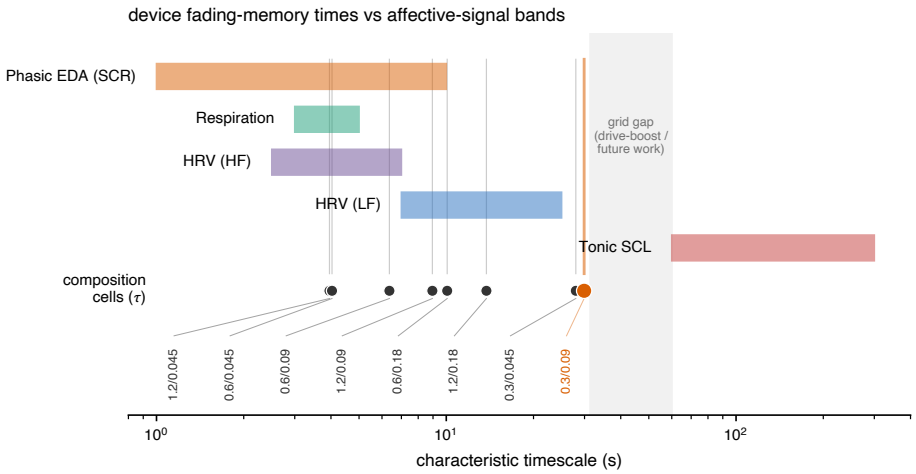


Figure 5.5: Each replicated PEO/LiOTf composition cell is placed at its derived relaxation time τ (markers, lower track) on the same logarithmic axis as characteristic timescale bands of peripheral affective signals (coloured bars; table 5.2). The grid ($\tau \approx 4$–$30\,\mathrm{s}$, anchored to measured $t_{1/2}$) overlaps the phasic-EDA, respiration, and HRV bands, with the lead cell PEO 0.3/0.09 (orange) near the slow LF-HRV band. Composition alone does not reach the tonic, minutes-scale band (shaded gap); testing that range would require a measured slower cell or matched drive-amplitude experiments.

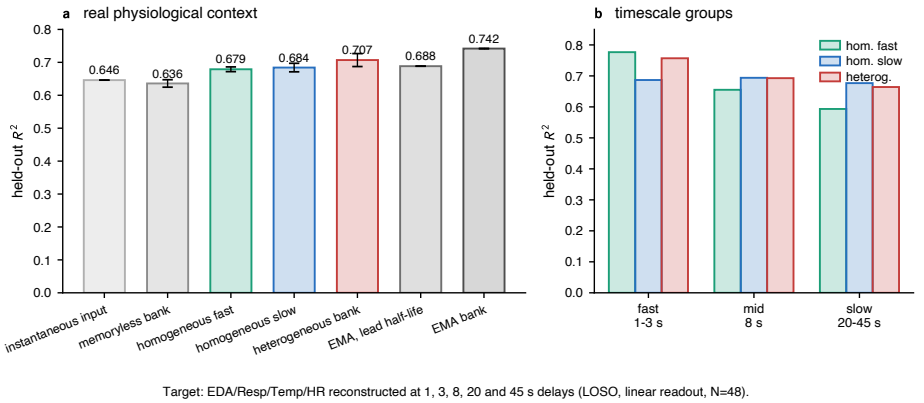


Figure 5.6: Multi-lag physiological context reconstruction on real WESAD streams (chest EDA, respiration, temperature, and causally derived heart rate). A linear read-out reconstructs each channel at delays of 1, 3, 8, 20 and 45 s (leave-one-subject-out, $N = 48$). Among device banks, the heterogeneous bank gives the highest overall subject-mean R^2 (0.707), ahead of the slow homogeneous control (0.684) and instantaneous input (0.646); a linear EMA bank at the same eight half-lives reaches 0.742. The zero-based axes avoid magnifying these modest differences.

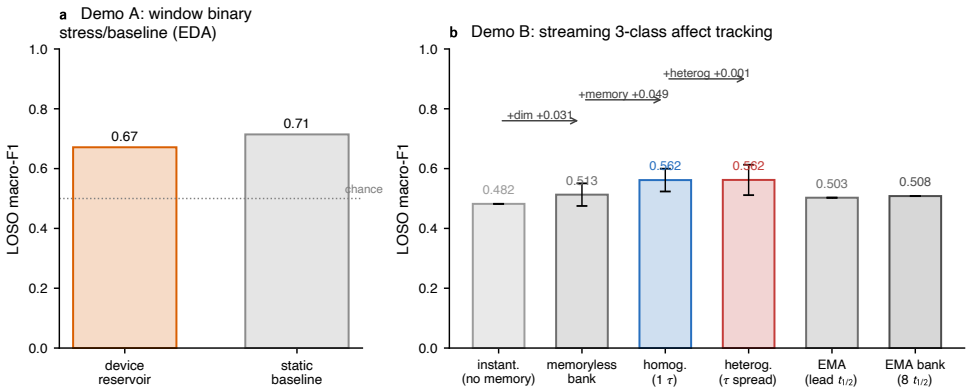


Figure 5.7: Affective classification with in-silico device reservoirs (leave-one-subject-out, linear read-out). (a) Window-level binary stress-versus-baseline from EDA: the device model scores 0.671, below the static-feature control (0.714). (b) Streaming three-class tracking. Fading memory improves on instantaneous input; the homogeneous and heterogeneous banks tie (0.562 ± 0.038 and 0.562 ± 0.051) and both exceed linear EMA controls (0.503 and 0.508). Error bars are ± 1 SD over ten bank seeds; both bar-chart axes begin at zero.

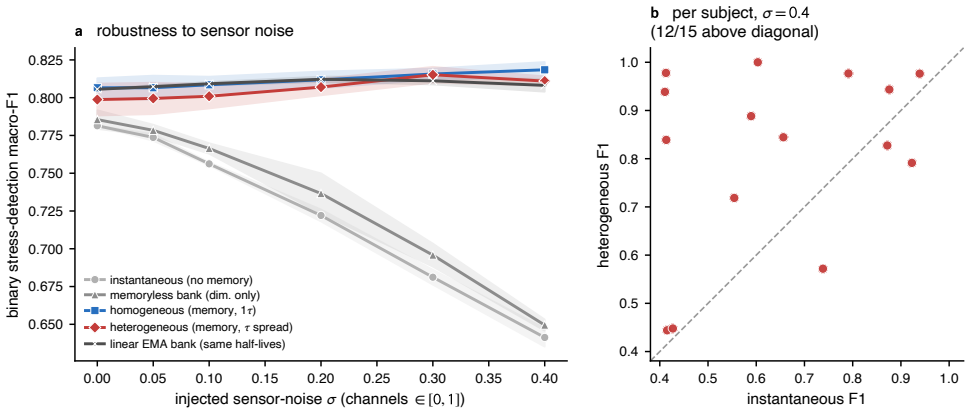


Figure 5.8: Continuous stress monitoring (binary stress vs. baseline-plus-amusement, per-step, leave-one-subject-out, linear read-out). (a) Detection macro-F1 against injected corruption. At $\sigma = 0.4$, instantaneous input falls to 0.641, while homogeneous and heterogeneous memory banks retain 0.818 and 0.811 and a linear EMA bank at the same half-lives retains 0.808; bands are ± 1 SD over seeds. (b) The heterogeneous bank exceeds instantaneous input in 12/15 subjects ($p = 0.015$). The homogeneous bank's score shows that timescale spread is not required.

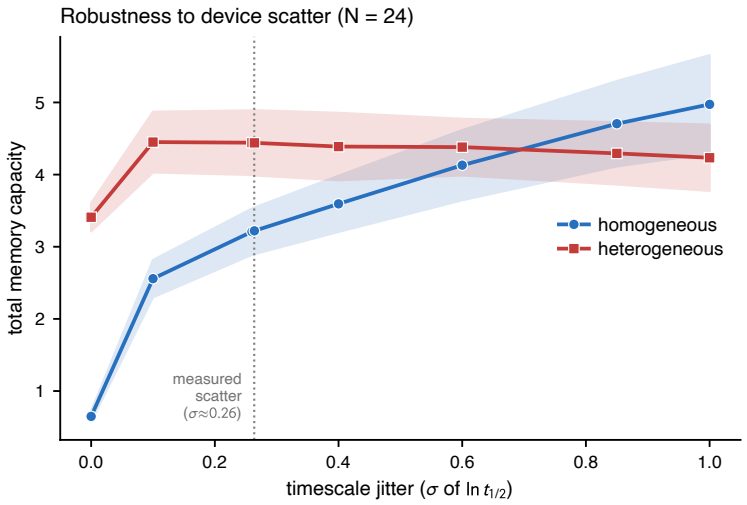


Figure 5.9: Total memory capacity versus injected log-normal timescale jitter for homogeneous and heterogeneous banks ($N = 24$, ten seeds; shaded bands ± 1 SD). The dotted line is the primary screened, substrate-level estimate $\sigma(\ln t_{1/2}) = 0.264$ (section D.4). Only the timescale is perturbed; α and β retain their measured cell-card values. Larger jitter values, including the legacy fit-based value near 0.85, are stress tests. The curves describe algorithmic sensitivity and do not establish variability as beneficial.

Chapter 6

Conclusions and Outlook

The thesis asked whether the volatility of an organic polymer-electrolyte memristive device can be controlled and put to use. Chapters 2 to 4 establish the device response, diagnose its reproducibility, and measure how composition sets its relaxation time. Chapter 5 tests what that time constant is good for, first as a designed filter element and then as a node of a reservoir, each against linear references. The conclusions below separate the measured device evidence from the model-based results and identify the experiments needed for a hardware test.

6.1 Summary of Contributions

The experimental and computational contributions differ in evidentiary strength because only the composition grid is replicated across devices.

A characterised synaptic platform (chapter 2). A reversible two-terminal organic memristive platform combines *Super Yellow*, *Hybrane*, and LiOTf in a solution-processed active layer between ITO and Ag electrodes. Across assay-specific junctions sharing that formulation and

architecture, it supports analogue I–V hysteresis (an 85 % conductance increase from the first to the second sweep at 1 V, growing ∼15-fold over ten cycles), reversible potentiation and depression exceeding 200 %, and two distinguishable retention regimes. Recalculation of the archived positive-pulse train gives 0.084 µJ–1.55 µJ per event as the state changes (median 1.08 µJ). Standard-exponential fits reported in the published SI give $\tau_S \approx 2.5$ s–3 s for short-term memory and $\tau_L \approx 4.7$ s for long-term memory. The formulation also supports one numerically separated four-state excitatory post-synaptic current sequence and an asymmetric Hebbian STDP window; direct re-fitting of the five-junction mean gives branch time constants of 89 ms and 150 ms. A fast-triflate/slow-lithium account is a plausible candidate interpretation, but ion motion was not measured and the later cation survey does not corroborate its simple ordering. The assays were independently peer reviewed [7], but they do not constitute a population study; table A.1 gives the specimen mapping.

A reproducibility diagnosis and a fabrication-record methodology (chapter 3). Successive batches of nominally identical *Hybrane* devices changed from low-current, high-area, multilevel behaviour to a high-conductivity response without a usable switching window. Per-device weighting, restriction to a standard protocol, sweep-amplitude matching, and recovery-tail sensitivity confirmed a multi-feature change: the on–off ratio fell from 2.44 to 1.21, normalised hysteresis area from 0.235 to 0.027, and potentiation at 3 V changed from +17.9 % to −2.1 %. Conduction also became more ohmic at a matched 1.2 V. The change followed fabrication date rather than device age, because finished *Hybrane* devices retained their characteristics over weeks. The evidence points to ageing of the *Hybrane* stock, with *Super Yellow* excluded, under a moisture-driven hydrolysis hypothesis supported by an annealing-recovery test. Replacing *Hybrane* with poly(ethylene oxide) increased the normalised window area from 0.26 to 0.42 at 3 V and maintained usable switching across roughly

two and a half years. The study also produced the fabrication-record standard, normalised characterisation procedure, automated feature pipeline, and trend-analysis tools used in the later chapters.

Compositional and chemical control of the dynamics (chapter 4). The replicated *Super Yellow*/PEO/LiOTf/Ag grid shows that composition tunes fading-memory time over roughly 2 s–20 s; higher ion-transport-polymer content gives a weaker, faster-relaxing response. Host (PEO versus TMPE) and anion (triflate versus TFSI) extend the range in the illustrative chemistry survey. Cation identity also shifts the dynamics, but the triflate series runs opposite to the hypothesised $Li^+ > Na^+ > K^+$ order and changes again with the anion. The available data therefore do not establish a transferable cation law. Drive amplitude changes the apparent timescale, so quantitative comparisons require a matched protocol. Equilibrium impedance falls along the composition axis and tracks fading-memory time, and its low-frequency capacitance exceeds the geometric capacitance by a median factor of about 69, consistent with seconds-scale relaxation arising from ionic storage within the film. ATR, XRD, and UV-Vis associate the host effect with short-range ether order in an X-ray-amorphous, salt-dissolved film without a shift in the *Super Yellow* optical gap. Because the PEO fraction changes both fading memory and input nonlinearity, each composition defines a paired operating point. Read as circuit elements, the replicated cells give two to five time constants that can be selected by formulation and told apart despite a within-cell tolerance of $\sigma(\ln t_{1/2}) \approx 0.26$.

Component cards and two uses of the time constant (chapter 5). Each replicated cell was summarised as a component card: half-life, stretch, the *RC* time constant with the same half-life, and the corner frequency of its measured kernel. The cards span time constants of

about 4 to 28 s and corners of 0.002 Hz–0.04 Hz; stretched cells have lower corners than their half-life suggests. Compact behavioural models ($\phi \otimes \lambda \otimes f$) that follow the full measured relaxation were then evaluated in simulation in two roles. As a designed filter, the fastest cell removes the tonic level of electrodermal signals as well as a digital filter (synthetic SCR detection F1 0.713 against 0.701; WESAD chest stress-versus-baseline AUC 0.952 against 0.951) and behaves as an RC element with the same half-life; the measured stretch never improves on that element, and two-composition band-passes fall below the phasic band. As reservoir banks, mixed compositions raise random-input linear memory from 3.22 to 4.44 and information-processing capacity from 5.18 to 9.77 (all ten seeds), and add 0.023 R^2 on multi-lag WESAD reconstruction over the best single-composition bank. A linear exponential-moving-average (EMA) bank at the same half-lives does better on that reconstruction (0.742 against 0.707) and matches the device banks under injected noise. The device banks exceed the linear controls on the sustained three-class labels (0.562 against 0.508), where they remain well below a 2018 engineered-feature pipeline. Seeds quantify algorithmic sensitivity rather than experimental replication, and all task results are simulations of measured device models.

6.2 Answers to the Thesis Questions

Realising and understanding an exemplar. *Can a reversible two-terminal organic memristive platform made from a solution-processed polymer-electrolyte composite support analogue switching, potentiation and depression, distinct retention regimes, EPSC, and STDP responses, and can cation–anion mobility asymmetry account for the retention regimes?* The common formulation and architecture of chapter 2, fabricated by solution processing and thermal evaporation, support these responses across assay-specific junctions. Differential displacement and relaxation of tri-

flate and lithium are consistent with the two regimes but are not identified by the electrical measurements. The reverse and anion-dependent cation ordering in chapter 4 means that the proposed simple mobility hierarchy is not corroborated; species-selective measurements remain necessary.

Composition as the quantitative control. *How do the dynamics depend on composition, and does moving the cation along the Li^+, Na^+, K^+ series shift the timescale as coordination chemistry suggests?* The replicated PEO/LiOTf grid shows a monotonic composition dependence: the ratio tunes fading-memory time over roughly an order of magnitude, with switching window and potentiation strength moving in the same direction. Cation identity shifts the dynamics, but the triflate retention series runs opposite to the predicted $Li^+ > Na^+ > K^+$ order and changes again with the anion. Sample size and protocol/electrode confounds prevent a transferable cation law. Coordination chemistry remains a qualitative interpretation rather than a quantitative result.

Utility for temporal processing. *What can the composition-set time constant be used for, how does a device compare with an RC element of the same half-life, and does a bank of mixed compositions compute anything that linear filtering does not?* In the simulations of chapter 5, the time constant is usable as a component value: a cell chosen by its corner frequency performs baseline removal on electrodermal signals at the accuracy of a digital filter and is equivalent to an RC element of the same half-life. Mixed-composition banks broaden random-input memory and nonlinear capacity, but on the physiological tasks a linear filter bank at the same time constants matches or exceeds them except on the sustained three-class labels. The time constants themselves carry most of the value.

6.3 Benchmarking Against the Right Reference Class

Because the device state relaxes over seconds, two reference classes apply: conventional filter components, for the designed-filter role, and temporal and event-driven neuromorphic systems, for the reservoir role [5, 27, 28]. Non-volatile DRAM, SRAM, and crossbar ReRAM serve different storage functions. Current physical-reservoir comparisons must also separate the physical node, input encoding, output layer, and network architecture [123]; task scores from different layers and protocols are not a common leaderboard.

Against conventional filters. As a filter, a composition-selected device performs like an *RC* element of the same half-life and like a digital filter, so it offers no accuracy advantage. Its possible advantage lies in how the time constant is obtained: a $0.0825\,\mathrm{cm}^2$ two-terminal film reaches seconds with no separate capacitor, whereas time constants of hundreds of milliseconds already require large capacitors or very small currents in analogue CMOS [6], and printed circuits have relied on printed low-pass filters for temporal processing, including printed electrodermal stress detection [81, 158]. Against these references the device has measured costs: write voltages of 3 V–4 V in the PEO protocols, a write energy of about 1 µJ per event in the proof-of-concept junction, a within-cell tolerance of $\sigma(\ln t_{1/2}) \approx 0.26$, and unencapsulated stability measured over hundreds of hours.

Against physical reservoirs. Physical reservoir computing has been demonstrated in dynamic memristors [33, 35], organic electrochemical devices [36], and delay-coupled single nodes [159]. More recent systems add direct electrical tuning in two-terminal oxide devices [126], gate- and

wavelength-controlled organic transistor dynamics [124], and an organic electrochemical reservoir with a physical device read-out [125]. These systems require fading memory and nonlinearity [5, 19]. The present devices have measured fading memory, while the fitted models have nonzero nonlinear information-processing capacity. Their sub-kilohertz response overlaps many physiological and environmental signals without time rescaling. Composition changes the fading-memory time over 2 s–20 s; in the simulations, the heterogeneous model bank raises linear and nonlinear capacity, but linear filters at the same time constants match it on most physiological tasks.

Against organic neuromorphic devices. Much of organic neuromorphic electronics targets linear, symmetric, non-volatile conductance updates for weight storage, as in the ENODe family [22–24]. The devices studied here use ion-relaxation timescales for temporal processing and are compared with ionic and electrolyte-gated organic neuromorphic systems [36, 115].

Scope of the claims. The experimentally supported claim is that composition sets seconds-scale fading memory in this two-terminal organic platform, with a measured tolerance. The computational claims are that the measured relaxation, used as a designed filter, performs slow-signal baseline removal as well as a conventional filter, and that banks of behavioural models add little to linear filtering at the same time constants on the physiological tasks studied. The application scores are in silico and do not establish the performance of a wired circuit; no priority is claimed for a first tunable two-terminal reservoir, and no affective-computing advantage is claimed. The simulated reservoir contains independent, non-recurrent nodes; its absolute NARMA-10 errors are high and its capacities remain below the node-count bound. A fabricated device filtering a

Table 6.1: Recent directly relevant reference points for tunable physical reservoirs. Temporal ranges are the values reported under each study's own extraction protocol; tasks, read-outs, and metrics differ, so the table calibrates scope rather than ranking performance. "Physical" means that measured device states formed the reservoir; it does not imply that every input/output operation was analogue or on-chip.

Study	Node and terminals	Tuning handle	Reported temporal range	Reservoir evidence and task scope
This thesis	*Super Yellow*/PEO/LiOTf organic composite; two-terminal	PEO/salt composition; drive amplitude	2 s–20 s across the replicated composition grid	Single-node dynamics measured; designed filters and reservoir banks simulated on WESAD/PhysioNet tasks against linear controls; no wired circuit
Gao et al. (2024) [124]	Organic vertical field-effect transistor; three-terminal optoelectronic	Input wavelength and gate bias	0.005 s–13.2 s with optical input and gate bias	Physical grouped-reservoir states; image recognition and time-series prediction
Han et al. (2025) [125]	Solid-electrolyte OECT; three-terminal	Ethylene-glycol content selects volatile or non-volatile operation	No common fitted-τ range reported	Physical reservoir and physical device read-out; three-class human-activity classification
Lee et al. (2025) [126]	Pt/HfO_2/TiN memristor; two-terminal	Inter-pulse base voltage	Fitted τ: 0.92 ms to 123 ms; inputs 0.14 ms–100 ms	Physical reservoir states; N-MNIST classification and Mackey–Glass prediction

recorded physiological signal remains the missing end-to-end test.

6.4 Limitations of the Evidence Base

The conclusions are limited by the evidence available for each part of the thesis.

1. **Extended protocols are proof-of-concept.** EPSC, separated short-term memory and long-term memory retention, and STDP exist only for the chapter 2 *Hybrane* formulation; some specimen identifiers are not resolved in the recovered archive, and the available multi-junction data share one substrate. They function as formulation-level lithium priors and checks in the later chapters and were not extended to population-level comparison across the PEO family.

2. **Only composition is replicated.** The quantitative core is the *Super Yellow*/PEO/LiOTf/Ag composition grid. The host, anion, and cation comparisons rest on one or two screened devices per matched cell and are illustrative. The electrode contrast is also illustrative and partly confounded with fabrication generation; its negative-polarity measurements are single-composition and uncurated, and its frequency-domain measurements rest on one batch with substrate-confounded cations. No powered material law, including a $Li^+ > Na^+ > K^+$ ordering, is claimed.

3. **Protocol-specific timescales.** Because the drive amplitude cosets the apparent fading-memory time, absolute retention values are protocol-specific, and the supra-threshold read used in the comparative corpus differs from the sub-threshold read of the proof-of-concept device.

4. **The composition assumption behind the models.** Pulse integration and state relaxation were characterised separately, at different write amplitudes (potentiation at 3 V, decay at 4 V) and one inter-pulse cadence. Combining them in the behavioural models assumes that the two regimes are compatible. The nominal composition cards already have an across-card write–retention association stronger than the central estimate transferred from the gold/TMPE corpus ($\rho = +0.46$), so that estimate maps to $\kappa^\star = 0$ in the sensitivity model. Across the tested coupling range, the heterogeneous-to-homogeneous linear-capacity ratio stays between 1.29 and 1.42. The designed-filter results further assume that the measured large-signal relaxation also governs small signals around a written state. A matched protocol and a small-signal relaxation measurement are still needed for the lead composition.

5. **The application results are in silico and only partly cross-corpus.** Every temporal-processing result is a simulation of measured device models with a linear read-out; no wired filter or reservoir was built. The physiological results use WESAD as the main corpus, with synthetic ground truth for the filters and the PhysioNet Non-EEG corpus for noise robustness. Linear filters at the same time constants match or exceed the device banks on most of these tasks, and the device models remain below an engineered-feature pipeline on the WESAD labels.

6. **Raw-data availability is incomplete.** The repository verifies the committed analysis code, derived tables and figures, and complete thesis build, but the experimental archive is separately held and unversioned. A SHA-256 manifest binds the principal processed database snapshot without covering every raw measurement. WESAD and PhysioNet Non-EEG must also be obtained independently; CASE is excluded because its raw corpus was unavailable and no

CASE result is reported. The exact boundary and regeneration commands are recorded in section D.3.

7. **Material stability remains open.** Devices were characterised unencapsulated on a ∼300 h ambient shelf scale; cycling endurance, read-disturb under continuous pulsing, and long-run drift under encapsulation were not systematically studied, and device ageing was not modelled in the comparative corpus.

6.5 Outlook and Future Directions

The immediate experimental work follows from the filter role identified in chapter 5. The model assumptions, the chemistry comparisons, stability, and integration follow.

Filter a recorded signal with a fabricated device. Recorded electrodermal signals, converted to the write protocol of chapter 4, could be replayed through a lead-row or fast-row device and read sub-threshold. Comparing the measured output with the simulated filter and with a digital reference on the same recordings would test the designed-filter result end to end.

Measure the small-signal relaxation. A small write superposed on a written state, at several operating points, would show whether small signals relax with the measured large-signal kernel or with the faster kernel predicted by the composed model. The answer fixes the corner frequency of a device used as a filter and decides whether the fastest cells already reach the sympathetic band.

Compare with a printed *RC* element. A device and a printed *RC* element realising the same time constant on the same substrate would settle the practical question: footprint, drive voltage, energy per update, tolerance, and drift over time.

Matched pulse protocol. A matched-amplitude pulse train should vary pulse count and inter-pulse interval together for one composition, electrode, and read protocol. The gold/TMPE negative-polarity corpus of chapter 4 already records integration and relaxation within single runs and finds interval-dependent potentiation and drive-depth-dependent relaxation. Repeating this measurement on the lead silver composition with a sub-threshold read would supply the missing joint parameters for the behavioural model.

Replicated chemistry survey. A cation series in one host and anion, at fixed protocol and electrode, with a sub-threshold read and at least three replicates per cell, would test the $Li^+ > Na^+ > K^+$ hypothesis at adequate power. The silver TMPE retention follows the opposite order, K>Na>Li; the gold steady-state window is flat; and the gold negative-polarity data resolve no order. The lock-in corpus contains one cation-ordered candidate: a $\approx 1.4\times$ K>Na>Li junction-capacitance trend at one substrate per salt (chapter 4). Profilometry of the existing substrates and a replicated series could test whether this trend is chemical or geometric. Chemistry- and bias-resolved impedance and a generation-controlled silver-versus-gold split would strengthen the microscopic and electrode comparisons.

Test the degradation chemistry. The moisture-driven hydrolytic-aging hypothesis for the *Hybrane* stock (chapter 3) cannot be tested directly because the aged consumable no longer exists in its degraded state. FTIR, GPC, Karl-Fischer/TGA, and EIS on retained aged-versus-

fresh stock would test the hypothesis in future material campaigns. The fabrication-record methodology now records reagent identity and history.

Extend the range of time constants. The replicated grid offers corners below about 0.04 Hz. The faster host and anion variants of chapter 4 (TMPE near 4 s, TFSI near 1 s) point to elements fast enough to set the upper edge of the sympathetic band, and modified polyether hosts could extend memory toward the minutes-scale tonic band. Replicated series would be needed to turn these survey values into component values, and to separate memory time from the input nonlinearity that composition currently changes with it.

Stabilise for long-run inference. Encapsulation and systematic measurements of ageing, endurance, and read disturb are required to assess continuous operation and remove uncontrolled ageing from later comparisons.

Build a reservoir only where linear filtering falls short. A wired bank of composition-diverse devices with an on-chip linear read-out would test the reservoir simulations directly. The simulations suggest that such a bank is worth building for tasks like the sustained WESAD labels, where device banks exceeded linear filters, and not for reconstruction or noise suppression, where a linear filter bank at the same time constants was as good or better.

Physiological sensor interfaces. Solution processing, mechanical flexibility, biocompatible chemistry, and seconds-scale dynamics make these devices candidates for skin-conformal sensing and inference. In the frequency-domain measurements of chapter 4, a DC-biased junction acts as a voltage-programmable low-pass amplifier below 10 Hz. A hardware

experiment could test whether one junction can combine pre-amplification and temporal filtering at a physiological sensor interface.

6.6 Closing Remarks

The replicated experiments identify composition as a control of seconds-scale fading memory in these polymer-electrolyte composites; the cation trend expected from coordination chemistry is not transferable across the available data. Read as component values, the compositions give a printable two-terminal element whose time constant is chosen at formulation. In simulation, that element filters slow physiological signals as well as a conventional filter, while banks of mixed compositions add little beyond linear filtering at the same time constants. Filtering a recorded signal with a fabricated device, measuring its small-signal relaxation, and comparing it with a printed RC element would test the practical value of the chemical control.

Appendix A

Supporting Information for Chapter 2

This appendix gathers the detailed experimental information and supporting evidence for the proof-of-concept platform of chapter 2. The main chapter retains the evidence needed to assess the formulation-and-architecture claim; the material collected here records the specimen mapping, supporting methods, protocol details, and auxiliary measurements.

A.1 Materials and Fabrication Detail

A.1.1 Component Notes

The semiconducting component is *Super Yellow*, the trivial name for the high-molecular-weight PPV copolymer sold as PDY-132 by Merck. The formal copolymer name is:

poly[{2,5-di(3′,7′-dimethyloctyloxy)-1,4-phenylenevinylene}-*co*-
{3-(4′-(3″,7″-dimethyloctyloxy)phenyl)-1,4-phenylenevinylene}-

co-
{3-(3′-(3′,7′-dimethyloctyloxy)phenyl)-1,4-phenylenevinylene}].

The conjugated phenylene-vinylene backbone supplies electronic transport, while the branched alkoxy side chains support solubility and reduce crystallisation-driven phase separation in blends [7, 147]. In ion-containing *Super Yellow*/*Hybrane* systems, related LEC devices have shown electroluminescence lifetimes exceeding 1600 h, supporting the chemical compatibility of this formulation family [106].

Hybrane is a hyperbranched polyester-amide produced by Polymer Factory (Sweden) [130]. The hyperbranched architecture creates an irregular, amorphous macromolecule rich in ether and ester oxygens. These oxygen atoms provide hard Lewis-base coordination sites for alkali-metal cations. Because ion motion is coupled to polymer segmental motion, transport in this class of host follows the polymer-electrolyte picture described in section 1.6.1, including non-Arrhenius temperature dependence near the glass-transition regime [90, 168].

LiOTf is a polymer-electrolyte salt widely used in batteries and LECs [100, 105]. Its processing role is to dissolve with *Super Yellow* and *Hybrane* in cyclohexanone without visible phase separation at the selected ratio. It provides a small, hard Li^+ cation (0.59 Å–0.76 Å for four- to six-fold oxygen coordination [146]) and a charge-delocalised triflate anion. This contrast motivates the candidate STM/LTM interpretation in section 2.5.1; it does not by itself determine ionic mobility.

A.1.2 Solution Preparation and Film Deposition

The three stock solutions were prepared separately in cyclohexanone at 15 mg mL^{-1} (*Super Yellow*), 10 mg mL^{-1} (*Hybrane*), and 5 mg mL^{-1} (LiOTf). The *Super Yellow* solution was stirred at 50 °C overnight.

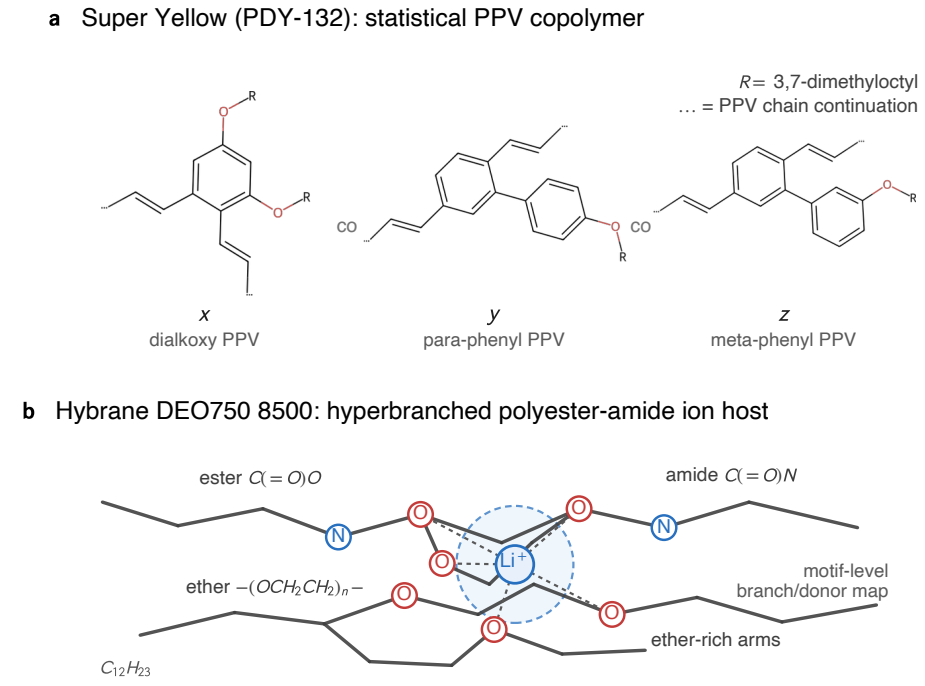


Figure A.1: Chemical components of the proof-of-concept composite. **(a)** *Super Yellow* is the commercial PDY-132 PPV-family statistical copolymer; R denotes the branched 3,7-dimethyloctyl side chain. **(b)** *Hybrane* DEO750 8500 is represented as an oxygen-rich hyperbranched polyester-amide motif, highlighting ether, ester, and amide donor sites rather than a monodisperse molecular formula. **(c)** LiOTf is shown as the lithium salt of triflate, $LiCF_3SO_3$. The selected active-layer formulation is *Super Yellow*:*Hybrane*:LiOTf = 1:0.30:0.09 by mass.

The solutions were then mixed to give the final mass ratio *Super Yellow*:*Hybrane*:LiOTf = 1:0.30:0.09, corresponding to final concentrations of $8.72\,\mathrm{mg\,mL^{-1}}$, $2.62\,\mathrm{mg\,mL^{-1}}$, and $0.78\,\mathrm{mg\,mL^{-1}}$, respectively. Solution handling was performed under dry nitrogen to limit moisture uptake.

ITO-patterned glass substrates were ultrasonicated successively in aqueous Mucasol detergent solution, deionised water, and 2-propanol, each for 15 min. After drying under nitrogen, substrates were transferred immediately into a nitrogen glovebox (oxygen and water below 1 ppm). The active layer was spin-coated at 2000 rpm for 60 s, then annealed at 75 °C for 3 h. The anneal removes residual cyclohexanone, allows chain relaxation, and stabilises the mixed polymer-electrolyte morphology.

The top Ag electrode was thermally evaporated in the glovebox through a shadow mask defining the $0.0825\,\mathrm{cm^2}$ junction. Ag was deposited at $0.1\,\mathrm{nm\,s^{-1}}$–$0.5\,\mathrm{nm\,s^{-1}}$ to a total thickness of 100 nm, monitored by quartz crystal microbalance. The slow initial rate was used to limit metal penetration into the soft polymer layer, which would otherwise risk filamentary shorts [169].

A.2 Electrical Protocol Detail

Electrical measurements used a Keithley 2450 SourceMeter connected to a custom probe station and controlled through Keithley TestScript Builder. Unless noted otherwise, measurements were carried out under nitrogen, with voltage applied to the Ag top electrode relative to the grounded ITO bottom electrode.

A.2.1 Assay-to-Specimen Map

The proof-of-concept claim concerns one formulation and device architecture, not one physical junction carrying every assay. Table A.1 records the finest specimen identity recoverable from the local archive and the published supporting information. A crossbar "junction" or "pixel" is one ITO/active-layer/Ag overlap; several junctions share a fabricated substrate and therefore are not independent device or batch replicates.

Table A.1: Assay-to-specimen map for the Chapter 2 evidence. "Published summary" indicates that the numerical result is recoverable from Ref. [7], but the local archive does not resolve the physical specimen identifier.

Assay	**Recovered evidence unit**	**Supported inference**
I–V hysteresis	One junction, NM_v025	Representative analogue hysteresis of the common formulation; no population estimate.
Pulse potentiation/depression	One junction (L5), NM_v055	Reversible pulse-protocol demonstration.
Pulse-number response	Five junctions, NM_v055, Day 8	Within-substrate junction spread; not independent-device or batch reproducibility.
STM/LTM retention	Three published fit traces; specimen ID unresolved	Existence and timescale of two protocol-defined retention regimes; no population estimate.
EPSC state sequence	Published state ratios; specimen ID unresolved	Four-state protocol demonstration; no population estimate.
STDP timing kernel	Five junctions, NM_v055, Day 19	Within-substrate mean timing response and junction spread.
Shelf stability	Published time series; specimen roster unresolved	Calendar stability over the reported interval; not cycling endurance or batch stability.

A.2.2 Read–Write–Wait Measurement

The variable-pulse and variable-delay measurements use a common read–write–wait sequence (fig. A.2). First, a low reading pulse, $V_{\mathrm{read}} = 0.5\,\mathrm{V}$,

establishes the initial conductance G_0. This voltage is below the independently determined ionic-redistribution threshold of approximately 0.7 V [7]. Second, a train of N_{pulses} write pulses of duration 0.1 s perturbs the ionic configuration. Third, after a waiting time t_{wait} under open-circuit conditions, a second read pulse measures G_f. The reported response is $\Delta G = (G_f/G_0) \times 100\%$. Between experiments, the device was reset by shorting the junction for 300 s, and G_0 was checked before the next measurement.

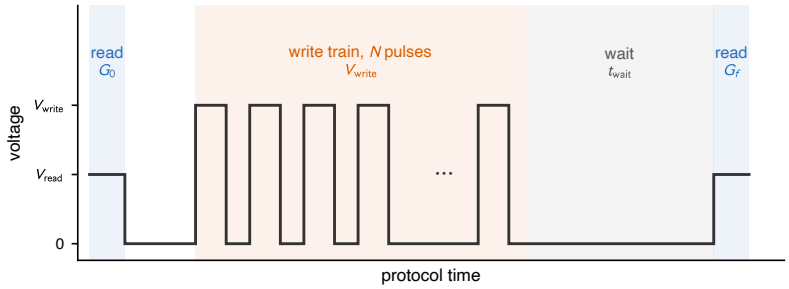


Figure A.2: Read–write–wait protocol used for the pulse-number and retention measurements. A low read pulse establishes G_0, a train of write pulses perturbs the ionic configuration, a controlled open-circuit waiting time allows relaxation, and a second read pulse measures G_f. Generated from the protocol reported in Ref. [7].

A.2.3 Extended Synaptic Protocols

The EPSC experiment defines states operationally by their signed current response under a common 1 V read-out bias. The device is first held at 1 V for 1 s to define S_0. Two 2 V, 50 ms pulses generate excited states S_1 and S_2, and two −2.5 V, 50 ms pulses generate inhibitory states S_3 and S_4. The signed ratios $R_n = I_{S_n}/I_{S_0}$ are reported in section 2.4.4.

For STDP, pre- and postsynaptic spikes are emulated by triangular voltage waveforms applied to the two-terminal device. The experimentally applied

voltage is the algebraic sum

$$V_{\mathrm{device}}(t, \Delta t) = V_{\mathrm{pre}}(t) + V_{\mathrm{post}}(t - \Delta t), \tag{A.1}$$

with $\Delta t = t_{\mathrm{post}} - t_{\mathrm{pre}}$. Each signed spike template has maximum amplitude $V_{sp} = 1.55\,\mathrm{V}$ and duration 1.2 s; Δt was varied from −600 ms to 600 ms. Template overlap produces a measured resultant as large as 3.1 V at $|\Delta t| = 50\,\mathrm{ms}$. Negative Δt produces potentiation and positive Δt produces depression under the sign convention used in section 2.4.5. The applied-voltage logic of both protocols is shown in fig. A.3.

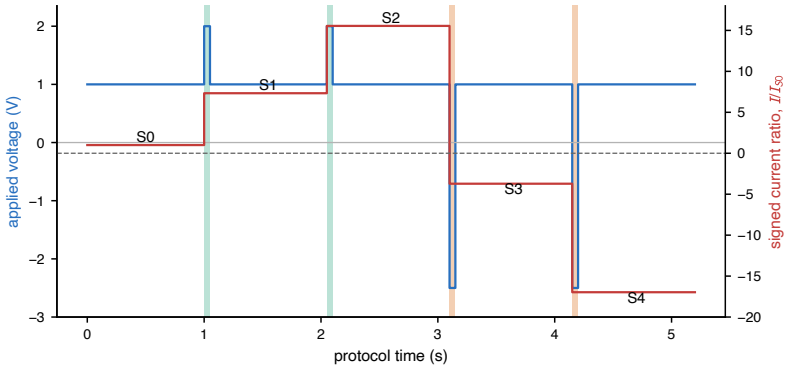
The archived branch-fit file is not used for the thesis figure because its depression curve does not fit the five-junction mean in the master table. Both branches were therefore re-fitted directly to that mean using the offset exponential of eq. (2.3), excluding the zero-delay point. The resulting constants are $\tau_+ = 89\,\mathrm{ms}$ and $\tau_- = 150\,\mathrm{ms}$; junction-resampling 95% intervals are 78 ms–95 ms and 134 ms–172 ms. These intervals describe within-substrate junction variation only. The fitted parameters and audit metrics are recorded in `handouts/ch2_stdp_fit.csv` and regenerated by `scripts/ch2_figures.py`.

A.2.4 Retention-Fit Parameters

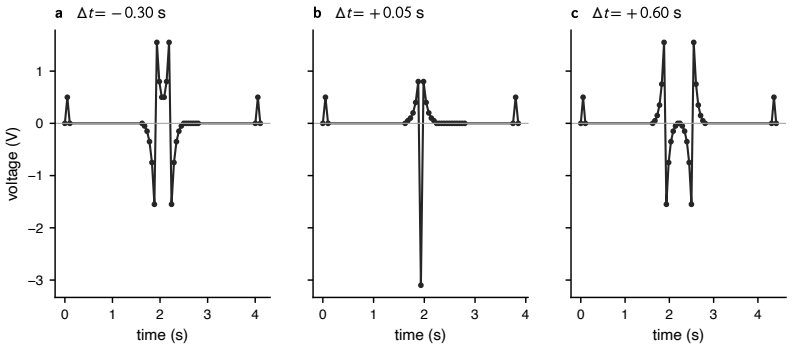
Table A.2 lists the standard-exponential fit parameters reported for the proof-of-concept retention traces. The fits use $G_f/G_0 = R_\infty + A\exp(-t_{\mathrm{wait}}/\tau)$, with $\beta = 1$ fixed.

A.3 Morphology and Stability Support

The spin speed was selected from a 500 rpm–3000 rpm optimisation [7]. Low speeds produced films that were too thick and less uniform; high



(a) EPSC protocol trace.



(b) STDP waveform superposition.

Figure A.3: Supporting protocol panels for the extended synaptic measurements. **(a)** Reconstructed EPSC trace showing the applied-voltage sequence and the signed current ratios reported for states S_0–S_4. **(b)** Measured total voltage across junction L5 for three timing offsets, read directly from the NM_v055 Day19 archive. These panels document the protocol logic; the compact evidence used in the main chapter is reported in fig. 2.5.

speeds made the film thin enough for substrate roughness to matter. The selected fabrication condition used 2000 rpm, giving a 209 nm active layer measured by contact profilometry across a masked step edge with an Ambios XP-1 profilometer.

Surface topography was characterised by tapping-mode AFM using a Bruker Dimension Icon instrument, a silicon probe with resonant frequency

Table A.2: Published standard-exponential fit parameters for the Chapter 2 retention traces. No free stretch exponent, confidence interval, or goodness-of-fit statistic was reported for these proof-of-concept fits.

Condition	τ (s)	R_∞	A
1 V, 10 pulses (STM)	2.57	1.01	0.374
1 V, 50 pulses (STM)	3.01	1.01	0.91
3 V, 10 pulses (LTM)	4.73	1.05	3.30

300 kHz and force constant 40 N m^{-1}, and a scan rate of 0.6 Hz. The AFM images showed smooth films with RMS roughness of 1 nm–3 nm over the investigated spin-speed range [7]. The absence of large phase-separated features supports the reading of the active layer as a homogeneous polymer-electrolyte composite at the scale relevant to the vertical junction.

Ambient shelf stability was assessed for approximately two weeks ($\sim$300 h) on unencapsulated devices stored in air at room temperature [7]. The monitored quantities were baseline conductance G_0, maximum conductance change $\Delta G_{\max}$, and STM retention time τ_S. All three remained within 20 % of their initial values over the measurement window. This is a shelf/calendar stability result, not a cycling-endurance measurement. Figure A.4 summarises the film thickness, AFM roughness, and shelf-stability metrics together.

Encapsulation outlook. The ambient stability is expected to improve further with encapsulation, standard practice in organic electronics: a thin SiO_2 or Si_3N_4 barrier deposited by atomic-layer or chemical-vapour deposition would suppress the oxygen and water ingress that are the two primary degradation pathways [170]. That even unencapsulated devices survive two weeks establishes the baseline robustness of the composite.

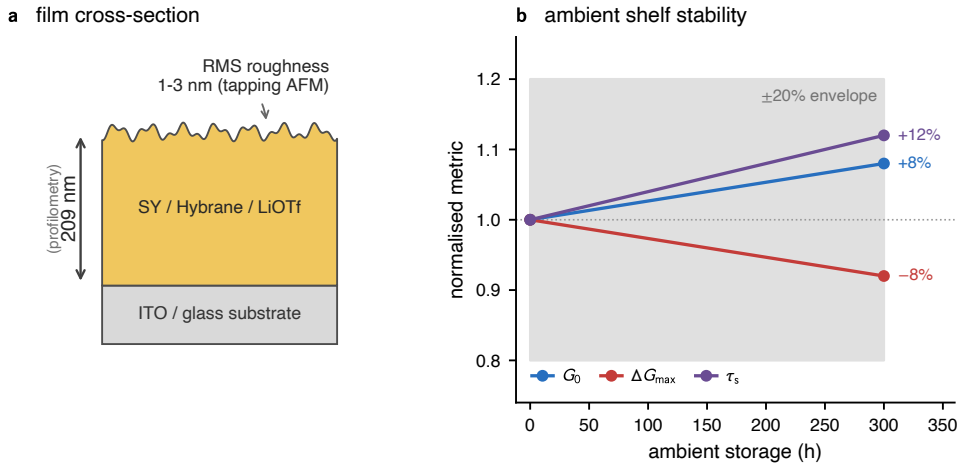


Figure A.4: Supporting morphology and shelf-stability summary for the Chapter 2 device family. The selected processing condition gives a 209 nm active layer; tapping-mode AFM reports 1 nm–3 nm RMS roughness across the investigated spin-speed window; and the ambient shelf-stability metrics remain within a 20 % envelope over approximately 300 h. Compiled from the supporting measurements reported in Ref. [7]; this is not a cycling-endurance result.

A.4 Ion-Transport Models for the Composite

This section details the injection and bulk-transport descriptions summarised in section 2.5. The two limiting-case models are inherited from the polymer light-emitting electrochemical cell (LEC) literature and reconciled by van Reenen *et al.* [104] as two regimes of the same ion redistribution.

A.4.1 Equilibrium Ion Distribution and Response to an Applied Field

In the idealised zero-bias starting state inherited from LEC models, the film is macroscopically electroneutral and any field is limited to built-in or interfacial contributions [100, 101, 104]. At the molecular scale, Li^+ may occupy coordination environments in the Hybrane matrix and triflate may exist in free, paired, or aggregated states; those populations were not measured in this blend.

When a voltage is applied across the device, an electric field $\mathcal{E} = V/d$ (where d is the film thickness) is established. In the absence of free charge carriers that could screen this field, it would be uniform throughout the film. However, the mobile ions respond to the field by migrating: cations drift toward the cathode (ITO at positive bias on Ag), and anions drift toward the anode (Ag). This migration leads to the build-up of a charged double layer at each electrode interface on a timescale determined by the ionic mobility and the film thickness. The resulting non-uniform ionic charge distribution concentrates the field in a thin screening length near each electrode, while the bulk of the film is field-free [101, 102], with consequences for charge injection described by the two models below.

A.4.2 Electrodynamic Model

In the electrodynamic (ED) model, first formulated by deMello *et al.* [101, 102] to describe LECs, charge injection at the electrode–organic interface is the rate-limiting step for current flow. In this injection-limited regime, the current density is governed by thermionic emission over the Schottky barrier at the metal–polymer interface,

$$J \propto T^2 \exp\left(-\frac{\Phi_B}{k_B T}\right), \tag{A.2}$$

where Φ_B is the effective barrier height, T the absolute temperature, and k_B the Boltzmann constant. The ionic double layer at the electrode interface reduces Φ_B by bending the polymer bands: positive ionic charge near ITO lowers the conduction-band edge there, while negative ionic charge near Ag raises the valence-band edge, both reducing the effective injection barrier. In the memristive context, conductance changes therefore arise from changes in the interface ion concentration: a higher concentration of positive ions near ITO means lower Φ_B, higher hole injection, and higher conductance, and the ions relax back toward equilibrium by thermal diffusion once the field is removed, the physical basis of the memory decay.

A.4.3 Electrochemical Doping Model: Ohmic Injection

The electrochemical doping (ECD) model, formulated by Pei *et al.* [100] and developed by subsequent authors [103, 140], instead takes charge injection to be ohmic (not rate-limiting), so the majority of the voltage drop occurs in the bulk and the current is limited by bulk conductivity. The defining feature is that ion accumulation *electrochemically dopes* the conjugated polymer: cations accumulate near the cathode, stabilising injected electrons (n-doping), and anions near the anode, stabilising injected holes (p-doping); the doped regions are far more conductive than the undoped polymer. The degree of doping (and hence the conductance) is set by the extent of ionic migration, which is governed by the history of applied voltage, so the conductance records the cumulative ionic-displacement history.

A.4.4 Resolution of the Two Models

The prolonged LEC-community debate over the relative validity of the ED and ECD models was reconciled by van Reenen *et al.* [104], who showed that the limiting behaviour depends on injection conditions: ED when injection is strongly barrier-limited and ECD when injection is closer to ohmic. A device may move between these limits as ionic redistribution develops. Applying that framework here is physically plausible, but the present measurements do not identify an ED-to-ECD transition or justify assigning ED uniquely to triflate and ECD uniquely to Li^+; section 2.5 states the assignment and its failed simple cation-ordering prediction explicitly.

A.5 Historical Comparison at the Time of Publication

Table A.3 places the device alongside representative organic and hybrid memristive devices available at the time of the 2022 publication, on the axes of section 1.3.4: device structure, accessible conductance states, short- and long-term memory, STDP, and reversibility. It is a historical positioning, not a 2026 state-of-the-art claim.

Two combinations not represented in the comparison set at the time stand out. The first is that the same 2-T formulation and architecture support both short-term memory- and long-term memory-labelled regimes while remaining fully reversible. Of the prior reports, Liu and co-workers [96] comes closest, with tunable short- to long-term transitions in a 2-T structure; the redox route, however, modifies the active layer irreversibly at each cycle, precluding practical cyclic operation [118]. The present platform reaches comparable short-term memory/long-term memory func-

Table A.3: Historical comparison of key performance metrics of the present work with representative organic and hybrid memristive devices available at the time of the 2022 publication. "2-T" = two-terminal; "3-T" = three-terminal; "STM" = short-term memory; "LTM" = long-term memory; "STDP" = spike-timing dependent plasticity; "—" = not reported or not applicable.

Reference	Material	Structure	States	STM	LTM	STDP	Reversible
Zakhidov 2010 [99]	$[Ru(bpy)_3]^{2+}/PF_6^-$	2-T	2	—	—	No	Partial
Lei 2014 [95]	PVA film	2-T	2	—	—	No	No
Liu 2016 [96]	$[EV(ClO_4)]/(BTPA\text{-}F)$	2-T	cont.	Yes (tunable)	Yes	No	No
Gkoupidenis 2015 [107]	PEDOT:PSS (OECT)	3-T	cont.	Yes	No	Yes	Yes
Xu 2016 [116]	PEDOT:PSS (OECT)	3-T	cont.	Yes	Yes	No	Yes
Kim 2018 [92]	Elastic organic composite	3-T	cont.	Yes	No	Yes	Yes
Goswami 2017 [97]	Molecular switch	2-T	2	—	Yes	No	Yes
Seo 2019 [117]	P3HT organic FET	3-T	cont.	Yes	No	Yes	Yes
This work [7]	**SY/Hybrane/LiOTf**	**2-T**	**cont.**	**Yes (10–15 s)**	**Yes (>45 s)**	**Yes**	**Full**

tionality with full reversibility (conductance returns to G_0 after the reset protocol). The second is STDP in a fully reversible 2-T architecture: the earlier organic STDP demonstrations [92, 107, 117] were all 3-T with a gate supplying the timing rule, whereas here the rule emerges from the superposition of pre- and postsynaptic pulses at the Ag terminal.

The remaining axes summarise the operating regime rather than a competitive claim. The ambient shelf lifetime (about 300 h unencapsulated) exceeds most comparable unencapsulated organic memristive devices by roughly an order of magnitude [121]; cycling endurance, not characterised here, is where mature inorganic platforms reach $> 10^{10}$ cycles [17]. Recalculation from the archived positive-pulse waveform gives 0.084 µJ–1.55 µJ per event as the junction potentiates (median 1.08 µJ), higher than the picojoule budgets of dedicated CMOS neuromorphic hardware [1, 28] and mature inorganic memristors [69]. The large junction area and long pulse contribute, but proportional energy scaling with area has not been demonstrated and is not claimed.

Appendix B

Supporting Information for Chapter 3

This appendix records the methodological backbone of the degradation analysis of section 3.3. The bridge chapter retains the result (the window collapse and the ohmic drift) and the figures; the four controls that make that result defensible against the confounds peculiar to this dataset are stated in full here. The analysis is reproduced by `scripts/bridge_hybrane_peo_reproducibility.py`.

B.1 The Four Methodological Controls

The degradation analysis is built on a methodological backbone of four controls; each was verified to change the conclusion if dropped.

1. **Per-device weighting (mandatory).** The number of pixels measured per device collapses from sixteen to two over the campaign and is strongly anti-correlated with date (Spearman $\rho = -0.80$, $p \approx 1 \times 10^{-15}$, $n = 65$; fig. 3.2a inset). Pixel-pooled statistics are

therefore biased toward the early, densely measured devices. Every feature is reduced to *one value per device* (the median over that device's freshest-day curves) before any test.

2. **Standard-protocol corpus only** ($n \approx 69$ devices, span 2021-02 to 2022-05): *Super Yellow*/*Hybrane*/LiOTf on silver, annealed at 75 °C with no second stage, at the lead mass ratios (*Hybrane* 0.3, LiOTf 0.09). This excludes the invasive experiments (above all the high-temperature anneal batches that partially recovered devices a year in) which would otherwise contaminate the feature-versus-date story.

3. **Sweep-amplitude matching, with each feature read where it is expressed.** A $0 \to +X \to 0$ loop excites the device more at larger X, so it conducts more even when read back at the same voltage; amplitude is confounded with both date and material (fig. 3.2b). The *window* features (on–off ratio, normalized area) only open up at high amplitude (at $\sim$1.2 V they sit near unity for every device, early or late, and *cannot* show a collapse), so they are read in the $\sim$3 V stratum, while the *conductivity* features are read in a tightly matched $\sim$1.2 V band.

4. **Pre/post-inflection contrast and recovery-tail sensitivity.** Because the notes locate the behavioural inflection at NM_v026 (2021-04-22), the window collapse is tested as a step, early ($\leq$Apr 2021) versus later (Mann–Whitney), instead of as a whole-campaign correlation. Every conductivity trend is additionally reported with and without the deliberately-reverted recovery batches ($\geq$2022-03).

The quantitative outcome of applying these controls (the window collapse at $\sim$3 V and the matched-amplitude ohmic drift) is reported in table 3.2 and discussed in section 3.3.

Appendix C

Supporting Information for Chapter 4

This appendix gathers the analysis traceability, the confound audits, the fit methodology, and the per-composition-cell quantitative tables that underlie chapter 4. The main chapter retains the result and the figures; the material here records how the numbers were produced and lists the per-cell values behind the heatmaps and design-space plots.

C.1 Analysis Traceability and Aggregation

Data-quality control. Two screening layers are applied before any number is reported, both recorded so that the analysis is reproducible. Individual measurements that are erratic or non-physical are flagged and excluded, including the archive's "Y/N" broken flags after normalisation. In addition, a visual review of the recorded curves was carried out; the resulting verdicts, whether a curve is used as measured, cleaned by removing identified anomalous points, or discarded, together with the exact points retained, are stored under the full device–measurement–day–

Table C.1: Raw-to-summary trail for the three common measurements used in chapter 4. The same recorded trail underlies the figures of section 4.4 and the behavioural models of chapter 5.

Stage	Artifact	Role in the analysis
Raw measurement	`DEVICES_LAB_DATA/.../D1_all.txt` or split `D1_I/V/T.txt`	Keithley current, voltage, and time traces for each device day, measurement type, and crossbar pixel.
Fabrication metadata	`DATABASE/UPDATED_DEVICES_LIBRARY.csv`	Canonical device composition, electrode, spin speed, annealing, storage, measurement atmosphere, and operator metadata.
Extracted features	`DEVICES_HYST_*`, `DEVICES_PULSES_*`, `DEVICES_DELAYTIME_*`	Per-curve, per-pixel, and per-device feature tables generated by the archive extraction pipeline.
Exclusion and curation	`FILTERED_DEVICES.csv`; `handouts/ch4_png_qa_curation.csv`	Red-flag exclusion list plus per-curve visual QA verdicts; retained points are applied by script, not selected separately for each figure.
Recomputed descriptors	`handouts/ch4_decay_fits.csv`; `handouts/ch4_pulse_descriptors.csv`	Post-curation descriptors: $t_{1/2}$, identified τ, β where stable, pulse-growth exponent α, peak ratio, and turnover.
Cell summaries and figures	`handouts/ch4_decay_by_cell.csv`; `handouts/ch4_pulses_by_cell.csv`; `scripts/ch4_comparative_figures.py`	Cell medians, substrate counts, representative curves, heatmaps, swarm plots, and design-space figures.

pixel key in a curation registry. The most specific matching verdict is applied automatically by the analysis scripts, instead of being made ad hoc. The fitted timescale obtained directly from the instrument-side pipeline was found not to be robust to such anomalous points, so all timescales reported in chapter 4 are recomputed after this screening. Table C.1 records the full raw-to-summary trail underlying these screened descriptors.

Table C.2: Figure-level sources for the Chapter 4 figures whose detailed file paths are not repeated in the main captions. Script names are relative to the thesis repository; raw measurements reside in the experimental archive paths listed by the scripts.

Output	Reproduced by	Principal inputs and audit trail
Figure 4.1	`scripts/thickness_rpm_audit.py`	`DEVICES_PROFILOMETRY_STATS.csv` joined to the screened Chapter 4 descriptors; audit in `handouts/14_thickness_rpm_confound_audit.md`.
Figures 4.2 to 4.5 and 4.14	`scripts/ch4_comparative_figures.py`	`handouts/ch4_decay_fits.csv`, `handouts/ch4_pulse_descriptors.csv`, `handouts/ch4_decay_by_cell.csv`, and `handouts/ch4_pulses_by_cell.csv`; curve curation in `handouts/ch4_png_qa_curation.csv`.
Figures 4.7 and 4.9	`scripts/ch4_comparative_figures.py`	Chemistry-landscape values from the screened descriptor tables; the protocol overlay uses the same-device v114 matched-amplitude comparison recorded in the Chapter 4 audit trail.
Figure 4.6	`scripts/ch4_iratr.py`; `scripts/ch4_xrd.py`	ATR band extraction in `handouts/ch4_iratr_bands.csv`; assessments in `handouts/16_iratr_assessment.md` and `handouts/17_xrd_assessment.md`.
Figure 4.8	`scripts/ch4_uvvis.py`	2025 *Super Yellow*/TMPE cation-series UV-Vis spectra; assessment in `handouts/18_uvvis_assessment.md`.
Figure 4.10	`scripts/ch4_eis.py`	0 V-DC impedance descriptors from the experimental database, joined to `handouts/ch4_decay_by_cell.csv`; per-cell output in `handouts/ch4_eis_by_cell.csv`.
Figure 4.11	`scripts/ch4_electrode.py`	Silver-versus-gold metrics in `handouts/ch4_electrode_by_cell.csv`; audit in `handouts/19_electrode_au_assessment.md`.
Figure C.2	`scripts/ch4_rigor.py`	On-state conductance from `DEVICES_HYST_PIXEL_INFO`; gradient/factorial/bootstrap, EIS timescale bridge, reproducibility and heterogeneity-resource statistics in `handouts/ch4_gradient_stats.csv`, `ch4_cell_ci.csv`, `ch4_factorial.csv`, `ch4_percolation.csv`, `ch4_eis_timescale.csv`, `ch4_cycling.csv`, `ch4_heterogeneity_resource.csv`.

The aggregation ladder. The aggregation is deliberately conservative. HYST raw points are first reduced to sweep-level features in `DEVICES_HYST_CURVE_INFO.csv`; broken or FILTERED rows are removed; surviving sweeps are reduced first to junction medians and then to one median per fabricated substrate; composition-cell values are medians across substrates. PULSES and DELAYTIME follow the same curve/junction/substrate/cell logic, but with protocol-specific descriptors: the pulse curves are summarised by α, peak ratio, and turnover, while the decay curves are summarised primarily by $t_{1/2}$, with τ and β retained only when the fit passes the identification thresholds in `scripts/ch4_dynamics_fits.py`. The representative curves in fig. 4.2 are selected by the figure script from substrates near the relevant cell medians, so they are illustrative examples of the same summarized population, not hand-picked evidence outside the pipeline.

Stretched-exponential identification. The depotentiation decays are fitted with the Kohlrausch form of eq. (4.1). The eight-to-thirteen-point decay curves often under-constrain β and τ jointly, so the fitted τ is reported only where the fit is well identified, and the model-free half-enhancement time $t_{1/2}$ is used as the primary timescale; where both are available they agree (table C.3). The stretch exponent is constrained to the physical Kohlrausch range $0 < \beta \leq 1$ of eq. (4.1): an unconstrained fit occasionally returns $\beta > 1$ (a *compressed* exponential, which does not correspond to a distribution of activation barriers), and such cells are instead pinned at $\beta = 1$, i.e. a near-undispersed simple-exponential relaxation, not reported as a spurious stretched fit. Fitted β values then lie predominantly in 0.6–0.9, i.e. genuinely stretched relaxation consistent with a distribution of relaxation times. The most ion-rich cells (high PEO, high salt) approach $\beta = 1$, but the sparse fits do not identify a unique microscopic origin for that pattern. The dimensionless quantity $1 - \beta$ describes within-curve stretching under this model; it is not commensurate

with the between-substrate statistic $\sigma(\ln t_{1/2})$. Their numerical similarity is therefore reported only as a descriptive comparison (section C.13), not as evidence that one barrier-width parameter causes both.

C.2 Film-Thickness and Fabrication Confound Audits

Film thickness as a measured covariate. In most batches, the spin-coating speed was raised for higher-PEO cells to partly equalise thickness (spanning 1500 rpm–3500 rpm). The compensation was deliberate but incomplete: median film thickness still rises from ≈ 227 nm at PEO 0.3 to ≈ 298 nm at PEO 1.2, giving a Pearson $r \approx 0.65$ between r_{PEO} and thickness (fig. 4.1). Four observations constrain, but do not eliminate, a thickness contribution:

1. Conductance ratios reduce sensitivity to absolute geometry, but the timescales need not be thickness-independent; this is why a direct covariate audit is required.

2. The activation voltage has no resolved thickness trend ($r \approx -0.06$) across a 2.6× thickness range. This is inconsistent with simple inverse-thickness threshold scaling, but it does not prove an intrinsic mechanism.

3. Within the single lead cell (PEO 0.3/salt 0.09) a 38% device-to-device thickness spread (196 nm–271 nm) leaves the fading-memory time essentially unchanged ($t_{1/2} \approx 18$ s–22 s).

4. In the 23 substrates with $t_{1/2}$, the partial correlation of thickness with $\log t_{1/2}$ at fixed PEO is $+0.28$, while the PEO association at fixed thickness remains -0.51.

The composition association therefore persists after adjustment for measured thickness, but this sample does not exclude a secondary thickness contribution. The full join and statistics are reproduced by `scripts/thickness_rpm_audit.py` and `handouts/14_thickness_rpm_confound_audit.md`.

Other fabrication variables are held constant or crossed. The full per-device fabrication record was audited to confirm that the composition trends are not driven by some other processing variable. Across the silver composition grid the principal process variables are held *constant*: the annealing temperature (75 °C, no second stage), the electrode (100 nm silver, evaporated by the same operator), the semiconductor loading, the solvent, and, which matters for a hygroscopic polyether, storage *and* measurement under a glovebox atmosphere, so ambient humidity is controlled out. The variables that do change across the grid (fabrication batch and date, the operator who prepared the solutions, the semiconductor lot, and the device age at measurement) are *crossed* with composition, not confounded with it: each composition cell is populated from several independent batches, and within any single batch every composition level was fabricated together and measured on the same day. The composition trends are accordingly reproduced *within* individual single-day batches, including one in which the spin speed was held fixed across all compositions, so that the trend appears with no thickness compensation at all. The full column-by-column audit is recorded in `handouts/15_fabrication_confound_audit.md` and reproduced by `scripts/fabrication_confound_audit.py`.

Table C.3: Fading-memory (DELAYTIME) descriptors per composition cell, silver electrode, matched 4 V/2 V protocol. $t_{1/2}$: model-free half-enhancement time (s). R_{60}: retention at 60 s. τ, β: Kohlrausch parameters (median over the n_{id} substrates with an identified fit). Em-dash: not identified.

Cation	PEO/salt	n	$t_{1/2}$ (s)	R_{60}	n_{id}	τ (s)	β
Li^+	0.15/0.18	1	22.4	0.27	0	—	—
Li^+	0.3/0.045	4	15.8	0.20	3	25.8	0.64
Li^+	0.3/0.09	3	19.2	0.28	2	23.9	0.83
Li^+	0.6/0.045	2	2.8	0.06	1	3.0	0.97
Li^+	0.6/0.09	3	3.5	0.02	1	3.0	0.61
Li^+	0.6/0.18	2	5.5	0.04	2	4.6	0.61
Li^+	1.2/0.045	3	2.7	0.13	0	—	—
Li^+	1.2/0.09	3	6.0	0.09	2	6.7	0.91
Li^+	1.2/0.18	3	9.0	0.13	3	10.0	0.87
Na^+	0.3/0.09	1	24.0	0.42	0	—	—
K^+	0.3/0.09	1	8.0	0.22	0	—	—

C.3 Per-Composition-Cell Summaries

Tables C.3 and C.4 list the per-cell values behind the heatmaps of section 4.4 and the design space of fig. 4.14. Each row is a composition cell; values are medians over screened substrates in the cell, with substrate count n given so that the replicated Li^+ composition series is distinguished from the single-substrate Na^+/K^+ chemistry points.

C.4 Realised Device Palette

The quantitative analyses of chapter 4 are deliberately confined to the replicated silver PEO/LiOTf composition series. It is nonetheless useful to see the full set of behaviours the device family actually realises, because the temporal chapter draws on this spread, both as a set of selectable time constants and as diversity for reservoir banks. Figure C.1 places

Table C.4: Potentiation (PULSES) descriptors per composition cell, silver electrode, 3 V/1.5 V protocol. α: log–log growth exponent. Peak: peak conductance ratio (×). N_{peak}: pulse count at peak. Turnover: fraction of substrates whose response turns over within $N \leq 1000$.

Cation	PEO/salt	n	α	Peak (×)	N_{peak}	Turnover
Li^+	0.15/0.045	1	0.06	1.4	100	100%
Li^+	0.3/0.045	4	1.11	482.6	300	100%
Li^+	0.3/0.09	4	0.90	149.9	1000	0%
Li^+	0.3/0.18	2	0.18	3.8	1000	0%
Li^+	0.6/0.045	3	0.57	14.8	100	100%
Li^+	0.6/0.09	3	0.83	202.3	300	100%
Li^+	0.6/0.18	3	0.63	76.9	1000	0%
Li^+	1.2/0.045	3	0.33	4.4	100	100%
Li^+	1.2/0.09	2	0.49	42.6	1000	0%
Li^+	1.2/0.18	3	0.37	8.0	1000	0%
Na^+	0.3/0.09	1	0.27	10.0	1000	0%
K^+	0.3/0.09	1	0.86	103.2	300	100%

every silver device that carries both a measured switching window and a measured fading timescale ($n = 54$, all hosts, cations and anions; FILTERED-flagged and broken pixels removed; device medians) in a single window-versus-retention map. The map is descriptive only: the chemistry axes are illustrative (section 4.5) and no comparative claim is drawn from it. Two features are worth noting. The populated region spans more than an order of magnitude in both axes, so the substrate offers a broad palette of (window, τ) operating points. And the lead composition is a cluster of replicate devices spread over $\tau \approx 0.4$ s–20 s, not an isolated point, the visual counterpart of the within-cell device-to-device scatter quantified in section D.4.



Figure C.1: Measured window-versus-retention palette of the silver device library ($n = 54$ devices with both a clean hysteresis window and a fading-memory fit, all chemistries). Colour encodes the ion-conductor host, marker encodes the salt cation; the lead PEO 0.3/0.09 cell is ringed. The cloud spans over a decade in each axis, and the lead-cell replicates alone span $\tau \approx 0.4\,\mathrm{s}$–$20\,\mathrm{s}$. Descriptive overview only; the quantitative claims of chapter 4 rest on the replicated composition series, not on this map.

C.5 Impedance: Equivalent Circuit and Per-Cell Values

Equivalent-circuit model. Beyond the model-free Nyquist-apex resistance used in section 4.7, each spectrum was also summarised by the bulk ionic resistance R_{ion} of an equivalent circuit adapted from the polymer light-emitting electrochemical cell analysis of Munar *et al.* [145], in which a series cable term is placed in parallel with (i) the semiconductor electronic branch (effectively open at 0 V), (ii) a geometric constant-phase element, and (iii) an ionic branch of two interfacial double layers in series with R_{ion}. Two cautions apply. First, the archive holds two automated fits of this circuit; the freely-fitted one (with per-parameter error estimates) tracks

Table C.5: Equilibrium (0 V-DC) impedance per composition cell, silver *Super Yellow*/PEO/LiOTf. $Z_{\mathrm{real}}^{\mathrm{apex}}$: model-free real impedance at the Nyquist apex. R_{ion}: freely-fitted bulk ionic resistance (corroborative only). $t_{1/2}$: independently measured fading-memory time, repeated from table C.3 for the co-decline of fig. 4.10b. Em-dash: no matched decay.

PEO	salt	$Z_{\mathrm{real}}^{\mathrm{apex}}$ (Ω)	R_{ion} (Ω)	$t_{1/2}$ (s)
0.15	0.045	2.5×10^8	2.3×10^7	—
0.15	0.09	1.7×10^8	8.7×10^6	—
0.15	0.18	1.0×10^8	8.4×10^6	22.4
0.3	0.045	5.3×10^7	4.0×10^5	15.8
0.3	0.09	1.3×10^8	1.0×10^6	19.2
0.3	0.18	9.9×10^7	4.9×10^5	—
0.6	0.045	5.3×10^6	1.4×10^5	3.3
0.6	0.09	2.6×10^7	4.6×10^4	3.5
0.6	0.18	1.9×10^7	8.9×10^4	5.5
1.2	0.045	4.1×10^5	1.9×10^2	2.7
1.2	0.09	5.6×10^5	1.1×10^4	6.0
1.2	0.18	3.8×10^7	1.5×10^5	9.0

composition appreciably better than a more tightly bounded variant, and only the former is used. Second, per-spectrum circuit fits are genuinely ill-conditioned, an independent refit confirms that several do not converge robustly, so R_{ion} is reported only as corroboration of the model-free quantity, never as a precise number. Both observables nonetheless fall steeply and monotonically with r_{PEO} (table C.5): the model-free $Z_{\mathrm{real}}^{\mathrm{apex}}$ at rank correlation -0.83 against r_{PEO} on cell medians, the fitted R_{ion} at -0.84.

C.6 Electrode-Contrast Per-Cell Values

Table C.6 lists the per-cell behavioural metrics behind the silver-versus-gold contrast of section 4.8, including the host-free and salt-free negative controls that confirm both the ion-transport polymer and the dissolved salt are required for the memristive response.

Table C.6: Per-cell behavioural metrics by top electrode (active silver vs inert gold), at the lead composition 0.3/0.09, under the canonical positive-polarity protocols. On–off and normalised loop area from HYST; V_{act} activation voltage (V); peak ratio and α from PULSES; $t_{1/2}$ from DELAYTIME (s). Em-dash: no clean decay/fit at that cell. †: no positive-protocol decay exists, but the negative-polarity corpus supplies the missing retention measurement under a different protocol (−5 V write, −2.25 V sparse read): model-free half-decay medians Li^+ 2.2 s ($n = 5$), Na^+ 2.4 s ($n = 11$), K^+ 0.7 s ($n = 12$); see section C.7. "none" rows are host-free or salt-free negative controls.

Chemistry	Elec.	on–off	area	V_{act}	peak	α	$t_{1/2}$
PEO/Li/OTf	Ag	3.25	0.33	2.38	123.5	0.89	19.2
PEO/Li/OTf	Au	2.00	0.12	2.74	29.4	0.45	87.2
TMPE/Li/OTf	Ag	3.82	0.32	4.18	53.7	0.33	3.7
TMPE/Li/OTf	Au	1.29	0.07	3.61	1.9	0.17	†
TMPE/Na/OTf	Ag	5.54	0.30	3.26	497.9	1.23	3.9
TMPE/Na/OTf	Au	1.47	0.11	6.39	—	—	†
TMPE/K/OTf	Ag	7.01	0.41	3.10	237.5	1.19	5.2
TMPE/K/OTf	Au	1.43	0.06	5.86	—	—	†
TMPE/Na/TFSI	Ag	1.76	0.10	2.38	25.1	0.46	—
Hy/Li/OTf	Ag	1.41	0.06	1.61	25.2	0.39	—
none/Li/OTf	Ag	0.99	0.00	1.07	—	—	—
TMPE/none	Ag	2.95	0.20	3.18	73.0	0.96	—
none/none	Au	1.03	0.01	1.29	2.8	—	—

C.7 The Gold-Electrode Negative-Polarity Dynamics Corpus

This section documents the origin, validation, and per-protocol results of the negative-polarity measurement campaigns behind the polarity findings of section 4.8 and the single-run drive–retention measurements cited in section 4.6. Reproduction: `scripts/uncat_au_tmpe_audit.py` (parser, waveform classifier, per-file inventory) and `scripts/uncat_au_tmpe_findings.py` (analyses); per-file tables in `handouts/uncat_inventory.csv` and `handouts/uncat_findings_*.csv`; a full audit record in

`handouts/28_uncategorized_au_tmpe_audit.md`.

Provenance. The corpus comprises eight dated campaign folders (2025-01-23 to 2025-02-21) stored outside the curated archive layout, holding 356 raw files of which 349 parse as standard Keithley source-meter dumps (three comma-separated rows: current, voltage, time; acquisition cadences matching the instrument's NPLC settings; 1 mA compliance). All campaigns sit on the Au/TMPE(0.3)/triflate(0.09) generation, one salt per campaign (Li, Na, K triflate). The January campaign is identified at file level: its contents are byte-identical to the catalogued folder of device NM_v316 (TMPE/LiOTf/Au, 3000 rpm, 90 °C 3 h, glovebox-stored and measured), which it duplicates and extends. The February campaigns exist only in this corpus and are assigned to the same generation by chemistry, recipe, and measurement date (most plausibly the catalogued NaTr, KTr, and LiTr substrates of late 2024, then $\sim$2–3 months old); their exact substrate identities are not documented. One recovered file is the measurement script itself, which fixes the relaxation protocol exactly: -5 V/0.02 s pulse packets at a -2.25 V inter-pulse level, applied until a $\sim$600 s stabilisation window elapses, then single-point -2.25 V reads after exponentially spaced delays (0.075 s–30 s) with the source at high impedance between reads.

Validation. Four consistency checks tie the corpus to the catalogued archive. (i) File format, acquisition cadences, and compliance match the instrument configuration of the curated raw data. (ii) Parsing the catalogued positive- and negative-going sweeps of the same generation (90 sweeps, v296–v316) with the same code reproduces the negative-over-positive response asymmetry in direction and magnitude, and the uncategorized sweeps agree with the catalogued on–off values at matched (fast) sweep rate, the large windows appear only at slow rate, which

Table C.7: Per-cation summary of the gold/TMPE/triflate negative-polarity corpus (medians; illustrative tier). Window: on–off ratio of the slowest −3 V sweeps versus the fastest. Potentiation: read-current enhancement at the matched protocol (−5 V pulses, −2.25 V read), compliance-saturated runs excluded. t_{50}: model-free half-decay of the sparse-read relaxation (drop > 30%, non-zero read). Kohlrausch β range over clean fits.

Cation	window (slow/fast, −3 V)	potentiation	n	t_{50} (s)	β
Li^+	≈22/≈23	233×	6	2.2 ($n = 5$)	0.33–0.44
Na^+	≈15/≈8	48×	12	2.4 ($n = 11$)	0.33–0.43
K^+	≈1.9/≈0.9	40×	15	0.7 ($n = 12$)	0.15–1.0

the catalogued protocol never probed. (iii) Driven positively, the corpus reproduces the weak gold response of table C.6. (iv) The only polarity-inverted measurements are the two sessions whose folder notes document physically reversed wiring; they are excluded from all asymmetry statistics.

Per-protocol results. Table C.7 summarises the corpus by protocol class and cation. Beyond the headline numbers of section 4.8: constant-bias holds bracket the activation threshold between −1.5 and −2.5 V in every campaign (current decaying below it, growing 10–3000× above it, with 63%-rise times of tens to hundreds of seconds); potentiation–depression runs show deep, fast depression by positive pulses (final-to-peak read ratios of 0.002–0.05 at ±4–5 V); and 17 files touch the 1 mA compliance ceiling (concentrated in the strongest K trains) and are flagged in the inventory and excluded from the potentiation statistics. The decay fits bound the stretch exponent to $\beta \in (0, 1]$ and are accompanied by a model-free half-decay time; the table reports the latter.

Bounds. The corpus is single-composition (0.3/0.09 throughout), spans one or two substrates per cation, is uncurated, and mixes protocol variants; the February substrate identities are inferred; the supra-threshold

negative read makes the decay times protocol-specific (comparable to the silver delay-time values in order of magnitude only); and device age at measurement differs between the January (days 1–6) and February (months-old) campaigns. Accordingly, every statement drawn from it in the main text is illustrative, and no cation ordering is claimed.

C.8 The Gold-Electrode Lock-in Frequency-Sweep Corpus

This section documents the origin, measurement decode, and quantitative findings of the lock-in frequency-sweep corpus behind the frequency-domain results of section 4.8. Reproduction: `scripts/lockin_au_tmpe_audit.py` (filename parser, per-file metrics, QA panels) and `scripts/lockin_au_tmpe_findings.py` (analyses L1–L5); per-file tables in `handouts/lockin_manifest.csv`, `handouts/lockin_metrics.csv`, and `handouts/lockin_findings_*.csv`; a full audit record in `handouts/29_lockin_au_tmpe_audit.md`.

Provenance. The corpus comprises 549 lock-in sweeper exports (plus 439 binary siblings) taken between 2025-03-04 and 2025-03-26 on a single Au/TMPE/triflate batch: the device-age field in every filename equals the measurement date minus 2025-03-03, so the batch has a single, self-consistent fabrication (top-electrode) date and is tracked from day 1 to day 23. Twelve pixels on three substrates were measured, one substrate per triflate salt (Li, Na, K). The substrates are not catalogued in the device library, the experimenter's own labels carry an "unknown" prefix, but by chemistry, recipe, and date they are most plausibly the electrode-free UV-Vis trio of 2025-02-26 (fig. 4.8) after Au evaporation, or unlogged

siblings of that batch; this identification is unproven, the same caveat class as the February campaigns of section C.7.

Measurement and decode. A Zurich Instruments MFLI drives $V_{\mathrm{off}} + V_{\mathrm{pk}} \sin(2\pi f t)$ through the junction into an external series capacitor to ground ($C_{\mathrm{ext}} = 4\,\mathrm{nF}$ standard; a 0.286 nF–2 nF series on one pixel; a "2.5 pF" day-1/2 configuration with no external capacitor, loaded by the instrument's 10 MΩ input); the voltage at the junction–capacitor node is demodulated at the drive frequency, and the transfer ratio is $|H| = V_{\mathrm{out,RMS}}/(V_{\mathrm{pk}}/\sqrt{2})$, the normalisation used by the corpus' own visualisation script. Sweeps descend from 500 kHz to 1 Hz in three captures (100+97+200 points, ∼1 s per point); the 110 "500k–1Hz" files are *derived* merges of the three captures (produced by the corpus' `combine_freq.py`; frequency-sorted, timer running backwards) and are excluded from all statistics to avoid double counting. Of the four recorded demodulator channels, demodulator 4 is the output fundamental used throughout; demodulator 2 duplicates it through a different demodulator filter; demodulator 3 behaves as a DC monitor of the output node (magnitude = DC value, sign carried by its phase); demodulator 1 is an auxiliary current channel, unused here. A 50 Hz mains glitch is visible in most captures and lies on none of the probe frequencies used below.

L1, divider validation. On unkK2-L5 (day 3, 0.5 Vpk, 10 kHz), stepping C_{ext} through 0.286, 0.5, 0.667, 1, 1.3, 1.66, 2 and 4 nF moves $|H|$ from 0.830 to 0.423, and the implied device capacitance $C_{\mathrm{dev}} = C_{\mathrm{ext}}|H|/(1-|H|)$ stays within 1.4 nF–2.9 nF (median 2.35 nF). The residual monotone trend of C_{dev} with C_{ext} is the expected signature of a lossy (constant-phase-like) dielectric, not an ideal one; the divider model is therefore taken as validated at the factor level, not as an exact calibration.

L2, the passive band is linear. At zero offset and $f \gtrsim 10\,\mathrm{Hz}$ the junction is a passive dielectric: across the 0.5–2.4 Vpk amplitude series the plateau $|H|$ at 1 kHz and 10 kHz changes by a median 0.17 %–0.29 % (maximum 8 %; 9 device-days). The effective capacitance at 1 kHz is 2.32 nF (Li), 2.93 nF (Na), and 3.19 nF (K) by substrate, $\approx$28 nF cm^{-2}–39 nF cm^{-2} at the $8.25 \times 10^{-2}\,\mathrm{cm}^2$ junction area, i.e. $\varepsilon_r \approx 5$–7 at the 150 nm–190 nm that profilometry measures on sibling substrates of the same recipe, with a pixel-to-pixel spread *within* each substrate of $\approx$1 %.

A candidate cation-ordered capacitance. The capacitance is itself cation-ordered, K > Na > Li with a span of $\approx 1.4\times$, two orders of magnitude above the within-substrate pixel spread, on three substrates fabricated in a single batch. With one substrate per salt this cannot be fully separated from substrate-level film variation, but the archive's profilometry of the same recipe (TMPE 0.3/salt 0.09/3000 rpm; `handouts/lockin_findings_thickness.csv`) bounds the one plausible nuisance channel. Explaining the span by thickness alone would require the Li film $\approx$ 37 % thicker than the K film; the two profiled same-day cation trios differ by only 6 %–12 % across salts, and in the anion-matched 2024-10-29 pair the K film is the *thicker* one, the direction that would weaken, not produce, the observed ordering. Salt-uncorrelated solution-to-solution scatter does reach $\approx$30 % in the archive, so a thickness origin is not excluded, merely unprecedented in cation-correlated form. The ordering is therefore recorded as a *candidate* cation dielectric effect, an effective permittivity rising from Li^+ to K^+, the direction qualitative coordination-strength reasoning would suggest (the most weakly coordinated cation the most polarizable), and it is the one cleanly cation-ordered observable in this archive. It is not in tension with the driven retention, which also moves with the cation, the silver TMPE/triflate series runs in the same K>Na>Li direction, but which cannot be fixed to a host- and anion-independent ordering at one or two devices per cell. The flat probes

(ion association by ATR, the optical gap by UV-Vis) address different physics (ion pairing, electronic structure) and are expected to be flat, so there is no contradiction. It is not advanced as demonstrated. The decisive, cheap test is profilometry on these three substrates, which still exist, followed by the replicated cation series of section 4.10. The gain magnitudes of the next paragraph are deliberately *not* given the same reading: unlike the capacitance plateau (offset-insensitive, week-stable, 1 % uniform), the low-frequency gain is sensitive to conditioning history and session protocol, both of which differ across the three substrates.

L3, bias-activated low-frequency gain. Table C.8 gives the day-23 offset series on unkLi2-L3 (11 offsets, 0.25 Vpk): the fundamental-frequency transfer at 1 Hz–2 Hz rises monotonically with $|V_{\mathrm{off}}|$, from $|H| \approx 0.5$ unbiased to $\approx$18 at $\pm$2.5 V, while $|H|$ at 10 Hz stays near unity and the $\geq$1 kHz plateau does not move (offset and zero-offset files give plateau medians 0.422 and 0.423). Across the corpus the prevalence of gain ($|H|_{2\,\mathrm{Hz}} > 1$) rises from 7 % of zero-offset captures ($n = 46$; the band is noisy but not amplifying) through 72 % at $|V_{\mathrm{off}}| = 1.4$ V ($n = 18$) to 100 % at $|V_{\mathrm{off}}| \geq 2.4$ V ($n = 20$), with medians 0.46, 1.25, 2.78, 6.69, and 17.3 at $|V_{\mathrm{off}}| = 0$, 1.4, 2.0, 2.4, 2.5 V. The 200 Hz boundary point, captured twice per condition minutes apart, repeats to a median 1.2 % ($n = 95$), so the gain values are not acquisition noise. Because $|H| > 1$ at the drive frequency is impossible for a passive divider into a passive load, the biased junction is an *active* element at these frequencies: the drive slowly modulates the ionic conductance, and that modulation mixes the DC bias up to the fundamental, transferring power from the bias into the signal band. The conclusion is robust to the demodulator amplitude convention (a worst-case $\sqrt{2}$ would reduce ≈ 18 to ≈ 13). Across 32 matched $\pm$ pairs the gain is close to polarity-symmetric, with a median 11 % positive-offset excess (Wilcoxon $p = 0.013$), far weaker than the DC potentiation asymmetry of section C.7.

Table C.8: Day-23 offset series on unkLi2-L3 (0.25 Vpk, $C_{\text{ext}} = 4\,\text{nF}$, 200–1 Hz capture): fundamental-frequency transfer ratio $|H|$ at four probe frequencies and the DC monitor at the output node. Offsets were measured in the order $0 \to +2.5$ then $-0.5 \to -2.5$; the later-measured negative branch is not systematically elevated, so the near-symmetry is not a session-drift artefact.

V_{off} (V)	$\lvert H\rvert_{1\,\text{Hz}}$	$\lvert H\rvert_{2\,\text{Hz}}$	$\lvert H\rvert_{5\,\text{Hz}}$	$\lvert H\rvert_{10\,\text{Hz}}$	DC out (V)
−2.5	17.2	16.7	7.5	1.28	−3.46
−2.0	13.1	13.4	5.9	1.01	−2.82
−1.5	12.9	9.3	3.9	1.23	−1.93
−1.0	6.1	6.6	3.1	1.07	−1.38
−0.5	5.4	2.1	1.6	0.90	−0.45
0.0	1.7	0.5	1.0	0.97	+0.02
+0.5	3.7	1.9	2.3	1.12	+0.41
+1.0	8.6	6.9	1.9	1.07	+1.44
+1.5	9.7	10.0	5.2	1.03	+2.10
+2.0	16.1	13.8	4.6	0.83	+2.89
+2.5	17.1	17.9	6.1	1.82	+3.71

L4, ageing. The passive plateau is stable within pixel where days repeat: K pixels move from 0.444 (day 3) to 0.450–0.451 (day 9), Na pixels stay within ±2 % over the same week, while the Li pixels rise by ≈15 % from day 4 (0.35–0.37) to day 10 (≈0.42) and the day-23 pixel sits at 0.39. One pixel-day deviates (unkK2-L5 day 4, plateau 0.55 at −17 to −21° phase): that pixel had served as the capacitor-series workhorse the previous day, so its elevated, resistive plateau is read as a conditioned (potentiated) state, not an artefact, it is excluded from the plateau statistics above. Bias-activated gain is present at every age at which offsets were applied (days 4, 9/10, and 23), but the offset–amplitude protocol changed between sessions, so no matched-condition ageing comparison of the gain itself is possible.

L5, DC monitor. The DC component at the output node follows the offset linearly in magnitude and sign across the day-23 series (table C.8, last column; ≈ 1.4× the offset in the export's convention), confirming

that under bias the junction DC-conducts well below the 10 MΩ input impedance, a ∼MΩ-scale biased state, the same order as the constant-bias end states of the DC corpus (section C.7). At zero offset the rectified DC is not significantly different from zero (median −0.18 V, 61 % of $n = 46$ captures negative, binomial $p = 0.18$).

Bounds. One batch, one substrate per salt (cation and substrate fully confounded), and one pixel for the day-23 series. The sweeps descend in frequency, so every low-frequency point follows minutes of conditioning under the same offset, the gain states are conditioned steady states, not pristine ones, and the low-frequency points at *zero* offset are visibly non-stationary (state drift during the point). Absolute $|H|$ and DC values inherit the demodulator amplitude convention (at most a factor $\sqrt{2}$; all gain and symmetry statements are ratios at fixed channel and unaffected); C_{dev} is correspondingly approximate. The substrate identities are inferred, as stated under provenance.

C.9 Composition-Gradient Significance

The composition trends of section 4.4 are reported in the main text as cell medians. This section attaches significance tests and non-parametric uncertainty to them on the replicated Ag/Li *Super Yellow*/PEO/LiOTf series. The fabricated substrate is the experimental unit: repeated curves are reduced to junction medians and junctions to one substrate median before inference. For each descriptor, a tie-aware Spearman correlation with r_{PEO} is tested using 20 000 two-sided permutations. Holm correction controls the family-wise error rate across the four reported descriptors; a sensitivity test restricts relabelling within acquisition quarter. All four associations remain below $p = 0.05$ after Holm correction, and the within-quarter permutations give the same conclusion (table C.9). These

Table C.9: Per-substrate tie-aware Spearman rank correlation of each descriptor with the PEO-to-*Super Yellow* mass ratio r_{PEO} (replicated Ag/Li grid). Raw permutation values use 20 000 two-sided relabellings, p_{Holm} controls the four-descriptor family, and p_{batch} restricts permutations within acquisition quarter.

Descriptor	n	ρ	raw p	p_{Holm}	p_{batch}
Fading-memory time $t_{1/2}$	23	−0.481	0.0224	0.0224	0.0157
Growth exponent α	27	−0.546	0.0038	0.0113	0.0047
Peak conductance ratio	27	−0.542	0.0043	0.0113	0.0021
On–off ratio	43	−0.529	0.0007	0.0026	0.0020

tests establish an association across the replicated grid; they do not by themselves assign a microscopic cause.

A non-parametric bootstrap (20 000 substrate-level resamples per cell) gives a 95 % interval for cells with at least three substrates; the full set is in `handouts/ch4_cell_ci.csv`. For $n = 2$, resampling a median yields an uninformative discrete result, so the table reports the observed range; for $n = 1$, no interval is assigned. For example, the PEO 0.3/salt 0.09 cell gives $t_{1/2} = 19.2\,[18.6, 22.0]$ s. The PEO 0.3/salt 0.045 cell is visibly right-skewed and its wider interval makes that limitation explicit.

The two composition axes interact for the pulse response. Each descriptor was regressed on centred $\log r_{PEO}$, centred $\log r_{\mathrm{salt}}$, their interaction, and acquisition-quarter fixed effects. Term significance was estimated with 20 000 Freedman–Lane residual permutations within acquisition quarter (table C.10). For $t_{1/2}$, the PEO term is resolved and the interaction does not clear $p = 0.05$. For growth exponent and peak ratio, both the PEO term and the PEO×salt interaction are resolved. The earlier near-orthogonality interpretation is therefore not supported: the effect of PEO on pulse magnitude depends on salt level. Turnover remains a separately reported categorical descriptor because a log-linear magnitude model does not encode it.

Table C.10: Batch-adjusted factorial decomposition on substrate medians from the replicated Ag/Li grid. Predictors are centred log mass ratios; acquisition-quarter fixed effects are included, and p-values use 20 000 Freedman–Lane residual permutations within quarter. Pulse descriptors show resolved PEO×salt interactions.

Descriptor	b_{PEO} (p)	b_{salt} (p)	b_{int} (p)	R^2	n
$\log t_{1/2}$	−0.800 (0.0054)	+0.255 (0.428)	+0.899 (0.091)	0.526	23
α	−0.278 (0.0010)	−0.205 (0.067)	+0.453 (0.0020)	0.701	27
log(peak)	−1.543 (0.0007)	−0.830 (0.178)	+2.272 (0.0067)	0.688	27

C.10 Absolute Conductance Tests Simple Dilution

The working mechanism of section 4.9 initially read the window contraction at high PEO as a *dilution* of the *Super Yellow* electronic-percolation network. That picture makes a direct, falsifiable prediction on a quantity the peak-normalised dynamics discard: if added ion-transport polymer simply interrupted the electronic network, the *absolute* on-state conductance should *fall* with PEO. It does the opposite. Within the replicated $r_{PEO} = 0.3,\ 0.6$ and 1.2 grid, the on-state conductance (HYST conductance at the sweep maximum, substrate medians, $n = 43$) has a positive overall association with PEO: Spearman $\rho = +0.61$ ($p < 0.001$) and a log–log slope of approximately +2.5 (fig. C.2). The cell-median conductance is not strictly monotone between PEO 0.6 and 1.2. The result therefore rejects the direction predicted by simple dilution, but it does not identify a unique conductance law. Thickness also rises with PEO (section C.2), so geometry remains a measured covariate rather than an explanation assumed away.

The literal dilution reading is therefore inconsistent with the observed sign. A parallel, weakly switchable path in the ion-rich phase that raises the conductance floor is one candidate interpretation carried into section 4.9.

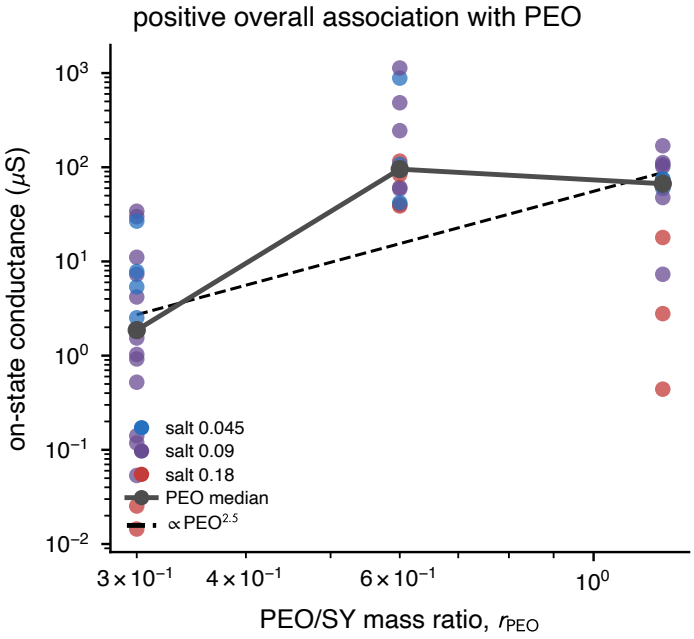


Figure C.2: Absolute on-state conductance (HYST, conductance at the sweep maximum; substrate medians from the replicated Ag series, coloured by r_{salt}) against r_{PEO}, log–log. The positive overall association (statistics printed in the panel) has the opposite sign to simple electronic-percolation dilution. Cell medians are not strictly monotone, and the raised-floor account in section 4.9 remains a candidate mechanism.

Since the on–off ratio falls with PEO, this candidate requires off-state conductance to rise faster than on-state conductance. The present data do not distinguish it from every alternative mixed-conduction mechanism.

C.11 Impedance–Memory Timescale Check

Section 4.7 notes that the EIS resistance *scale* tracks the fading-memory time across composition, while the small-signal apex *frequency* does not:

for the $n = 23$ substrates with matched EIS and decay data, the median Nyquist-apex time is $\tau_{RC} = 1/(2\pi f_{\mathrm{apex}}) \approx 0.089\,\mathrm{s}$, more than an order of magnitude shorter than the seconds-long $t_{1/2}$. The following is an order-of-magnitude consistency check, not an identification of a common physical clock.

The apex RC is set by the geometric capacitance $C_{\mathrm{geo}} = 1/(2\pi f_{\mathrm{apex}} Z_{\mathrm{real}}^{\mathrm{apex}})$. The forgetting of a *written* state, by contrast, is the relaxation of a displaced ionic profile and is governed by the much larger chemical/double-layer capacitance C_{chem} of the ion-rich film. Measuring C_{chem} independently from the low-frequency branch of the same spectra ($C_{\mathrm{lf}} = -1/(2\pi f Z_{\mathrm{imag}})$ at the lowest measured frequency) gives a median capacitance enhancement $C_{\mathrm{lf}}/C_{\mathrm{geo}} \approx 69$ ($n = 23$). Multiplying this ratio by the median apex time gives $\approx 6.2\,\mathrm{s}$, close to the measured median $t_{1/2} \approx 6.0\,\mathrm{s}$. Because the low-frequency branch is finite-bandwidth, the capacitance estimate is model-dependent, and the EIS and large-signal decay probe different operating states, this numerical agreement is supporting consistency only. It neither proves that one capacitance sets the decay nor upgrades the resistance correlation to a unique mechanism.

C.12 Repeatability Checks

A physical reservoir requires elements that are diverse *across* the bank but *individually reproducible*. The chapter documents the diversity at length; this section quantifies the reproducibility. The fading-memory protocol measured a single junction per device, so a population-level repeat-*decay* reproducibility cannot be estimated, only one series device carries two junctions, and there the model-free $t_{1/2}$ agrees to $\sigma(\ln t_{1/2}) \approx 0.19$; broader repeat-decay statistics are a stated gap (section 4.10). The steady-state switching window, however, is recorded over several sweeps per junction (median ≈ 5), giving a genuine junction-level sweep reproducibility: the

sweep-to-sweep scatter of the on–off ratio is $\sigma(\ln R_{\mathrm{on/off}}) \approx 0.14$ across 142 series junctions, i.e. each junction reproduces its switching window to about 14 % within that protocol. This statistic does not establish repeat-decay reproducibility. It is smaller than the across-composition timescale span ($\Delta \ln t_{1/2} \approx 1.95$, section C.13), but the comparison crosses two different observables and is therefore descriptive.

C.13 Heterogeneous Timescales

The chapter treats the spread of fading-memory times as a possible resource. The cell-median $t_{1/2}$ spans $\Delta \ln t_{1/2} \approx 1.95$ across the composition series, while the robust between-substrate scatter within a cell is $\sigma_{\mathrm{within}}(\ln t_{1/2}) \approx 0.264$. If, as a design heuristic, two bands are considered distinguishable when separated by between one and two within-cell standard deviations, then $N_{\mathrm{eff}} = 1 + \Delta \ln t_{1/2}/(k\sigma_{\mathrm{within}})$ gives about 2.15–4.69 for $k = 2$ to 1. This is not an information-capacity estimate: it ignores task, noise correlations, readout, and finite sampling. It only indicates that the measured composition span can plausibly supply a small number of separated fading-memory bands for the modelling in chapter 5.

Comparison of within-curve and between-substrate dispersion. The median identified stretch gives $1 - \beta \approx 0.387$, while the robust between-substrate statistic is $\sigma(\ln t_{1/2}) \approx 0.264$. These numbers are of similar order, but they are not commensurate: $1 - \beta$ is a dimensionless shape parameter of one fitted decay, whereas $\sigma(\ln t_{1/2})$ is a population dispersion. Their similarity neither demonstrates a common activation-barrier distribution nor excludes independent sources of variation.

Appendix D

Supporting Information for Chapter 5

This appendix documents how the behavioural-model parameter cards of section 5.2 are sourced from the measured archive, and lists the simulation and read-out settings used across the three evidence tiers. Together with the per-cell tables of section C.3, it defines the inputs and controls needed to audit the in-silico results; the executable workflow is in the repository (`scripts/ch5_figures.py`, `scripts/ch5_physio_context.py`, `scripts/ch5_onset.py`, `scripts/ch5_deployment.py`, `scripts/ch5_noneeg.py`).

D.1 Parameter-Card Sourcing

Each replicated composition cell is reduced to a parameter card $(t_{1/2,c}, \beta_c, \alpha_c, R_{\text{peak},c}, N_{\text{peak},c})$ that drives one reservoir node type. The values are taken directly from the comparative summaries, with no free re-tuning:

- the fading-memory law $\lambda_c(\Delta t) = \exp[-(\Delta t/\tau_c)^{\beta_c}]$ is anchored to the model-free cell-median $t_{1/2,c}$ in table C.3. Where β_c is identified, $\tau_c = t_{1/2,c}/(\ln 2)^{1/\beta_c}$; otherwise $\beta_c = 1$ and $\tau_c = t_{1/2,c}/\ln 2$. Each node relaxes along the whole sampled kernel, so the simulated decay matches λ_c at every delay and every sampling interval (tested in `tests/test_ch5_model.py`);

- the write envelope uses the per-cell growth exponent, peak enhancement, and pulse count at the peak from table C.4; it is normalised to $G/G_0 = 1$ at zero effective pulses and capped at the measured peak;

- the injected device-to-device jitter acts only on the timescale and uses the screened substrate-level within-cell scatter in section D.4. No common jitter is assumed for α or β.

The heterogeneous banks of sections 5.5 and 5.7 span the replicated PEO/LiOTf composition cells of table C.3 (τ from ≈ 4 to ≈ 30 s); the unreplicated PEO 0.15 cell is excluded. Homogeneous controls are jittered copies of the lead cell PEO 0.3/0.09 alone.

D.2 Simulation and Read-out Settings

Table D.1 collects the settings used throughout chapter 5. All read-outs are a single linear ridge regression; no nonlinear decoder is used, so any demonstrated computation is attributable to the device nodes rather than the read-out. The read-out ridge penalty ($\lambda = 10^{-3}$), the causal label-smoothing horizon (15 s), the bank size N, and the reservoir step Δt are fixed *a priori* and held identical across all banks and all held-out subjects; none is selected on the held-out subject, so the leave-one-subject-out protocol carries no hyperparameter leakage.

Table D.1: Simulation and read-out settings across the three evidence tiers of chapter 5. All banks are spatial banks of independent (non-recurrent) leaky nodes driven by randomly-masked input; all read-outs are linear ridge regression.

Setting	Value
Node model	pulse progress is the causal sum of the clipped drive $\mathrm{clip}(w_i u_n, 0, 1)/N_{\mathrm{peak},i}$ weighted by the measured kernel λ_i^{1/α_i}, capped at the measured peak, then mapped through $x = p^{\alpha_i}$ (section 5.2). At the peak, the stored history is held as a non-negative sum of exponential modes fitted to the kernel and rescaled
Linear controls	exponential moving averages of the input channels at $\tau = t_{1/2}/\ln 2$ for the lead cell and for all eight cells, with the same read-out
Read-out	single linear ridge regression
Bank size N	24 (benchmarks); 48 (physiology/affect)
Algorithmic variability	10 random input/bank seeds; means ± SD and sign consistency reported; seeds are not inferential replicates
Inferential unit	independent held-out subject for physiological corpora; independently fabricated device for experimental comparisons
Memory capacity	MC_k summed over lags, random uniform input [160]
NARMA	NARMA-10 system identification, NRMSE [161]
Information-processing capacity	degree-1 (linear) + degree-2 (nonlinear), bounded by N [82]
Affective corpus	WESAD chest signals, leave-one-subject-out over 15 subjects [166]
Streaming cadence	$\Delta t = 1\,\mathrm{s}$; reconstruction delays $1, 3, 8, 20, 45\,\mathrm{s}$
Subject calibration	first 60 s only; 5th/95th percentiles mapped monotonically to 0.25/0.75; no later-sample or label information
Heart rate	causal band-pass/integration; completed RR intervals assigned by past-value hold
Timeline handling	reservoir and causal smoother run over the full session; undefined-label samples masked only for fit/score
Ridge convention	mean-squared-error objective plus $\lambda = 10^{-3}$ (Gram matrix divided by sample count)
Affect scoring	class-balanced macro-F1

D.3 Computational Reproducibility and Data Availability

The thesis repository records the tested Python environment, regression tests, the ordered analysis scripts, a hash manifest for the principal processed experimental tables, and an automated audit of the complete LaTeX build. Running `makeverify` tests the calculations and behavioural-model invariants covered by the regression suite, rebuilds the complete thesis from the committed source and derived artifacts, and rejects LaTeX errors, undefined citations or references, and overfull boxes. This is a verification of the deposited snapshot; it is not, by itself, a regeneration of every figure from raw measurement files.

Raw-to-result regeneration is split into explicit workflows. The command `makereproduce-core` requires the separately held, unversioned `Nanomem_Devices_Library` as a sibling of the thesis repository. Its processed database is checked against the SHA-256 manifest in `provenance/database_inputs.sha256`; that manifest binds the principal tables used by Chapters 3–5 but does not cover every raw Keithley, spectroscopy, lock-in, or Chapter 2 source file. The raw experimental archive is not distributed with this repository, so the deposited repository alone cannot independently reconstruct the experimental results from raw data.

The command `makereproduce-physio` requires independently obtained WESAD and PhysioNet Non-EEG corpora under the ignored `data/` directory. CASE is not part of the reproduction graph: its raw corpus was unavailable for this audited snapshot, cached CASE outputs were removed, and no CASE result is reported. The combined `makereproduce-all` workflow checks the required inputs, executes the available analysis stages sequentially, stops at the first failure, and finishes with the test and thesis-build audit. Exact placement, scope, and limitations are documented in `docs/reproducibility.md`.

For all physiological corpora, the raw-stream caches store a schema identifier and a signature of the source files; a preprocessing or source mismatch forces regeneration. ECG and wrist-BVP filters are causal, and heart rate at time t is the most recent completed RR interval rather than a linear interpolation involving the next beat. The calibration transform is a Cauchy cumulative map centred on the initial-prefix robust range, so later out-of-range values retain ordering rather than collapsing at a hard clipping boundary.

D.4 Measured Device-to-Device Scatter

The robustness analysis of section 5.10 (fig. 5.9) injects variation as multiplicative log-normal jitter on each node's timescale, $\tau_i = \tau\, e^{\varepsilon}$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. Because chapter 4 defines $t_{1/2}$ as the robust primary timescale, the matching estimate is the within-cell spread of $\ln t_{1/2}$, after junctions have been collapsed to one median per fabricated substrate.

Across the eight replicated cells with finite estimates (23 substrates), the median cell standard deviation is $\sigma(\ln t_{1/2}) = 0.2637$; individual cell estimates range from 0.0894 to 1.0179. The median is used as the default jitter, and the range motivates the sensitivity sweep. The value is descriptive rather than a precise population parameter because each cell contains few substrates and batch effects cannot be eliminated. A secondary audit of the older, unscreened instrument-side exponential fits gives $\sigma(\ln \tau) = 0.846$ within cells ($n = 27$) and is retained only as an upper stress-test reference. Pooling the entire composition sweep gives 1.59 but is invalid as a within-cell estimate because it includes the deliberately designed composition differences. The audit is reproduced by `scripts/ch5_scatter_audit.py`.

Table D.2: Primary and sensitivity estimates for timescale jitter in the silver PEO/LiOTf series. The primary value uses screened, model-free substrate $t_{1/2}$ values on the replicated grid; fit-based values are secondary diagnostics.

Estimate	Value
Median within-cell $\sigma(\ln t_{1/2})$; 23 substrates, 8 cells	**0.2637**
Range of cell-level $\sigma(\ln t_{1/2})$	0.0894–1.0179
Legacy within-cell $\sigma(\ln \tau)$; instrument fits, $n = 27$	0.846
Across-group $\sigma(\ln \tau)$; conflates designed sweep	1.59
Previously annotated undocumented value	0.12

D.5 Continuous-Monitoring Stress Test

The deployment-robustness result of section 5.9 (`scripts/ch5_onset.py`) is a continuous binary stress detector: at every $\Delta t = 1\,\mathrm{s}$ step the read-out predicts stress (WESAD label 2) versus not-stress (baseline and amusement pooled), leave-one-subject-out, with the same single linear ridge read-out and 15-second causal label smoothing used in the streaming task. Five matched feature sets of $N = 24$ nodes are compared on identical input masks: the instantaneous input (no memory), a memoryless bank (leak set to zero, dimensionality only), the homogeneous lead-only bank, the heterogeneous composition bank, and a linear EMA bank at the eight cell half-lives.

Sensor corruption is modelled by adding, to the per-subject scaled streams (channels in $[0, 1]$), independent Gaussian noise of standard deviation σ at every sample, plus sparse motion-artefact bursts (short windows of amplified noise at a 1% per-sample rate), clipped back to $[0, 1]$; the same realisation is applied to the training and held-out streams so the comparison isolates each architecture's intrinsic noise tolerance. Robustness curves are seed-averaged over three bank/noise seeds; the per-subject significance test at $\sigma = 0.4$ is a paired Wilcoxon signed-rank over the fifteen subjects. Reported metrics are the binary macro-F1, the pooled

false-alarm rate (fraction of scored not-stress steps called stress), and, for completeness, the transition-window macro-F1 and onset latency. Both are computed on the full physical timeline: transition windows lie within ±20 s of entry to or exit from a contiguous valid-label segment, and each stress onset is the start of a contiguous label-2 segment. Undefined intervals are never concatenated before smoothing or event detection.

The cross-corpus replication (table 5.6, `scripts/ch5_noneeg.py`) applies the *same* detector and noise model to the PhysioNet Non-EEG corpus [167] (open access; obtainable with `wfdb.dl_database('noneeg','data/noneeg')`). The 8 Hz electrodermal and temperature record and the 1 Hz heart-rate record are resampled onto a common timeline and per-subject scaled exactly as the WESAD streams; the phase annotations map cognitive and emotional stress to the stress class and relaxation to not-stress, with the physical-stress (exercise) phase excluded.

Real-motion corruption. To confirm the robustness is not an artefact of the Gaussian noise model, the corruption is also driven by the *measured* chest accelerometer (`add_motion_noise`, `load_motion` in the scripts). A per-step motion index is the magnitude of the gravity-removed acceleration, causally RMS-smoothed over 1 s and scaled from the initial calibration prefix; the injected noise standard deviation at each step is scale × (0.05 + motion), with an amplified artefact burst where the index exceeds 0.6, so corruption lands where the wearer actually moves. At corruption scale 0.8, instantaneous, heterogeneous, and EMA scores are 0.702, 0.811, and 0.809. The heterogeneous-minus-instantaneous subject difference is +0.111 ($p = 0.055$, not resolved); EMA-minus-instantaneous is also +0.111 ($p = 0.035$). Thus the measured-motion control supports temporal filtering, not a device-specific effect.

D.6 Control Experiments

Three controls underpin the claims of chapter 5 and are tabulated here. They are reproduced by `scripts/ch5_reservoir.py` (the generic-ESN and coupling controls) and `scripts/ch5_onset.py` (the classical-filter control). For seed-only controls, matched means, standard deviations, effect differences, and sign consistency describe algorithmic sensitivity; no population-level p-value is assigned. Paired Wilcoxon tests and rank-biserial effects are retained only where the pairs are independent held-out subjects.

Classical-filter control. Table D.3 is the control behind the deployment-robustness reframing of section 5.9: an ordinary exponential moving average of the raw channels at the device half-lives, fed to the same linear read-out with no reservoir, recovers the noise robustness, so the robustness is the generic property of a fading-memory low-pass at these time constants rather than anything specific to the device nonlinearity. The device's contribution is to *be* that filter, passively and at the budget of section D.8. The cross-corpus and wrist robustness results (tables 5.6 and 5.7) share this character: there too the gain is the low-pass advantage of temporal memory over an instantaneous read. On the wrist the device bank and the EMA bank do not differ ($p = 0.38$); on the Non-EEG corpus the device bank exceeds the EMA bank by 0.023 (16/20 subjects, $p = 0.017$).

Generic-ESN control (matching thesis). Table D.4 asks whether the *particular* measured timescales matter or only their spread, by replacing the device bank with an echo-state network of the same size whose leak times are drawn at random. A random bank over the same range ties the device bank, and a broader random range exceeds it: the resource is the

Table D.3: Classical-filter control for the continuous stress monitor (binary macro-F1, leave-one-subject-out). Paired at $\sigma = 0.4$ over fifteen subjects: EMA(single)−instantaneous = +0.170 ($p = 0.022$, $r_{\mathrm{rb}} = +0.67$); EMA(bank)−EMA(single) = +0.003 ($p = 0.35$); device reservoir−EMA(bank) = −0.002 ($p = 0.64$, $r_{\mathrm{rb}} = -0.14$). The linear filter at the device half-lives matches the device bank.

Detector (linear read-out on . . .)	clean	$\sigma = 0.4$
Instantaneous channels (no memory)	0.781	0.641
EMA at the lead half-life	0.803	0.804
EMA bank at the eight half-lives (linear)	**0.806**	0.808
Device reservoir (nonlinear, heterogeneous)	0.799	**0.811**

Table D.4: Total linear memory capacity ($N = 24$, ten matched seeds, means ± SD) of the measured-card heterogeneous bank versus generic random leaky banks of equal size. The same-range control exceeds the device bank by 0.53 on average (device higher in 4/10 seeds); the broad-range control exceeds it by 1.12 (device higher in 1/10). These are algorithmic-sensitivity comparisons, not population-level tests.

Reservoir ($N = 24$)	total MC
Measured-card heterogeneous bank	4.44 ± 0.45
Generic bank, same τ range (3–30 s)	4.97 ± 0.78
Generic bank, broad τ range (0.5–60 s)	5.56 ± 0.74

timescale *spread*, which the composition grid supplies intrinsically, and extending the range (the drive-amplitude variable of chapter 4) is where further capacity lies.

Coupling-sensitivity control. Table D.5 bounds the composition assumption of section 5.2: each node's retention time is rescaled with its own write-nonlinearity exponent by a factor $e^{\kappa z}$ (z the standardised exponent), modelling the drive–retention coupling the separate-amplitude measurement cannot exclude. The heterogeneous bank keeps the higher capacity across the sweep (ratio 1.29–1.42, higher in every seed pair). The sweep is then *anchored to data*. The gold/TMPE negative-polarity

Table D.5: Memory capacity under an injected drive–retention coupling κ ($N = 24$, ten seeds). The heterogeneous bank stays above the homogeneous one throughout. The nominal cards already yield $\rho \approx +0.57$, above the transferred Au/TMPE central estimate of +0.46, so the data-anchored central point is $\kappa^\star = 0$ and the upper confidence point is +0.31. Seed consistency describes algorithmic sensitivity, not experimental replication.

κ	het MC	hom MC	ratio	het higher
-0.50	4.57	3.22	1.42	10/10
-0.25	4.55	3.22	1.41	10/10
$0.00^\dagger$	4.44	3.22	1.38	10/10
$+0.25$	4.32	3.22	1.34	10/10
$+0.50$	4.17	3.22	1.29	10/10
$+0.31^\ddagger$	4.29	3.22	1.33	10/10

corpus of chapter 4 supplies the only direct measurement of this coupling: across 28 single-run potentiation + relaxation traces the write-depth and the forgetting time are rank-correlated at $\rho = +0.46$ ($p = 0.015$; 95% CI 0.10–0.71 by Fisher's z). Calibrating κ so the simulated across-node write–retention correlation matches that figure fixes the data-anchored operating point. The nominal composed cards already carry an across-node rank correlation of about $\rho = +0.57$, so the central external estimate maps to $\kappa^\star = 0$; the upper confidence limit maps to about +0.31. The heterogeneous-to-homogeneous ratio is 1.38 at $\kappa^\star = 0$ and 1.33 at the upper limit, within the 1.29–1.42 span of the sweep. The figure is transferred as an order-of-magnitude anchor only – a different host, electrode, and polarity – but it converts the bound from an arbitrary range into a data-referenced point and independently confirms the *sign* of the coupling the composed model assumes. The calibration (κ-for-ρ bisection) is reproduced by `scripts/ch5_reservoir.py`.

Chance baselines. Under the streaming three-class protocol of section 5.8, a majority-class predictor scores macro-F1 0.23 and a stratified-random predictor 0.33 (leave-one-subject-out, per step), so the reservoir's

0.562 seed-mean score is above both floors. The ten node-mask seeds quantify algorithmic sensitivity and are not treated as inferential replicates.

Drive-boosted tonic-node extension. The minutes-scale tonic gap of fig. 5.5 is probed by extending the heterogeneous bank with drive-boosted nodes at $\tau = 120$ and 300 s, a labelled extrapolation of the drive-amplitude variable, copying the lead cell's write nonlinearity and stretch and overriding only τ (`tonic_cards` in `scripts/ch5_model.py`). On memory capacity (DT = 5 s, k to 40) the extension raises the long-lag tail beyond 75 s only slightly ($MC_{k>15}$ from 0.82 to 0.97, positive in 9/10 seed pairs) and leaves total capacity unchanged (4.67 and 4.64). On the physiological-context reconstruction, with the target delays extended to 90 and 120 s so the tonic context is present, the slow-band held-out R^2 moves from 0.634 to 0.637. The matched-seed gain is $+0.003 \pm 0.022$ and positive in 5/10 draws; the overall R^2 changes by -0.001 (`tonic_extension` in `scripts/ch5_physio_context.py`). The stretched tails of the measured cells already carry much of the long-lag memory. Seeds quantify algorithmic sensitivity and are not used for a population-level test. The extension is not part of the measured composition banks anywhere else in the chapter.

D.7 Designed-Filter Analysis

The designed-filter results of section 5.3 are reproduced by `scripts/ch5_filterbank.py`, with regression tests in `tests/test_ch5_filterbank.py`. All filters are causal and start in steady state at the first sample.

Kernels. A device low-pass is the measured relaxation $\exp[-(t/\tau_c)^{\beta_c}]$ sampled at 4 Hz, normalised to unit DC gain and truncated where it falls

below 10^{-4}. Every kernel reproduces the cell-median half-life to within one sample. The *RC* reference is a single exponential with $\tau = t_{1/2}/\ln 2$. The digital references are causal second-order Butterworth band-pass filters at 0.05 Hz–1 Hz (phasic) and 0.045 Hz–0.25 Hz (sympathetic band [156]).

Synthetic sessions. Each 40 min session alternates 5 min calm and aroused blocks. SCR onsets follow a Poisson process at nominal rates of 2 and 8 per minute, with a 1 s refractory gap (realised rates about 2 and 7 per minute). Each SCR is a Bateman function [157] with rise constant drawn from 0.5 s–1.2 s, decay constant from 2 s–7 s, and a log-normal amplitude (median 0.2 µS); responses below 0.05 µS are not scored. The tonic level starts between 2 and 12 µS, drifts with a 300 s correlation time, and rises by 0.2 µS–1.0 µS in aroused blocks with a 60 s time constant. The clean condition adds white noise of 0.005 µS, the noisy condition 0.03 µS, and the artefact condition adds one rectangular artefact per minute (0.25 s–1.5 s, ±0.2 µS–1.5 µS).

Scoring. Candidate peaks at least 1 s apart are ranked by prominence. For each filter, the lag is the median lag of maximum correlation with the true phasic signal on 12 calibration sessions, and the prominence threshold maximises pooled F1 on the same sessions. Detections are then matched one-to-one to true SCR peaks within 2 s of the expected time on 30 separate test sessions. Detection on the noise-free phasic trace itself scores below the high-pass filters (F1 0.662 clean), because high-pass filtering separates overlapping responses; it is therefore not a ceiling. On WESAD, per-minute phasic power is the variance of the filter output within non-overlapping 60 s windows inside each protocol segment, and the AUC compares stress with baseline windows within each subject. Synthetic sessions are summarised by means, standard deviations, and sign counts; subjects are the inferential unit for WESAD.

Table D.6: Composed nonlinear node versus the measured-kernel filter (synthetic, clean condition, 30 test sessions). Drive levels of 0.05 and 1 give identical results for both read-outs.

Design	Model	SCR F1	r with true phasic
High-pass, PEO 1.2/0.045	measured kernel (linear)	0.713	0.56
	composed, level-matched gain	0.712	0.61
	composed, small-signal gain	0.748	0.53
	kernel $h^{1/\alpha}$, linear	0.746	0.33
Band-pass, PEO 1.2/0.045 − 0.3/0.045	composed, level-matched gain	0.636	0.00
	composed, small-signal gain	0.638	0.67

Composed nonlinear node. In the composed node of section 5.2, pulse progress relaxes as $h^{1/\alpha}$ and the enhancement is $x = p^{\alpha}$. Because a power law has no intrinsic scale, the drive amplitude cancels from the filter output unless the node saturates, which it does not at the drives tested (peak progress 0.06 for the high-pass and 0.59 for the band-pass at full drive). A read-out gain fixed on the calibration prefix must match the small-signal slope, which is $1/\alpha$ times the level-matched gain; with a level-matched gain, baseline subtraction is incomplete. Table D.6 lists the results. If the composed model holds, the small-signal kernel is $h^{1/\alpha}$, shorter than h when $\alpha < 1$; for the fastest cell ($\alpha = 0.33$) its time constant is about 1.3 s and its corner about 0.12 Hz, inside the sympathetic band. This prediction rests on the composition assumption and is tested by measuring the relaxation of a small write superposed on a written state.

D.8 Deployment Envelope

Table D.8 collects the operating reference quoted in section 5.10 (`scripts/ch5_deployment.py`). The trainable-parameter count is exact: the reser-

Table D.7: Single-device high-pass built from each cell minus the *RC* element with the same half-life (nominal devices, so only kernel shape differs). Synthetic: mean F1 difference and sessions with a positive difference (of 30). WESAD: mean AUC difference of per-minute phasic power, subjects with a positive difference (of 15), and paired Wilcoxon p.

Cell	$t_{1/2}$ (s)	β	ΔF1 clean	ΔF1 noisy	ΔF1 artefacts	ΔAUC chest	ΔAUC wrist
PEO 1.2/0.045	2.74	1.00	0.000 (0)	0.000 (0)	0.000 (0)	0.000 (0; –)	0.000 (0; –)
PEO 0.6/0.045	2.76	0.97	0.000 (6)	0.000 (8)	0.000 (11)	−0.001 (1; 0.26)	0.000 (2; 0.28)
PEO 0.6/0.09	3.50	0.61	−0.012 (7)	−0.015 (4)	−0.006 (10)	−0.009 (1; 0.01)	−0.004 (6; 1.00)
PEO 0.6/0.18	5.54	0.61	−0.009 (6)	−0.012 (7)	−0.002 (16)	−0.009 (2; 0.04)	−0.002 (5; 0.91)
PEO 1.2/0.09	5.98	0.91	0.000 (10)	0.000 (13)	+0.001 (14)	−0.001 (0; 0.11)	−0.002 (3; 0.16)
PEO 1.2/0.18	9.04	0.87	+0.003 (18)	−0.006 (8)	+0.004 (17)	−0.003 (2; 0.09)	+0.001 (5; 0.35)
PEO 0.3/0.045	15.83	0.64	−0.001 (13)	−0.003 (13)	−0.007 (10)	−0.005 (3; 0.19)	−0.001 (5; 0.72)
PEO 0.3/0.09	19.20	0.83	0.000 (13)	−0.001 (11)	0.000 (11)	−0.001 (4; 0.68)	+0.001 (5; 0.67)

voir dynamics are fixed physical devices, and the read-out is a single $(N{+}1) \times K$ ridge-regression weight matrix. The number of read-out operations, NK multiply–accumulates per step, is also exact for that abstraction; their energy depends on an unselected implementation and is not estimated.

The only available write-energy reference is the proof-of-concept *Super Yellow*/*Hybrane*/LiOTf junction of chapter 2. For each of its first 50 positive events, the archived voltage and current sample are multiplied by the measured energised dwell, $E_k \simeq |V_k I_k| \Delta t_k$. The energy rises with state from 0.084 µJ to 1.55 µJ; its median is 1.08 µJ. The N-fold products in table D.8 simply show the consequence of charging every node once per update at that reference value. They are not bounds: the simulated PEO cards come from 3 V–4 V protocols on another host, their write current has not been measured in the proposed bank, and read, addressing, sensor, conversion, and control energy are absent.

Table D.8: Operating-budget reference for an N-node device reservoir read out by a linear classifier of $K = 3$ classes. Parameter and operation counts are exact for the simulated architecture. Energy rows are N-fold transfers of one measured *Hybrane* junction's positive-write sequence and are not predictions for a PEO bank.

Quantity	$N = 24$	$N = 48$
Trainable read-out weights	75	147
Training method	one closed-form ridge solve; physical dynamics untrained	
Read-out operations per step	72 MACs	144 MACs
Reference core energy/update, median	25.9 µJ	51.8 µJ
Reference range across source train	2.0 µJ–37.2 µJ	4.0 µJ–74.4 µJ
Reference core power at 1 Hz	25.9 µW	51.8 µW
Full-system power	not measured	

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